

**Dubai Metro, UAE
Green Line – Al-Qusais Depot**

**Heat Exchanger Facility
(Building 27)**

**Structural Design Calculations
(DCP4 Submission)**

Dec 2007

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Rev B1	2

Table

1	Introduction.....	04
2	Building Form and Structural System	04
3	Drawings.....	04
4	3D- Frame Analysis and Design	06
5	Foundation Design	245
6	Slab Design	270
7	Base Plate Design	322
8	Connection Design	331
9	Cable Pit Design	346

1 Introduction

The Dubai Metro Project is under construction by the Dubai Rapid Link Consortium (DURL). The Civil Works are contracted to the Japan Turkey Metro Joint Venture (JTMJV), and the Rail Systems Works are contracted to Mitsubishi Heavy Industries (MHI). The rolling stock will be provided by Kinki Sharyo.

The Heat Exchanger Facility is located adjacent to the Siding & Cleaning Building.

This submission for Heat Exchanger Facility (Building 27) in Al-Qusais Depot is made by the Designers for the JTMJV, and as such concentrates on the scope of the Civil Works Contract, which includes the Architecture, Structure and M&E Design.

2 Building Form and Structural System

Heat Exchanger Facility Shed is 19m long, 17m wide and 8.41m high. This structure will provide pumping station area, equipment area & AHU.

For a structurally efficient solution together with fulfilling functional requirement, a 3x3 bay RCC frame spaced 6.5m, 6m & 6.5m respectively along grid A, B, C & D and spaced at 5.04m, 6.85m & 5.31m respectively along grid 1, 2, 3 & 4 is proposed. This RCC structure is covered with insulated metal panelled roof which is supported by steel frame spaced at 6.5m, 6m & 6.5m respectively along grid 1, 2, 3 & 4. This frame made up of I beam supported by 2 column between span. The building is to be clad with insulated standing seam aluminium sheeting, fixed to cold formed Z purlins which span on to structural frame.

A 225mm thick non-suspended ground level slab monolithic with RC Beams is proposed to resist various forces from equipments.

3 Drawings

To complement the purpose of this submission, the following Structural drawings are submitted along with this report.

	<u>Drawing Title</u>	<u>Drawing No.</u>
Structural Drawings		
1	Al-Qusais-Depot - General Notes And Typical Details Sheet 1 of 3	DM001-D-GDP-CDP-DD-DCC-551990
2	Al-Qusais-Depot - General Notes And Typical Details Sheet 2 of 3	DM001-D-GDP-CDP-DD-DCC-551991

	<u>Drawing Title</u>	<u>Drawing No.</u>
3	Al-Qusais-Depot - General Notes And Typical Details Sheet 3 of 3	DM001-D-GDP-CDP-DD-DCC-551992
4	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - Foundation Layout Plan	DM001-D-GBC-WHE-DD-DCC-551825
5	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - Ground Slab Layout Plan	DM001-D-GBC-WHE-DD-DCC-551826
6	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - RC Roof Level Layout Plan	DM001-D-GBC-WHE-DD-DCC-551827
7	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - Steel Roof Level Layout Plan	DM001-D-GBC-WHE-DD-DCC-551828
8	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - Section A-A	DM001-D-GBC-WHE-DD-DCC-551829
9	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - Section B-B	DM001-D-GBC-WHE-DD-DCC-551830
10	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - Elevations	DM001-D-GBC-WHE-DD-DCC-551832
11	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - Footing Details	DM001-D-GBC-WHE-DD-DCC-551834
12	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - Column Schedule	DM001-D-GBC-WHE-DD-DCC-551835
13	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - Ground Slab Level Reinforcement Details	DM001-D-GBC-WHE-DD-DCC-551837
14	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - RC Roof Slab Reinforcement Detail	DM001-D-GBC-WHE-DD-DCC-551838
15	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - Pit Reinforcement Details	DM001-D-GBC-WHE-DD-DCC-551840
16	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - Ground Slab Level Beam Schedule	DM001-D-GBC-WHE-DD-DCC-551842
17	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - RC Roof Level Beam Schedule	DM001-D-GBC-WHE-DD-DCC-551845
18	Al-Qusais Depot, Building - 27 - Heat Exchanger Facility - Roof Setting Out - Plan & Section	DM001-D-GBC-WHE-DD-DCC-551850

4 3D- Frame Analysis and Design

Roof loading :

Dead load

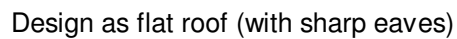
W_{DL} :

Sheeting	0.1 kPa				
Purlins	0.1 kPa				
Insulation	0.3 kPa				
M&E Finishes	0.3 kPa				
	<u>0.8 kPa</u>	x	1	=	0.8 kN/m

Live load

		0.6 kPa	x	1	=	0.6 kN/m
$P1_{LL}$:	A/C Equipment S/W					
	= 5 kN					(point load at mid span)

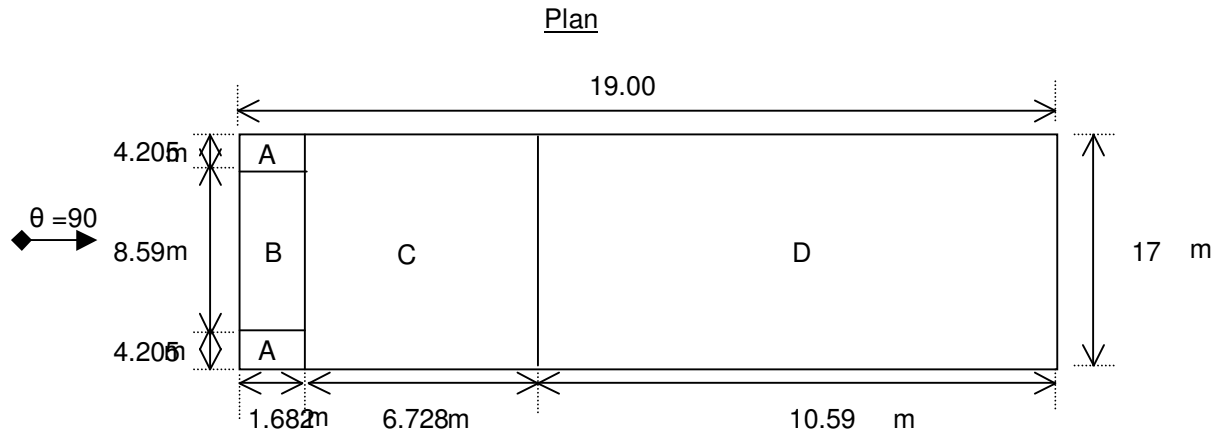
Plan



B	=	19.00	m
H	=	8.41	m
b	=	smaller of (B,2H)	
	=	2 x 8.41	
	=	16.82	m

Zone	Windward		Leeward		
	Area (m ²)	C _{pe}	Area	C _{pe1}	C _{pe2}
A	14.1	-2			
B	17.8	-1.4			
C	127.8	-0.7	0.0	-0.7	-0.07
D	1.7	-0.2	161.5	-0.2	0.2
Total	161.5		161.5		
Average		-0.78		-0.20	0.20

Average C_{pe} for roof (Wind direction normal to gable end)



Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 17 \text{ m} \\
 H &= 8.41 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 8.41 \\
 &= 16.82 \text{ m}
 \end{aligned}$$

For max^m uplift, assume zone C coefficient=>

$$C_{pe} = -0.7$$

For max^m downward force, assume zone D coefficient=>

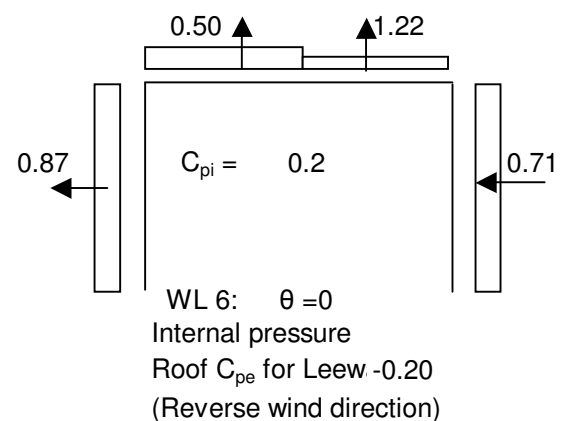
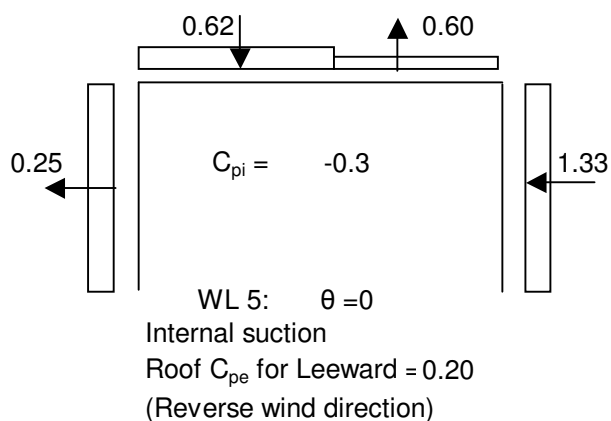
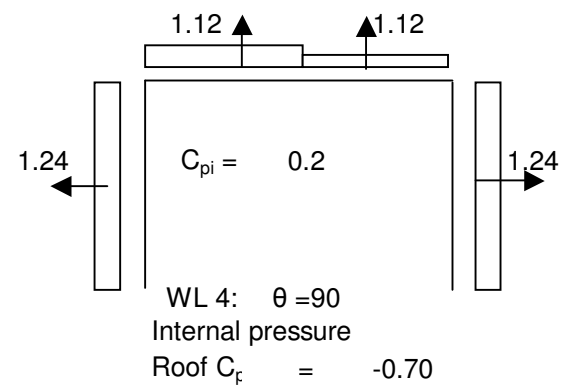
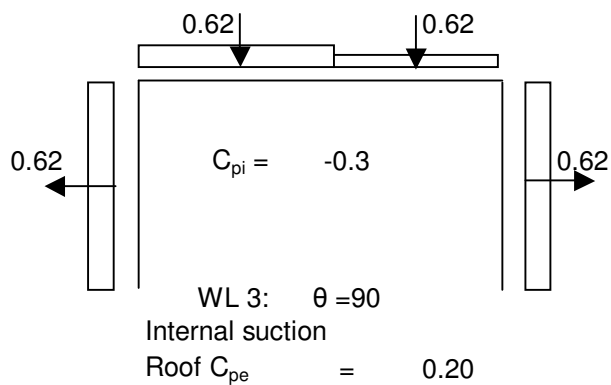
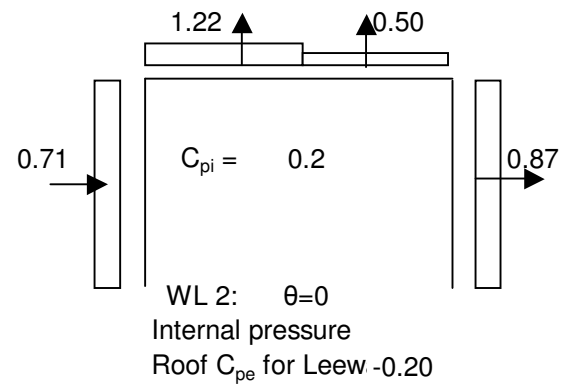
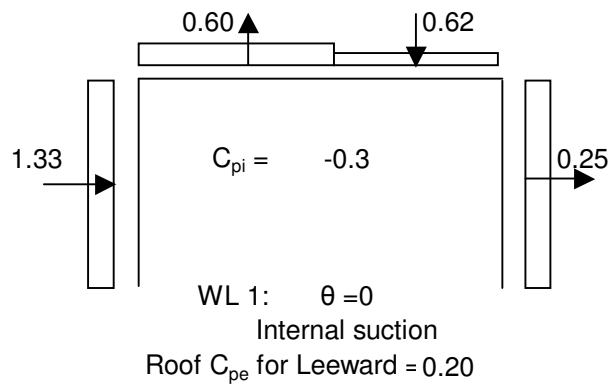
$$C_{pe} = 0.2$$

Wind Load Computation

$$\begin{aligned}
 D &= 17.00 \text{ m} \\
 H &= 8.41 \text{ m} \\
 D/H &= 2.02
 \end{aligned}$$

Element	Wind Speed V_e m/s	Dynamic Wind Pr. q kN/m ²	External Pressure Coefficient, C_{pe}			
			$\theta = 0$ (normal to eave)		$\theta = 90$ (normal to gable end)	
			Windward	Leeward		
Roof	45	1.24	-0.78	-0.2	0.2	-0.7 (Zone C) 0.2 (Zone D)
Wall	45	1.24	0.77	-0.5	-0.5	-0.8 (Zone B) -0.8 (Zone B)

Wind Load = $1 \times q \times (C_{pe} - C_{pi})$ kN/m (For unit width)



Parameters taken In STAAD For EQ Calculations

Design Information:

1) Seismic Zone	=	2A	(from Design Brief)
2) Occupancy Category	=	1	(Tanle-16-k, Conservatively adopt 1)
3) Lateral Force procedure	=	Static Procedure	(Section 1629.8 seismic zone 2A, structures not more than 5 stories)
4)Importance Factor	I	= 1	(Table 16-K)
5)Numeric. Coef: Strength & Ductility	R	= 4.5	(Table 16-N for OMRF - steel only)
	R	= 3.5	(Table 16-N for OMRF - Concrete only)
6)Soil Type		= S _D	(from Design Brief)
7) C_t (as per clause 1630.2.2)		= 0.0853	(for steel moment resisting frame)
		= 0.0731	(for RC moment resisting frame)
8) Near Source Factor	N_a	= 1	(From Table 16-S)
9)Near Source Factor	N_v	= 1	(From Table 16-T)

In STAAD :

Z	=	0.15	(For Zone 2A)
I	=	1	
RWX =RWZ	=	4.5	(for steel structure)
	=	3.5	(for concrete structure)
STYP	=	4	(Type of Soil S _D)
C_t	=	0.0853	(for steel moment resisting frame)
	=	0.0731	(for RC moment resisting frame)
P_x	=	0.381	(for steel structure)
	=	0.48	(for concrete structure)
P_z	=	0.381	(for steel structure)
	=	0.48	(for concrete structure)
N_a	=	1	
N_v	=	1	

3 hour Fire rating considerations as per BS 8110-1:1997 (Tb 8.4 Pg24, Fig 3.2 Pg 25)

		Code Requirement	Provided	Units
Concrete Beams →	Width	240	300/400	m
	Cover	40	40	m
Concrete Columns →	Width	400	500/550	m
	Cover	25	50	m
Concrete Slabs →	Thickness	150	175/225	m
	Cover	35	35	m

Load Cases For R.C. Structure

Basic Load Cases

Number	Name
21	EQX (EARTHQUAKE LOAD)
22	EQZ (EARTHQUAKE LOAD)
1	SELFWEIGHT
2	DL
3	LL
4	PH(X-DIR)
5	PH(Z-DIR)
11	WL1 (0 DEG ,INTERNAL SUCTION -0.3)
12	WL2 (0 DEG ,INTERNAL PRESSURE +0.2)
13	WL3 (90 DEG ,INTERNAL SUCTION -0.3, WW&LW
14	WL4 (90 DEG ,INTERNAL PRESSURE +0.2, WW&LW -
15	WL5 (0 DEG ,INTERNAL SUCTION -0.3, LEE WARD -
16	WL6 (0 DEG ,INTERNAL PRESSURE +0.2, LEE WARD -

Combination Load Cases

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
101	1.0DL+1.0LL	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	1.00
102	1.0DL+0.5LL	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.50
103	0.8DL+0.8LL+0.8WL1	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		11	WL1 (0 DEG ,INTERNAL SUCTION -0.3)	0.80
104	1.0DL+1.0WL1	1	SELFWEIGHT	1.00
		2	DL	1.00
		11	WL1 (0 DEG ,INTERNAL SUCTION -0.3)	1.00
105	0.8DL+0.8LL+0.8WL2	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		12	WL2 (0 DEG ,INTERNAL PRESSURE	0.80
106	1.0DL+1.0WL2	1	SELFWEIGHT	1.00
		2	DL	1.00
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.00
107	0.8DL+0.8LL+0.8WL3	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		13	WL3 (90 DEG ,INTERNAL SUCTION -0.3,	0.80
108	1.0DL+1.0WL3	1	SELFWEIGHT	1.00
		2	DL	1.00
		13	WL3 (90 DEG ,INTERNAL SUCTION -0.3,	1.00
109	0.8DL+0.8LL+0.8WL4	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		14	WL4 (90 DEG ,INTERNAL PRESSURE	0.80
110	1.0DL+1.0WL4	1	SELFWEIGHT	1.00
		2	DL	1.00

Document No.: DM001-D-GAC-WLR-DR-DCC-544959

Rev B1

Date: 31 Dec 2007

		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.00
111	0.8DL+0.8LL+0.8WL5	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		15	WL5 (0 DEG ,INTERNAL SUCTION -0.3,	0.80
112	1.0DL+1.0WL4	1	SELFWEIGHT	1.00
		2	DL	1.00
		15	WL5 (0 DEG ,INTERNAL SUCTION -0.3,	1.00
113	0.8DL+0.8LL+0.8WL6	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		16	WL6 (0 DEG ,INTERNAL PRESSURE	0.80
114	1.0DL+1.0WL6	1	SELFWEIGHT	1.00
		2	DL	1.00
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.00
115	DL+0.72EQX+0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
116	DL+0.72EQX-0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22
117	DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
118	DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22
119	DL+0.72EQZ+0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
120	DL+0.72EQZ-0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
121	DL-0.72EQZ+0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
122	DL-0.72EQZ-0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
123	0.9DL+0.72EQX+0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
124	0.9DL+0.72EQX-0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22
125	0.9DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
126	0.9DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22

127	0.9DL+0.72EQZ+0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
128	0.9DL+0.72EQZ-0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
129	0.9DL-0.72EQZ+0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
130	0.9DL-0.72EQZ-0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
131	1.0DL+0.75LL+0.54EQX+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	0.54
		22	EQZ (EARTHQUAKE LOAD)	0.17
132	1.0DL+0.75LL+0.54EQX-0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	0.54
		22	EQZ (EARTHQUAKE LOAD)	-0.17
133	1.0DL+0.75LL-0.54EQX+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	-0.54
		22	EQZ (EARTHQUAKE LOAD)	0.17
134	1.0DL+0.75LL-0.54EQX-0.17 EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	-0.54
		22	EQZ (EARTHQUAKE LOAD)	-0.17
135	1.0DL+0.75LL+0.54EQZ+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	0.54
		21	EQX (EARTHQUAKE LOAD)	0.17
136	1.0DL+0.75LL+0.54EQZ-0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	0.54
		21	EQX (EARTHQUAKE LOAD)	-0.17
137	1.0DL+0.75LL-0.54EQZ+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	-0.54
		21	EQX (EARTHQUAKE LOAD)	0.17
138	1.0DL+0.75LL-0.54EQZ-0.17 EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	-0.54
		21	EQX (EARTHQUAKE LOAD)	-0.17
200	1.4DL+1.6LL	1	SELFWEIGHT	1.40
		2	DL	1.40
		3	LL	1.60
201	1.2DL+1.2LL+1.2WL1	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		11	WL1 (0 DEG ,INTERNAL SUCTION -0.3)	1.20

202	1.4DL+1.4WL1	1	SELFWEIGHT	1.40
		2	DL	1.40
		11	WL1 (0 DEG ,INTERNAL SUCTION -0.3)	1.40
203	1DL+1.4WL1	1	SELFWEIGHT	1.00
		2	DL	1.00
		11	WL1 (0 DEG ,INTERNAL SUCTION -0.3)	1.40
204	1.2DL+1.2LL+1.2WL2	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.20
205	1.4DL+1.4WL2	1	SELFWEIGHT	1.40
		2	DL	1.40
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.40
206	1DL+1.4WL2	1	SELFWEIGHT	1.00
		2	DL	1.00
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.40
207	1.2DL+1.2LL+1.2WL3	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		13	WL3 (90 DEG ,INTERNAL SUCTION -0.3,	1.20
208	1.4DL+1.4WL3	1	SELFWEIGHT	1.40
		2	DL	1.40
		13	WL3 (90 DEG ,INTERNAL SUCTION -0.3,	1.40
209	1DL+1.4WL3	1	SELFWEIGHT	1.00
		2	DL	1.00
		13	WL3 (90 DEG ,INTERNAL SUCTION -0.3,	1.40
210	1.2DL+1.2LL+1.2WL4	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.20
211	1.4DL+1.4WL4	1	SELFWEIGHT	1.40
		2	DL	1.40
		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.40
212	1DL+1.4WL4	1	SELFWEIGHT	1.00
		2	DL	1.00
		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.40
213	1.2DL+1.2LL+1.2WL5	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		15	WL5 (0 DEG ,INTERNAL SUCTION -0.3,	1.20
214	1.4DL+1.4WL5	1	SELFWEIGHT	1.40
		2	DL	1.40
		15	WL5 (0 DEG ,INTERNAL SUCTION -0.3,	1.40
215	1DL+1.4WL5	1	SELFWEIGHT	1.00
		2	DL	1.00
		15	WL5 (0 DEG ,INTERNAL SUCTION -0.3,	1.40
216	1.2DL+1.2LL+1.2WL6	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.20
217	1.4DL+1.4WL6	1	SELFWEIGHT	1.40
		2	DL	1.40
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.40
218	1DL+1.4WL6	1	SELFWEIGHT	1.00
		2	DL	1.00
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.40
219	1.32DL+1.1LL+1.1EQX+0.33 EQZ	1	SELFWEIGHT	1.32
		2	DL	1.32
		3	LL	1.10
		21	EQX (EARTHQUAKE LOAD)	1.10
		22	EQZ (EARTHQUAKE LOAD)	0.33
220	1.32DL+1.1LL+1.1EQX-0.33 EQZ	1	SELFWEIGHT	1.32
		2	DL	1.32

		3	LL	1.10
		21	EQX (EARTHQUAKE LOAD)	1.10
		22	EQZ (EARTHQUAKE LOAD)	-0.33
221	1.32DL+1.1LL-1.1EQX+0.33 EQZ	1	SELFWEIGHT	1.32
		2	DL	1.32
		3	LL	1.10
		21	EQX (EARTHQUAKE LOAD)	-1.10
		22	EQZ (EARTHQUAKE LOAD)	0.33
222	1.32DL+1.1LL-1.1EQX-0.33 EQZ	1	SELFWEIGHT	1.32
		2	DL	1.32
		3	LL	1.10
		21	EQX (EARTHQUAKE LOAD)	-1.10
		22	EQZ (EARTHQUAKE LOAD)	-0.33
223	1.32DL+1.1LL+1.1EQZ+0.33 EQX	1	SELFWEIGHT	1.32
		2	DL	1.32
		3	LL	1.10
		22	EQZ (EARTHQUAKE LOAD)	1.10
		21	EQX (EARTHQUAKE LOAD)	0.33
224	1.32DL+1.1LL+1.1EQZ-0.33 EQX	1	SELFWEIGHT	1.32
		2	DL	1.32
		3	LL	1.10
		22	EQZ (EARTHQUAKE LOAD)	1.10
		21	EQX (EARTHQUAKE LOAD)	-0.33
225	1.32DL+1.1LL-1.1EQZ+0.33 EQX	1	SELFWEIGHT	1.32
		2	DL	1.32
		3	LL	1.10
		22	EQZ (EARTHQUAKE LOAD)	-1.10
		21	EQX (EARTHQUAKE LOAD)	0.33
226	1.32DL+1.1LL-1.1EQZ-0.33 EQX	1	SELFWEIGHT	1.32
		2	DL	1.32
		3	LL	1.10
		22	EQZ (EARTHQUAKE LOAD)	-1.10
		21	EQX (EARTHQUAKE LOAD)	-0.33
227	0.99DL+1.1EQX+0.33EQZ	1	SELFWEIGHT	0.99
		2	DL	0.99
		21	EQX (EARTHQUAKE LOAD)	1.10
		22	EQZ (EARTHQUAKE LOAD)	0.33
228	0.99DL+1.1EQX-0.33EQZ	1	SELFWEIGHT	0.99
		2	DL	0.99
		21	EQX (EARTHQUAKE LOAD)	1.10
		22	EQZ (EARTHQUAKE LOAD)	-0.33
229	0.99DL-1.1EQX+0.33EQZ	1	SELFWEIGHT	0.99
		2	DL	0.99
		21	EQX (EARTHQUAKE LOAD)	-1.10
		22	EQZ (EARTHQUAKE LOAD)	0.33
230	0.99DL-1.1EQX-0.33EQZ	1	SELFWEIGHT	0.99
		2	DL	0.99
		21	EQX (EARTHQUAKE LOAD)	-1.10
		22	EQZ (EARTHQUAKE LOAD)	-0.33
231	0.99DL+1.1EQZ+0.33EQX	1	SELFWEIGHT	0.99
		2	DL	0.99
		22	EQZ (EARTHQUAKE LOAD)	1.10
		21	EQX (EARTHQUAKE LOAD)	0.33
232	0.99DL+1.1EQZ-0.33EQX	1	SELFWEIGHT	0.99
		2	DL	0.99
		22	EQZ (EARTHQUAKE LOAD)	1.10
		21	EQX (EARTHQUAKE LOAD)	-0.33
233	0.99DL-1.1EQZ+0.33EQX	1	SELFWEIGHT	0.99
		2	DL	0.99
		22	EQZ (EARTHQUAKE LOAD)	-1.10
		21	EQX (EARTHQUAKE LOAD)	0.33
234	0.99DL-1.1EQZ-0.33EQX	1	SELFWEIGHT	0.99
		2	DL	0.99

		22	EQZ (EARTHQUAKE LOAD)	-1.10
		21	EQX (EARTHQUAKE LOAD)	-0.33
235	1.0DL+1.0LL+1.0PHX	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	1.00
		4	PH(X-DIR)	1.00
236	1.0DL+1.0LL+1.0PHZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	1.00
		5	PH(Z-DIR)	1.00
301	SELF+DL	1	SELFWEIGHT	1.00
		2	DL	1.00
302	LL	3	LL	1.00
303	0.8(LL+WL1)	3	LL	0.80
		11	WL1 (0 DEG ,INTERNAL SUCTION -0.3)	0.80
304	0.8(LL+WL2)	3	LL	0.80
		12	WL2 (0 DEG ,INTERNAL PRESSURE	0.80
305	0.8(LL+WL3)	3	LL	0.80
		13	WL3 (90 DEG ,INTERNAL SUCTION -0.3,	0.80
306	0.8(LL+WL4)	3	LL	0.80
		14	WL4 (90 DEG ,INTERNAL PRESSURE	0.80
307	0.8(LL+WL5)	3	LL	0.80
		15	WL5 (0 DEG ,INTERNAL SUCTION -0.3,	0.80
308	0.8(LL+WL6)	3	LL	0.80
		16	WL6 (0 DEG ,INTERNAL PRESSURE	0.80
309	0.8(LL+EQX/1.4+0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	0.57
		22	EQZ (EARTHQUAKE LOAD)	0.17
310	0.8(LL+EQX/1.4-0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	0.57
		22	EQZ (EARTHQUAKE LOAD)	-0.17
311	0.8(LL-EQX/1.4+0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	-0.57
		22	EQZ (EARTHQUAKE LOAD)	0.17
312	0.8(LL-EQX/1.4-0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	-0.57
		22	EQZ (EARTHQUAKE LOAD)	-0.17
313	0.8(LL+EQZ/1.4+0.3EQX/1.4)	3	LL	0.80
		22	EQZ (EARTHQUAKE LOAD)	0.57
		21	EQX (EARTHQUAKE LOAD)	0.17
314	0.8(LL+EQZ/1.4-0.3EQX/1.4)	3	LL	0.80
		22	EQZ (EARTHQUAKE LOAD)	0.57
		21	EQX (EARTHQUAKE LOAD)	-0.17
315	0.8(LL-EQZ/1.4+0.3EQX/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	-0.57
		22	EQZ (EARTHQUAKE LOAD)	0.17
316	0.8(LL-EQZ/1.4-0.3EQX/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	-0.57
		22	EQZ (EARTHQUAKE LOAD)	-0.17

Load Cases For Steel Structure

Basic Load Cases

Number	Name
21	EQX (EARTHQUAKE LOAD)
22	EQZ (EARTHQUAKE LOAD)
1	SELFWEIGHT
2	DL
3	LL
4	PH(X-DIR)
5	PH(Z-DIR)
11	WL1 (0 DEG,INTERNAL SUCTION (CPI=-0.3)
12	WL2 (0 DEG ,INTERNAL PRESSURE CPI=+0.2)
13	WL3 (90 DEG ,INTERNAL SUCTION CPI=-0.3, WW&LW CPE=+0.2)
14	WL4 (90 DEG ,INTERNAL PRESSURE CPI=+0.2, WW&LW CPE=-0.7)
15	WL5 (0 DEG ,INTERNAL SUCTION CPI=-0.3, LEE WARD -0.7, REVERSE)
16	WL6 (0 DEG ,INTERNAL PRESSURE CPI=+0.2, LEE WARD -0.2, REVERSE)

Combination Load Cases

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
101	1.0DL+1.0LL	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	1.00
102	1.0DL+0.5LL	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.50
103	0.8DL+0.8LL+0.8WL1	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		11	WL1 (0 DEG,INTERNAL SUCTION (CPI=-	0.80
104	1.0DL+1.0WL1	1	SELFWEIGHT	1.00
		2	DL	1.00
		11	WL1 (0 DEG,INTERNAL SUCTION (CPI=-	1.00
105	0.8DL+0.8LL+0.8WL2	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		12	WL2 (0 DEG ,INTERNAL PRESSURE	0.80
106	1.0DL+1.0WL2	1	SELFWEIGHT	1.00
		2	DL	1.00
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.00
107	0.8DL+0.8LL+0.8WL3	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		13	WL3 (90 DEG ,INTERNAL SUCTION CPI=-	0.80
108	1.0DL+1.0WL3	1	SELFWEIGHT	1.00
		2	DL	1.00
		13	WL3 (90 DEG ,INTERNAL SUCTION CPI=-	1.00
109	0.8DL+0.8LL+0.8WL4	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		14	WL4 (90 DEG ,INTERNAL PRESSURE	0.80
110	1.0DL+1.0WL4	1	SELFWEIGHT	1.00
		2	DL	1.00

Document No.: DM001-D-GBC-WHE-DR-DCC-545034

Rev B1

Date: 31 Dec 2007

		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.00
111	0.8DL+0.8LL+0.8WL5	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		15	WL5 (0 DEG ,INTERNAL SUCTION CPI=-	0.80
112	1.0DL+1.0WL4	1	SELFWEIGHT	1.00
		2	DL	1.00
		15	WL5 (0 DEG ,INTERNAL SUCTION CPI=-	1.00
113	0.8DL+0.8LL+0.8WL6	1	SELFWEIGHT	0.80
		2	DL	0.80
		3	LL	0.80
		16	WL6 (0 DEG ,INTERNAL PRESSURE	0.80
114	1.0DL+1.0WL6	1	SELFWEIGHT	1.00
		2	DL	1.00
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.00
115	DL+0.72EQX+0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
116	DL+0.72EQX-0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22
117	DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
118	DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22
119	DL+0.72EQZ+0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
120	DL+0.72EQZ-0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
121	DL-0.72EQZ+0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
122	DL-0.72EQZ-0.22EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
123	0.9DL+0.72EQX+0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
124	0.9DL+0.72EQX-0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22
125	0.9DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	0.22
126	0.9DL-0.72EQX+0.22EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	-0.72
		22	EQZ (EARTHQUAKE LOAD)	-0.22

127	0.9DL+0.72EQZ+0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
128	0.9DL+0.72EQZ-0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
129	0.9DL-0.72EQZ+0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	0.22
130	0.9DL-0.72EQZ-0.22EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	-0.72
		21	EQX (EARTHQUAKE LOAD)	-0.22
131	1.0DL+0.75LL+0.54EQX+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	0.54
		22	EQZ (EARTHQUAKE LOAD)	0.17
132	1.0DL+0.75LL+0.54EQX-0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	0.54
		22	EQZ (EARTHQUAKE LOAD)	-0.17
133	1.0DL+0.75LL-0.54EQX+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	-0.54
		22	EQZ (EARTHQUAKE LOAD)	0.17
134	1.0DL+0.75LL-0.54EQX-0.17 EQZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		21	EQX (EARTHQUAKE LOAD)	-0.54
		22	EQZ (EARTHQUAKE LOAD)	-0.17
135	1.0DL+0.75LL+0.54EQZ+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	0.54
		21	EQX (EARTHQUAKE LOAD)	0.17
136	1.0DL+0.75LL+0.54EQZ-0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	0.54
		21	EQX (EARTHQUAKE LOAD)	-0.17
137	1.0DL+0.75LL-0.54EQZ+0.17	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	-0.54
		21	EQX (EARTHQUAKE LOAD)	0.17
138	1.0DL+0.75LL-0.54EQZ-0.17 EQX	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	0.75
		22	EQZ (EARTHQUAKE LOAD)	-0.54
		21	EQX (EARTHQUAKE LOAD)	-0.17
200	1.4DL+1.6LL	1	SELFWEIGHT	1.40
		2	DL	1.40
		3	LL	1.60
201	1.2DL+1.2LL+1.2WL1	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		11	WL1 (0 DEG,INTERNAL SUCTION (CPI=-	1.20

Document No.: DM001-D-GBC-WHE-DR-DCC-545034

Rev B1

Date: 31 Dec 2007

202	1.4DL+1.4WL1	1	SELFWEIGHT	1.40
		2	DL	1.40
		11	WL1 (0 DEG,INTERNAL SUCTION (CPI=-	1.40
203	1DL+1.4WL1	1	SELFWEIGHT	1.00
		2	DL	1.00
		11	WL1 (0 DEG,INTERNAL SUCTION (CPI=-	1.40
204	1.2DL+1.2LL+1.2WL2	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.20
205	1.4DL+1.4WL2	1	SELFWEIGHT	1.40
		2	DL	1.40
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.40
206	1DL+1.4WL2	1	SELFWEIGHT	1.00
		2	DL	1.00
		12	WL2 (0 DEG ,INTERNAL PRESSURE	1.40
207	1.2DL+1.2LL+1.2WL3	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		13	WL3 (90 DEG ,INTERNAL SUCTION CPI=-	1.20
208	1.4DL+1.4WL3	1	SELFWEIGHT	1.40
		2	DL	1.40
		13	WL3 (90 DEG ,INTERNAL SUCTION CPI=-	1.40
209	1DL+1.4WL3	1	SELFWEIGHT	1.00
		2	DL	1.00
		13	WL3 (90 DEG ,INTERNAL SUCTION CPI=-	1.40
210	1.2DL+1.2LL+1.2WL4	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.20
211	1.4DL+1.4WL4	1	SELFWEIGHT	1.40
		2	DL	1.40
		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.40
212	1DL+1.4WL4	1	SELFWEIGHT	1.00
		2	DL	1.00
		14	WL4 (90 DEG ,INTERNAL PRESSURE	1.40
213	1.2DL+1.2LL+1.2WL5	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		15	WL5 (0 DEG ,INTERNAL SUCTION CPI=-	1.20
214	1.4DL+1.4WL5	1	SELFWEIGHT	1.40
		2	DL	1.40
		15	WL5 (0 DEG ,INTERNAL SUCTION CPI=-	1.40
215	1DL+1.4WL5	1	SELFWEIGHT	1.00
		2	DL	1.00
		15	WL5 (0 DEG ,INTERNAL SUCTION CPI=-	1.40
216	1.2DL+1.2LL+1.2WL6	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	1.20
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.20
217	1.4DL+1.4WL6	1	SELFWEIGHT	1.40
		2	DL	1.40
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.40
218	1DL+1.4WL6	1	SELFWEIGHT	1.00
		2	DL	1.00
		16	WL6 (0 DEG ,INTERNAL PRESSURE	1.40
219	1.2DL+0.5LL+1.0EQX+0.3 EQZ	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	0.50
		21	EQX (EARTHQUAKE LOAD)	1.00
		22	EQZ (EARTHQUAKE LOAD)	0.30
220	1.2DL+0.5LL+1.0EQX-0.3 EQZ	1	SELFWEIGHT	1.20
		2	DL	1.20

		3	LL	0.50
		21	EQX (EARTHQUAKE LOAD)	1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.30
221	1.2DL+0.5LL-1.0EQX+0.3 EQZ	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	0.50
		21	EQX (EARTHQUAKE LOAD)	-1.00
		22	EQZ (EARTHQUAKE LOAD)	0.30
222	1.2DL+0.5LL-1.0EQX-0.3 EQZ	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	0.50
		21	EQX (EARTHQUAKE LOAD)	-1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.30
223	1.2DL+0.5LL+1.0EQZ+0.3 EQX	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	0.50
		22	EQZ (EARTHQUAKE LOAD)	1.00
		21	EQX (EARTHQUAKE LOAD)	0.30
224	1.2DL+0.5LL+1.0EQZ-0.3 EQX	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	0.50
		22	EQZ (EARTHQUAKE LOAD)	1.00
		21	EQX (EARTHQUAKE LOAD)	-0.30
225	1.2DL+0.5LL-1.0EQZ+0.3 EQX	1	SELFWEIGHT	1.20
		2	DL	1.20
		3	LL	0.50
		22	EQZ (EARTHQUAKE LOAD)	-1.00
		21	EQX (EARTHQUAKE LOAD)	0.30
226	1.2DL+0.5LL-1.0EQZ-0.3 EQX	1	SELFWEIGHT	1.00
		2	DL	1.20
		3	LL	0.50
		22	EQZ (EARTHQUAKE LOAD)	-1.00
		21	EQX (EARTHQUAKE LOAD)	-0.30
227	0.9DL+1.0EQX+0.3EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	1.00
		22	EQZ (EARTHQUAKE LOAD)	0.30
228	0.9DL+1.0EQX-0.3EQZ2	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.30
229	0.9DL-1.0EQX+0.3EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	-1.00
		22	EQZ (EARTHQUAKE LOAD)	0.30
230	0.9DL-1.0EQX+0.3EQZ	1	SELFWEIGHT	0.90
		2	DL	0.90
		21	EQX (EARTHQUAKE LOAD)	-1.00
		22	EQZ (EARTHQUAKE LOAD)	-0.30
231	0.9DL+1.0EQZ+0.3EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	1.00
		21	EQX (EARTHQUAKE LOAD)	0.30
232	0.9DL+1.0EQZ-0.3EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	1.00
		21	EQX (EARTHQUAKE LOAD)	-0.30
233	0.9DL-1.0EQZ+0.3EQX	1	SELFWEIGHT	0.90
		2	DL	0.90
		22	EQZ (EARTHQUAKE LOAD)	-1.00
		21	EQX (EARTHQUAKE LOAD)	0.30
234	0.9DL-1.0EQZ-0.3EQX	1	SELFWEIGHT	0.90
		2	DL	0.90

		22	EQZ (EARTHQUAKE LOAD)	-1.00
		21	EQX (EARTHQUAKE LOAD)	-0.30
235	1.0DL+1.0LL+1.0PHX	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	1.00
		4	PH(X-DIR)	1.00
236	1.0DL+1.0LL+1.0PHZ	1	SELFWEIGHT	1.00
		2	DL	1.00
		3	LL	1.00
		5	PH(Z-DIR)	1.00
301	SELF+DL	1	SELFWEIGHT	1.00
		2	DL	1.00
302	LL	3	LL	1.00
303	0.8(LL+WL1)	3	LL	0.80
		11	WL1 (0 DEG,INTERNAL SUCTION (CPI=-	0.80
304	0.8(LL+WL2)	3	LL	0.80
		12	WL2 (0 DEG ,INTERNAL PRESSURE	0.80
305	0.8(LL+WL3)	3	LL	0.80
		13	WL3 (90 DEG ,INTERNAL SUCTION CPI=-	0.80
306	0.8(LL+WL4)	3	LL	0.80
		14	WL4 (90 DEG ,INTERNAL PRESSURE	0.80
307	0.8(LL+WL5)	3	LL	0.80
		15	WL5 (0 DEG ,INTERNAL SUCTION CPI=-	0.80
308	0.8(LL+WL6)	3	LL	0.80
		16	WL6 (0 DEG ,INTERNAL PRESSURE	0.80
309	0.8(LL+EQX/1.4+0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	0.57
		22	EQZ (EARTHQUAKE LOAD)	0.17
310	0.8(LL+EQX/1.4-0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	0.57
		22	EQZ (EARTHQUAKE LOAD)	-0.17
311	0.8(LL-EQX/1.4+0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	-0.57
		22	EQZ (EARTHQUAKE LOAD)	0.17
312	0.8(LL-EQX/1.4-0.3EQZ/1.4)	3	LL	0.80
		21	EQX (EARTHQUAKE LOAD)	-0.57
		22	EQZ (EARTHQUAKE LOAD)	-0.17
313	0.8(LL+EQZ/1.4+0.3EQX/1.4)	3	LL	0.80
		22	EQZ (EARTHQUAKE LOAD)	0.57
		21	EQX (EARTHQUAKE LOAD)	0.17
314	0.8(LL+EQZ/1.4-0.3EQX/1.4)	3	LL	0.80
		22	EQZ (EARTHQUAKE LOAD)	0.57
		21	EQX (EARTHQUAKE LOAD)	-0.17
315	0.8(LL-EQZ/1.4+0.3EQX/1.4)	3	LL	0.80
		22	EQZ (EARTHQUAKE LOAD)	-0.57
		21	EQX (EARTHQUAKE LOAD)	0.17
316	0.8(LL-EQZ/1.4-0.3EQX/1.4)	3	LL	0.80
		22	EQZ (EARTHQUAKE LOAD)	-0.57
		21	EQX (EARTHQUAKE LOAD)	-0.17

DISPLACEMENTS SUMMARY

	Node	L/C	Horizontal X mm	Vertical Y mm	Horizontal Z mm	Resultant mm	Rotational		
							rX deg	rY deg	rZ deg
Max X	42	310 0.8(LL+EQX/1.4-0.3EQZ/1.4)	1.707	0.007	0.002	1.707	-0.001	0.056	-0.033
Min X	3	312 0.8(LL-EQX/1.4-0.3EQZ/1.4)	-1.707	0.007	0.002	1.707	-0.001	-0.056	0.033
Max Y	59	310 0.8(LL+EQX/1.4-0.3EQZ/1.4)	0.772	2.028	0.024	2.17	-0.004	-0.001	-0.034
Min Y	61	301 SELF+DL	0.015	-4.872	0.008	4.872	-0.002	0	-0.077
Max Z	82	303 0.8(LL+WL1)	-0.001	0.253	2.949	2.96	0.001	0.027	-0.003
Min Z	77	307 0.8(LL+WL5)	0.001	0.253	-2.95	2.96	-0.001	0.027	0.003
Max rX	26	303 0.8(LL+WL1)	0	0	0	0	0.096	0.027	-0.003
Min rX	27	307 0.8(LL+WL5)	0	0	0	0	-0.096	0.027	0.003
Max rY	83	301 SELF+DL	0.003	-0.081	-0.193	0.209	0.001	0.249	-0.03
Min rY	77	301 SELF+DL	0.003	-0.081	0.192	0.209	-0.001	-0.249	-0.03
Max rZ	110	312 0.8(LL-EQX/1.4-0.3EQZ/1.4)	0	0	0	0	-0.003	0	0.416
Min rZ	104	310 0.8(LL+EQX/1.4-0.3EQZ/1.4)	0	0	0	0	-0.003	0	-0.415
Max Rst	61	301 SELF+DL	0.015	-4.872	0.008	4.872	-0.002	0	-0.077


 MAXIMUM DEFLECTION

ALLOWABLE DEFLECTION (IN MM)

Deflection in Y	Span/180	3200/180	17.77
Deflection in X	HEIGHT/300	6700/300	22.33
Deflection in Z	HEIGHT/300	4800/300	16

Column Summary

Ref. node on STAAD	Location on drawing	Size of column				Reinforcement Req.		Reinforcement Prov.		
		Stump to Level - 1		Level - 1 to level - 2		Stump to Level - 1 (cm ²)	Level - 1 to level - 2 (cm ²)	Stump to Level - 1 (cm ²)	Level - 1 to level - 2 (cm ²)	
		Width (b) (mm)	Depth (h) (mm)	Width (b) (mm)	Depth (h) (mm)					
1	A-1	550	550	500	500	28	50	79	79	
2	A-2	550	550	500	500	36	99	128	128	
3	A-3	550	550	500	500	58	97	128	128	
4	A-4	550	550	500	500	45	42	79	79	
5	B-1	550	550	500	500	64	76	96	96	
6	B-2	550	550	-	-	90	-	96	-	
7	B-3	550	550	-	-	92	-	96	-	
8	B-4	550	550	500	500	30	77	96	96	
9	C-1	550	550	500	500	46	58	96	96	
10	C-2	550	550	500	500	30	91	128	128	
11	C-3	550	550	500	500	51	93	128	128	
12	C-4	550	550	500	500	65	57	96	96	
13	D-1	550	550	500	500	30	38	59	59	
14	D-2	550	550	500	500	24	51	59	59	
15	D-3	550	550	500	500	49	52	79	79	
16	D-4	550	550	500	500	63	38	79	79	

Note : 12T-25 = 59 cm²
16T-25 = 79 cm²
12T-32 = 96 cm²
16T-32 = 128 cm²

Document No.: DM001-D-GBC-WHE-DR-DCC-545034

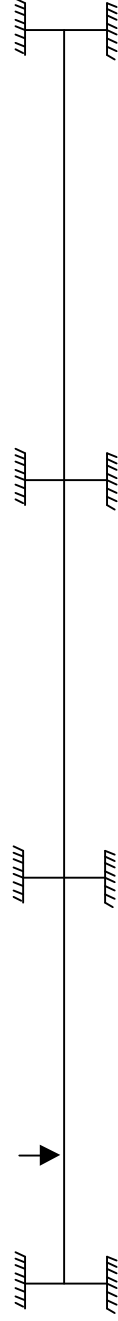
Rev B1

Date: 31 Dec. 2007

Bar Combination used

For 300mm x 600mm beams

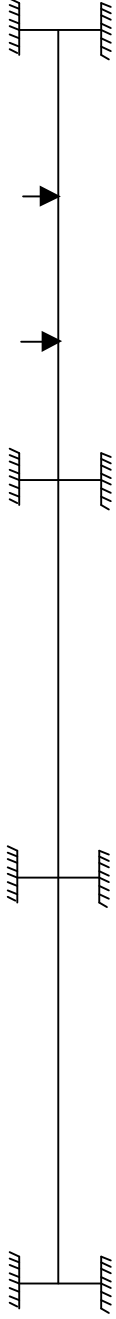
(No. + Bar dia)	A_{st} (cm ²)
3T20	10
3T20 + 2T16	13
3T25	15
3T20 + 2T20	16
3T20 + 3T20	19
3T25 + 3T20	24
3T32	24
3T25 + 3T25	30
3T32 + 3T25	39
3T32 + 3T32	48
3T32 + 3T25 + 3T25	54

[illegible]

Date: 31 Dec 2007

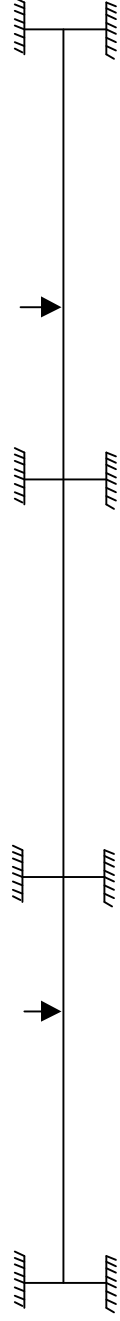


Ref Beam no in STAAD		10		11		12		
Beam marking on Drg.		1BC/1-2		1BC/2-3		1BC/3-4		
Reinforcement Req./Prov. (cm ²)	Top	34/39	3/39	35/48	41/48	44/54	52/54	38/39
	Bottom	14/15	10/15	12/15	9/15	13/15	20/24	23/24
Stirrups	Req.	T10-406	T10-541	T10-309	T10-138	T10-249	T10-131	T10-123
	Prov.	T10-150	T10-200	T10-150	T10-100	T10-200	T10-100	T10-100

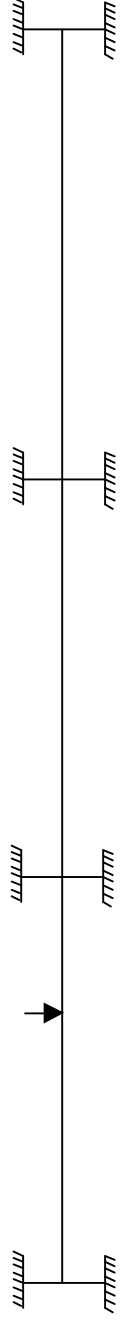


Ref Beam no in STAAD		14		15		16		17		18	
Beam marking on Drg.		1BD/1-2		1BD/2-3				1BD/3-4			
Reinforcement Req./Prov. (cm ²)	Top	27/30	0/30	28/39	35/39	36/48	43/48	0/48	0/48	41/48	41/48
	Bottom	9/10	9/10	6/10	6/10	8/10	11/24	12/24	12/24	20/24	20/24
Stirrups	Req.	T10-361	T10-270	T10-270	T10-181	T10-166	T10-139	T10-204	T10-204	T10-158	T10-158
	Prov.	T10-150	T10-200	T10-150	T10-150	T10-150	T10-100	T10-150	T10-150	T10-150	T10-150

Note : All beam sections are 300mm x 600mm

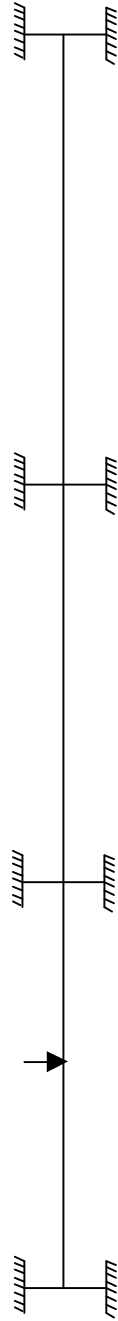


Ref Beam no in STAAD	19	20	21	22	23
Beam marking on Drg.	1B1/D-C		1B1/C-B		1B1/B-A
Reinforcement Req./Prov. (cm ²)	30/39 18/24	36/39 11/24	38/39 7/15	42/48 15/24	29/30 13/24
Stirrups	T10-293 2T10-100	T10-146 T10-100	T10-145 2T10-100	T10-85 2T10-100	T10-135 T10-100

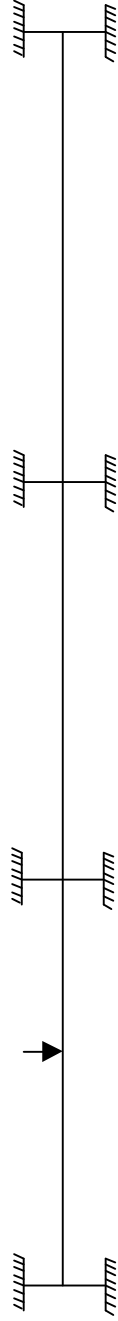


Ref Beam no in STAAD	26	27	28	29
Beam marking on Drg.	1B2/D-C		1B2/C-B	1B2/B-A
Reinforcement Req./Prov. (cm ²)	36/39 18/24	37/39 14/24	37/39 7/15	37/39 18/24
Stirrups	T10-176 T10-100	T10-189 T10-150	T10-107 T10-100	T10-105 T10-100

Note : All beam sections are 300mm x 600mm



Ref Beam no in STAAD	30	31	32	33
Beam marking on Drg.	1B3/D-C		1B3/C-B	1B3/B-A
Reinforcement Req./Prov. (cm ²)	Top 38/39	38/39	0/10	4/39
	Bottom 15/24	11/24	10/15	7/24
		13/24	8/15	20/24
Stirrups	Req. T10-126	T10-106	T10-554	T10-221
	Prov. T10-100	T10-150	T10-100	T10-100
			T10-277	T10-245
			T10-100	T10-100
				T10-276
				T10-100



Ref Beam no in STAAD	36	37	38	39
Beam marking on Drg.	1B4/D-C		1B4/C-B	1B4/B-A
Reinforcement Req./Prov. (cm ²)	Top 35/39	37/39	0/10	0/30
	Bottom 13/24	8/24	12/15	7/24
		10/24	4/15	14/24
Stirrups	Req. T10-87	T10-126	T10-88	T10-531
	Prov. 2T10-100	T10-100	2T10-100	T10-100
			T10-146	T10-100
			T10-100	T10-100
			T10-92	T10-147
			2T10-100	T10-100
				T10-221
				T10-100

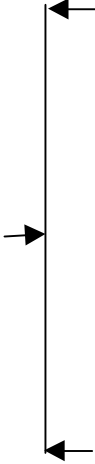
Note : All beam sections are 300mm x 600mm



Ref Beam no in STAAD	5			
Beam marking on Drg.	1B1/1-2a			
Reinforcement Req./Prov. (cm ²)	Top	0/10	0/10	0/10
	Bottom	0/10	3/10	0/10
Stirrups	Req.	T10-199	T10-199	T10-199
	Prov.	T10-150	T10-150	T10-150



Ref Beam no in STAAD	13			
Beam marking on Drg.	1BD2/1-2			
Reinforcement Req./Prov. (cm ²)	Top	0/10	0/10	0/10
	Bottom	0/10	5/10	0/10
Stirrups	Req.	T10-324	T10-324	T10-324
	Prov.	T10-200	T10-200	T10-200



Ref Beam no in STAAD	40	41	42
Beam marking on Drg.	1BD1/3-4		
Reinforcement Req./Prov. (cm ²)	Top	0/10	0/10
	Bottom	0/10	9/10
Stirrups	Req.	T10-370	T10-370
	Prov.	T10-200	T10-200

Ref Beam no in STAAD	24	25
Beam marking on Drg.	1B2a/B-A	
Reinforcement Req./Prov. (cm ²)	Top	0/10
	Bottom	0/10
Stirrups	Req.	T10-360
	Prov.	T10-200

A horizontal beam of length l is shown. It is supported by a pin support at the left end and a roller support at the right end. A uniformly distributed load of intensity q acts downwards along the entire length of the beam. Two point loads, each of magnitude P , act downwards on the beam at distances $\frac{l}{3}$ and $\frac{2l}{3}$ from the left end.

A horizontal beam of length l is shown. It is supported by a pin support at the left end and a roller support at the right end. A uniformly distributed load of intensity q acts downwards along the entire length of the beam. Two point loads, each of magnitude P , act downwards on the beam at distances $\frac{l}{3}$ and $\frac{2l}{3}$ from the left end.

mm	mm	mm	mm
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Ref Beam no in STAAD		107		108		109				
Beam marking on Drg.		RBC/1-2		RBC/2-3		RBC/3-4				
Reinforcement Req./Prov. (cm ²)	Top	18/24	0/24	19/24	15/24	0/10	15/24	18/24	0/19	17/19
	Bottom	11/15	8/15	6/15	5/10	4/10	5/10	5/10	7/10	9/10
Stirrups	Req.	T10-148	T10-148	T10-148	T10-399	T10-399	T10-399	T10-145	T10-145	T10-145
	Prov.	T10-100	T10-100	T10-100	T10-150	T10-200	T10-150	T10-100	T10-100	T10-100

mm	mm	mm	mm
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Ref Beam no in STAAD		110		111		112			
Beam marking on Drg.		RBD/1-2		RBD/2-3		RBD/3-4			
Reinforcement Req./Prov. (cm ²)	Top	13/19	0/19	12/19	11/19	0/10	10/19	11/19	11/19
	Bottom	8/15	4/15	6/15	5/10	3/10	5/10	3/10	7/10
Stirrups	Req.	T10-148	T10-148	T10-148	T10-388	T10-388	T10-388	T10-145	T10-145
	Prov.	T10-100	T10-100	T10-100	T10-150	T10-200	T10-150	T10-100	T10-100

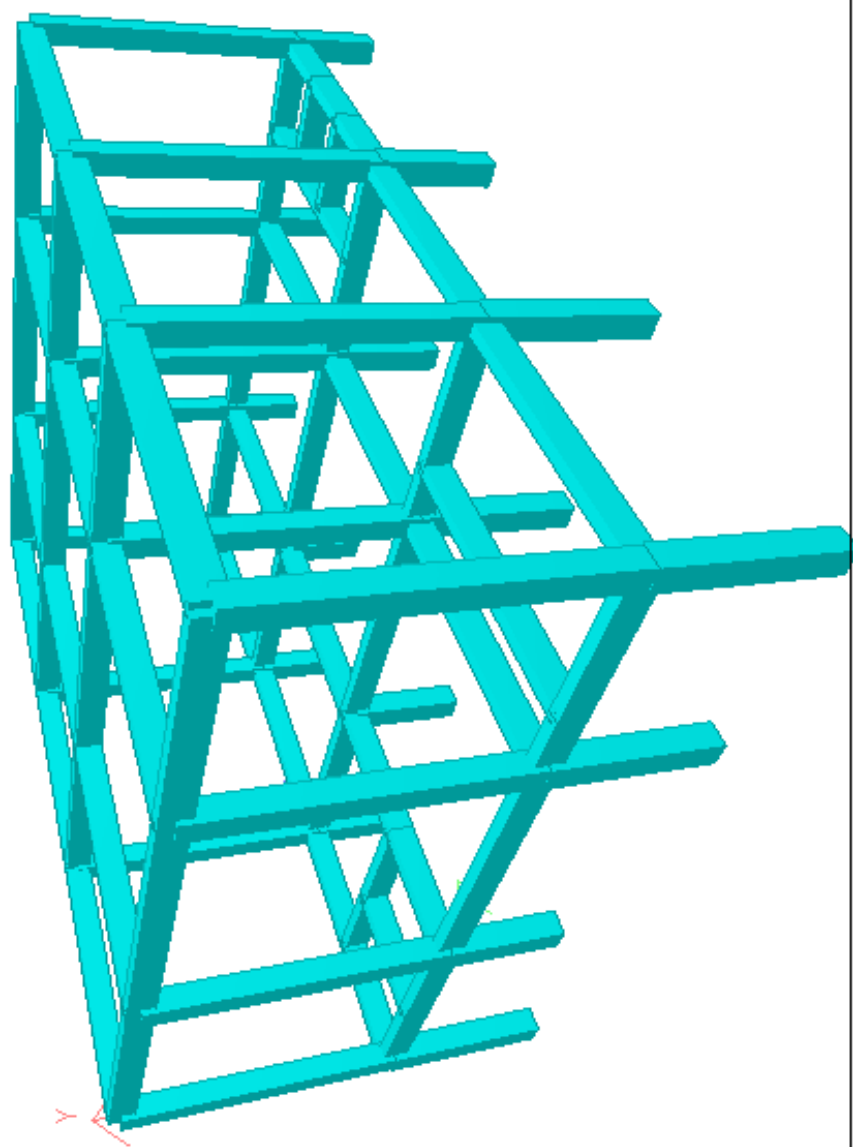
Ref Beam no in Staad		116		117	118
Beam marking on Drg.		RB2/D-C		RB2/C-A	RB2/C-A
Reinforcement Req./Prov. (cm ²)	Top	16/24	6/24	36/48	45/48
	Bottom	15/24	0/24	4/24	36/39
Stirrups	Req.	T10-359	T10-359	T10-193	T10-90
	Prov.	T10-150	T10-200	T10-150	2T10-100
				T10-111	T10-83
				T10-100	2T10-100



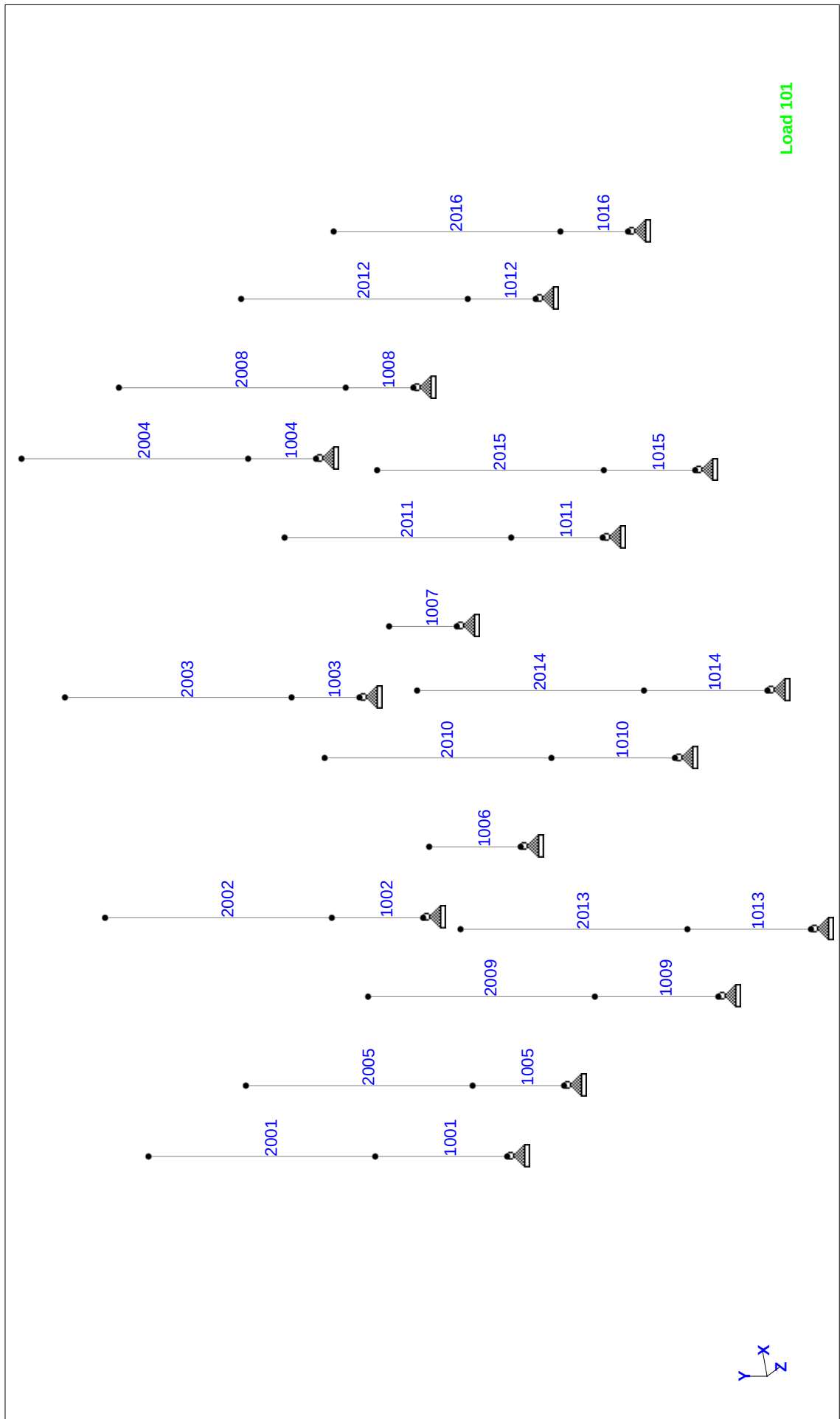
Ref Beam no in Staad		119		120		121	
Beam marking on Drg.		RB3/D-C		RB3/C-A		RB3/C-A	
Reinforcement Req./Prov. (cm ²)	Top	17/24	6/24	37/48	46/48	4/48	40/48
	Bottom	16/24	0/24	4/24	13/39	37/39	8/39
Stirrups	Req.	T10-359	T10-359	T10-193	T10-92	T10-115	T10-83
	Prov.	T10-150	T10-200	T10-150	2T10-100	T10-100	2T10-100



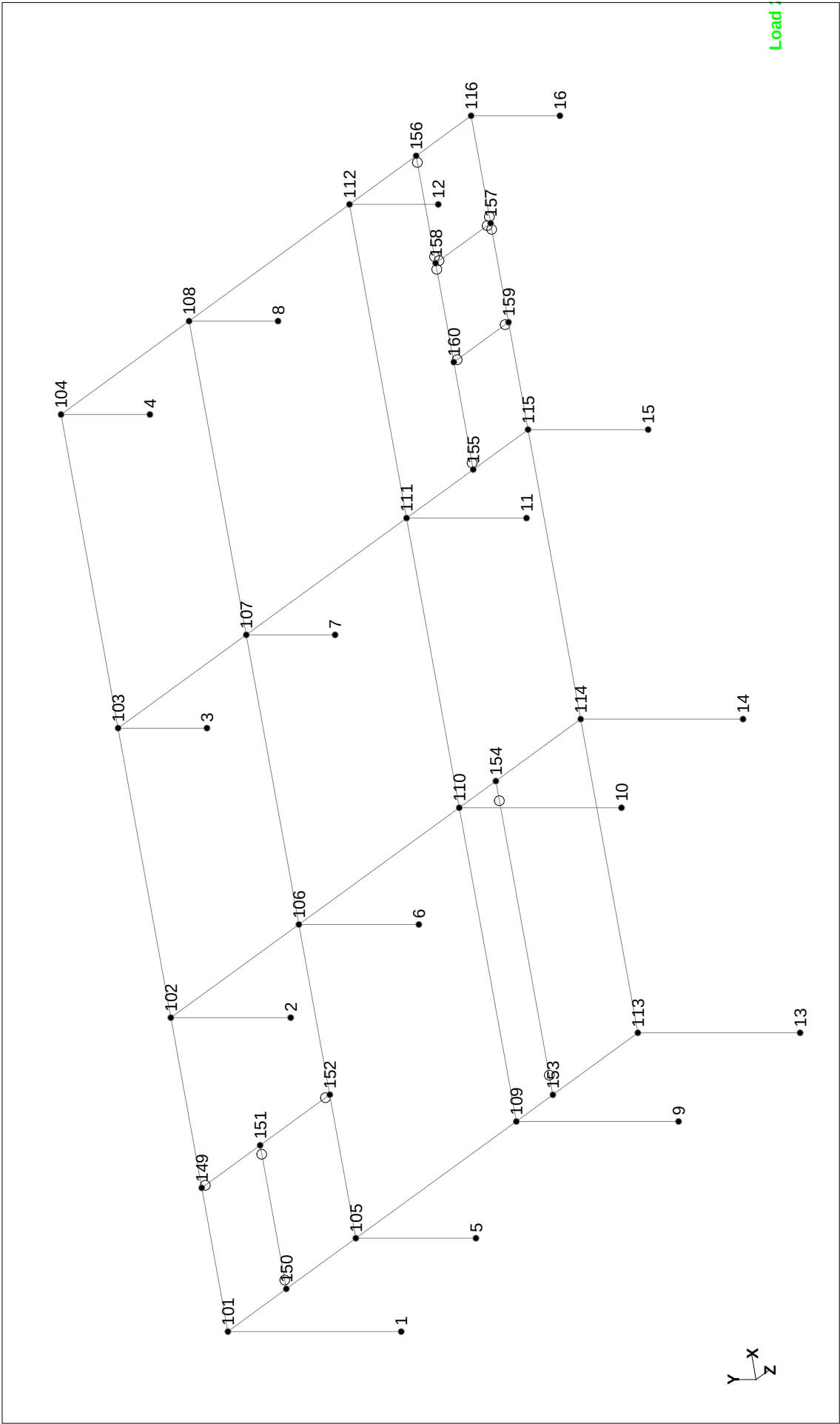
Ref Beam no in Staad		122		123		124	
Beam marking on Drg.		RB4/D-C		RB4/C-B		RB4/B-A	
Reinforcement Req./Prov. (cm ²)	Top	13/15	3/15	12/15	9/15	9/15	11/15
	Bottom	10/15	4/15	7/15	3/10	3/10	11/15
Stirrups	Req.	T10-359	T10-359	T10-359	T10-133	T10-133	T10-111
	Prov.	T10-100	T10-100	T10-100	T10-100	T10-100	T10-100



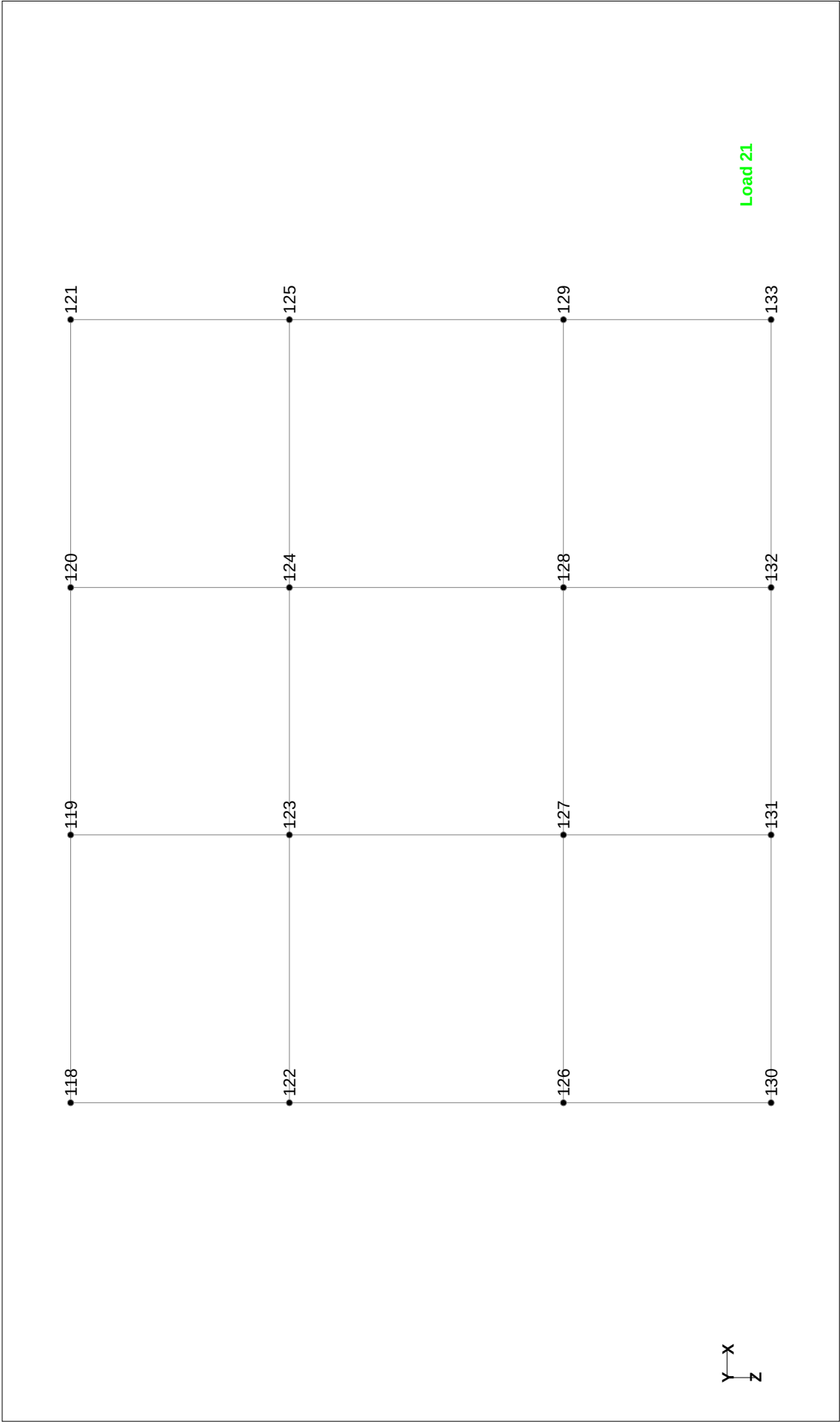
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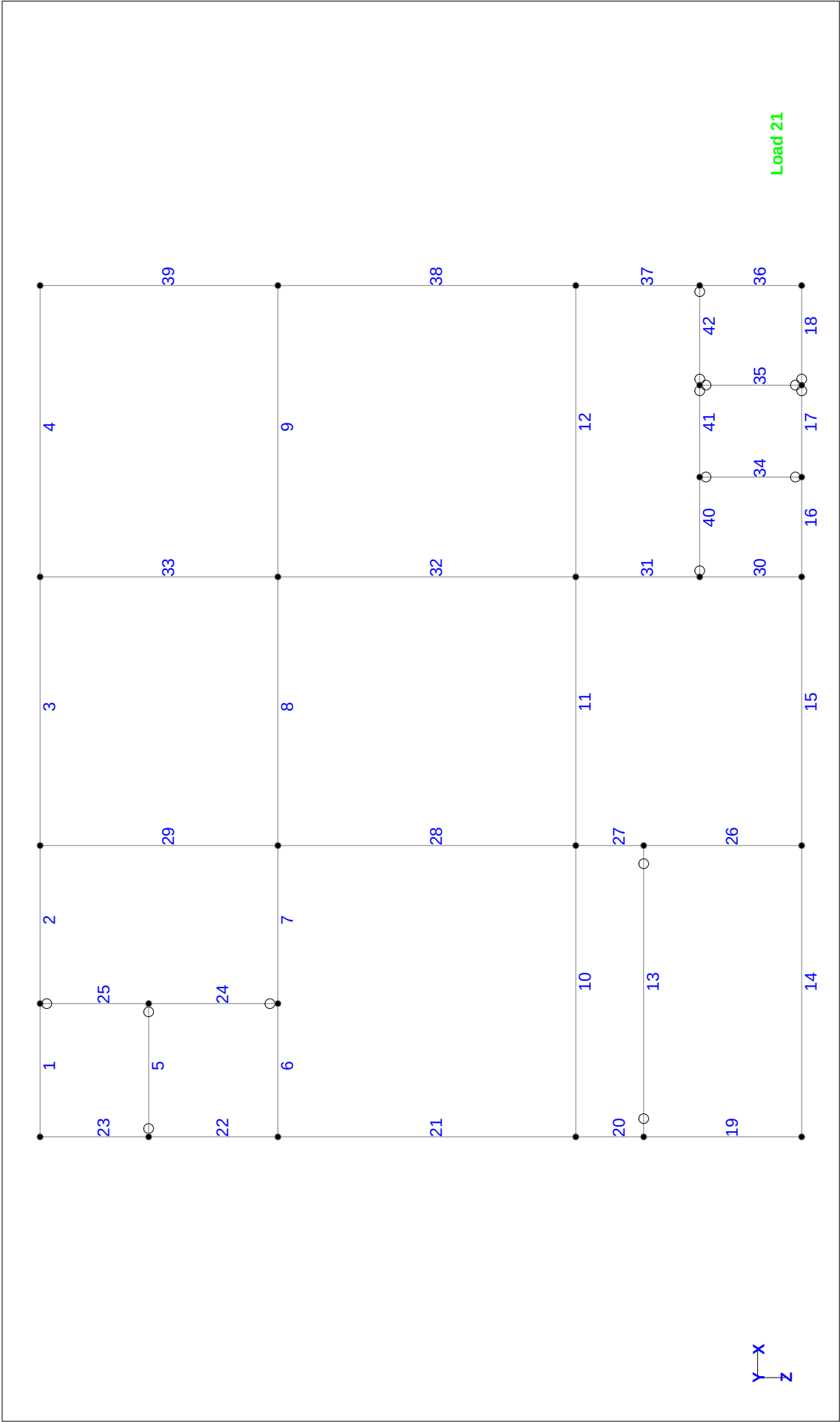


Whole Structure (Input data was modified after picture taken)

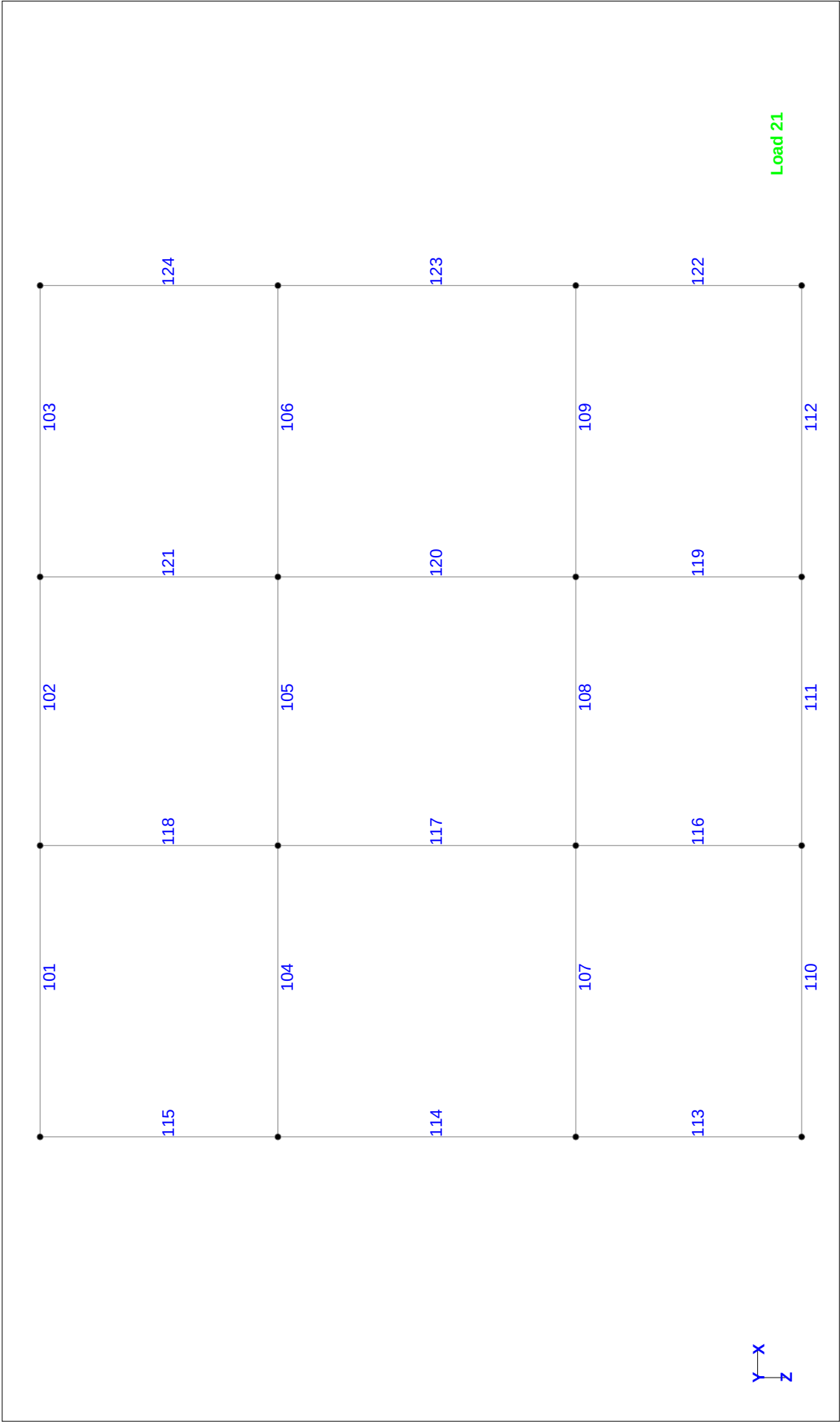


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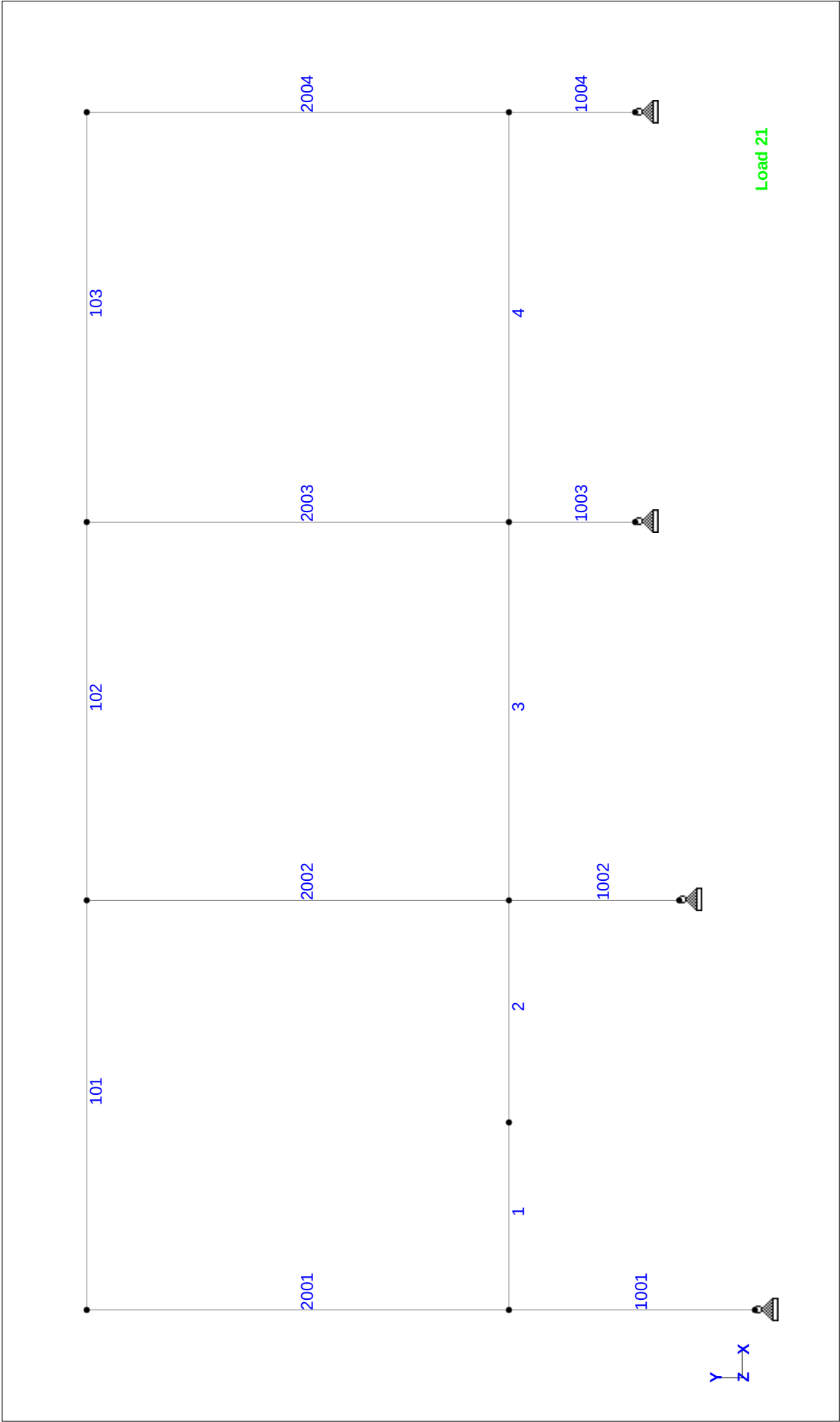


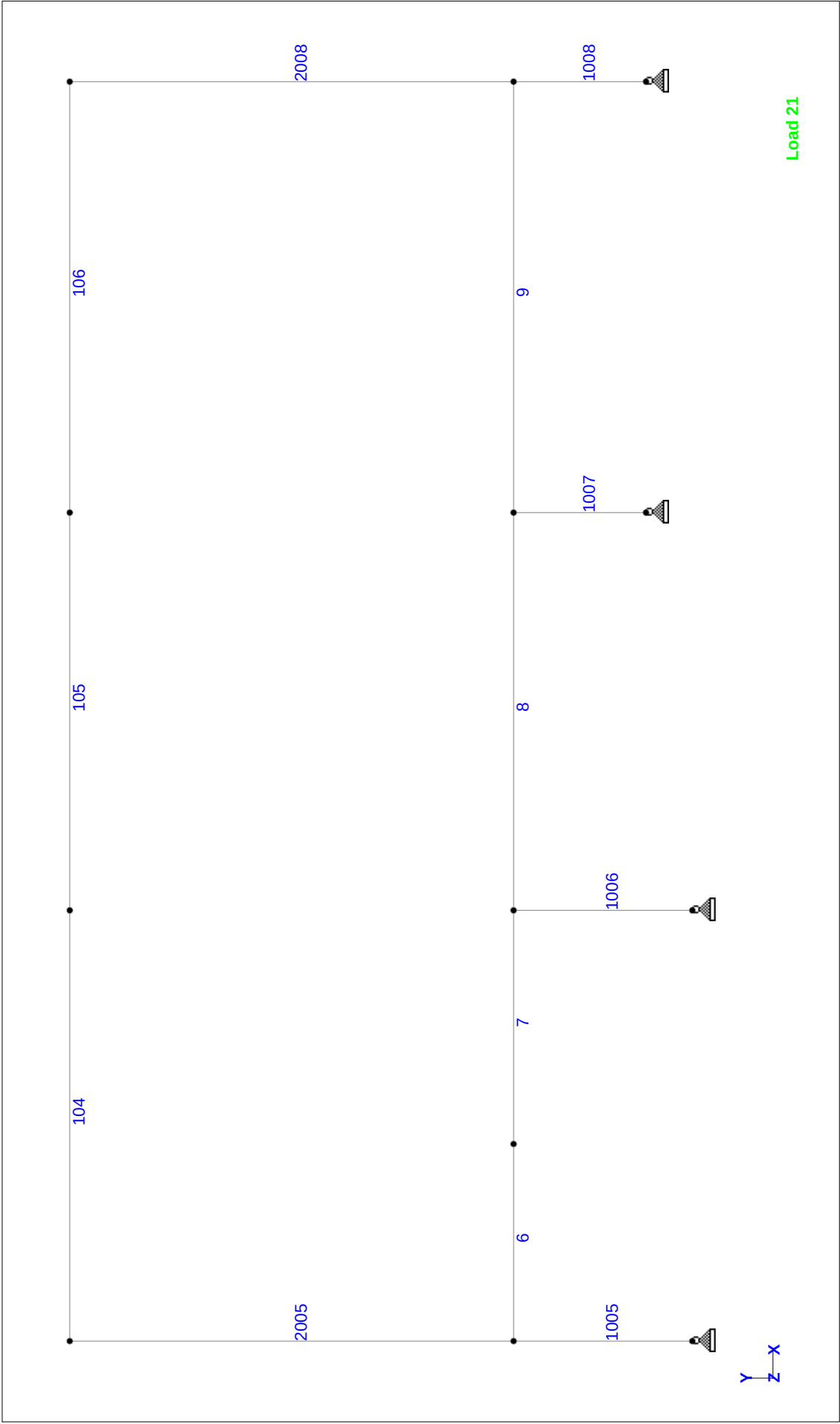


Load 21

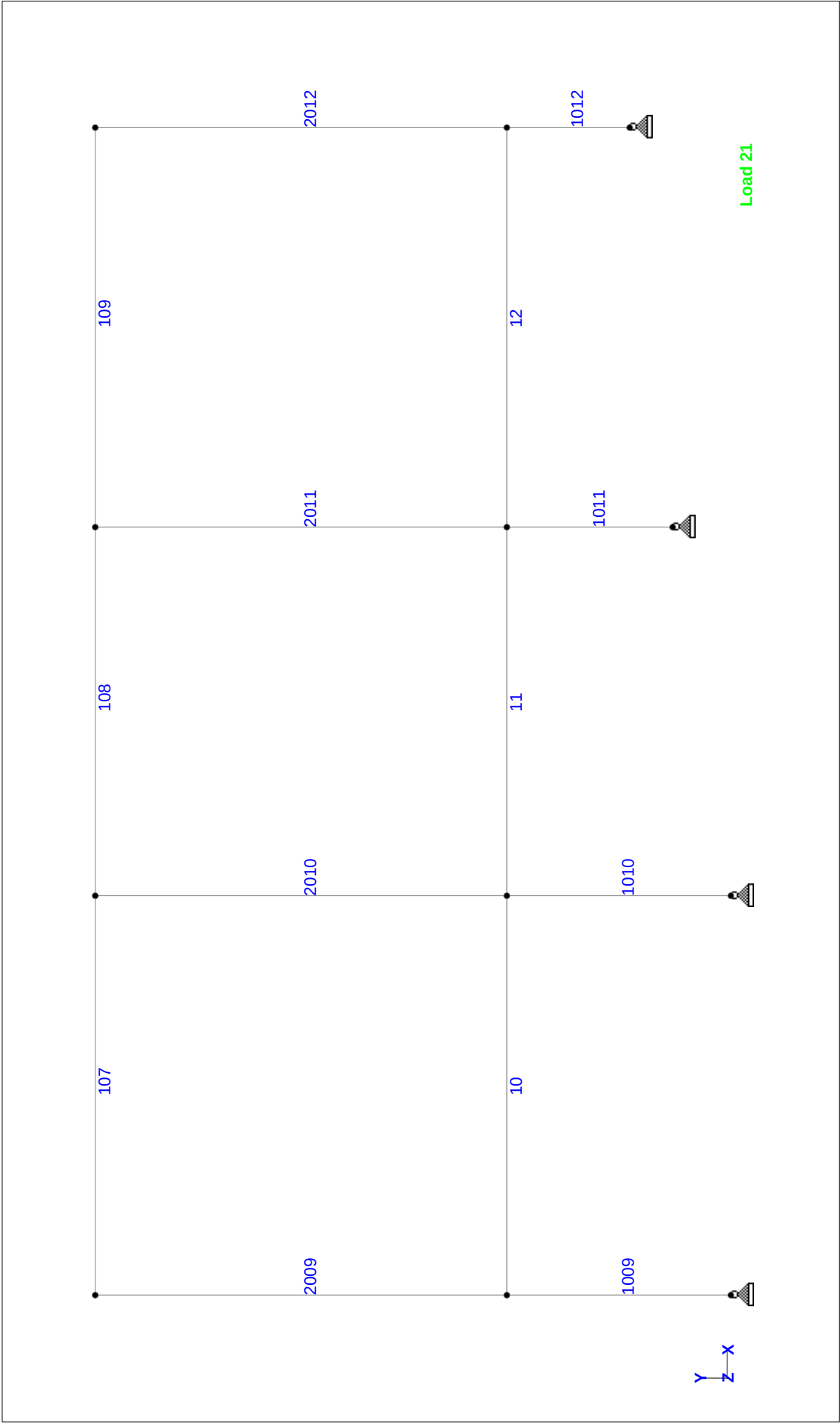


MEMBER NO (ROOF)

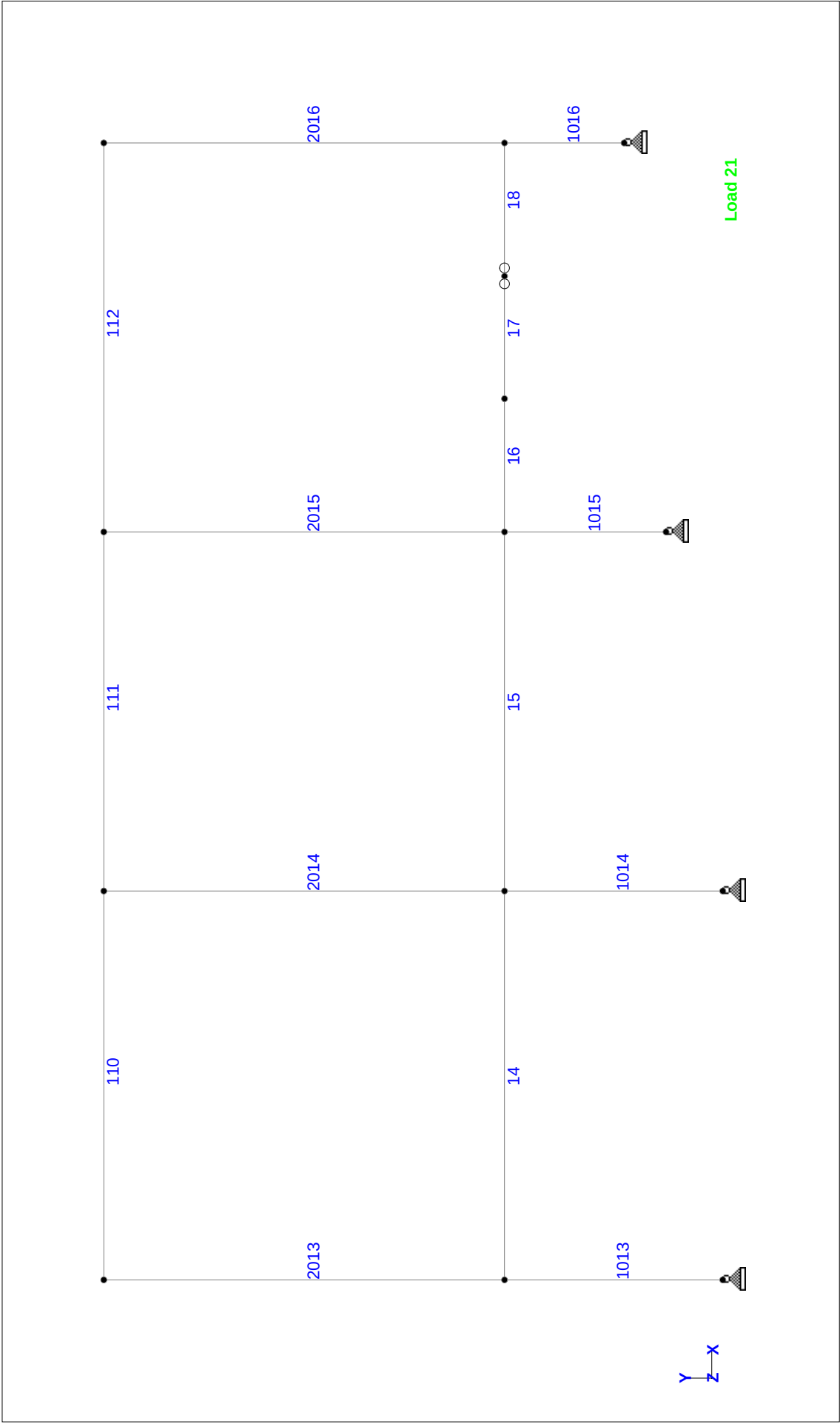




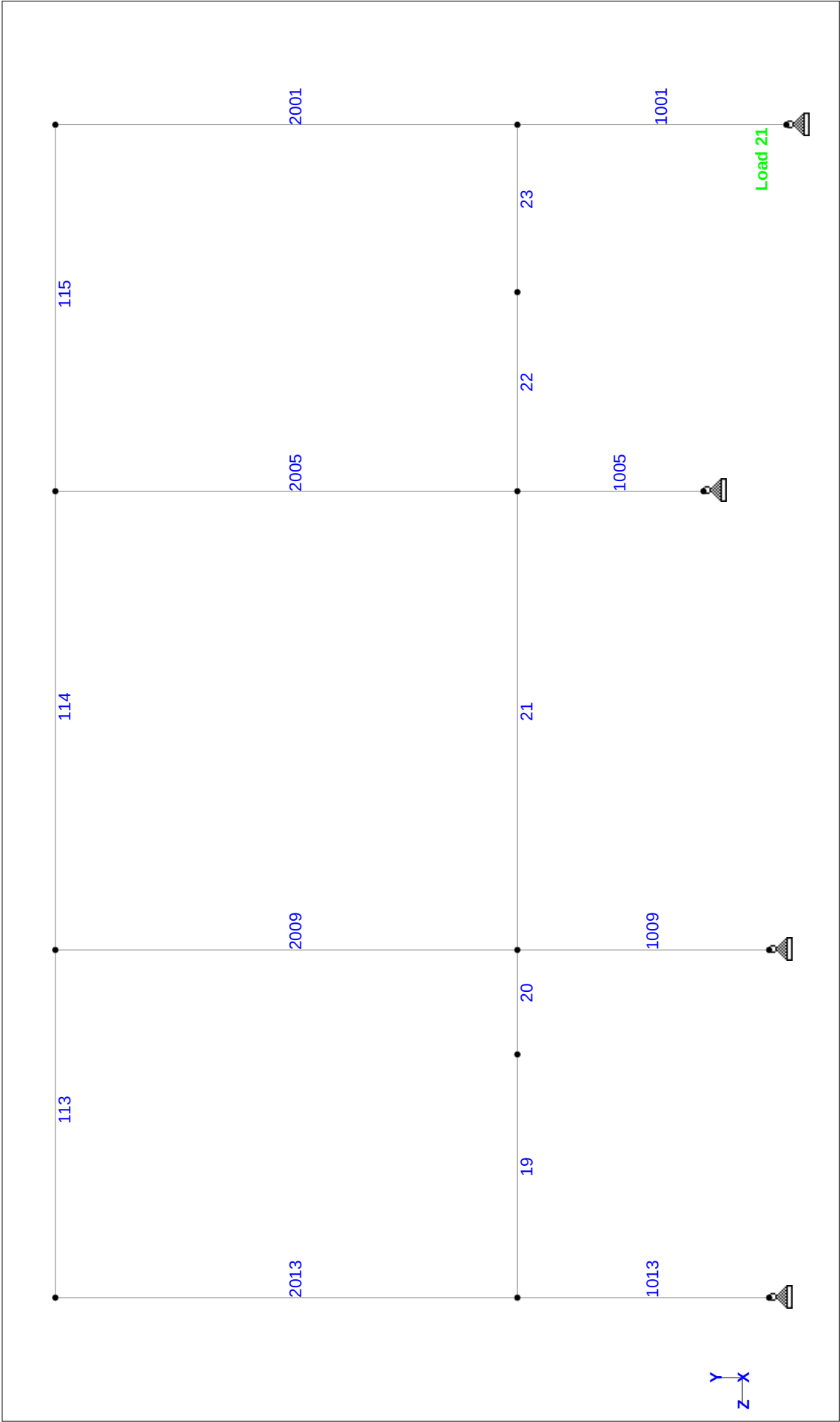
ELEVATION (GRID-27 B)



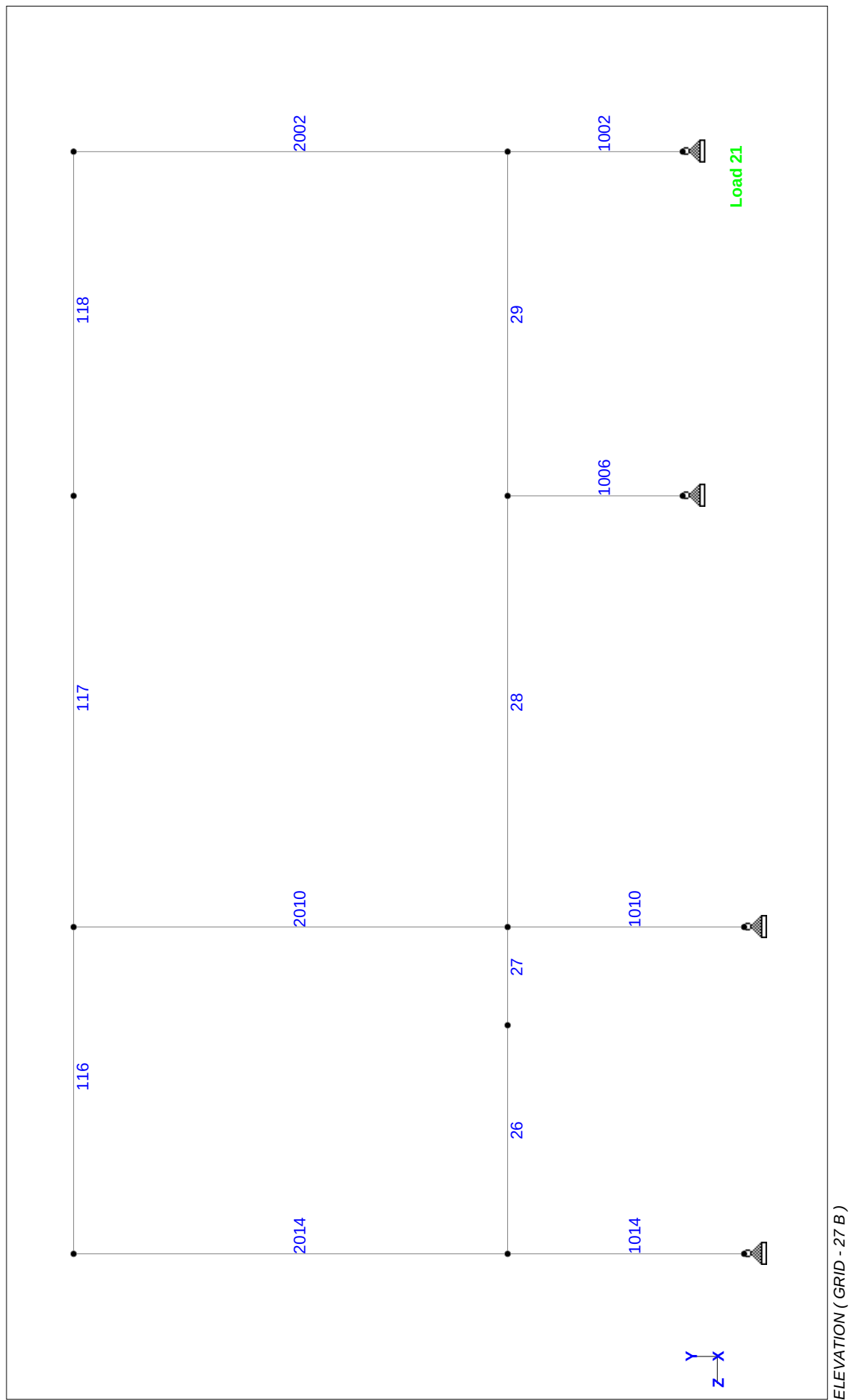
ELEVATION (GRID-27 C)

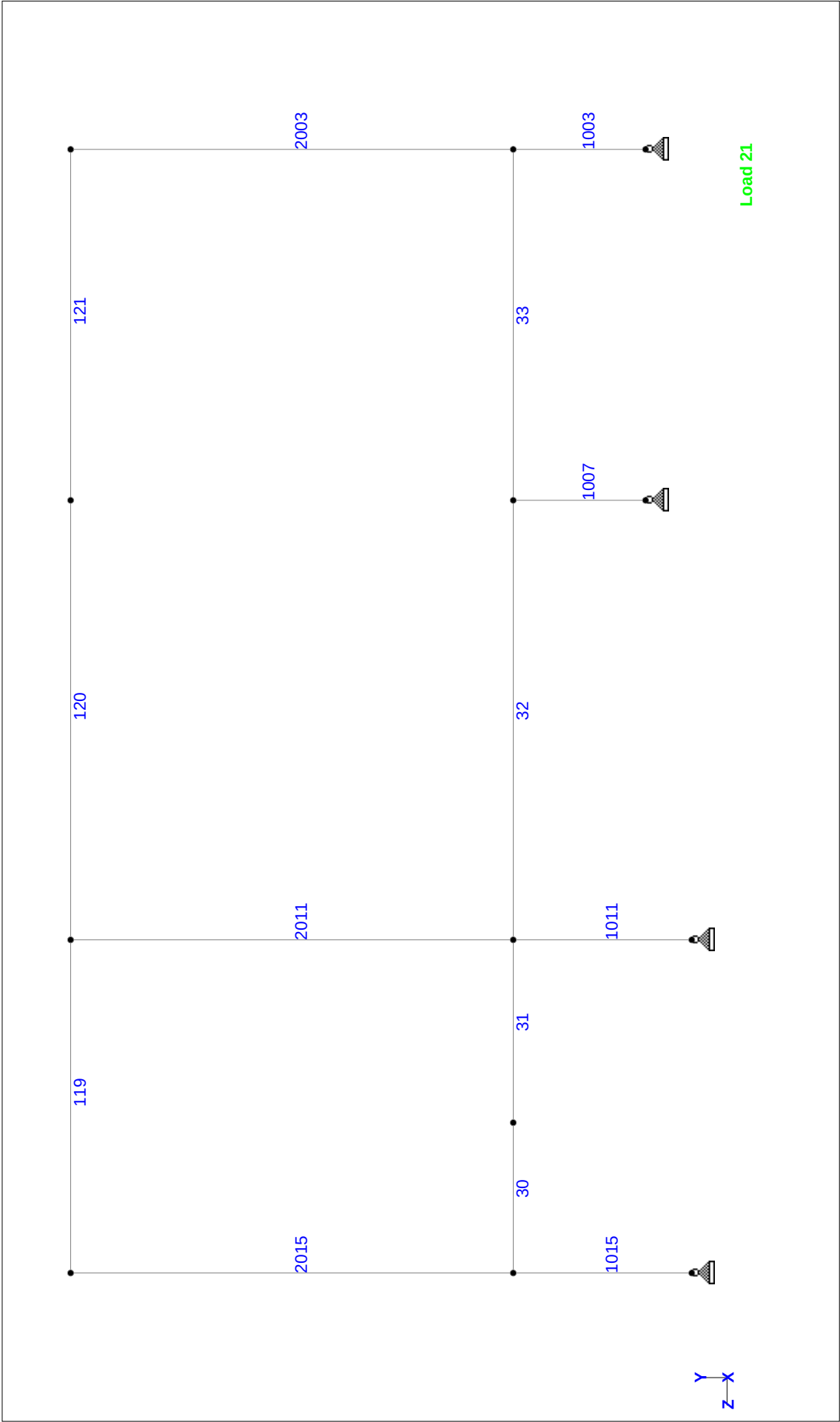


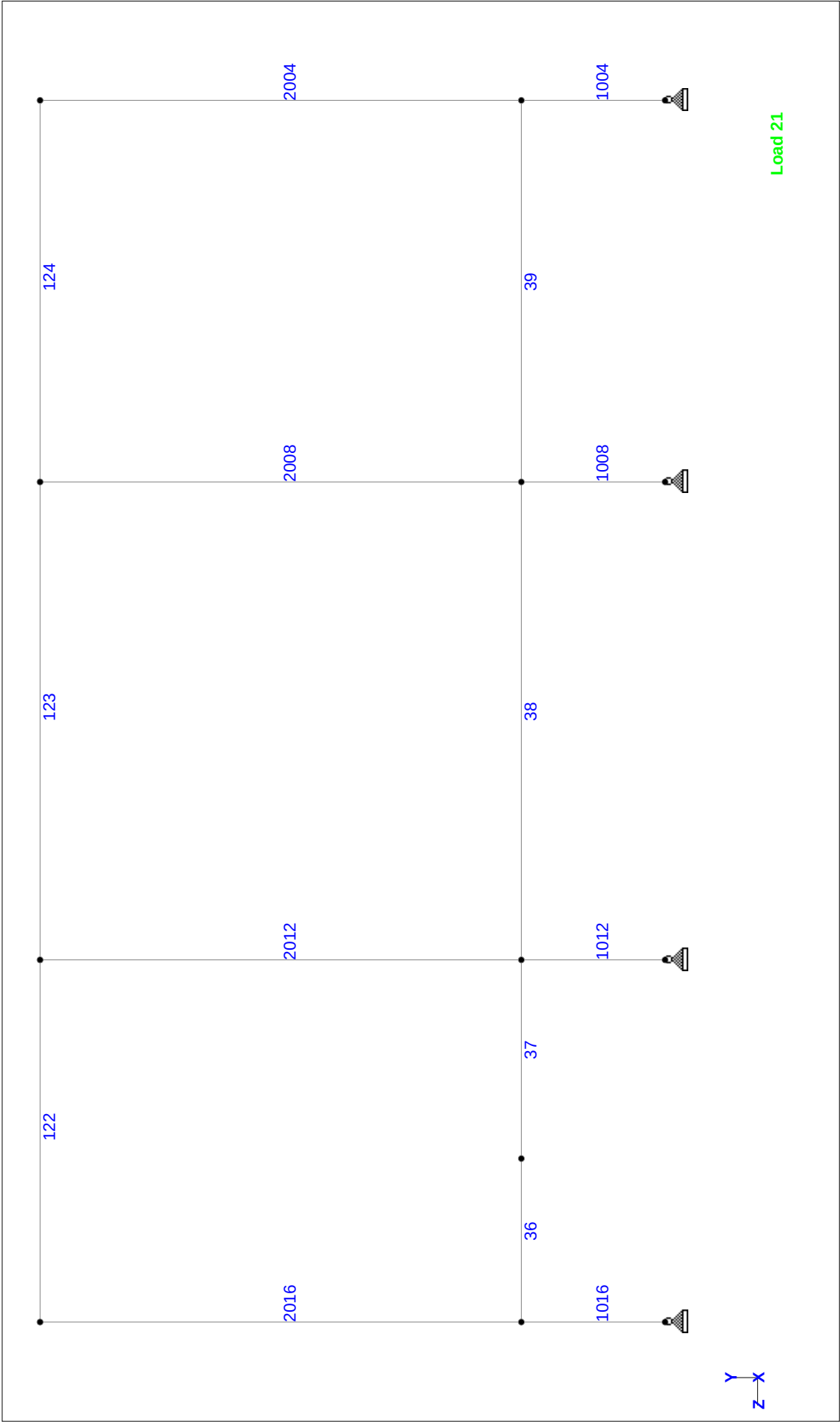
ELEVATION (GRID -27 D)

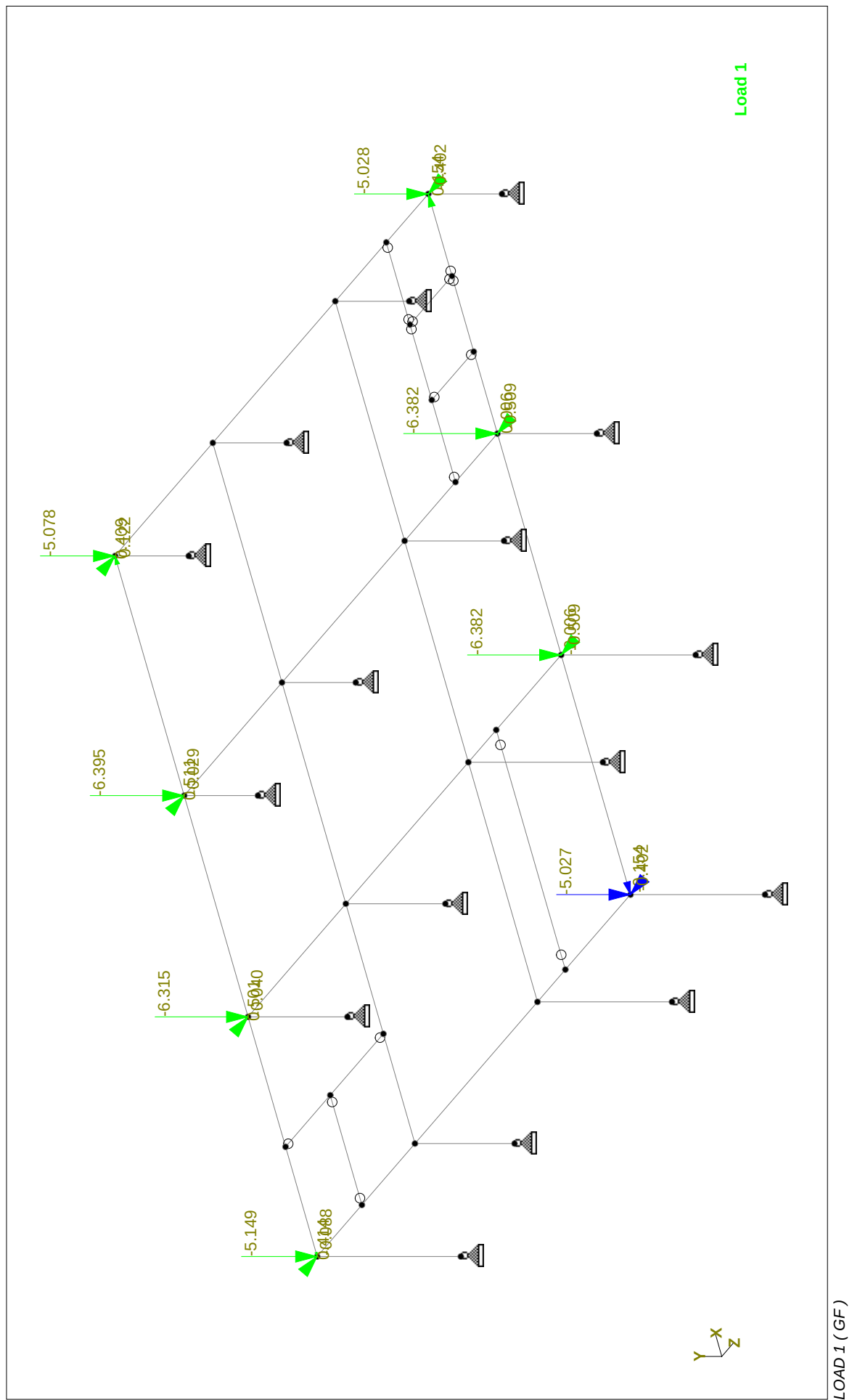


ELEVATION (GRID -27-1)

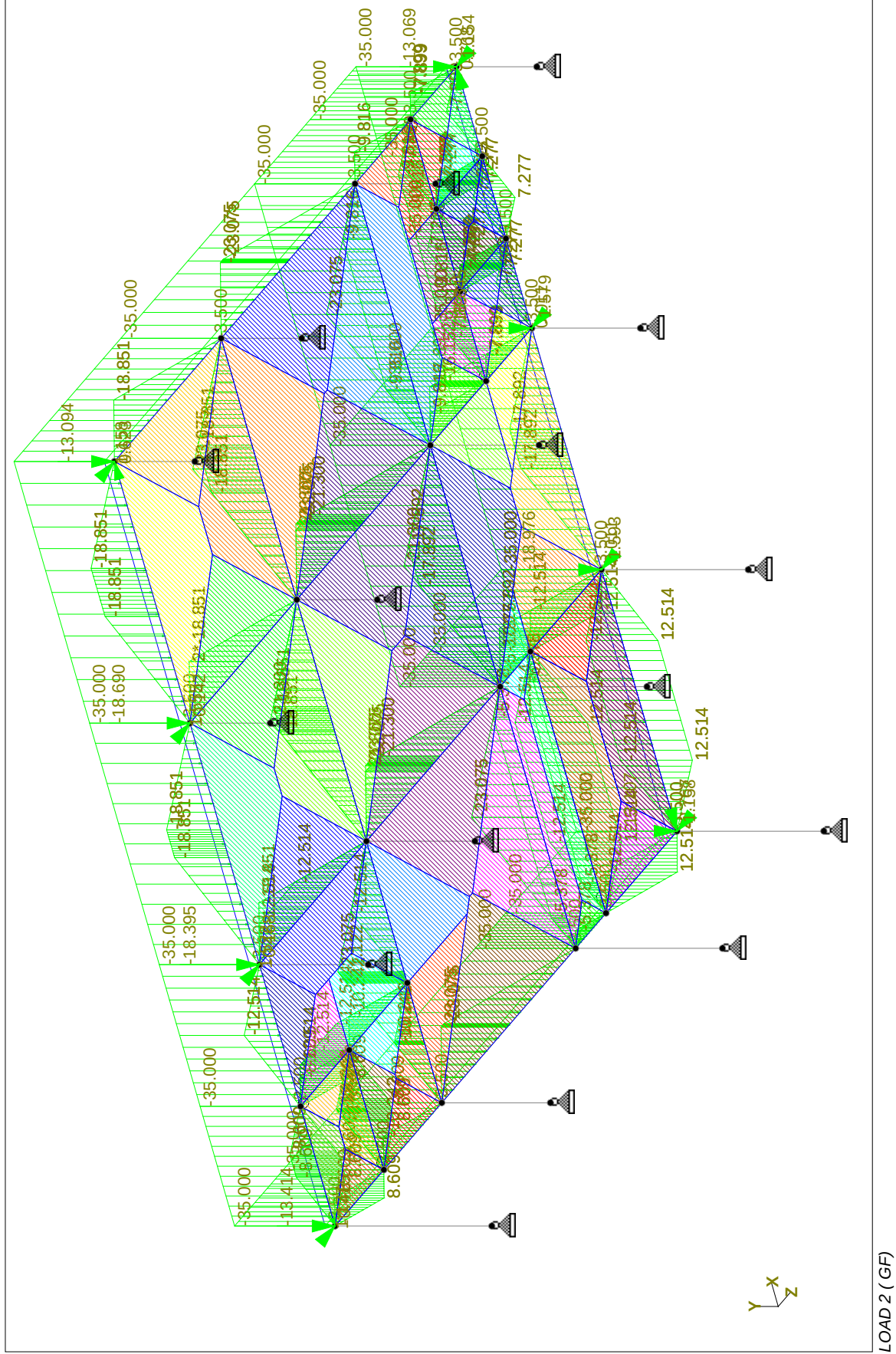


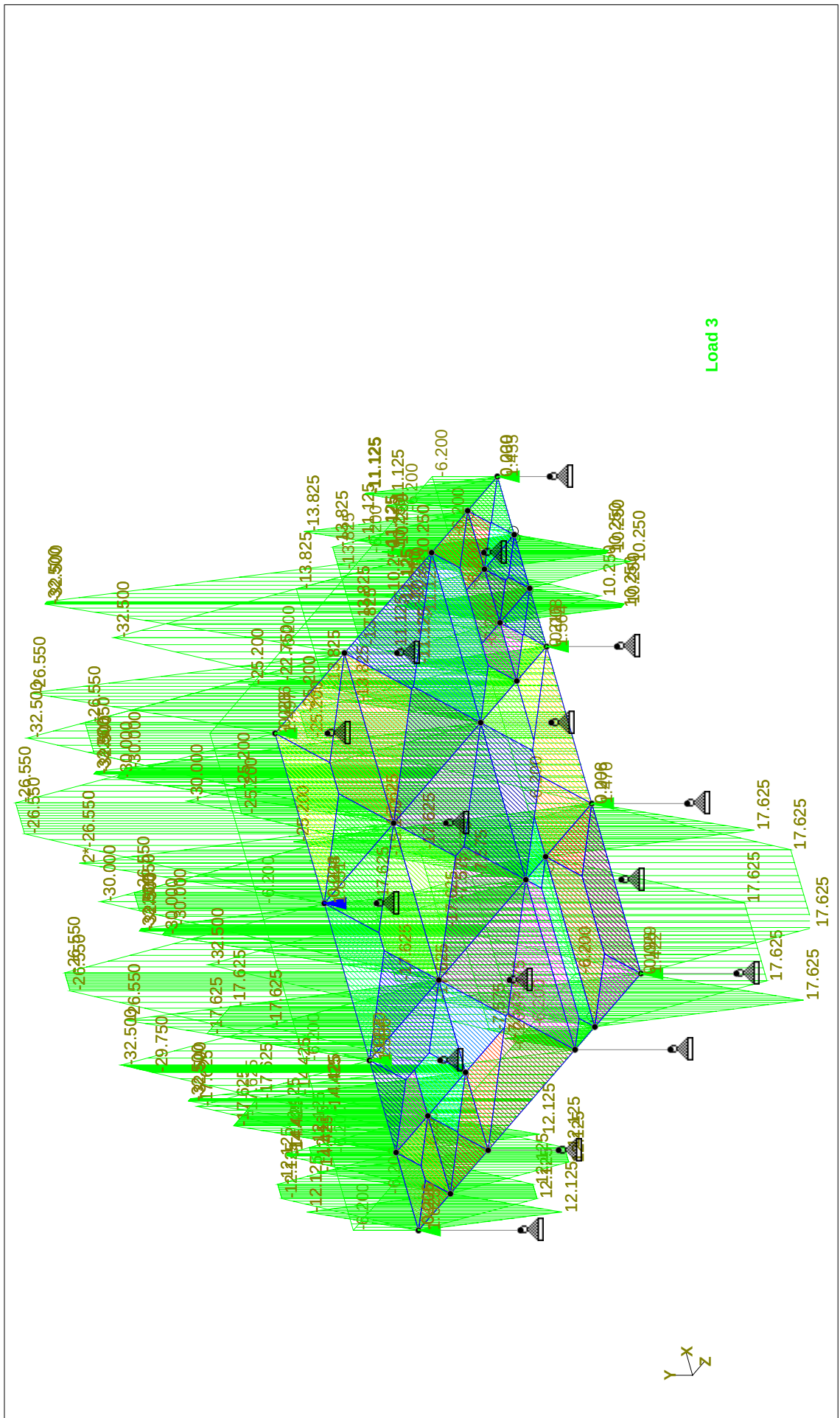


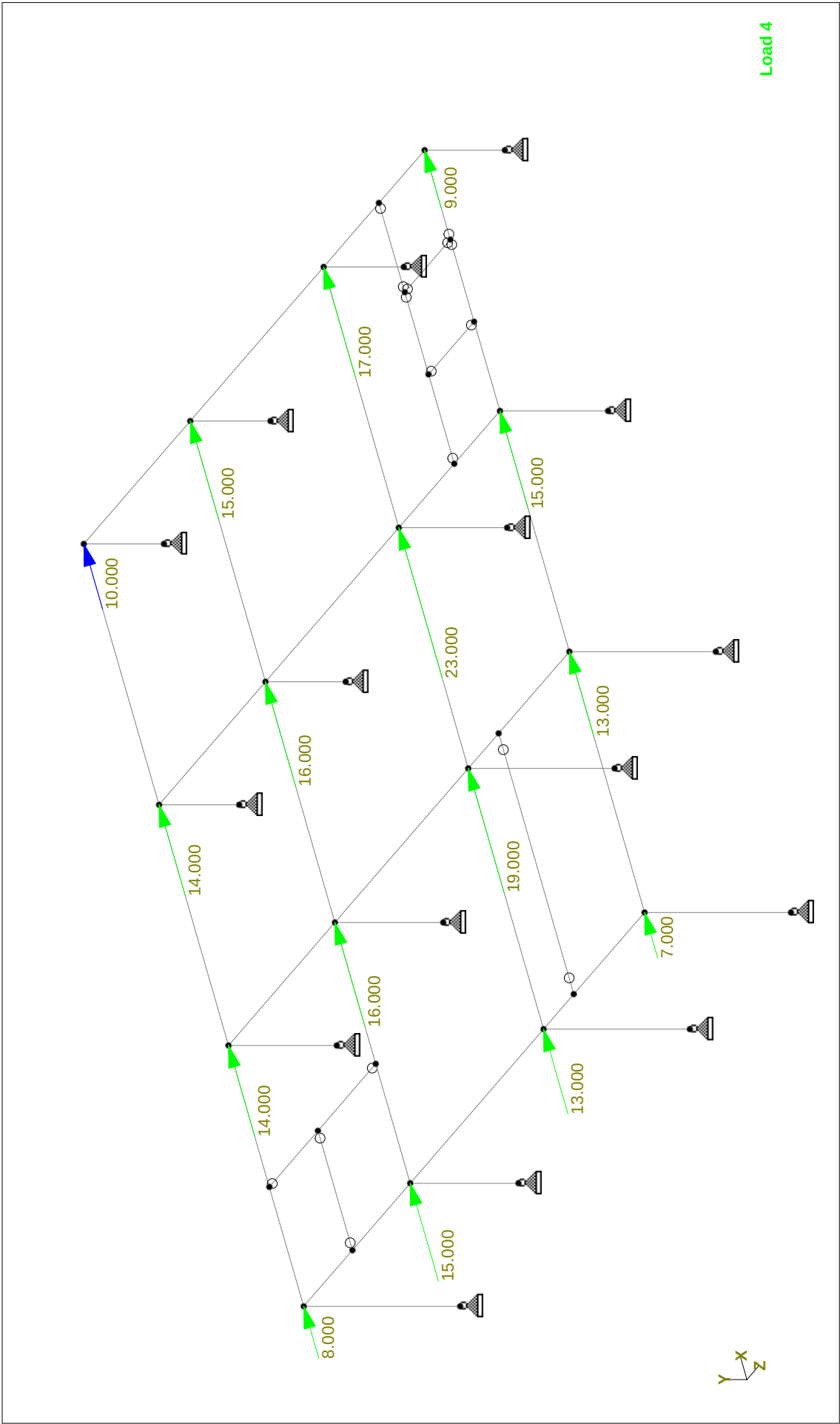




Load 1

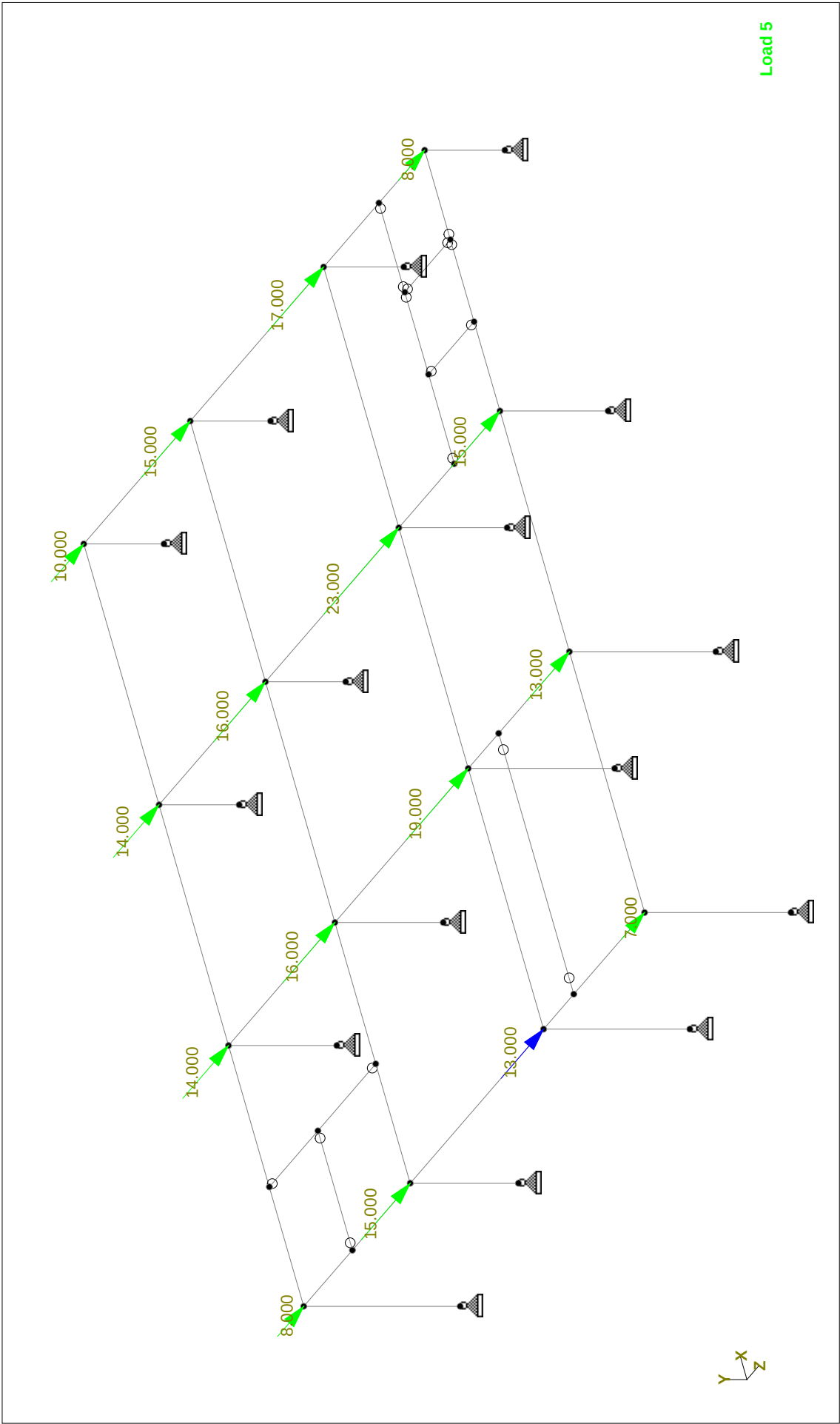






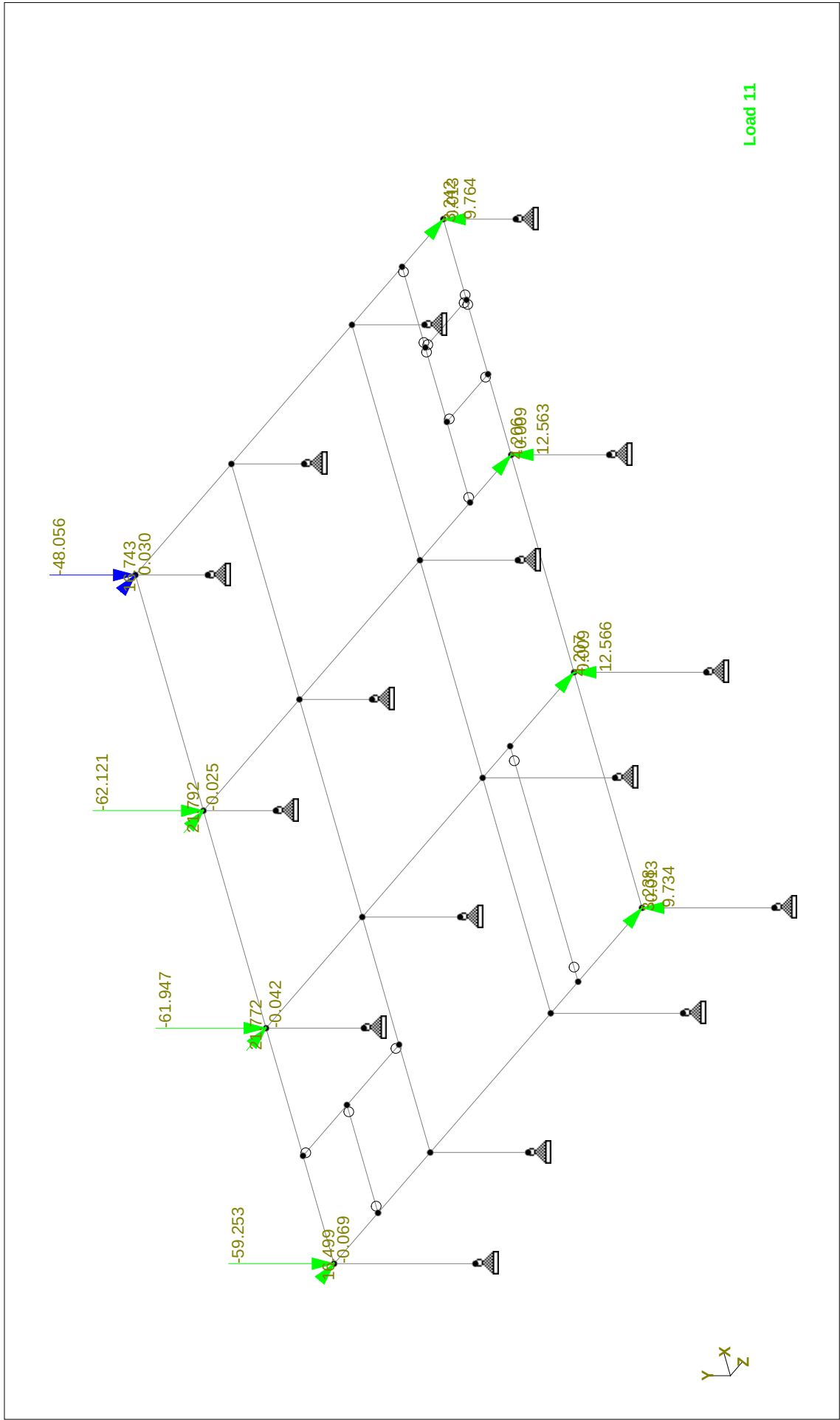
Load 4

LOAD 4 (GF)



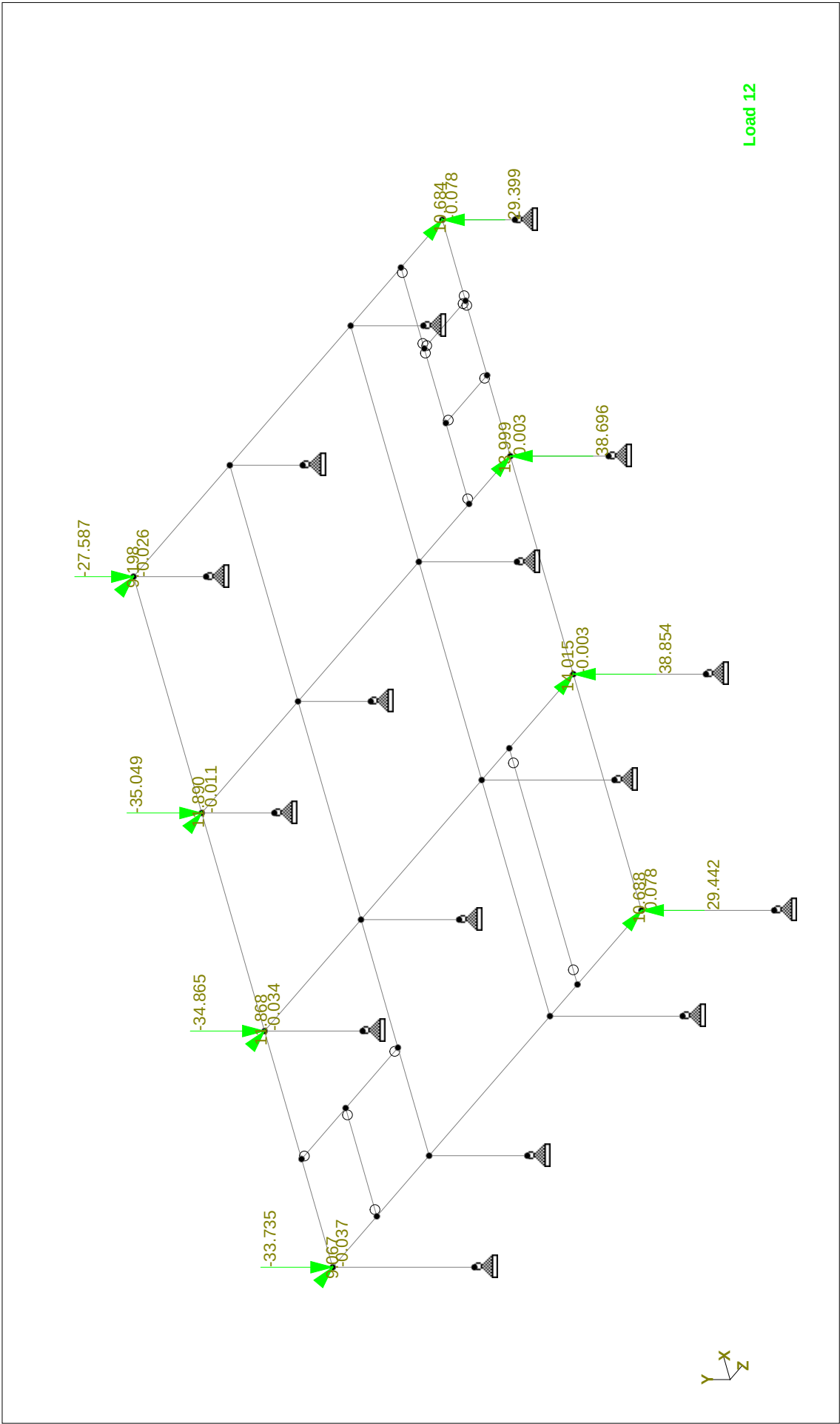
Load 5

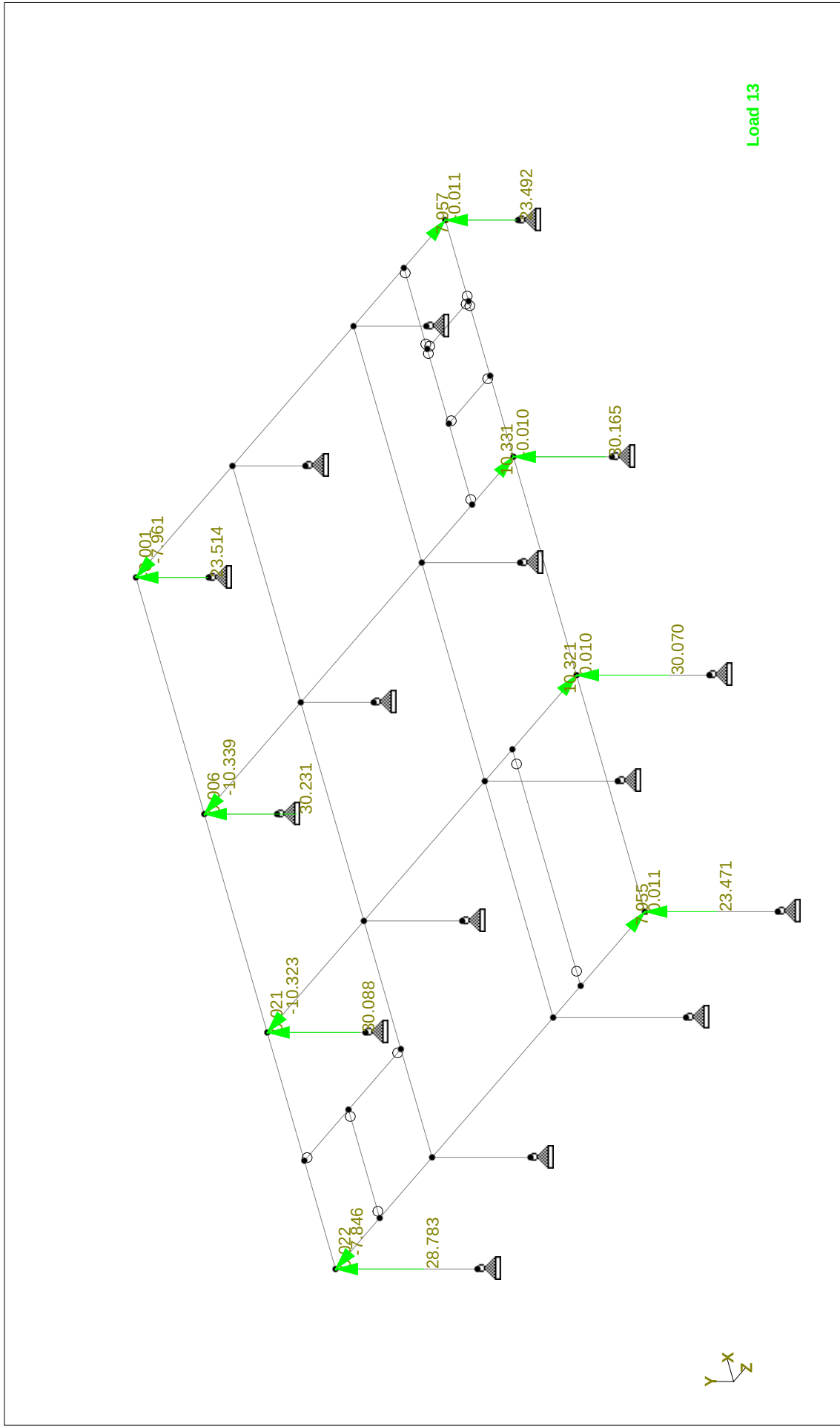
LOAD 5 (GF)



Load 11

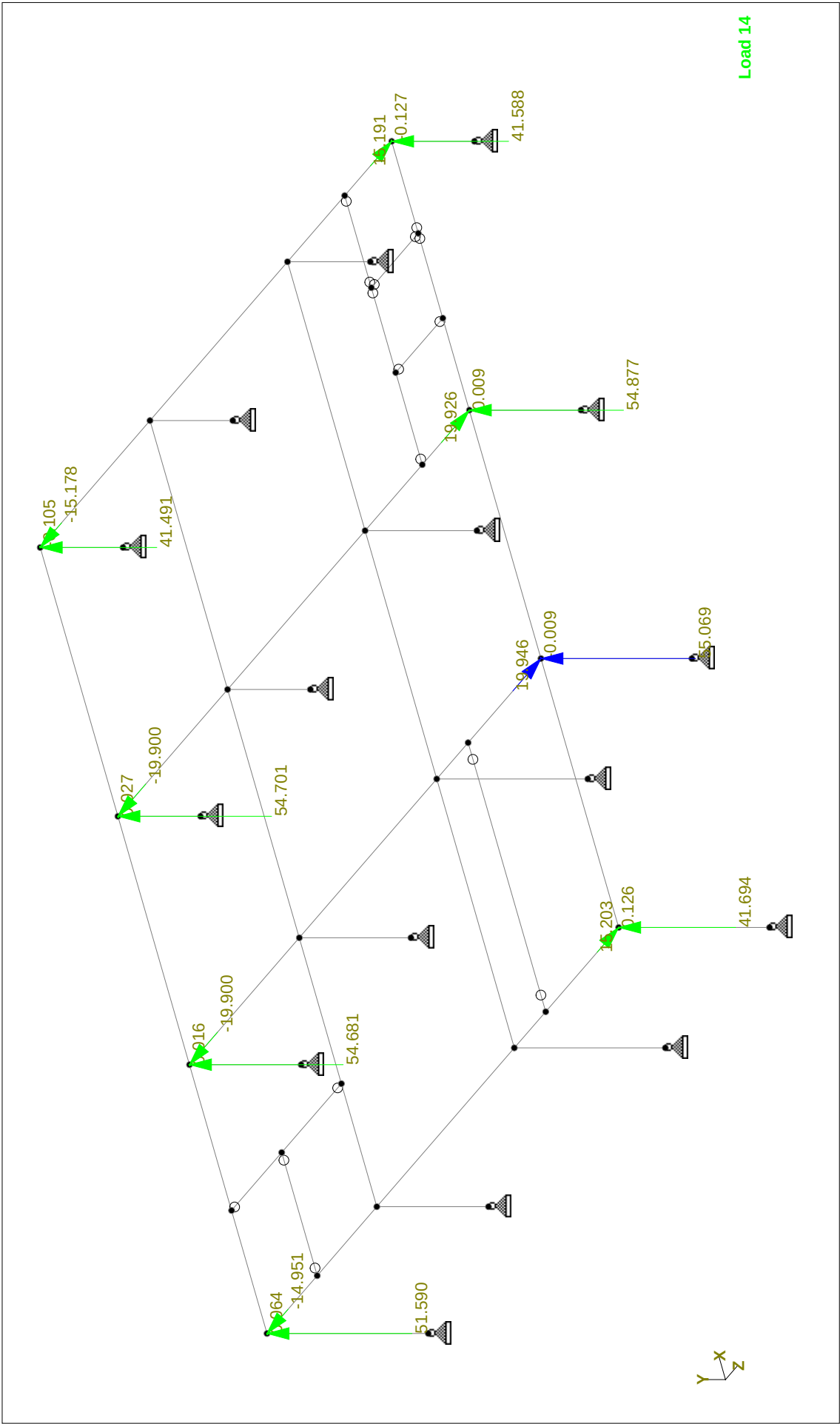
LOAD 11 (GF)





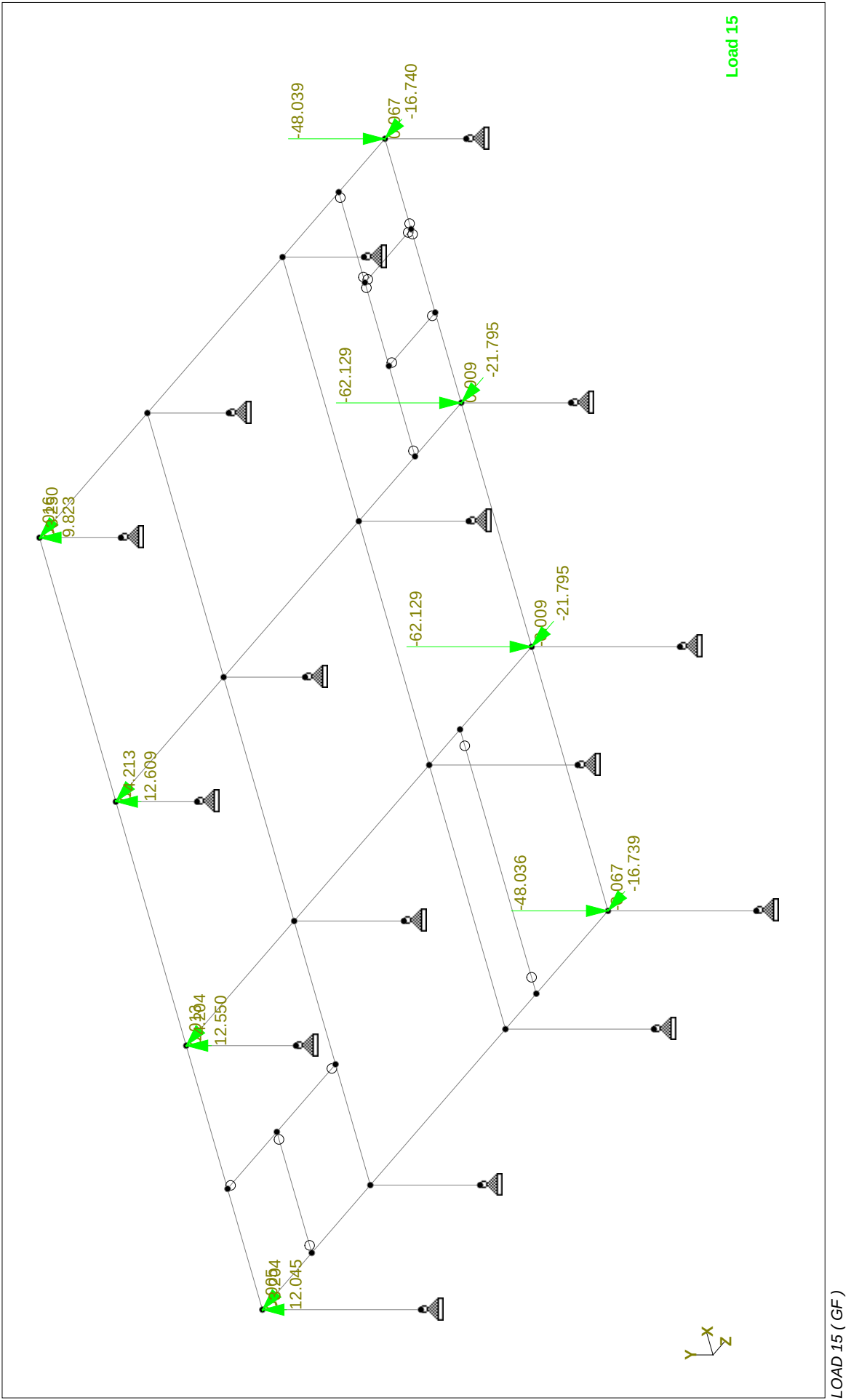
LOAD 13 (GF)

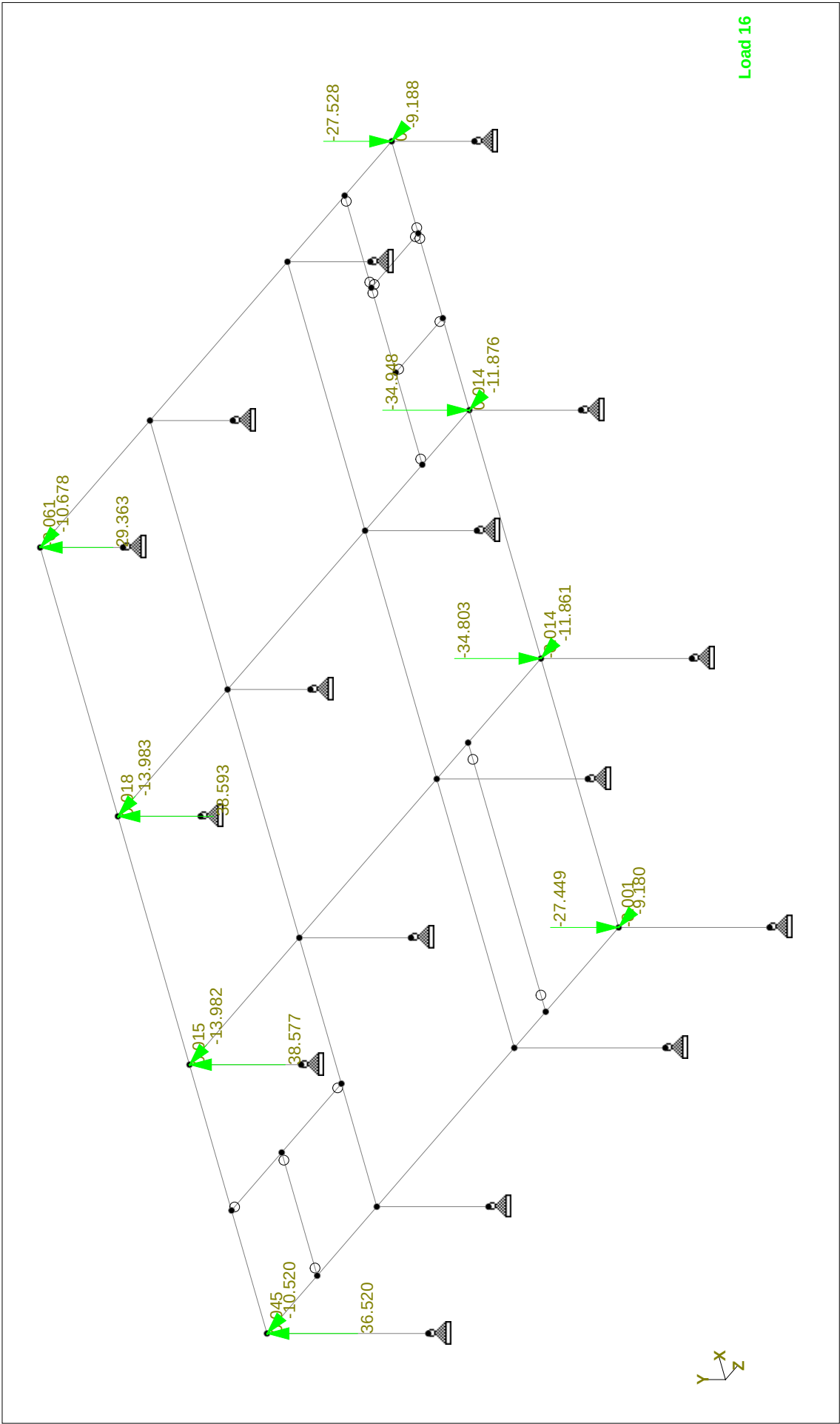
Load 13



LOAD 14 (GF)

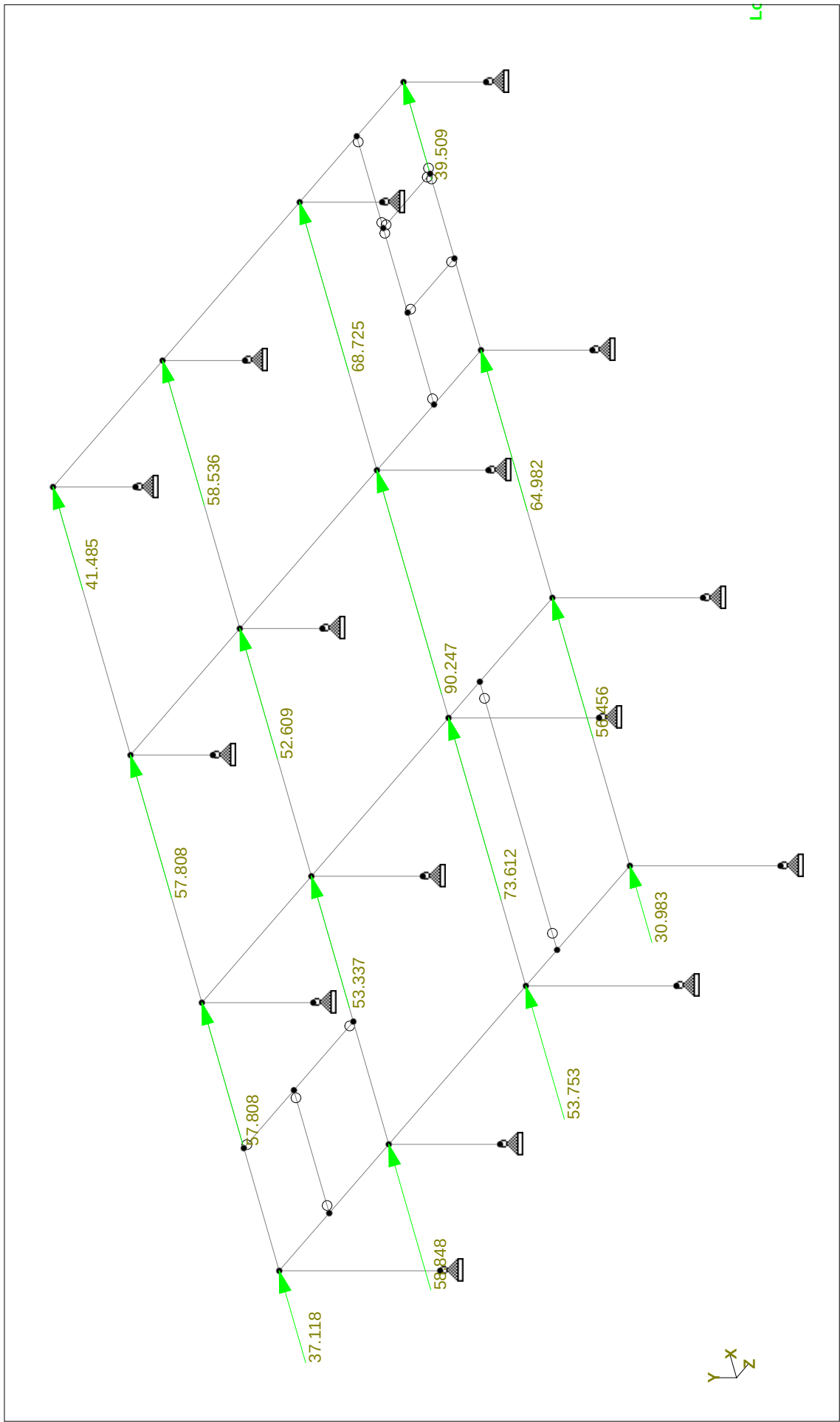
Load 14



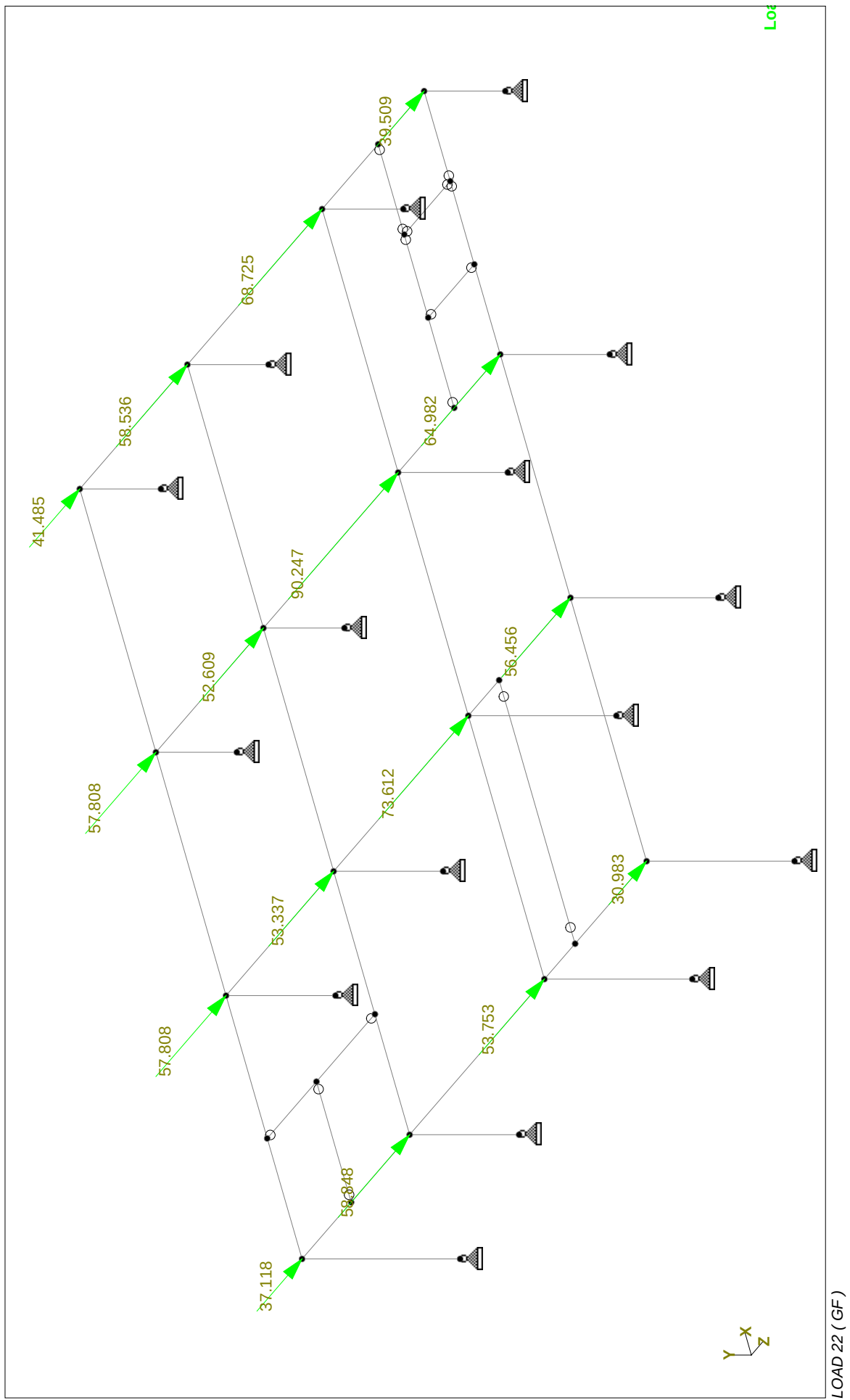


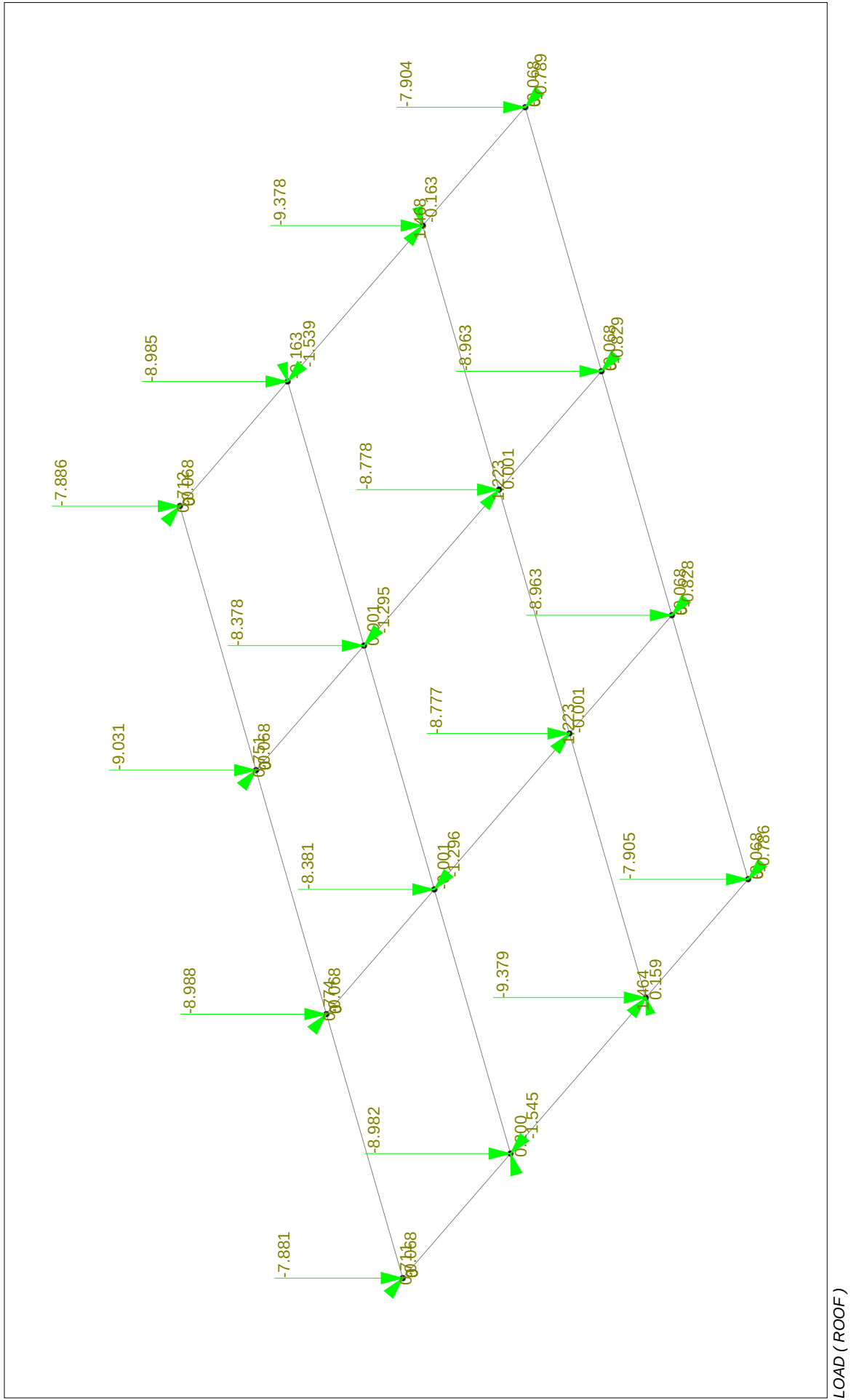
Load 16

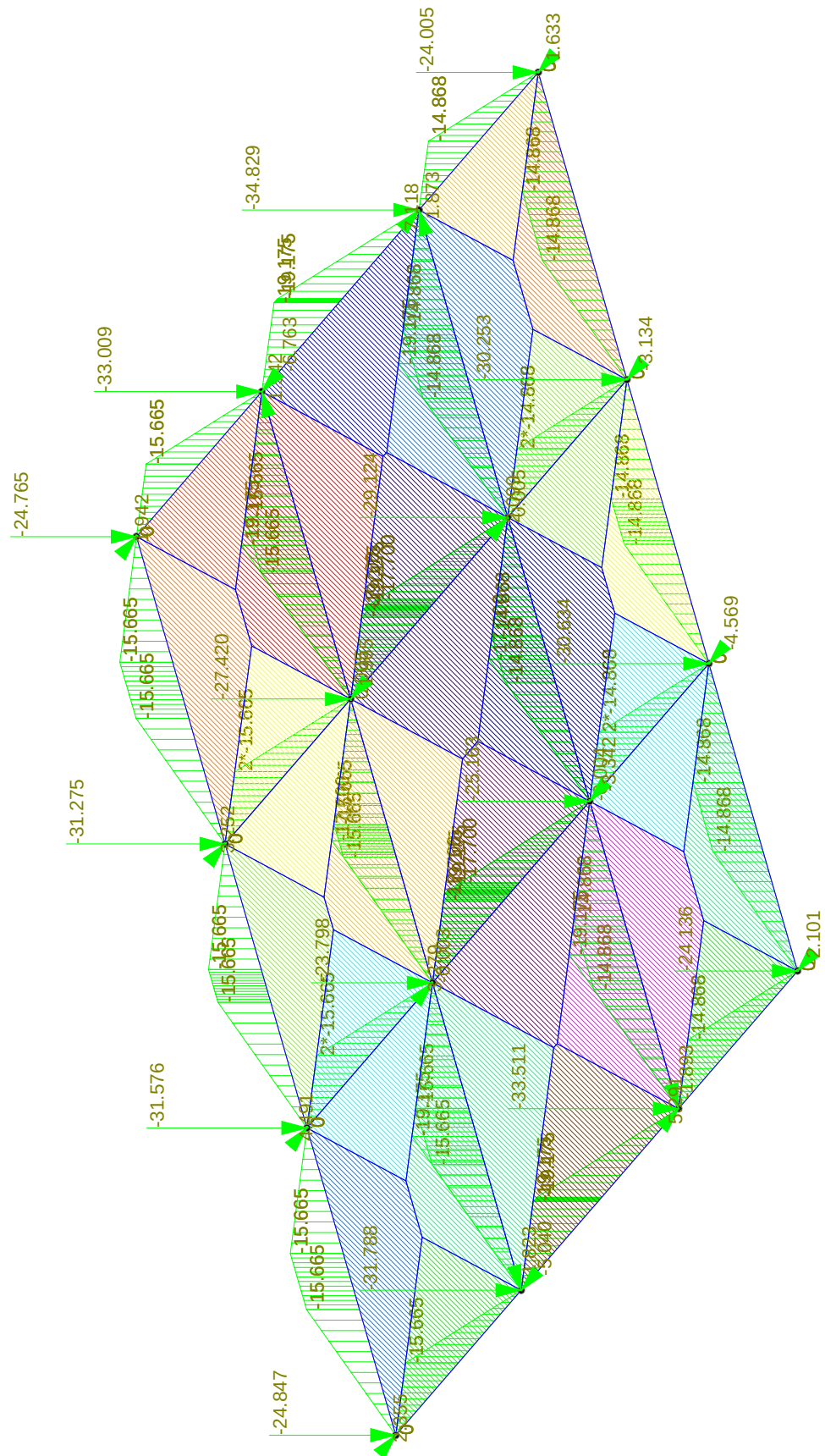
LOAD 16 (GF)

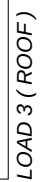


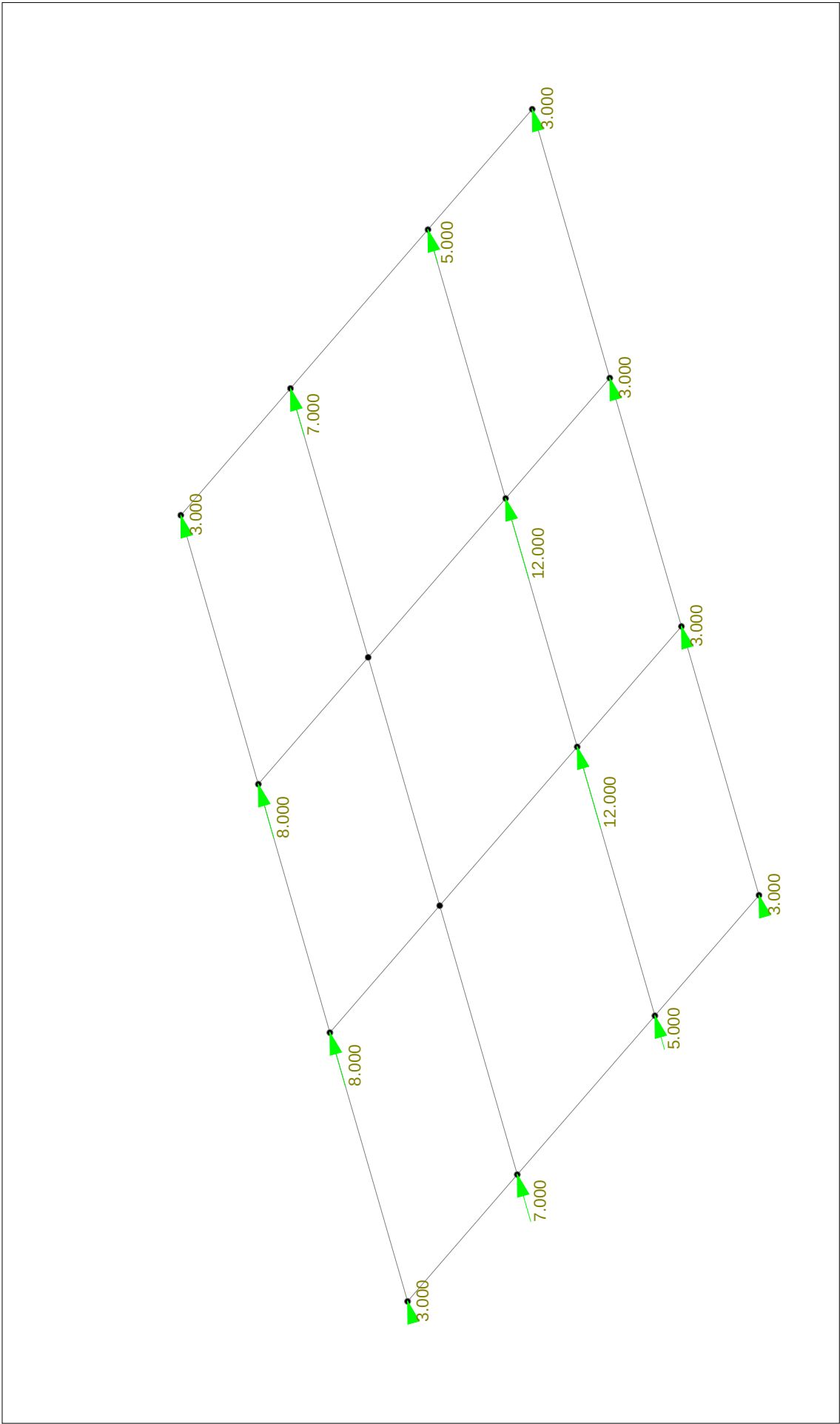
LOAD 21 (GF)



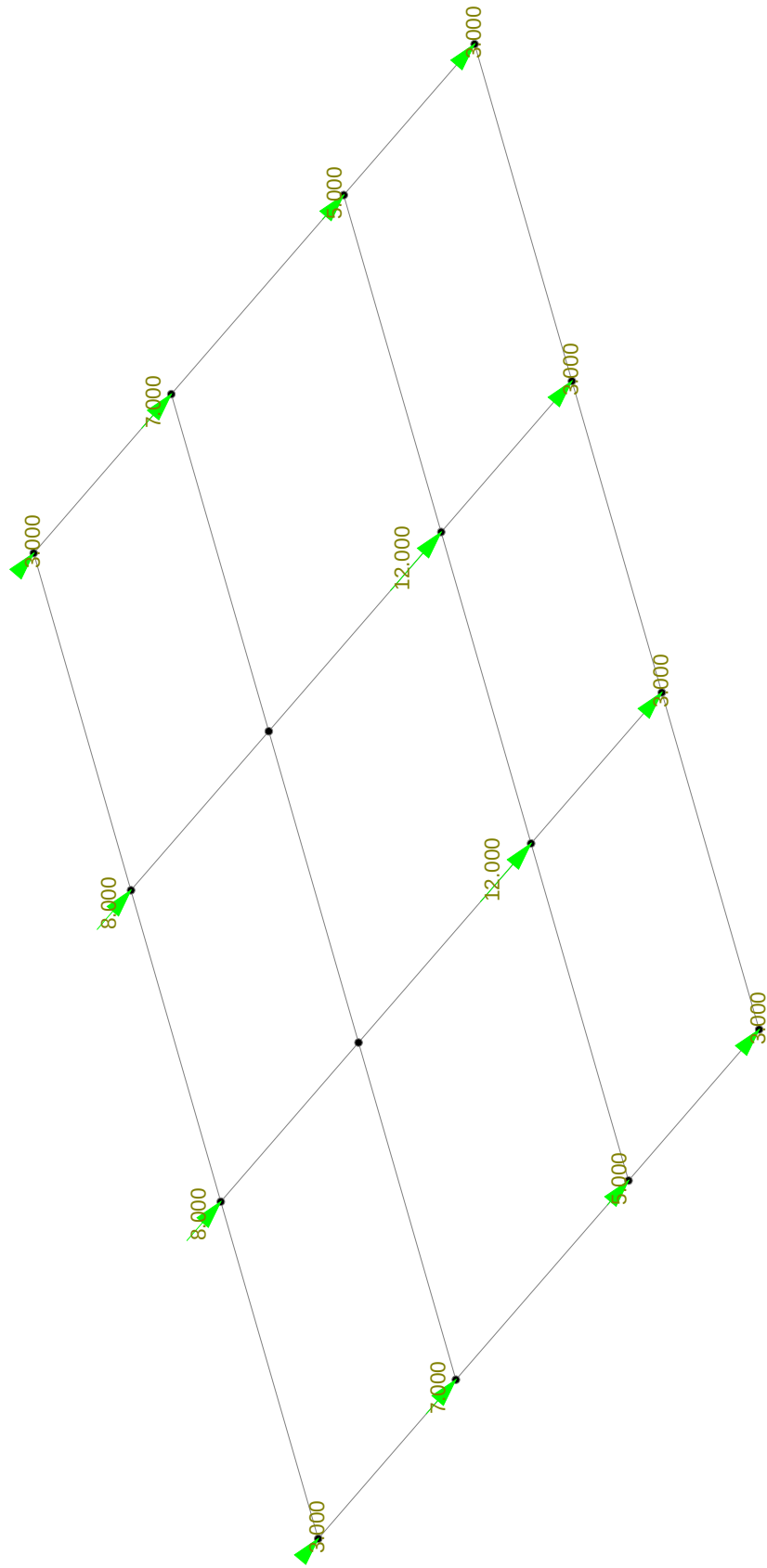




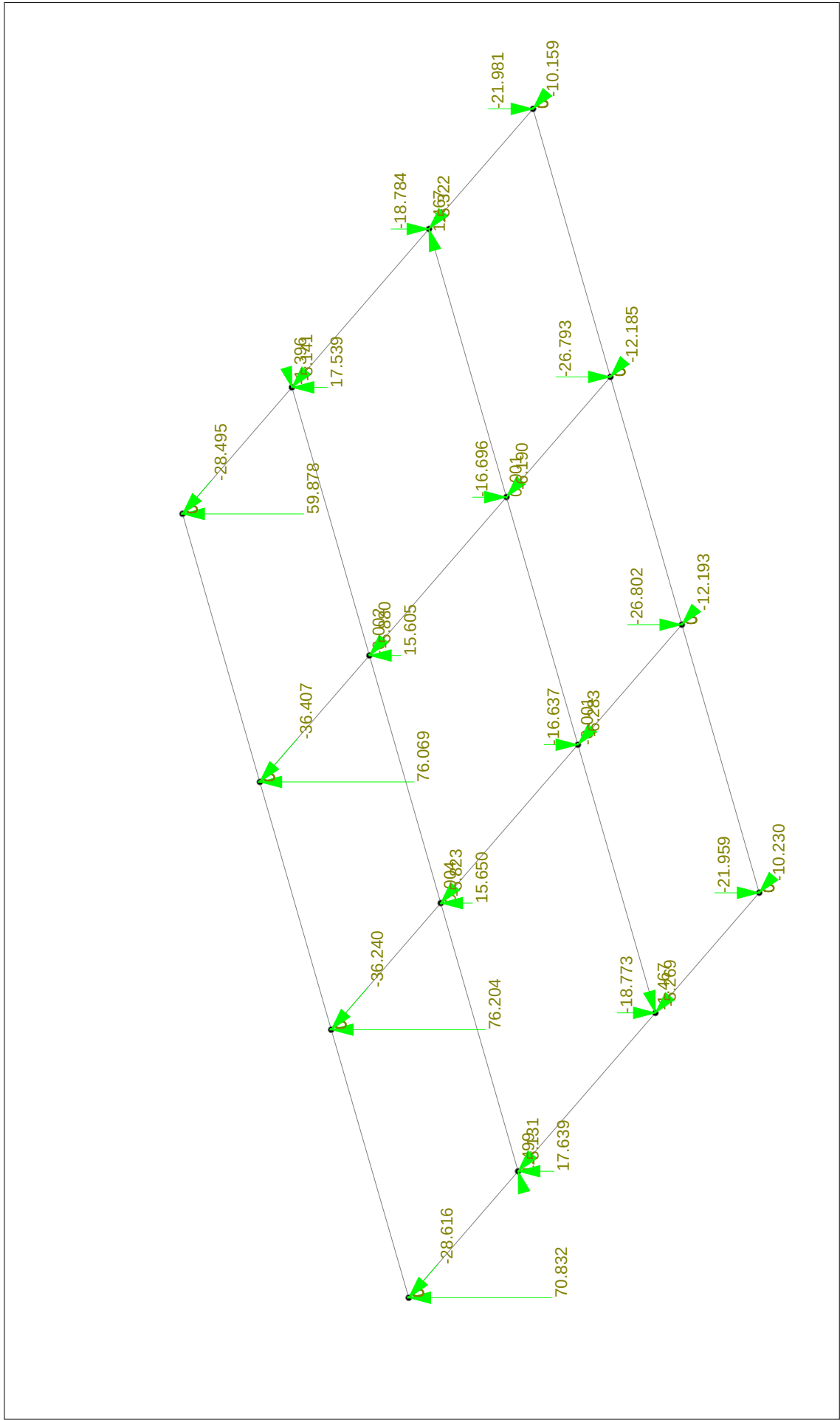




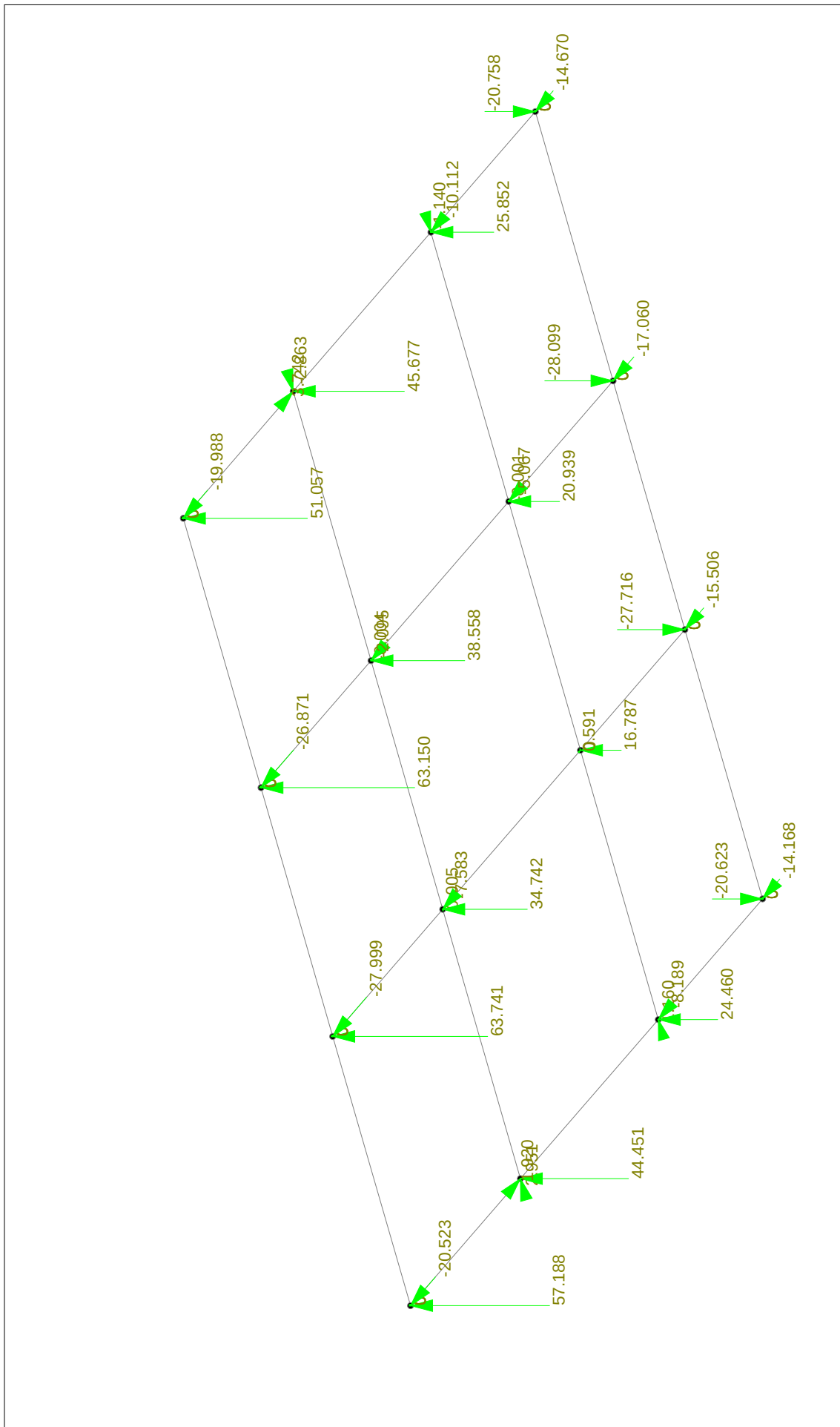
LOAD 4 (ROOF)



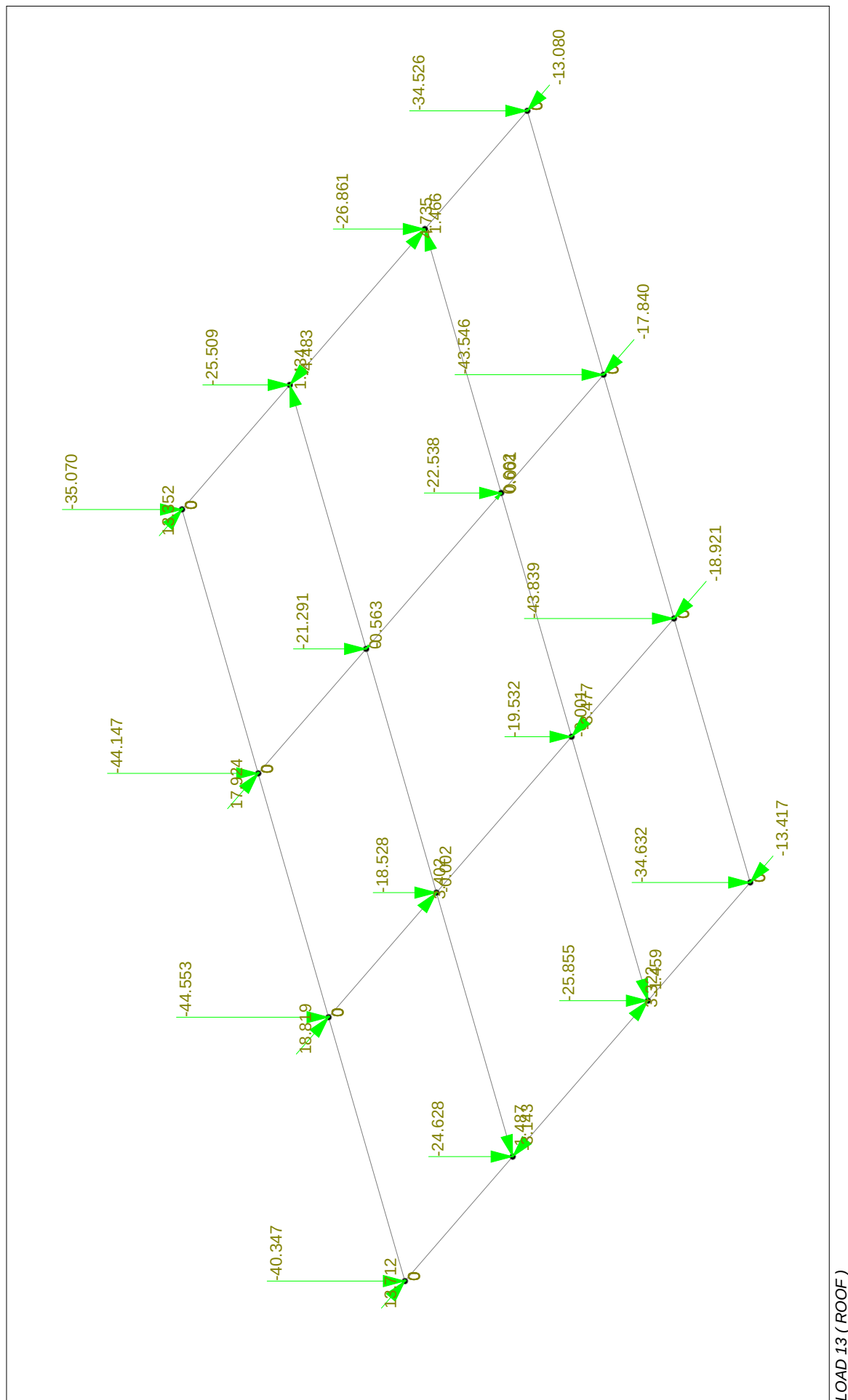
LOAD 5 (ROOF)



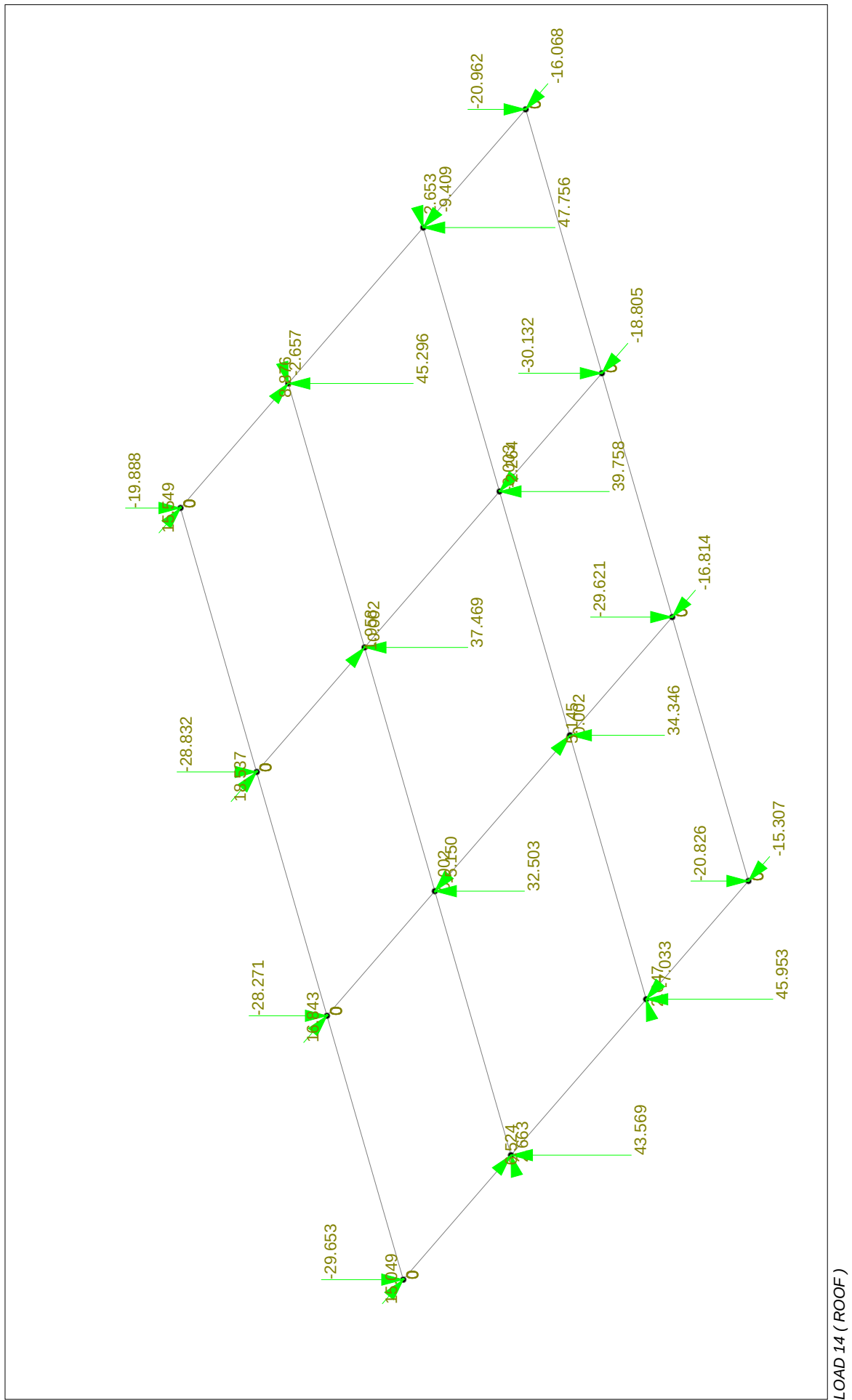
LOAD 11 (ROOF)



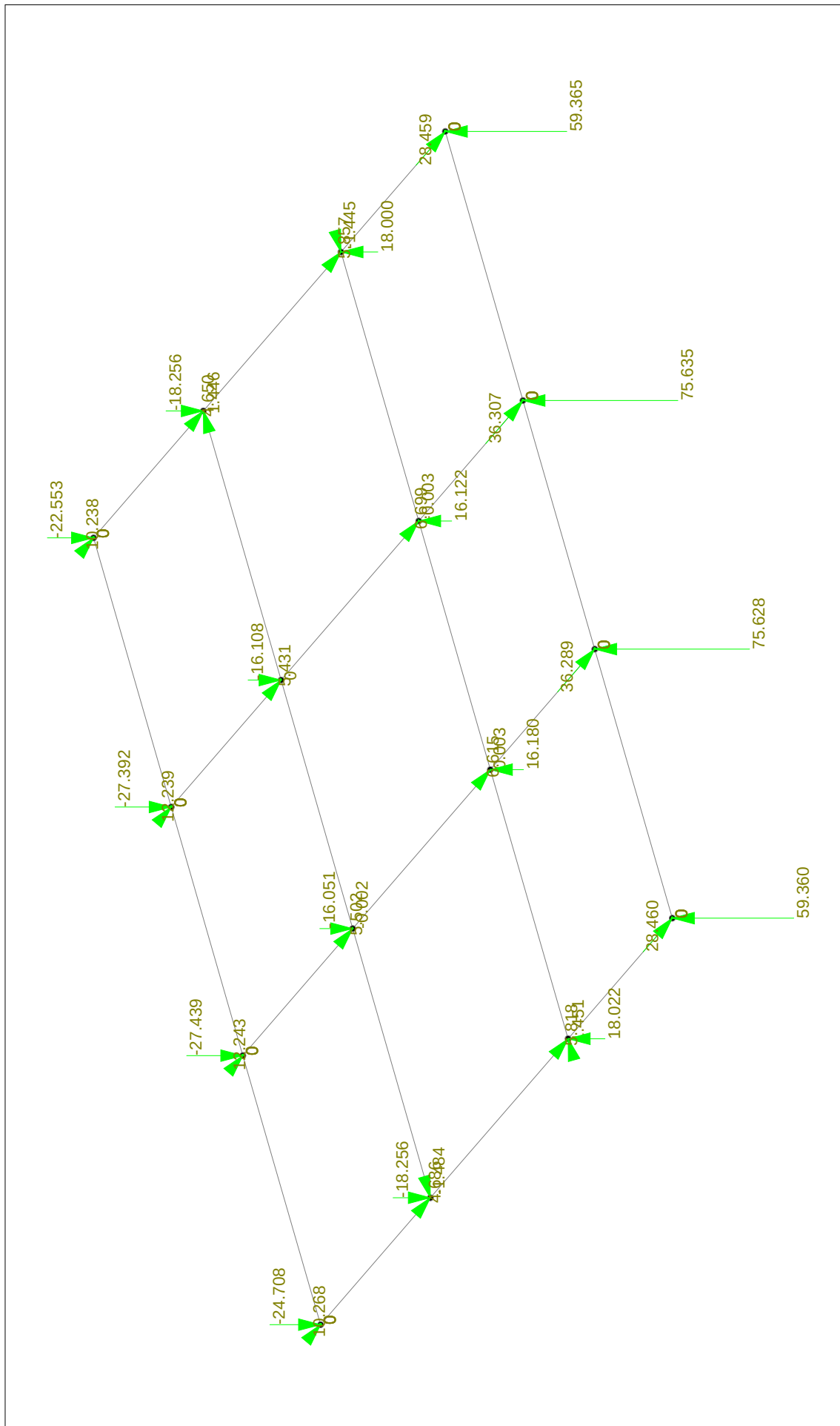
LOAD 12 (ROOF)



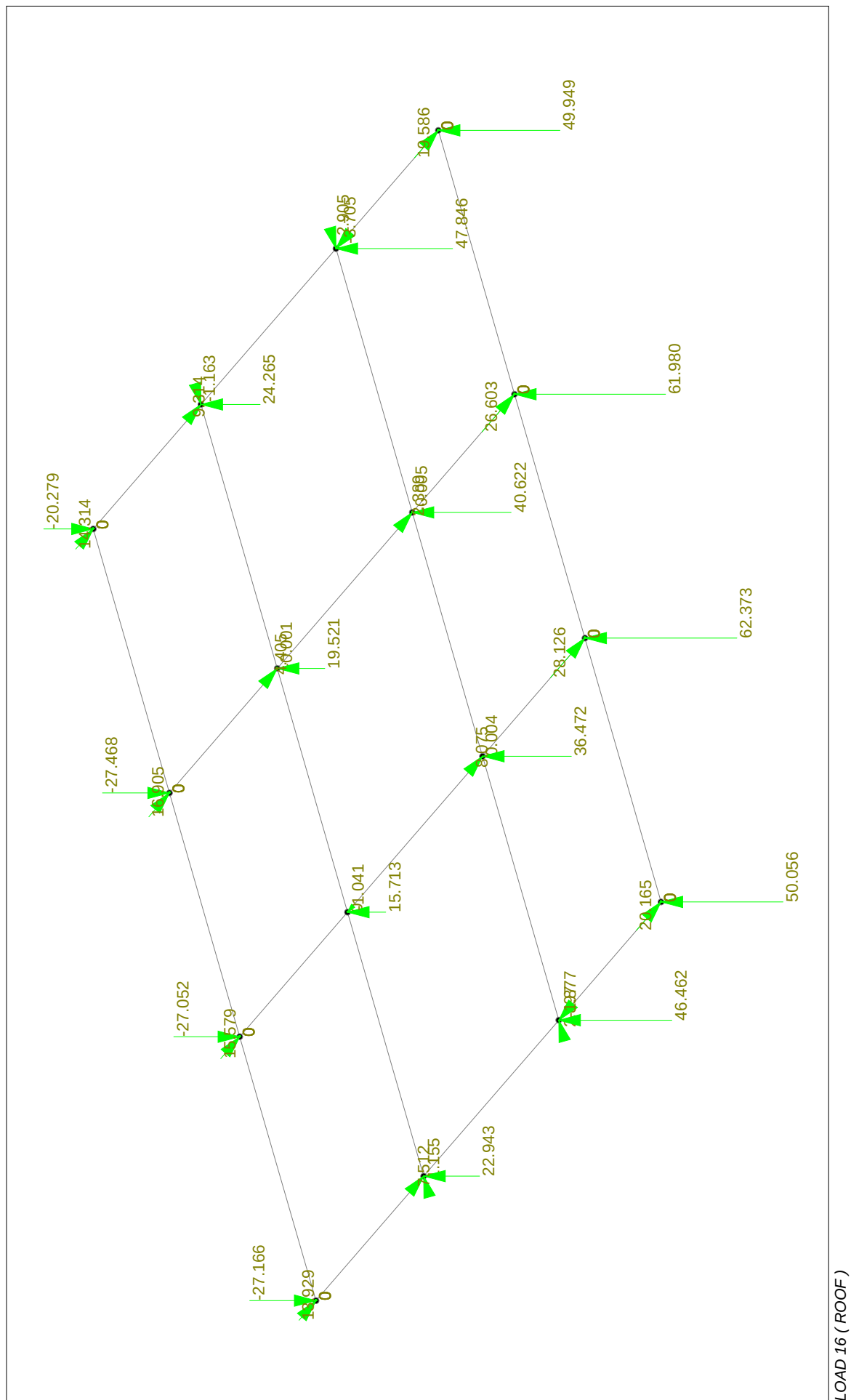
LOAD 13 (ROOF)

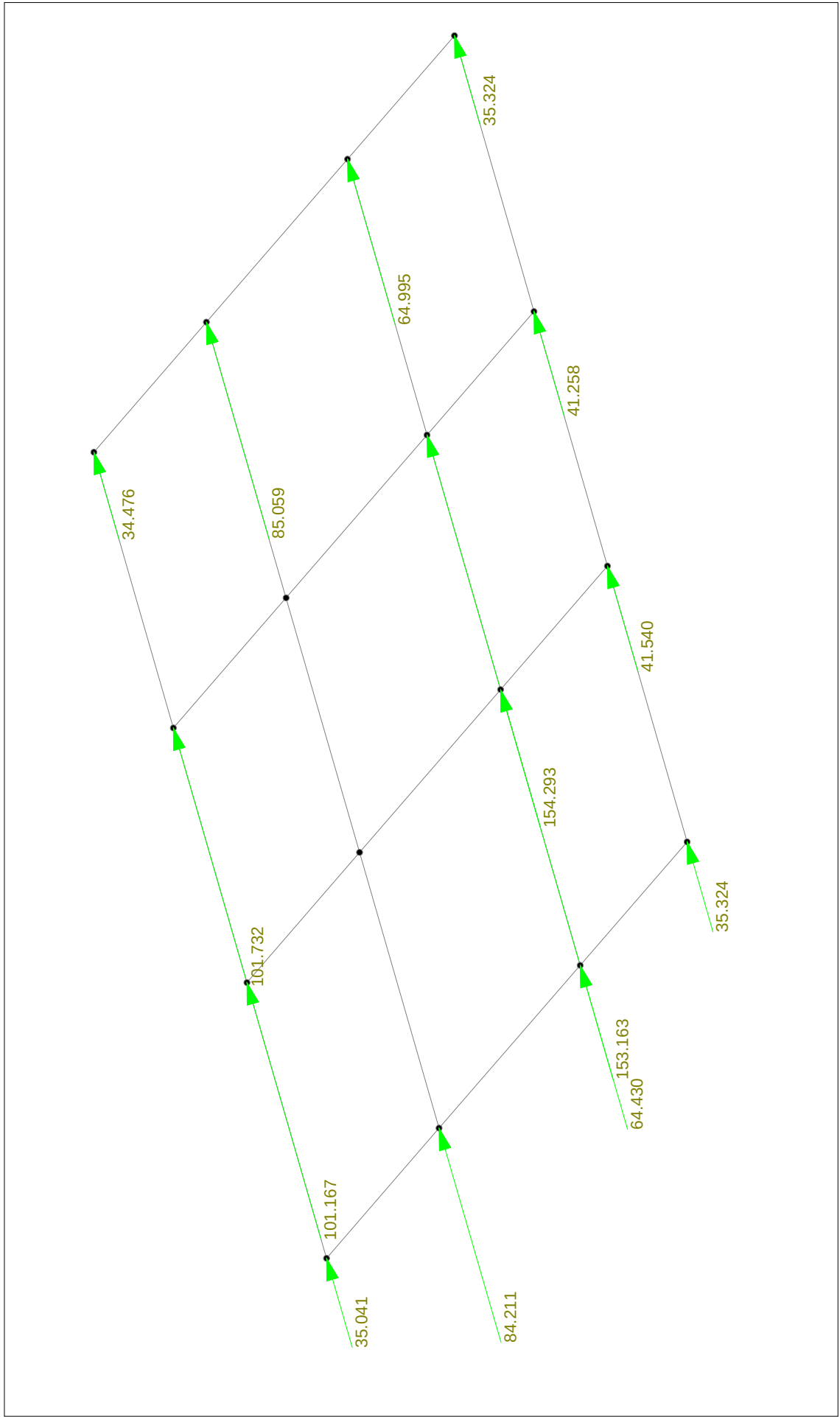


LOAD 14 (ROOF)

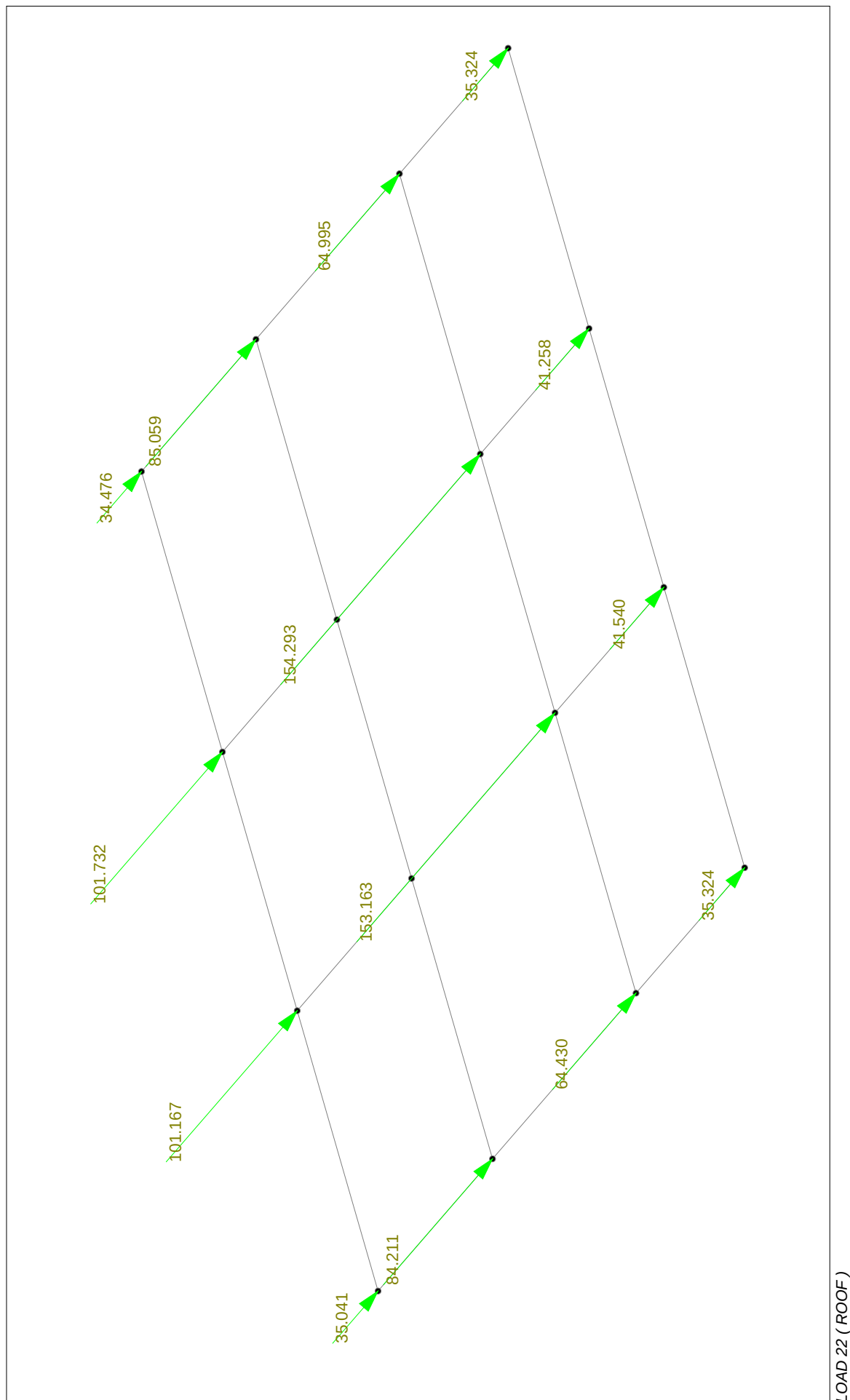


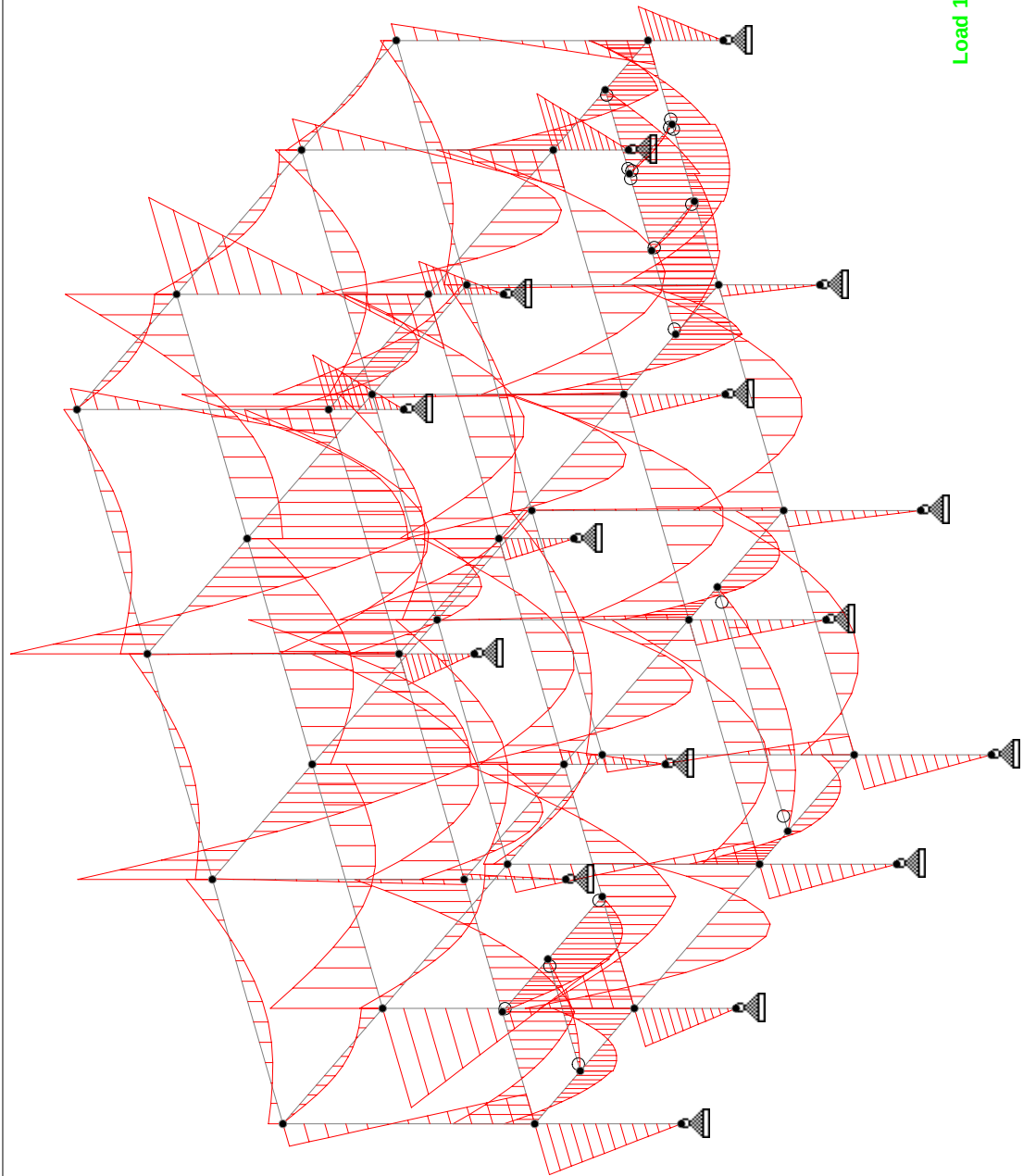
LOAD 15 (ROOF 0





LOAD 21 (ROOF)

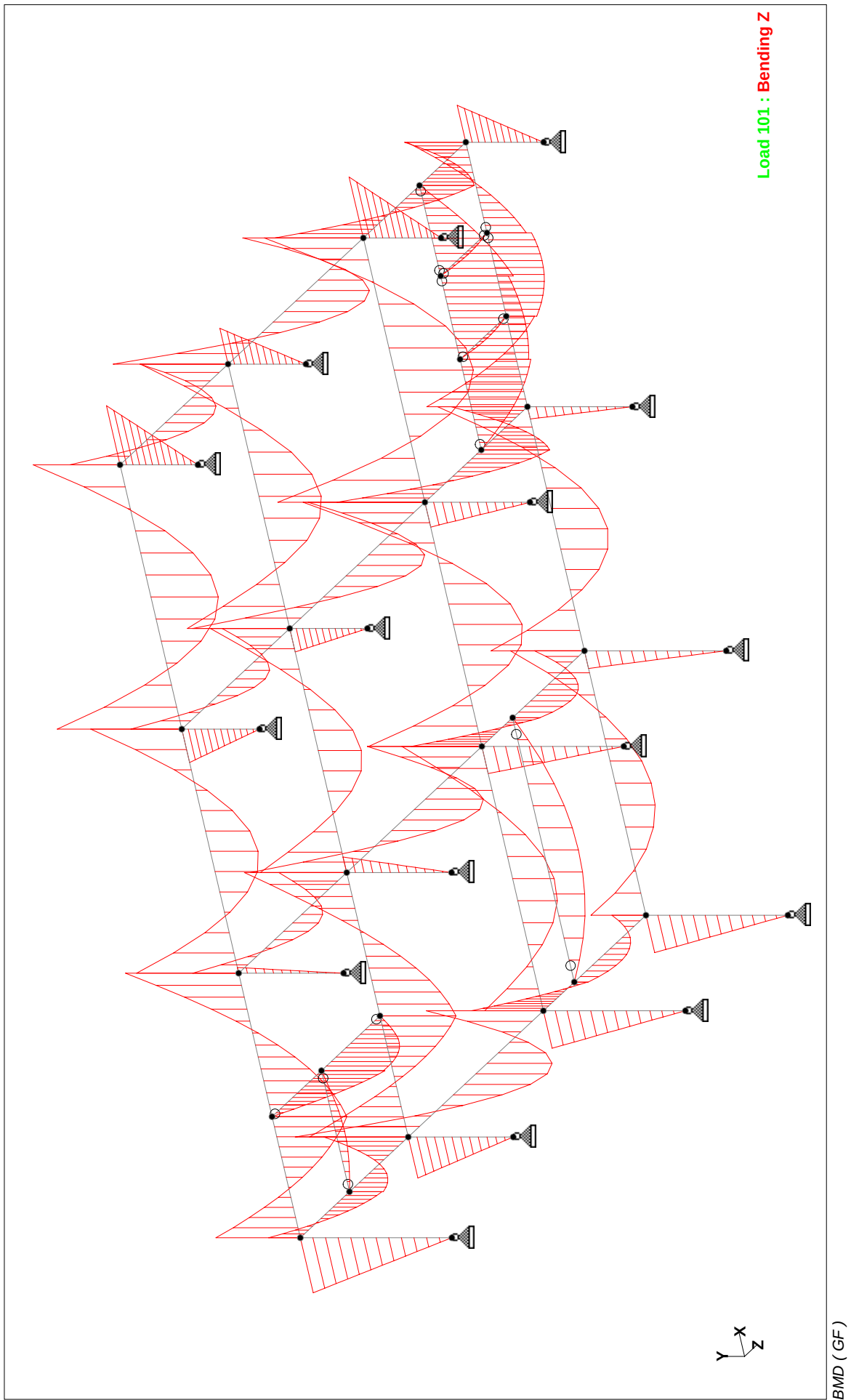


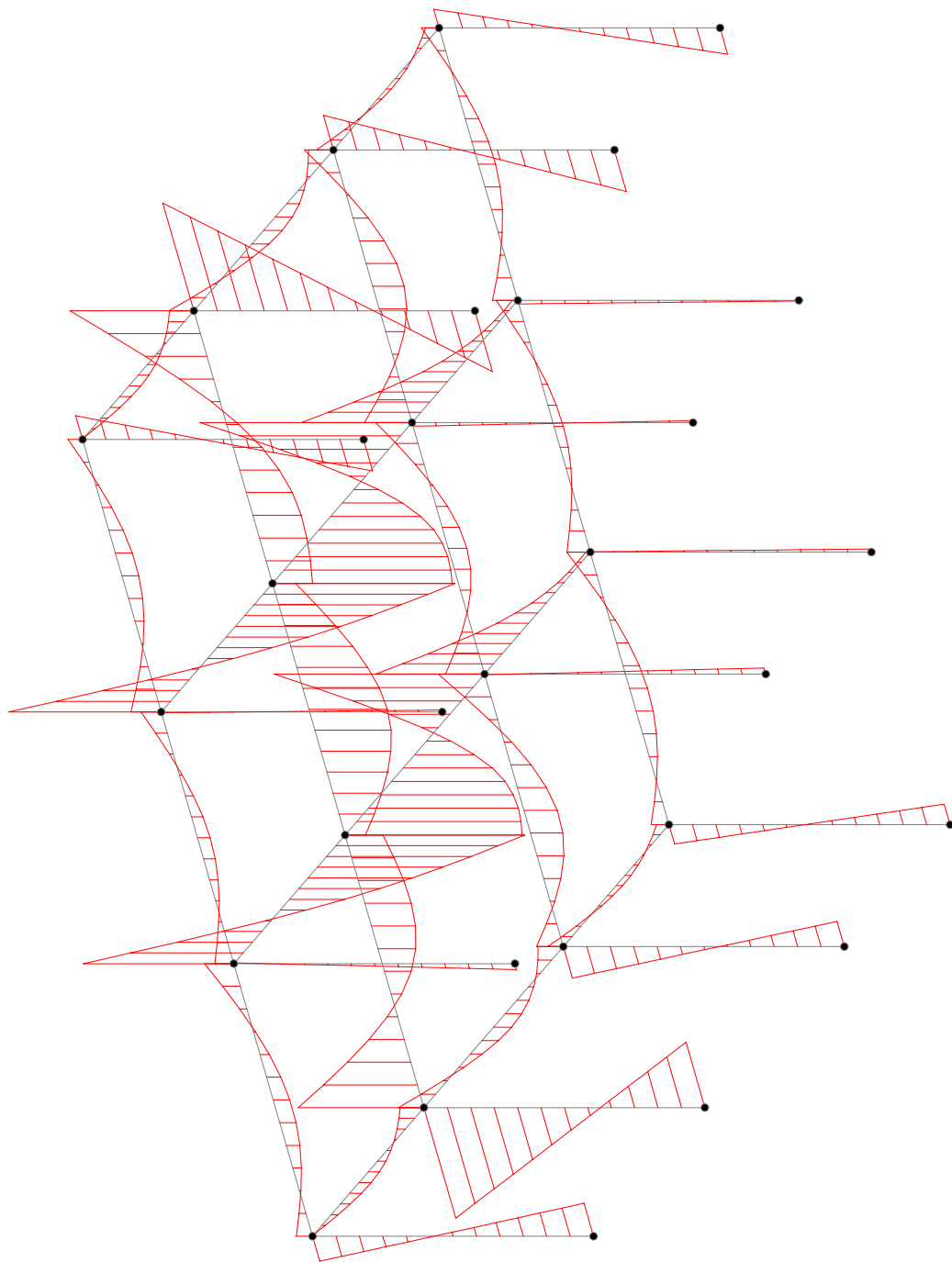


Load 101 : Bending Z



BMD (SPACE)





BMD (ROOF)

*
* STAAD.Pro
* Version 2006 Bld 1002.US
* Proprietary Program of
* Research Engineers, Intl.
* Date= DEC 26, 2007
* Time= 15:32:58
*
* USER ID: roark
*

1. STAAD SPACE BLDG-27-RCC (HEAT EXCHANGER FACILITY)
INPUT FILE: BLDG-27 (26-12-07).STD
2. START JOB INFORMATION
3. ENGINEER DATE 23-OCT-07
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. *****
7. UNIT METER KN
8. JOINT COORDINATES
9. 1 0 -3.9 0; 2 6.5 -2.7 0; 3 12.5 -2 0; 4 19 -2 0; 5 0 -2.7 5.31
10. 6 6.5 -2.7 5.31; 7 12.5 -2 5.31; 8 19 -2 5.31; 9 0 -3.65 11.96
11. 10 6.5 -3.65 11.96; 11 12.5 -2.7 11.96; 12 19 -2 11.96; 13 0 -3.65 17
12. 14 6.5 -3.65 17; 15 12.5 -2.7 17; 16 19 -2 17; 101 0 0 0; 102 6.5 0 0
13. 103 12.5 0 0; 104 19 0 0; 105 0 0 5.31; 106 6.5 0 5.31; 107 12.5 0 5.31
14. 108 19 0 5.31; 109 0 11.96; 110 6.5 0 11.96; 111 12.5 0 11.96
15. 112 19 0 11.96; 113 0 17; 114 6.5 0 17; 115 12.5 0 17; 116 19 0 17
16. 118 0 6.7 0; 119 6.5 6.7 0; 120 12.5 6.7 0; 121 19 6.7 0; 122 0 6.7 5.31
17. 123 6.5 6.7 5.31; 124 12.5 6.7 5.31; 125 19 6.7 5.31; 126 0 6.7 11.96
18. 127 6.5 6.7 11.96; 128 12.5 6.7 11.96; 129 19 6.7 11.96; 130 0 6.7 17
19. 131 6.5 6.7 17; 132 12.5 6.7 17; 133 19 6.7 17; 149 2.975 0 0; 150 0 0 2.425
20. 151 2.975 0 2.425; 152 2.975 0 5.31; 153 0 13.475; 154 6.5 0 13.475
21. 155 12.5 0 14.725; 156 19 0 14.725; 157 16.775 0 17; 158 16.775 0 14.725
22. 159 14.725 0 17; 160 14.725 0 14.725
23. *****
24. MEMBER INCIDENCES
25. 1 101 149; 2 149 102; 3 102 103; 4 103 104; 5 150 151; 6 105 152; 7 152 106
26. 8 106 107; 9 107 108; 10 109 110; 11 110 111; 12 111 112; 13 153 154
27. 14 113 114; 15 114 115; 16 115 159; 17 159 157; 18 157 116; 19 113 153
28. 20 153 109; 21 109 105; 22 105 150; 23 150 101; 24 152 151; 25 151 149
29. 26 114 154; 27 154 110; 28 110 106; 29 106 102; 30 115 155; 31 155 111
30. 32 111 107; 33 107 103; 34 159 160; 35 157 158; 36 116 156; 37 156 112
31. 38 112 108; 39 108 104; 40 155 160; 41 160 158; 42 158 156; 101 118 119
32. 102 119 120; 103 120 121; 104 122 123; 105 123 124; 106 124 125; 107 126 127
33. 108 127 128; 109 128 129; 110 130 131; 111 131 132; 112 132 133; 113 130 126
34. 114 126 122; 115 122 118; 116 131 127; 117 127 123; 118 123 119; 119 132 128
35. 120 128 124; 121 124 120; 122 133 129; 123 129 125; 124 125 121; 1001 1 101
36. 1002 2 102; 1003 3 103; 1004 4 104; 1005 5 105; 1006 6 106; 1007 7 107
37. 1008 8 108; 1009 9 109; 1010 10 110; 1011 11 111; 1012 12 112; 1013 13 113
38. 1014 14 114; 1015 15 115; 1016 16 116; 2001 101 118; 2002 102 119
39. 2003 103 120; 2004 104 121; 2005 105 122; 2008 108 125; 2009 109 126
40. 2010 110 127; 2011 111 128; 2012 112 129; 2013 113 130; 2014 114 131

41. 2015 115 132; 2016 116 133
42. *****
43. DEFINE MATERIAL START
44. ISOTROPIC CONCRETE
45. E 2.17185E+007
46. POISSON 0.17
47. DENSITY 24
48. ALPHA 1E-005
49. DAMP 0.05
50. ISOTROPIC STEEL
51. E 2.05E+008
52. POISSON 0.3
53. DENSITY 78.5
54. ALPHA 1.2E-005
55. DAMP 0.03
56. END DEFINE MATERIAL
57. MEMBER PROPERTY BRITISH
58. *****
59. 1001 TO 1016 PRIS YD 0.55 ZD 0.55
60. 2001 TO 2005 2008 TO 2016 PRIS YD 0.5 ZD 0.5
61. 1 TO 42 PRIS YD 0.6 ZD 0.3
62. 101 TO 124 PRIS YD 0.6 ZD 0.3
63. *****STEEL*****
64. *****
65. CONSTANTS
66. MATERIAL CONCRETE ALL
67. *****
68. SUPPORTS
69. 1 TO 16 PINNED
70. *101 TO 116 118 TO 122 125 TO 133 PINNED
71. *****
72. *MEMBER RELEASE
73. *105 106 START MZ
74. *104 105 END MZ
75. *****
76. UNIT MMS NEWTON
77. MEMBER RELEASE
78. 5 13 24 34 35 40 START MZ
79. 5 13 25 34 35 42 END MZ
80. 18 42 START MX
81. 17 41 END MX
82. UNIT METER KN
83. DEFINE UBC LOAD
84. ZONE 0.15 I 1 RWX 3.5 RWZ 3.5 STYP 4 CT 0.0731 PX 0.514 PZ 0.514 NA 1 NV 1
85. JOINT WEIGHT
86. 101 WEIGHT 357
87. 102 WEIGHT 556
88. 103 WEIGHT 556
89. 104 WEIGHT 399
90. 105 WEIGHT 566
91. 106 WEIGHT 513
92. 107 WEIGHT 506
93. 108 WEIGHT 563
94. 109 WEIGHT 517
95. 110 WEIGHT 708
96. 111 WEIGHT 868

80

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

147. 133 FX 0 FY -7.904 FZ 0
 148. 121 FX 0 FY -7.886 FZ 0
 149. 130 FX 0 FY -0.068 FZ -0.786
 150. 118 FX 0 FY -0.068 FZ 0.711
 151. 131 FX 0 FY -0.068 FZ -0.828
 152. 119 FX 0 FY -0.068 FZ 0.774
 153. 132 FX 0 FY -0.068 FZ -0.829
 154. 120 FX 0 FY -0.068 FZ 0.751
 155. 133 FX 0 FY -0.068 FZ -0.789
 156. 121 FX 0 FY -0.068 FZ 0.712
 157. LOAD 2 DL
 158. MEMBER LOAD
 159. *****
 160. *****WALL(T=250 MM,DENSITY=23 KN/M^3,H=6.1 M)=0.25X23X6.1=35.0 KN/M
 161. **PROJECTION OF SLAB (0.62X0.225)=0.62X0.225X25=3.5 KN/M
 162. **FLOOR DL (225 THICK+FIN+M & E)= 7.1 KN/M^2
 163. **ROOF DL (175 THICK+FIN+M & E)= 5.9 KN/M^2
 164. *****
 165. 1 TO 4 14 TO 18 UNI GY -3.5
 166. 19 TO 23 36 TO 39 UNI GY -3.5
 167. *****WALL LOAD *****
 168. 1 TO 4 11 12 14 TO 23 26 27 30 31 36 TO 39 UNI GY -35
 169. FLOOR LOAD
 170. *****G.L.*****
 171. YRANGE -1 1 FLOAD -7.1
 172. YRANGE -1 1 FLOAD 7.1 XRANGE 0 3 ZRANGE 0 2.5
 173. YRANGE -1 1 FLOAD 7.1 XRANGE 0 6.5 ZRANGE 13 17
 174. YRANGE -1 1 FLOAD 7.1 XRANGE 14.5 17 ZRANGE 14.5 17
 175. *****ROOF*****
 176. YRANGE 6 7 FLOAD -5.9
 177. *****
 178. JOINT LOAD
 179. *****DEAD LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 180. 113 FX -0.767 FY -13.107 FZ -1.158
 181. 101 FX -0.415 FY -13.414 FZ 1.177
 182. 114 FX -0.011 FY -18.976 FZ -1.593
 183. 102 FX -0.168 FY -18.395 FZ 1.522
 184. 115 FX 0.011 FY -18.845 FZ -1.579
 185. 103 FX -0.142 FY -18.69 FZ 1.555
 186. 116 FX 0.768 FY -13.069 FZ -1.154
 187. 104 FX 0.623 FY -13.094 FZ 1.158
 188. 126 FX -1.893 FY -33.511 FZ 5.291
 189. 122 FX -1.823 FY -31.788 FZ -5.04
 190. 127 FX -0.004 FY -25.163 FZ -3.342
 191. 123 FX -0.003 FY -23.798 FZ 3.279
 192. 128 FX 0.005 FY -29.124 FZ 2.099
 193. 124 FX 0.005 FY -27.42 FZ -1.965
 194. 129 FX 1.873 FY -34.829 FZ 7.118
 195. 125 FX 1.942 FY -33.009 FZ -6.763
 196. 130 FX 0 FY -24.136 FZ 0
 197. 118 FX 0 FY -24.847 FZ 0
 198. 131 FX 0 FY -30.634 FZ 0
 199. 119 FX 0 FY -31.576 FZ 0

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

97. 112 WEIGHT 661
 98. 113 WEIGHT 298
 99. 114 WEIGHT 543
 100. 115 WEIGHT 625
 101. 116 WEIGHT 380
 102. 118 WEIGHT 124
 103. 119 WEIGHT 358
 104. 120 WEIGHT 360
 105. 121 WEIGHT 122
 106. 122 WEIGHT 298
 107. 125 WEIGHT 301
 108. 126 WEIGHT 228
 109. 127 WEIGHT 542
 110. 128 WEIGHT 546
 111. 129 WEIGHT 230
 112. 130 WEIGHT 125
 113. 131 WEIGHT 147
 114. 132 WEIGHT 146
 115. 133 WEIGHT 125
 116. LOAD 21 EQX (EARTHQUAKE LOAD)
 117. UBC LOAD X
 118. LOAD 22 EQZ (EARTHQUAKE LOAD)
 119. UBC LOAD Z
 120. *****
 121. LOAD 1 SELFWEIGHT
 122. SELFWEIGHT Y -1
 123. *****SELFWEIGHT TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 124. JOINT LOAD
 125. 113 FX -0.154 FY -5.027 FZ -0.402
 126. 101 FX -0.088 FY -5.149 FZ 0.414
 127. 114 FX -0.006 FY -6.382 FZ -0.509
 128. 102 FX -0.04 FY -6.315 FZ 0.501
 129. 115 FX 0.006 FY -6.382 FZ -0.509
 130. 103 FX -0.029 FY -6.395 FZ 0.511
 131. 116 FX 0.154 FY -5.028 FZ -0.402
 132. 104 FX 0.122 FY -5.078 FZ 0.409
 133. 126 FX 0.159 FY -9.379 FZ 1.464
 134. 122 FX 0.2 FY -8.982 FZ -1.545
 135. 127 FX -0.001 FY -8.777 FZ 1.223
 136. 123 FX -0.001 FY -8.381 FZ -1.296
 137. 128 FX 0.001 FY -8.778 FZ 1.223
 138. 124 FX 0.001 FY -8.378 FZ -1.295
 139. 129 FX -0.163 FY -9.378 FZ 1.468
 140. 125 FX -0.163 FY -8.985 FZ -1.539
 141. 130 FX 0 FY -7.905 FZ 0
 142. 118 FX 0 FY -7.881 FZ 0
 143. 131 FX 0 FY -8.963 FZ 0
 144. 119 FX 0 FY -8.988 FZ 0
 145. 132 FX 0 FY -8.963 FZ 0
 146. 120 FX 0 FY -9.031 FZ 0

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

253. 133 FX 0 FY -12.724 FZ 0
 254. 121 FX 0 FY -13.295 FZ 0
 255. 130 FX 0 FY 0 FZ -3.021
 256. 118 FX 0 FY 0 FZ 3.235
 257. 131 FX 0 FY 0 FZ -5.267
 258. 119 FX 0 FY 0 FZ 5.192
 259. 132 FX 0 FY 0 FZ -4.211
 260. 120 FX 0 FY 0 FZ 4.31
 261. 133 FX 0 FY 0 FZ -2.663
 262. 121 FX 0 FY 0 FZ 2.933
 263. LOAD 4 (NOTIONAL HORIZONTAL LOAD X-DIR)
 264. **PH=1.5% (1.4DL+1.6LL)
 265. JOINT LOAD
 266. 101 FX 8
 267. 102 FX 14
 268. 103 FX 14
 269. 104 FX 10
 270. 105 FX 15
 271. 106 FX 16
 272. 107 FX 16
 273. 108 FX 15
 274. 109 FX 13
 275. 110 FX 19
 276. 111 FX 23
 277. 112 FX 17
 278. 113 FX 7
 279. 114 FX 13
 280. 115 FX 15
 281. 116 FX 9
 282. 118 FX 3
 283. 119 FX 8
 284. 120 FX 8
 285. 121 FX 3
 286. 122 FX 7
 287. 125 FX 7
 288. 126 FX 5
 289. 127 FX 12
 290. 128 FX 12
 291. 129 FX 5
 292. 130 FX 3
 293. 131 FX 3
 294. 132 FX 3
 295. 133 FX 3
 296. LOAD 5 (NOTIONAL HORIZONTAL LOAD (Z-DIR)
 297. **PH=1.5% (1.4DL+1.6LL)
 298. JOINT LOAD
 299. 101 FZ 8
 300. 102 FZ 14
 301. 103 FZ 14
 302. 104 FZ 10
 303. 105 FZ 15
 304. 106 FZ 16
 305. 107 FZ 16
 306. 108 FZ 15
 307. 109 FZ 13
 308. 110 FZ 19

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

200. 132 FX 0 FY -30.253 FZ 0
 201. 120 FX 0 FY -31.275 FZ 0
 202. 133 FX 0 FY -24.005 FZ 0
 203. 121 FX 0 FY -24.765 FZ 0
 204. 130 FX 0 FY 0 FZ -2.101
 205. 118 FX 0 FY 0 FZ 2.355
 206. 131 FX 0 FY 0 FZ -4.569
 207. 119 FX 0 FY 0 FZ 4.491
 208. 132 FX 0 FY 0 FZ -3.134
 209. 120 FX 0 FY 0 FZ 3.252
 210. 133 FX 0 FY 0 FZ -1.633
 211. 121 FX 0 FY 0 FZ 1.942
 212. *****
 213. LOAD 3 LL
 214. MEMBER LOAD
 215. **PROJECTION OF SLAB =0.62X10=6.2 KN/M
 216. **FLOOR LL2 = 10 KN/M^2
 217. **ROOF LL3 = 0.6 KN/M^2
 218. *****
 219. 1 TO 4 TO 14 TO 18 UNI GY -6.2
 220. 19 TO 23 36 TO 39 UNI GY -6.2
 221. FLOOR LOAD
 222. *****G.L.*****
 223. YRANGE -1 1 FLOAD -10
 224. YRANGE -1 1 FLOAD 10 XRANGE 0 3 ZRANGE 0 2.5
 225. YRANGE -1 1 FLOAD 10 XRANGE 0 6.5 ZRANGE 13.4 17
 226. YRANGE -1 1 FLOAD 10 XRANGE 14.5 17 ZRANGE 14.5 17
 227. *****ROOF*****
 228. YRANGE 6 7 FLOAD -0.6
 229. JOINT LOAD
 230. *****LIVE LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 231. 113 FX -0.029 FY 1.422 FZ 0.196
 232. 101 FX -0.006 FY 1.623 FZ -0.206
 233. 114 FX 0.008 FY 1.47 FZ 0.206
 234. 102 FX 0.007 FY 1.586 FZ -0.22
 235. 115 FX -0.008 FY 1.564 FZ 0.216
 236. 103 FX -0.004 FY 1.651 FZ -0.228
 237. 116 FX 0.03 FY 1.455 FZ 0.2
 238. 104 FX 0.028 FY 1.497 FZ -0.206
 239. 126 FX -1.415 FY -25.094 FZ 3.298
 240. 122 FX -1.43 FY -23.837 FZ -3.079
 241. 127 FX 0 FY -18.917 FZ -3.336
 242. 123 FX -0.001 FY -17.912 FZ 3.288
 243. 128 FX 0.001 FY -21.85 FZ 0.701
 244. 124 FX 0 FY -20.605 FZ -0.583
 245. 129 FX 1.418 FY -26.072 FZ 4.641
 246. 125 FX 1.402 FY -24.73 FZ -4.379
 247. 130 FX 0 FY -12.817 FZ 0
 248. 118 FX 0 FY -13.507 FZ 0
 249. 131 FX 0 FY -15.389 FZ 0
 250. 119 FX 0 FY -16.127 FZ 0
 251. 132 FX 0 FY -15.104 FZ 0
 252. 120 FX 0 FY -15.768 FZ 0

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

365. LOAD 12 WL2 (0 DEG ,INTERNAL PRESSURE +0.2)
 366. JOINT LOAD
 367. 113 FX 0.078 FY 29.442 FZ 10.688
 368. 101 FX -0.037 FY -33.735 FZ 9.067
 369. 114 FX -0.003 FY 38.854 FZ 14.015
 370. 102 FX -0.034 FY -34.865 FZ 11.868
 371. 115 FX 0.003 FY 38.696 FZ 13.999
 372. 103 FX -0.011 FY -35.049 FZ 11.89
 373. 116 FX -0.078 FY 29.399 FZ 10.684
 374. 104 FX -0.026 FY -27.587 FZ 9.198
 375. 126 FX 1.16 FY 24.46 FZ -8.189
 376. 122 FX 2.951 FY 44.451 FZ 1.92
 377. 127 FX 0 FY 16.787 FZ 0.591
 378. 123 FX 0.005 FY 34.742 FZ -7.583
 379. 128 FX -0.001 FY 20.939 FZ -5.067
 380. 124 FX -0.004 FY 38.558 FZ -2.095
 381. 129 FX -1.14 FY 25.852 FZ -10.112
 382. 125 FX -2.863 FY 45.677 FZ 3.742
 383. 130 FX 0 FY -20.623 FZ 0
 384. 118 FX 0 FY 57.188 FZ 0
 385. 131 FX 0 FY -27.716 FZ 0
 386. 119 FX 0 FY 63.741 FZ 0
 387. 132 FX 0 FY -28.099 FZ 0
 388. 120 FX 0 FY 63.15 FZ 0
 389. 133 FX 0 FY -20.758 FZ 0
 390. 121 FX 0 FY 51.057 FZ 0
 391. 130 FX 0 FY 0 FZ -14.168
 392. 118 FX 0 FY 0 FZ -20.523
 393. 131 FX 0 FY 0 FZ -15.506
 394. 119 FX 0 FY 0 FZ -27.999
 395. 132 FX 0 FY 0 FZ -17.06
 396. 120 FX 0 FY 0 FZ -26.871
 397. 133 FX 0 FY 0 FZ -14.67
 398. 121 FX 0 FY 0 FZ -19.988
 399. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 400. LOAD 13 WL3 (90 DEG ,INTERNAL SUCTION -0.3, WM&LW +0.2)
 401. JOINT LOAD
 402. 113 FX 0.011 FY 23.471 FZ 7.955
 403. 101 FX 0.022 FY 28.783 FZ -7.846
 404. 114 FX 0.01 FY 30.07 FZ 10.321
 405. 102 FX 0.021 FY 30.088 FZ -10.323
 406. 115 FX -0.01 FY 30.165 FZ 10.331
 407. 103 FX 0.006 FY 30.231 FZ -10.339
 408. 116 FX -0.011 FY 23.492 FZ 7.957
 409. 104 FX -0.001 FY 23.514 FZ -7.961
 410. 126 FX -1.459 FY -25.855 FZ 3.322
 411. 122 FX -1.487 FY -24.628 FZ -3.143
 412. 127 FX -0.001 FY -19.532 FZ -3.477
 413. 123 FX -0.002 FY -18.528 FZ 3.402
 414. 128 FX 0.001 FY -22.538 FZ 0.662
 415. 124 FX 0 FY -21.291 FZ -0.563
 416. 129 FX 1.466 FY -26.861 FZ 4.735
 417. 125 FX 1.434 FY -25.509 FZ -4.483
 418. 130 FX 0 FY -34.632 FZ 0
 419. 118 FX 0 FY -40.347 FZ 0
 420. 131 FX 0 FY -43.839 FZ 0

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

309. 111 FZ 23
 310. 112 FZ 17
 311. 113 FZ 7
 312. 114 FZ 13
 313. 115 FZ 15
 314. 116 FZ 8
 315. 118 FZ 3
 316. 119 FZ 8
 317. 120 FZ 8
 318. 121 FZ 3
 319. 122 FZ 7
 320. 125 FZ 7
 321. 126 FZ 5
 322. 127 FZ 12
 323. 128 FZ 12
 324. 129 FZ 5
 325. 130 FZ 3
 326. 131 FZ 3
 327. 132 FZ 3
 328. 133 FZ 3
 329. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 330. LOAD 11 WL1 (0 DEG ,INTERNAL SUCTION -0.3)
 331. JOINT LOAD
 332. 113 FX -0.013 FY 9.734 FZ 3.238
 333. 101 FX -0.069 FY -59.253 FZ 16.499
 334. 114 FX 0.009 FY 12.566 FZ 4.207
 335. 102 FX -0.042 FY -61.947 FZ 21.772
 336. 115 FX -0.009 FY 12.563 FZ 4.206
 337. 103 FX -0.025 FY -62.121 FZ 21.792
 338. 116 FX 0.013 FY 9.764 FZ 3.242
 339. 104 FX 0.03 FY -48.056 FZ 16.743
 340. 126 FX -1.467 FY -18.773 FZ -5.269
 341. 122 FX 1.499 FY 17.639 FZ -5.131
 342. 127 FX -0.001 FY -16.637 FZ -6.283
 343. 123 FX 0.004 FY 15.65 FZ -5.823
 344. 128 FX 0.001 FY -16.696 FZ -6.19
 345. 124 FX -0.002 FY 15.605 FZ -5.88
 346. 129 FX 1.467 FY -18.784 FZ -5.322
 347. 125 FX -1.396 FY 17.539 FZ -5.141
 348. 130 FX 0 FY -21.959 FZ 0
 349. 118 FX 0 FY 70.832 FZ 0
 350. 131 FX 0 FY -26.802 FZ 0
 351. 119 FX 0 FY 76.204 FZ 0
 352. 132 FX 0 FY -26.793 FZ 0
 353. 120 FX 0 FY 76.069 FZ 0
 354. 133 FX 0 FY -21.981 FZ 0
 355. 121 FX 0 FY 59.878 FZ 0
 356. 130 FX 0 FY 0 FZ -10.23
 357. 118 FX 0 FY 0 FZ -28.616
 358. 131 FX 0 FY 0 FZ -12.193
 359. 119 FX 0 FY 0 FZ -36.24
 360. 132 FX 0 FY 0 FZ -12.185
 361. 120 FX 0 FY 0 FZ -36.407
 362. 133 FX 0 FY 0 FZ -10.159
 363. 121 FX 0 FY 0 FZ -28.495
 364. *****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

421. 119 FX 0 FY -44.553 FZ 0
 422. 132 FX 0 FY -43.546 FZ 0
 423. 120 FX 0 FY -44.147 FZ 0
 424. 133 FX 0 FY -34.526 FZ 0
 425. 121 FX 0 FY -35.07 FZ 0
 426. 130 FX 0 FY 0 FZ -13.417
 427. 118 FX 0 FY 0 FZ 13.712
 428. 131 FX 0 FY 0 FZ -18.921
 429. 119 FX 0 FY 0 FZ 18.819
 430. 132 FX 0 FY 0 FZ -17.84
 431. 120 FX 0 FY 0 FZ 17.924
 432. 133 FX 0 FY 0 FZ -13.08
 433. 121 FX 0 FY 0 FZ 13.352
 434. ****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 435. LOAD 14 WL4 (90 DEG , INTERNAL PRESSURE +0.2, WW&LW -0.7)
 436. JOINT LOAD
 437. 113 FX 0.126 FY 41.694 FZ 15.203
 438. 101 FX 0.064 FY 51.59 FZ -14.951
 439. 114 FX -0.009 FY 55.069 FZ 19.946
 440. 102 FX 0.016 FY 54.681 FZ -19.9
 441. 115 FX 0.009 FY 54.877 FZ 19.926
 442. 103 FX 0.027 FY 54.701 FZ -19.9
 443. 116 FX -0.127 FY 41.588 FZ 15.191
 444. 104 FX -0.105 FY 41.491 FZ -15.178
 445. 126 FX 2.647 FY 45.953 FZ -7.033
 446. 122 FX 2.663 FY 43.569 FZ 6.524
 447. 127 FX 0.002 FY 34.346 FZ 5.145
 448. 123 FX 0.002 FY 32.503 FZ -5.15
 449. 128 FX -0.003 FY 39.758 FZ -2.264
 450. 124 FX -0.002 FY 37.469 FZ 1.958
 451. 129 FX -2.653 FY 47.756 FZ -9.409
 452. 125 FX -2.657 FY 45.296 FZ 8.876
 453. 130 FX 0 FY -20.826 FZ 0
 454. 118 FX 0 FY -29.653 FZ 0
 455. 131 FX 0 FY -29.621 FZ 0
 456. 119 FX 0 FY -28.271 FZ 0
 457. 132 FX 0 FY -30.132 FZ 0
 458. 120 FX 0 FY -28.832 FZ 0
 459. 133 FX 0 FY -20.962 FZ 0
 460. 121 FX 0 FY -19.888 FZ 0
 461. 130 FX 0 FY 0 FZ -15.307
 462. 118 FX 0 FY 0 FZ 15.049
 463. 131 FX 0 FY 0 FZ -16.814
 464. 119 FX 0 FY 0 FZ 16.843
 465. 132 FX 0 FY 0 FZ -18.805
 466. 120 FX 0 FY 0 FZ 18.537
 467. 133 FX 0 FY 0 FZ -16.068
 468. 121 FX 0 FY 0 FZ 15.549
 469. ****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 470. LOAD 15 WL5 (0 DEG , INTERNAL SUCTION -0.3, LEE WARD -0.7, REVERSE)
 471. JOINT LOAD
 472. 113 FX -0.067 FY -48.036 FZ -16.739
 473. 101 FX 0.005 FY 12.045 FZ -3.204
 474. 114 FX -0.009 FY -62.129 FZ -21.795
 475. 102 FX 0.013 FY 12.55 FZ -4.204
 476. 115 FX 0.009 FY -62.129 FZ -21.795

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

477. 103 FX 0 FY 12.609 FZ -4.213
 478. 116 FX 0.067 FY -48.039 FZ -16.74
 479. 104 FX 0.016 FY 9.823 FZ -3.25
 480. 126 FX 1.451 FY 18.022 FZ 5.818
 481. 122 FX -1.484 FY -18.256 FZ 4.686
 482. 127 FX 0.003 FY 16.18 FZ 6.615
 483. 123 FX -0.002 FY -16.051 FZ 5.502
 484. 128 FX -0.003 FY 16.122 FZ 6.699
 485. 124 FX 0 FY -16.108 FZ 5.431
 486. 129 FX -1.445 FY 18 FZ 5.857
 487. 125 FX 1.446 FY -18.256 FZ 4.65
 488. 130 FX 0 FY 59.36 FZ 0
 489. 118 FX 0 FY -24.708 FZ 0
 490. 131 FX 0 FY 75.628 FZ 0
 491. 119 FX 0 FY -27.439 FZ 0
 492. 132 FX 0 FY 75.635 FZ 0
 493. 120 FX 0 FY -27.392 FZ 0
 494. 133 FX 0 FY 59.365 FZ 0
 495. 121 FX 0 FY -22.553 FZ 0
 496. 130 FX 0 FY 0 FZ 28.46
 497. 118 FX 0 FY 0 FZ 10.268
 498. 131 FX 0 FY 0 FZ 36.289
 499. 119 FX 0 FY 0 FZ 12.243
 500. 132 FX 0 FY 0 FZ 36.307
 501. 120 FX 0 FY 0 FZ 12.239
 502. 133 FX 0 FY 0 FZ 28.459
 503. 121 FX 0 FY 0 FZ 10.238
 504. ****WIND LOAD TRANSFER BY STEEL STRUCTURE TO RCC STRUCTURE *****
 505. LOAD 16 WL6 (0 DEG , INTERNAL PRESSURE +0.2, LEE WARD -0.2, REVERSE)
 506. JOINT LOAD
 507. 113 FX -0.001 FY -27.449 FZ -9.18
 508. 101 FX 0.045 FY 36.52 FZ -10.52
 509. 114 FX -0.014 FY -34.803 FZ -11.861
 510. 102 FX 0.015 FY 38.577 FZ -13.982
 511. 115 FX 0.014 FY -34.948 FZ -11.876
 512. 103 FX 0.018 FY 38.593 FZ -13.983
 513. 116 FX 0 FY -27.528 FZ -9.188
 514. 104 FX -0.061 FY 29.363 FZ -10.678
 515. 126 FX 2.897 FY 46.462 FZ -1.877
 516. 122 FX 1.155 FY 22.943 FZ 7.512
 517. 127 FX 0.004 FY 36.472 FZ 8.075
 518. 123 FX 0 FY 15.713 FZ -1.041
 519. 128 FX -0.005 FY 40.622 FZ 2.389
 520. 124 FX -0.001 FY 19.521 FZ 4.405
 521. 129 FX -2.905 FY 47.846 FZ -3.705
 522. 125 FX -1.163 FY 24.265 FZ 9.314
 523. 130 FX 0 FY 50.056 FZ 0
 524. 118 FX 0 FY -27.166 FZ 0
 525. 131 FX 0 FY 62.373 FZ 0
 526. 119 FX 0 FY -27.052 FZ 0
 527. 132 FX 0 FY 61.98 FZ 0
 528. 120 FX 0 FY -27.468 FZ 0
 529. 133 FX 0 FY 49.949 FZ 0
 530. 121 FX 0 FY -20.279 FZ 0
 531. 130 FX 0 FY 0 FZ 20.165
 532. 118 FX 0 FY 0 FZ 13.929

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

533. 131 FX 0 FY 0 FZ 28.126
 534. 119 FX 0 FY 0 FZ 15.579
 535. 132 FX 0 FY 0 FZ 26.603
 536. 120 FX 0 FY 0 FZ 16.905
 537. 133 FX 0 FY 0 FZ 19.586
 538. 121 FX 0 FY 0 FZ 14.314
 539. *****
 540. ** SERVICEABILITY*
 541. *****
 542. LOAD COMB 101 1.0DL+1.0LL
 543. 1 1.0 2 1.0 3 1.0
 544. LOAD COMB 102 1.0DL+0.5LL
 545. 1 1.0 2 1.0 3 0.5
 546. LOAD COMB 103 0.8DL+0.8LL+0.8WL1
 547. 1 0.8 2 0.8 3 0.8 11 0.8
 548. LOAD COMB 104 1.0DL+1.0WL1
 549. 1 1.0 2 1.0 11 1.0
 550. LOAD COMB 105 0.8DL+0.8LL+0.8WL2
 551. 1 0.8 2 0.8 3 0.8 12 0.8
 552. LOAD COMB 106 1.0DL+1.0WL2
 553. 1 1.0 2 1.0 12 1.0
 554. LOAD COMB 107 0.8DL+0.8LL+0.8WL3
 555. 1 0.8 2 0.8 3 0.8 13 0.8
 556. LOAD COMB 108 1.0DL+1.0WL3
 557. 1 1.0 2 1.0 13 1.0
 558. LOAD COMB 109 0.8DL+0.8LL+0.8WL4
 559. 1 0.8 2 0.8 3 0.8 14 0.8
 560. LOAD COMB 110 1.0DL+1.0WL4
 561. 1 1.0 2 1.0 14 1.0
 562. LOAD COMB 111 0.8DL+0.8LL+0.8WL5
 563. 1 0.8 2 0.8 3 0.8 15 0.8
 564. LOAD COMB 112 1.0DL+1.0WL4
 565. 1 1.0 2 1.0 15 1.0
 566. LOAD COMB 113 0.8DL+0.8LL+0.8WL6
 567. 1 0.8 2 0.8 3 0.8 16 0.8
 568. LOAD COMB 114 1.0DL+1.0WL6
 569. 1 1.0 2 1.0 16 1.0
 570. *****EQX & EQZ *****
 571. **DL+EQ/1.4 = 1.0DL+0.72EQX+0.22 EQZ, EQ=EQX+0.3EQZ
 572. **0.9 DL+EQ/1.4 =0.9 DL+0.72 EQX+0.22EQZ
 573. **DL 0.75(LL+EQ/1.4) =1.0 DL+0.75 LL+0.54 EQX+0.17EQZ
 574. *****
 575. LOAD COMB 115 DL+0.72EQX+0.22EQZ
 576. 1 1.0 2 1.0 21 0.72 22 0.22
 577. LOAD COMB 116 DL+0.72EQX-0.22EQZ
 578. 1 1.0 2 1.0 21 0.72 22 -0.22
 579. LOAD COMB 117 DL-0.72EQX+0.22EQZ
 580. 1 1.0 2 1.0 21 -0.72 22 0.22
 581. LOAD COMB 118 DL-0.72EQX+0.22EQZ
 582. 1 1.0 2 1.0 21 -0.72 22 -0.22
 583. LOAD COMB 119 DL+0.72EQX+0.22EQX
 584. 1 1.0 2 1.0 22 0.72 21 0.22
 585. LOAD COMB 120 DL+0.72EQZ-0.22EQX
 586. 1 1.0 2 1.0 22 0.72 21 -0.22
 587. LOAD COMB 121 DL-0.72EQZ+0.22EQX
 588. 1 1.0 2 1.0 22 -0.72 21 0.22

BLDG-27-RCC (HEAT EXCHANGER FACILITY)

589. LOAD COMB 122 DL-0.72EQZ-0.22EQX
 590. 1 1.0 2 1.0 22 -0.72 21 -0.22
 591. LOAD COMB 123 0.9DL+0.72EQX+0.22EQZ
 592. 1 0.9 2 0.9 21 0.72 22 0.22
 593. LOAD COMB 124 0.9DL+0.72EQX-0.22EQZ
 594. 1 0.9 2 0.9 21 0.72 22 -0.22
 595. LOAD COMB 125 0.9DL-0.72EQX+0.22EQZ
 596. 1 0.9 2 0.9 21 -0.72 22 0.22
 597. LOAD COMB 126 0.9DL-0.72EQX+0.22EQZ
 598. 1 0.9 2 0.9 21 -0.72 22 -0.22
 599. LOAD COMB 127 0.9DL+0.72EQZ+0.22EQX
 600. 1 0.9 2 0.9 22 0.72 21 0.22
 601. LOAD COMB 128 0.9DL+0.72EQZ-0.22EQX
 602. 1 0.9 2 0.9 22 0.72 21 -0.22
 603. LOAD COMB 129 0.9DL-0.72EQZ+0.22EQX
 604. 1 0.9 2 0.9 22 -0.72 21 0.22
 605. LOAD COMB 130 0.9DL-0.72EQZ-0.22EQX
 606. 1 0.9 2 0.9 22 -0.72 21 -0.22
 607. LOAD COMB 131 1.0DL+0.75LL+0.54EQX+0.17 EQZ
 608. 1 1.0 2 1.0 3 0.75 21 0.54 22 0.17
 609. LOAD COMB 132 1.0DL+0.75LL+0.54EQX-0.17 EQZ
 610. 1 1.0 2 1.0 3 0.75 21 0.54 22 -0.17
 611. LOAD COMB 133 1.0DL+0.75LL-0.54EQX+0.17 EQZ
 612. 1 1.0 2 1.0 3 0.75 21 -0.54 22 0.17
 613. LOAD COMB 134 1.0DL+0.75LL-0.54EQX-0.17 EQZ
 614. 1 1.0 2 1.0 3 0.75 21 -0.54 22 -0.17
 615. LOAD COMB 135 1.0DL+0.75LL+0.54EQZ+0.17 EQX
 616. 1 1.0 2 1.0 3 0.75 22 0.54 21 0.17
 617. LOAD COMB 136 1.0DL+0.75LL+0.54EQZ-0.17 EQX
 618. 1 1.0 2 1.0 3 0.75 22 0.54 21 -0.17
 619. LOAD COMB 137 1.0DL+0.75LL-0.54EQZ+0.17 EQX
 620. 1 1.0 2 1.0 3 0.75 22 -0.54 21 0.17
 621. LOAD COMB 138 1.0DL+0.75LL-0.54EQZ-0.17 EQX
 622. 1 1.0 2 1.0 3 0.75 22 -0.54 21 -0.17
 623. *****
 624. ** CONCRETE DESIGN
 625. *****
 626. LOAD COMB 200 1.4DL+1.6LL
 627. 1 1.4 2 1.4 3 1.6
 628. *****WL1*****
 629. LOAD COMB 201 1.2DL+1.2LL+1.2WL1
 630. 1 1.2 2 1.2 3 1.2 11 1.2
 631. LOAD COMB 202 1.4DL+1.4WL1
 632. 1 1.4 2 1.4 11 1.4
 633. LOAD COMB 203 1DL+1.4WL1
 634. 1 1.0 2 1.0 11 1.4
 635. *****WL2*****
 636. LOAD COMB 204 1.2DL+1.2LL+1.2WL2
 637. 1 1.2 2 1.2 3 1.2 12 1.2
 638. LOAD COMB 205 1.4DL+1.4WL2
 639. 1 1.4 2 1.4 12 1.4
 640. LOAD COMB 206 1DL+1.4WL2
 641. 1 1.0 2 1.0 12 1.4
 642. *****WL3*****
 643. LOAD COMB 207 1.2DL+1.2LL+1.2WL3
 644. 1 1.2 2 1.2 3 1.2 13 1.2

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BLDG-27-RCC (HEAT EXCHANGER FACILITY )

645. LOAD COMB 208 1.4DL+1.4WL3
646. 1 1.4 2 1.4 13 1.4
647. LOAD COMB 209 1DL+1.4WL3
648. 1 1.0 2 1.0 13 1.4
649. *****WL4*****
650. LOAD COMB 210 1.2DL+1.2LL+1.2WL4
651. 1 1.2 2 1.2 3 1.2 14 1.2
652. LOAD COMB 211 1.4DL+1.4WL4
653. 1 1.4 2 1.4 14 1.4
654. LOAD COMB 212 1DL+1.4WL4
655. 1 1.0 2 1.0 14 1.4
656. *****WL5*****
657. LOAD COMB 213 1.2DL+1.2LL+1.2WL5
658. 1 1.2 2 1.2 3 1.2 15 1.2
659. LOAD COMB 214 1.4DL+1.4WL5
660. 1 1.4 2 1.4 15 1.4
661. LOAD COMB 215 1DL+1.4WL5
662. 1 1.0 2 1.0 15 1.4
663. *****WL6*****
664. LOAD COMB 216 1.2DL+1.2LL+1.2WL6
665. 1 1.2 2 1.2 3 1.2 16 1.2
666. LOAD COMB 217 1.4DL+1.4WL6
667. 1 1.4 2 1.4 16 1.4
668. LOAD COMB 218 1DL+1.4WL6
669. 1 1.0 2 1.0 16 1.4
670. *****EQX & EQZ *****
671. LOAD COMB 219 1.32DL+1.1LL+1.1EQX+0.33 EQZ
672. 1 1.32 2 1.32 3 1.1 21 1.1 22 0.33
673. LOAD COMB 220 1.32DL+1.1LL+1.1EQX-0.33 EQZ
674. 1 1.32 2 1.32 3 1.1 21 1.1 22 -0.33
675. LOAD COMB 221 1.32DL+1.1LL-1.1EQX+0.33 EQZ
676. 1 1.32 2 1.32 3 1.1 21 -1.1 22 0.33
677. LOAD COMB 222 1.32DL+1.1LL-1.1EQX-0.33 EQZ
678. 1 1.32 2 1.32 3 1.1 21 -1.1 22 -0.33
679. LOAD COMB 223 1.32DL+1.1LL+1.1EQZ+0.33 EQX
680. 1 1.32 2 1.32 3 1.1 22 1.1 21 0.33
681. LOAD COMB 224 1.32DL+1.1LL+1.1EQZ-0.33 EQX
682. 1 1.32 2 1.32 3 1.1 22 1.1 21 -0.33
683. LOAD COMB 225 1.32DL+1.1LL-1.1EQZ+0.33 EQX
684. 1 1.32 2 1.32 3 1.1 22 -1.1 21 0.33
685. LOAD COMB 226 1.32DL+1.1LL-1.1EQZ-0.33 EQX
686. 1 1.32 2 1.32 3 1.1 22 -1.1 21 -0.33
687. LOAD COMB 227 0.99DL+1.1EQX+0.33EQZ
688. 1 0.99 2 0.99 21 1.1 22 0.33
689. LOAD COMB 228 0.99DL+1.1EQX-0.33EQZ
690. 1 0.99 2 0.99 21 1.1 22 -0.33
691. LOAD COMB 229 0.99DL-1.1EQX+0.33EQZ
692. 1 0.99 2 0.99 21 -1.1 22 0.33
693. LOAD COMB 230 0.99DL-1.1EQX-0.33EQZ
694. 1 0.99 2 0.99 21 -1.1 22 -0.33
695. LOAD COMB 231 0.99DL+1.1EQZ+0.33EQX
696. 1 0.99 2 0.99 22 1.1 21 0.33
697. LOAD COMB 232 0.99DL+1.1EQZ-0.33EQX
698. 1 0.99 2 0.99 22 1.1 21 -0.33
699. LOAD COMB 233 0.99DL-1.1EQZ+0.33EQX
700. 1 0.99 2 0.99 22 -1.1 21 0.33

BLDG-27-RCC (HEAT EXCHANGER FACILITY )

701. LOAD COMB 234 0.99DL-1.1EQZ-0.33EQX
702. 1 0.99 2 0.99 22 -1.1 21 -0.33
703. LOAD COMB 235 1.0DL+1.0LL+1.0PHX
704. 1 1.0 2 1.0 3 1.0 4 1.0
705. LOAD COMB 236 1.0DL+1.0LL+1.0PHZ
706. 1 1.0 2 1.0 3 1.0 5 1.0
707. *****DEFLECTION*****
708. LOAD COMB 301 SELF+DL
709. 1 1.0 2 1.0
710. LOAD COMB 302 LL
711. 3 1.0
712. LOAD COMB 303 0.8 (LL+WL1)
713. 3 0.8 11 0.8
714. LOAD COMB 304 0.8 (LL+WL2)
715. 3 0.8 12 0.8
716. LOAD COMB 305 0.8 (LL+WL3)
717. 3 0.8 13 0.8
718. LOAD COMB 306 0.8 (LL+WL4)
719. 3 0.8 14 0.8
720. LOAD COMB 307 0.8 (LL+WL5)
721. 3 0.8 15 0.8
722. LOAD COMB 308 0.8 (LL+WL6)
723. 3 0.8 16 0.8
724. LOAD COMB 309 0.8 (LL+EQX/1.4+0.3EQZ/1.4)
725. 3 0.8 21 0.57 22 0.17
726. LOAD COMB 310 0.8 (LL+EQX/1.4-0.3EQZ/1.4)
727. 3 0.8 21 0.57 22 -0.17
728. LOAD COMB 311 0.8 (LL-EQX/1.4+0.3EQZ/1.4)
729. 3 0.8 21 -0.57 22 0.17
730. LOAD COMB 312 0.8 (LL-EQX/1.4-0.3EQZ/1.4)
731. 3 0.8 21 -0.57 22 -0.17
732. LOAD COMB 313 0.8 (LL+EQZ/1.4+0.3EQX/1.4)
733. 3 0.8 22 0.57 21 0.17
734. LOAD COMB 314 0.8 (LL+EQZ/1.4-0.3EQX/1.4)
735. 3 0.8 22 0.57 21 -0.17
736. LOAD COMB 315 0.8 (LL-EQZ/1.4+0.3EQX/1.4)
737. 3 0.8 21 -0.57 22 0.17
738. LOAD COMB 316 0.8 (LL-EQZ/1.4-0.3EQX/1.4)
739. 3 0.8 21 -0.57 22 -0.17
740. PERFORM ANALYSIS

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 60/ 96/ 16
ORIGINAL/FINAL BAND-WIDTH= 33/ 12/ 66 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 312
SIZE OF STIFFNESS MATRIX = 21 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.5/ 22160.8 MB
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**WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS & CHANGE.

ZERO STIFFNESS IN DIRECTION 4 AT JOINT 157 EQN.NO. 4
LOADS APPLIED OR DISTRIBUTED HERE FROM ELEMENTS WILL BE IGNORED.
THIS MAY BE DUE TO ALL MEMBERS AT THIS JOINT BEING RELEASED OR
EFFECTIVELY RELEASED IN THIS DIRECTION.

ZERO STIFFNESS IN DIRECTION 4 AT JOINT 158 EQN.NO. 22

*

* X DIRECTION : Ta = 0.429 Tb = 0.811 Tuser = 0.514 *

* T = 0.514, LOAD FACTOR = 1.000 *

* UBC TYPE = 97 *

* UBC FACTOR V = 0.1571 x 12268.00 = 1927.83 KN *

*

*

* Z DIRECTION : Ta = 0.429 Tb = 0.796 Tuser = 0.514 *

* T = 0.514, LOAD FACTOR = 1.000 *

* UBC TYPE = 97 *

* UBC FACTOR V = 0.1571 x 12268.00 = 1927.83 KN *

*

741. LOAD LIST 101 TO 138

742. PRINT SUPPORT REACTION

□ SUPPORT REACTION

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
1	101	34.73	536.15	11.57	0.00	0.00	0.00
	102	29.73	493.99	10.80	0.00	0.00	0.00
	103	27.67	446.99	11.53	0.00	0.00	0.00
	104	24.58	474.43	12.86	0.00	0.00	0.00
	105	27.73	432.60	10.87	0.00	0.00	0.00
	106	24.65	456.45	12.04	0.00	0.00	0.00
	107	27.76	437.43	9.29	0.00	0.00	0.00
	108	24.69	462.48	10.06	0.00	0.00	0.00
	109	27.82	411.91	9.54	0.00	0.00	0.00
	110	24.77	430.58	10.38	0.00	0.00	0.00
	111	27.90	412.06	6.82	0.00	0.00	0.00
	112	24.86	430.76	6.98	0.00	0.00	0.00
	113	27.90	400.41	7.83	0.00	0.00	0.00
	114	24.87	416.20	8.23	0.00	0.00	0.00
	115	20.55	299.01	-0.90	0.00	0.00	0.00
	116	20.06	395.15	23.49	0.00	0.00	0.00
	117	29.39	508.53	-3.44	0.00	0.00	0.00
	118	28.90	604.68	20.95	0.00	0.00	0.00
	119	24.17	262.51	-29.50	0.00	0.00	0.00
	120	26.87	326.53	-30.28	0.00	0.00	0.00
	121	22.57	577.15	50.32	0.00	0.00	0.00
	122	25.27	641.17	49.54	0.00	0.00	0.00
	123	18.07	253.82	-1.90	0.00	0.00	0.00
	124	17.59	349.96	22.49	0.00	0.00	0.00
	125	26.91	463.35	-4.45	0.00	0.00	0.00
	126	26.42	559.49	19.94	0.00	0.00	0.00
	127	21.70	217.32	-30.50	0.00	0.00	0.00
	128	24.40	281.34	-31.28	0.00	0.00	0.00
	129	20.10	531.97	49.32	0.00	0.00	0.00
	130	22.80	595.99	48.54	0.00	0.00	0.00
	131	29.10	399.35	2.71	0.00	0.00	0.00
	132	28.72	473.64	21.56	0.00	0.00	0.00
	133	35.73	556.50	0.81	0.00	0.00	0.00
	134	35.35	630.79	19.65	0.00	0.00	0.00
	135	31.78	372.34	-18.45	0.00	0.00	0.00
	136	33.87	421.81	-19.05	0.00	0.00	0.00
	137	30.58	608.33	41.42	0.00	0.00	0.00
	138	32.67	657.80	40.82	0.00	0.00	0.00
2	101	-5.30	1031.04	-5.33	0.00	0.00	0.00
	102	-5.09	916.99	-8.41	0.00	0.00	0.00
	103	-4.66	826.44	-1.97	0.00	0.00	0.00
	104	-5.39	804.93	-8.63	0.00	0.00	0.00
	105	-4.55	803.30	-2.11	0.00	0.00	0.00
	106	-5.25	776.01	-8.81	0.00	0.00	0.00
	107	-4.25	843.58	-4.84	0.00	0.00	0.00
	108	-4.88	826.36	-12.22	0.00	0.00	0.00
	109	-4.23	790.96	-2.25	0.00	0.00	0.00
	110	-4.86	760.58	-8.99	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
111	-3.82	823.99	-7.04	0.00	0.00	0.00	0.00
112	-4.35	801.88	-14.96	0.00	0.00	0.00	0.00
113	-3.91	794.40	-4.05	0.00	0.00	0.00	0.00
114	-4.46	764.89	-11.23	0.00	0.00	0.00	0.00
115	-78.07	768.71	-39.01	0.00	0.00	0.00	0.00
116	-79.96	852.47	11.12	0.00	0.00	0.00	0.00
117	70.22	753.39	-34.12	0.00	0.00	0.00	0.00
118	68.33	837.15	16.01	0.00	0.00	0.00	0.00
119	-24.44	668.21	-94.27	0.00	0.00	0.00	0.00
120	20.87	663.53	-92.78	0.00	0.00	0.00	0.00
121	-30.61	942.33	69.78	0.00	0.00	0.00	0.00
122	14.70	937.65	71.28	0.00	0.00	0.00	0.00
123	-77.58	688.42	-37.86	0.00	0.00	0.00	0.00
124	-79.47	772.18	12.27	0.00	0.00	0.00	0.00
125	70.70	673.09	-32.97	0.00	0.00	0.00	0.00
126	68.82	756.85	17.16	0.00	0.00	0.00	0.00
127	-23.95	587.92	-93.12	0.00	0.00	0.00	0.00
128	21.36	583.23	-91.63	0.00	0.00	0.00	0.00
129	-30.12	862.04	70.93	0.00	0.00	0.00	0.00
130	15.19	857.36	72.43	0.00	0.00	0.00	0.00
131	-60.07	947.40	-28.07	0.00	0.00	0.00	0.00
132	-61.53	1012.12	10.66	0.00	0.00	0.00	0.00
133	51.14	935.91	-24.40	0.00	0.00	0.00	0.00
134	49.69	1000.63	14.33	0.00	0.00	0.00	0.00
135	-20.39	873.03	-68.97	0.00	0.00	0.00	0.00
136	14.62	869.41	-67.81	0.00	0.00	0.00	0.00
137	-25.01	1078.62	54.07	0.00	0.00	0.00	0.00
138	10.00	1075.00	55.23	0.00	0.00	0.00	0.00
101	40.99	1040.89	-2.50	0.00	0.00	0.00	0.00
102	34.04	920.45	-8.45	0.00	0.00	0.00	0.00
103	31.77	834.26	-1.63	0.00	0.00	0.00	0.00
104	25.82	801.93	-13.92	0.00	0.00	0.00	0.00
105	31.95	810.83	-0.75	0.00	0.00	0.00	0.00
106	26.04	772.65	-12.83	0.00	0.00	0.00	0.00
107	32.86	851.66	-2.80	0.00	0.00	0.00	0.00
108	27.18	823.69	-15.39	0.00	0.00	0.00	0.00
109	32.70	798.22	1.26	0.00	0.00	0.00	0.00
110	26.98	756.89	-10.32	0.00	0.00	0.00	0.00
111	33.81	832.06	-3.26	0.00	0.00	0.00	0.00
112	28.37	799.19	-15.96	0.00	0.00	0.00	0.00
113	33.48	801.88	0.40	0.00	0.00	0.00	0.00
114	27.96	761.47	-11.38	0.00	0.00	0.00	0.00
115	-141.53	730.13	-34.83	0.00	0.00	0.00	0.00
116	-146.03	810.45	15.58	0.00	0.00	0.00	0.00
117	200.22	789.55	-44.35	0.00	0.00	0.00	0.00
118	195.72	869.87	6.05	0.00	0.00	0.00	0.00
119	-17.74	659.49	-95.41	0.00	0.00	0.00	0.00
120	86.68	677.65	-98.33	0.00	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
121	-32.49	922.35	69.55	0.00	0.00	0.00	0.00
122	71.93	940.51	66.64	0.00	0.00	0.00	0.00
123	-144.23	650.13	-33.39	0.00	0.00	0.00	0.00
124	-148.74	730.45	17.02	0.00	0.00	0.00	0.00
125	197.51	709.55	-42.92	0.00	0.00	0.00	0.00
126	193.01	789.87	7.49	0.00	0.00	0.00	0.00
127	-20.45	579.49	-93.98	0.00	0.00	0.00	0.00
128	83.97	597.65	-96.89	0.00	0.00	0.00	0.00
129	-35.20	842.35	70.99	0.00	0.00	0.00	0.00
130	69.22	860.51	68.08	0.00	0.00	0.00	0.00
131	-88.90	927.35	-21.38	0.00	0.00	0.00	0.00
132	-92.38	989.42	17.57	0.00	0.00	0.00	0.00
133	167.41	971.92	-28.52	0.00	0.00	0.00	0.00
134	163.93	1033.98	10.43	0.00	0.00	0.00	0.00
135	2.70	875.08	-66.21	0.00	0.00	0.00	0.00
136	83.39	889.11	-68.46	0.00	0.00	0.00	0.00
137	-8.36	1072.22	57.51	0.00	0.00	0.00	0.00
138	72.33	1086.25	55.26	0.00	0.00	0.00	0.00
101	-72.66	586.46	46.12	0.00	0.00	0.00	0.00
102	-60.47	521.89	38.92	0.00	0.00	0.00	0.00
103	-58.83	485.70	39.75	0.00	0.00	0.00	0.00
104	-49.15	478.00	35.27	0.00	0.00	0.00	0.00
105	-58.76	472.16	39.18	0.00	0.00	0.00	0.00
106	-49.07	461.07	34.57	0.00	0.00	0.00	0.00
107	-58.04	477.65	37.16	0.00	0.00	0.00	0.00
108	-48.17	467.93	32.04	0.00	0.00	0.00	0.00
109	-58.27	452.89	37.98	0.00	0.00	0.00	0.00
110	-48.45	436.98	33.07	0.00	0.00	0.00	0.00
111	-57.44	453.50	33.33	0.00	0.00	0.00	0.00
112	-47.42	437.75	27.26	0.00	0.00	0.00	0.00
113	-57.72	441.95	35.13	0.00	0.00	0.00	0.00
114	-47.77	423.31	29.50	0.00	0.00	0.00	0.00
115	-139.83	542.53	10.68	0.00	0.00	0.00	0.00
116	-142.60	629.29	47.28	0.00	0.00	0.00	0.00
117	46.04	285.35	16.15	0.00	0.00	0.00	0.00
118	43.26	372.12	52.75	0.00	0.00	0.00	0.00
119	-72.14	354.63	-29.01	0.00	0.00	0.00	0.00
120	-15.35	276.05	-27.34	0.00	0.00	0.00	0.00
121	-81.22	638.60	90.77	0.00	0.00	0.00	0.00
122	-24.42	560.02	92.44	0.00	0.00	0.00	0.00
123	-135.00	496.79	7.51	0.00	0.00	0.00	0.00
124	-137.77	583.56	44.11	0.00	0.00	0.00	0.00
125	50.87	239.62	12.98	0.00	0.00	0.00	0.00
126	48.09	326.39	49.58	0.00	0.00	0.00	0.00
127	-67.31	308.90	-32.18	0.00	0.00	0.00	0.00
128	-10.52	230.32	-30.51	0.00	0.00	0.00	0.00
129	-76.39	592.87	87.60	0.00	0.00	0.00	0.00
130	-19.60	514.29	89.27	0.00	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
131	-135.19	617.09	26.33	0.00	0.00	0.00	0.00
132	-137.34	684.14	54.61	0.00	0.00	0.00	0.00
133	4.20	424.21	30.43	0.00	0.00	0.00	0.00
134	2.06	491.26	58.71	0.00	0.00	0.00	0.00
135	-85.10	478.05	-3.04	0.00	0.00	0.00	0.00
136	-41.22	417.32	-1.75	0.00	0.00	0.00	0.00
137	-91.91	691.02	86.79	0.00	0.00	0.00	0.00
138	-48.03	630.30	88.08	0.00	0.00	0.00	0.00
5	101	37.44	1012.38	18.58	0.00	0.00	0.00
102	22.33	878.54	14.36	0.00	0.00	0.00	0.00
103	31.23	779.24	29.70	0.00	0.00	0.00	0.00
104	8.81	706.39	28.69	0.00	0.00	0.00	0.00
105	32.06	757.46	26.14	0.00	0.00	0.00	0.00
106	9.86	679.16	24.24	0.00	0.00	0.00	0.00
107	29.04	832.87	14.73	0.00	0.00	0.00	0.00
108	6.08	773.43	9.98	0.00	0.00	0.00	0.00
109	31.50	770.20	15.25	0.00	0.00	0.00	0.00
110	9.16	695.08	10.63	0.00	0.00	0.00	0.00
111	28.69	840.70	-0.04	0.00	0.00	0.00	0.00
112	5.64	783.21	-8.49	0.00	0.00	0.00	0.00
113	30.29	800.22	4.07	0.00	0.00	0.00	0.00
114	7.65	732.62	-3.35	0.00	0.00	0.00	0.00
115	5.59	687.72	-32.36	0.00	0.00	0.00	0.00
116	7.11	640.20	60.67	0.00	0.00	0.00	0.00
117	7.34	849.22	-40.38	0.00	0.00	0.00	0.00
118	8.85	801.70	52.66	0.00	0.00	0.00	0.00
119	4.48	797.81	-140.87	0.00	0.00	0.00	0.00
120	5.01	847.15	-143.32	0.00	0.00	0.00	0.00
121	9.44	642.27	163.62	0.00	0.00	0.00	0.00
122	9.97	691.62	161.17	0.00	0.00	0.00	0.00
123	4.87	613.25	-33.38	0.00	0.00	0.00	0.00
124	6.39	565.73	59.66	0.00	0.00	0.00	0.00
125	6.62	774.75	-41.40	0.00	0.00	0.00	0.00
126	8.13	727.23	51.64	0.00	0.00	0.00	0.00
127	3.76	723.33	-141.89	0.00	0.00	0.00	0.00
128	4.29	772.68	-144.34	0.00	0.00	0.00	0.00
129	8.72	567.80	162.60	0.00	0.00	0.00	0.00
130	9.25	617.15	160.15	0.00	0.00	0.00	0.00
131	28.65	903.26	-16.47	0.00	0.00	0.00	0.00
132	29.82	866.53	55.43	0.00	0.00	0.00	0.00
133	29.96	1024.38	-22.48	0.00	0.00	0.00	0.00
134	31.13	987.66	49.41	0.00	0.00	0.00	0.00
135	27.82	984.72	-96.76	0.00	0.00	0.00	0.00
136	28.23	1022.85	-98.66	0.00	0.00	0.00	0.00
137	31.54	868.07	131.60	0.00	0.00	0.00	0.00
138	31.95	906.20	129.71	0.00	0.00	0.00	0.00
6	101	-14.20	719.28	26.87	0.00	0.00	0.00
102	-10.37	526.82	20.38	0.00	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
103	-11.32	569.39	40.51	0.00	0.00	0.00	0.00
104	-6.49	326.83	37.65	0.00	0.00	0.00	0.00
105	-11.34	571.64	36.02	0.00	0.00	0.00	0.00
106	-6.52	329.64	32.03	0.00	0.00	0.00	0.00
107	-11.34	574.63	21.69	0.00	0.00	0.00	0.00
108	-6.51	333.38	14.13	0.00	0.00	0.00	0.00
109	-11.38	576.67	21.30	0.00	0.00	0.00	0.00
110	-6.56	335.93	13.64	0.00	0.00	0.00	0.00
111	-11.40	581.50	2.44	0.00	0.00	0.00	0.00
112	-6.59	341.96	-9.94	0.00	0.00	0.00	0.00
113	-11.41	581.24	6.58	0.00	0.00	0.00	0.00
114	-6.60	341.64	-4.76	0.00	0.00	0.00	0.00
115	-103.48	357.14	-38.42	0.00	0.00	0.00	0.00
116	-103.49	320.91	65.98	0.00	0.00	0.00	0.00
117	90.42	347.82	-38.21	0.00	0.00	0.00	0.00
118	90.40	311.60	66.19	0.00	0.00	0.00	0.00
119	-36.14	395.07	-156.98	0.00	0.00	0.00	0.00
120	23.11	392.22	-156.91	0.00	0.00	0.00	0.00
121	-36.18	276.51	184.69	0.00	0.00	0.00	0.00
122	23.07	273.67	184.75	0.00	0.00	0.00	0.00
123	-102.82	323.70	-39.81	0.00	0.00	0.00	0.00
124	-102.83	287.48	64.59	0.00	0.00	0.00	0.00
125	91.07	314.38	-39.60	0.00	0.00	0.00	0.00
126	91.06	278.16	64.80	0.00	0.00	0.00	0.00
127	-35.48	361.63	-158.37	0.00	0.00	0.00	0.00
128	23.76	358.78	-158.30	0.00	0.00	0.00	0.00
129	-35.52	243.08	183.30	0.00	0.00	0.00	0.00
130	23.72	240.23	183.36	0.00	0.00	0.00	0.00
131	-84.99	640.54	-16.79	0.00	0.00	0.00	0.00
132	-85.00	612.55	63.88	0.00	0.00	0.00	0.00
133	60.43	633.55	-16.63	0.00	0.00	0.00	0.00
134	60.42	605.56	64.04	0.00	0.00	0.00	0.00
135	-35.16	668.61	-104.52	0.00	0.00	0.00	0.00
136	10.62	666.41	-104.47	0.00	0.00	0.00	0.00
137	-35.19	579.69	151.73	0.00	0.00	0.00	0.00
138	10.59	577.49	151.78	0.00	0.00	0.00	0.00
7	101	28.86	709.83	38.37	0.00	0.00	0.00
102	19.76	516.68	29.59	0.00	0.00	0.00	0.00
103	23.65	562.14	56.37	0.00	0.00	0.00	0.00
104	11.35	316.38	52.91	0.00	0.00	0.00	0.00
105	23.91	564.24	50.55	0.00	0.00	0.00	0.00
106	11.69	319.00	45.63	0.00	0.00	0.00	0.00
107	22.78	567.03	31.06	0.00	0.00	0.00	0.00
108	10.27	322.50	21.28	0.00	0.00	0.00	0.00
109	23.62	569.01	30.45	0.00	0.00	0.00	0.00
110	11.32	324.97	20.51	0.00	0.00	0.00	0.00
111	22.52	573.70	4.88	0.00	0.00	0.00	0.00
112	9.94	330.84	-11.45	0.00	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
113	23.09	573.50	10.19	0.00	0.00	0.00	0.00
114	10.66	330.58	-4.81	0.00	0.00	0.00	0.00
115	-168.97	315.27	-44.32	0.00	0.00	0.00	0.00
116	-168.19	282.08	89.74	0.00	0.00	0.00	0.00
117	189.50	365.00	-48.11	0.00	0.00	0.00	0.00
118	190.28	331.80	85.96	0.00	0.00	0.00	0.00
119	-45.39	370.25	-197.98	0.00	0.00	0.00	0.00
120	64.14	385.45	-199.14	0.00	0.00	0.00	0.00
121	-42.84	261.63	240.77	0.00	0.00	0.00	0.00
122	66.69	276.82	239.62	0.00	0.00	0.00	0.00
123	-170.04	282.92	-46.40	0.00	0.00	0.00	0.00
124	-169.26	249.72	87.66	0.00	0.00	0.00	0.00
125	188.43	332.64	-50.19	0.00	0.00	0.00	0.00
126	189.21	299.45	83.87	0.00	0.00	0.00	0.00
127	-46.45	337.90	-200.06	0.00	0.00	0.00	0.00
128	63.08	353.09	-201.22	0.00	0.00	0.00	0.00
129	-43.90	229.27	238.69	0.00	0.00	0.00	0.00
130	65.63	244.47	237.53	0.00	0.00	0.00	0.00
131	-110.42	607.43	-16.40	0.00	0.00	0.00	0.00
132	-109.82	581.78	87.20	0.00	0.00	0.00	0.00
133	158.43	644.73	-19.24	0.00	0.00	0.00	0.00
134	159.04	619.08	84.36	0.00	0.00	0.00	0.00
135	-18.97	648.12	-130.11	0.00	0.00	0.00	0.00
136	65.67	659.86	-131.00	0.00	0.00	0.00	0.00
137	-17.05	566.65	198.96	0.00	0.00	0.00	0.00
138	67.58	578.39	198.07	0.00	0.00	0.00	0.00
8	101	-43.82	1008.24	27.79	0.00	0.00	0.00
102	-24.73	870.40	22.40	0.00	0.00	0.00	0.00
103	-36.04	776.89	32.89	0.00	0.00	0.00	0.00
104	-6.87	695.42	30.34	0.00	0.00	0.00	0.00
105	-37.14	752.99	30.63	0.00	0.00	0.00	0.00
106	-8.25	665.55	27.51	0.00	0.00	0.00	0.00
107	-33.99	830.70	22.26	0.00	0.00	0.00	0.00
108	-4.31	762.68	17.05	0.00	0.00	0.00	0.00
109	-36.90	764.33	22.97	0.00	0.00	0.00	0.00
110	-7.95	679.73	17.93	0.00	0.00	0.00	0.00
111	-34.07	836.74	11.22	0.00	0.00	0.00	0.00
112	-4.41	770.23	3.25	0.00	0.00	0.00	0.00
113	-35.86	794.54	14.39	0.00	0.00	0.00	0.00
114	-6.65	717.49	7.22	0.00	0.00	0.00	0.00
115	-41.95	852.50	-20.74	0.00	0.00	0.00	0.00
116	-42.18	810.38	46.49	0.00	0.00	0.00	0.00
117	30.89	654.73	-12.46	0.00	0.00	0.00	0.00
118	30.66	612.61	54.77	0.00	0.00	0.00	0.00
119	-16.40	831.69	-94.25	0.00	0.00	0.00	0.00
120	5.86	771.26	-91.72	0.00	0.00	0.00	0.00
121	-17.15	693.85	125.76	0.00	0.00	0.00	0.00
122	5.11	633.42	128.29	0.00	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
123	-41.38	779.24	-22.44	0.00	0.00	0.00	0.00
124	-41.61	737.13	44.79	0.00	0.00	0.00	0.00
125	31.45	581.48	-14.16	0.00	0.00	0.00	0.00
126	31.22	539.36	53.07	0.00	0.00	0.00	0.00
127	-15.83	758.43	-95.96	0.00	0.00	0.00	0.00
128	6.42	698.01	-93.43	0.00	0.00	0.00	0.00
129	-16.58	620.60	124.06	0.00	0.00	0.00	0.00
130	5.67	560.17	126.59	0.00	0.00	0.00	0.00
131	-61.50	1029.76	-3.98	0.00	0.00	0.00	0.00
132	-61.68	997.21	47.96	0.00	0.00	0.00	0.00
133	-6.87	881.43	2.23	0.00	0.00	0.00	0.00
134	-7.05	848.89	54.18	0.00	0.00	0.00	0.00
135	-42.59	1014.36	-58.39	0.00	0.00	0.00	0.00
136	-25.40	967.66	-56.43	0.00	0.00	0.00	0.00
137	-43.16	910.98	106.62	0.00	0.00	0.00	0.00
138	-25.96	864.29	108.58	0.00	0.00	0.00	0.00
9	101	25.16	870.72	-19.10	0.00	0.00	0.00
102	17.74	758.30	-14.64	0.00	0.00	0.00	0.00
103	20.09	724.92	-8.71	0.00	0.00	0.00	0.00
104	10.28	681.32	-1.97	0.00	0.00	0.00	0.00
105	20.14	688.56	-10.44	0.00	0.00	0.00	0.00
106	10.34	635.87	-4.14	0.00	0.00	0.00	0.00
107	20.10	716.97	-15.44	0.00	0.00	0.00	0.00
108	10.29	671.38	-10.39	0.00	0.00	0.00	0.00
109	20.18	660.85	-15.33	0.00	0.00	0.00	0.00
110	10.39	601.23	-10.25	0.00	0.00	0.00	0.00
111	20.16	668.57	-21.69	0.00	0.00	0.00	0.00
112	10.37	610.89	-18.21	0.00	0.00	0.00	0.00
113	20.19	649.05	-19.94	0.00	0.00	0.00	0.00
114	10.41	586.49	-16.01	0.00	0.00	0.00	0.00
115	-14.74	500.61	-30.26	0.00	0.00	0.00	0.00
116	-15.23	544.48	13.87	0.00	0.00	0.00	0.00
117	35.88	747.29	-34.25	0.00	0.00	0.00	0.00
118	35.39	791.16	9.88	0.00	0.00	0.00	0.00
119	3.39	536.42	-81.79	0.00	0.00	0.00	0.00
120	18.86	611.79	-83.01	0.00	0.00	0.00	0.00
121	1.80	679.99	62.64	0.00	0.00	0.00	0.00
122	17.26	755.36	61.41	0.00	0.00	0.00	0.00
123	-15.77	436.03	-29.24	0.00	0.00	0.00	0.00
124	-16.26	479.89	14.89	0.00	0.00	0.00	0.00
125	34.85	682.71	-33.23	0.00	0.00	0.00	0.00
126	34.36	726.57	10.90	0.00	0.00	0.00	0.00
127	2.36	471.83	-80.77	0.00	0.00	0.00	0.00
128	17.82	547.20	-81.99	0.00	0.00	0.00	0.00
129	0.77	615.40	63.65	0.00	0.00	0.00	0.00
130	16.23	690.77	62.43	0.00	0.00	0.00	0.00
131	2.66	705.06	-32.42	0.00	0.00	0.00	0.00
132	2.28	738.95	1.68	0.00	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
10	133	40.62	890.07	-35.42	0.00	0.00	0.00
	134	40.24	923.97	-1.32	0.00	0.00	0.00
	135	16.07	731.55	-70.56	0.00	0.00	0.00
	136	28.02	789.79	-71.50	0.00	0.00	0.00
	137	14.88	839.23	37.76	0.00	0.00	0.00
	138	26.83	897.47	36.82	0.00	0.00	0.00
	101	18.34	1440.01	3.30	0.00	0.00	0.00
	102	17.14	1240.11	9.07	0.00	0.00	0.00
	103	14.61	1182.02	4.35	0.00	0.00	0.00
	104	15.87	1077.72	16.97	0.00	0.00	0.00
11	105	14.66	1144.02	2.66	0.00	0.00	0.00
	106	15.92	1030.23	14.86	0.00	0.00	0.00
	107	14.65	1175.05	3.19	0.00	0.00	0.00
	108	15.92	1069.01	15.52	0.00	0.00	0.00
	109	14.71	1112.33	1.36	0.00	0.00	0.00
	110	15.99	990.61	13.24	0.00	0.00	0.00
	111	14.72	1122.16	1.10	0.00	0.00	0.00
	112	16.00	1002.90	12.91	0.00	0.00	0.00
	113	14.74	1097.84	0.89	0.00	0.00	0.00
	114	16.03	972.50	12.65	0.00	0.00	0.00
12	115	-39.47	1021.58	-1.33	0.00	0.00	0.00
	116	-40.08	1091.18	31.89	0.00	0.00	0.00
	117	71.96	989.24	-2.21	0.00	0.00	0.00
	118	71.35	1058.85	31.01	0.00	0.00	0.00
	119	-0.10	931.26	-39.39	0.00	0.00	0.00
	120	33.95	921.38	-39.66	0.00	0.00	0.00
	121	-2.07	1159.05	69.34	0.00	0.00	0.00
	122	31.98	1149.17	69.07	0.00	0.00	0.00
	123	-41.07	917.56	-2.82	0.00	0.00	0.00
	124	-41.67	987.16	30.41	0.00	0.00	0.00
13	125	70.36	885.22	-3.70	0.00	0.00	0.00
	126	69.76	954.83	29.53	0.00	0.00	0.00
	127	-1.69	827.23	-40.88	0.00	0.00	0.00
	128	32.36	817.35	-41.14	0.00	0.00	0.00
	129	-3.67	1055.03	67.86	0.00	0.00	0.00
	130	30.38	1045.15	67.59	0.00	0.00	0.00
	131	-23.82	1325.30	-6.32	0.00	0.00	0.00
	132	-24.28	1379.08	19.35	0.00	0.00	0.00
	133	59.76	1301.04	-6.98	0.00	0.00	0.00
	134	59.29	1354.83	18.69	0.00	0.00	0.00
14	135	5.32	1258.46	-34.48	0.00	0.00	0.00
	136	33.61	1250.82	-34.69	0.00	0.00	0.00
	137	3.84	1429.30	47.06	0.00	0.00	0.00
	138	30.15	1421.67	46.86	0.00	0.00	0.00
	139	22.53	1624.34	16.04	0.00	0.00	0.00
	140	18.48	1405.34	22.04	0.00	0.00	0.00
	141	17.85	1329.24	13.91	0.00	0.00	0.00
	142	14.22	1223.54	29.38	0.00	0.00	0.00
	143	14.22	1223.54	29.38	0.00	0.00	0.00
	144	14.22	1223.54	29.38	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
10	105	17.91	1287.20	11.65	0.00	0.00	0.00
	106	14.28	1170.99	26.56	0.00	0.00	0.00
	107	18.02	1325.38	13.73	0.00	0.00	0.00
	108	14.42	1218.72	29.16	0.00	0.00	0.00
	109	18.04	1254.51	10.63	0.00	0.00	0.00
	110	14.45	1130.13	25.29	0.00	0.00	0.00
	111	18.18	1270.03	12.08	0.00	0.00	0.00
	112	14.62	1149.53	27.09	0.00	0.00	0.00
	113	18.15	1241.59	11.08	0.00	0.00	0.00
	114	14.59	1113.97	25.84	0.00	0.00	0.00
11	115	-120.22	1105.28	9.68	0.00	0.00	0.00
	116	-121.64	1175.58	48.32	0.00	0.00	0.00
	117	150.49	1197.09	7.75	0.00	0.00	0.00
	118	149.07	1267.38	46.39	0.00	0.00	0.00
	119	-24.61	1057.28	-34.89	0.00	0.00	0.00
	120	58.11	1085.34	-35.48	0.00	0.00	0.00
	121	-29.25	1287.33	91.56	0.00	0.00	0.00
	122	53.46	1315.38	90.97	0.00	0.00	0.00
	123	-121.67	986.65	6.88	0.00	0.00	0.00
	124	-123.08	1056.94	45.52	0.00	0.00	0.00
12	125	149.05	1078.45	4.95	0.00	0.00	0.00
	126	147.63	1148.75	43.59	0.00	0.00	0.00
	127	-26.05	938.65	-37.70	0.00	0.00	0.00
	128	56.66	966.70	-38.29	0.00	0.00	0.00
	129	-30.70	1168.69	88.75	0.00	0.00	0.00
	130	52.02	1196.75	88.16	0.00	0.00	0.00
	131	-80.47	1453.25	4.84	0.00	0.00	0.00
	132	-81.56	1507.57	34.69	0.00	0.00	0.00
	133	122.57	1522.11	3.39	0.00	0.00	0.00
	134	121.47	1576.42	33.25	0.00	0.00	0.00
13	135	-9.72	1417.73	-28.15	0.00	0.00	0.00
	136	54.20	1439.41	-28.61	0.00	0.00	0.00
	137	-13.20	1590.27	66.69	0.00	0.00	0.00
	138	50.72	1611.94	66.23	0.00	0.00	0.00
	139	-74.85	1018.63	-24.20	0.00	0.00	0.00
	140	-60.74	890.40	-19.70	0.00	0.00	0.00
	141	-60.16	844.08	-8.00	0.00	0.00	0.00
	142	-46.98	798.63	-1.01	0.00	0.00	0.00
	143	-60.12	806.43	-10.77	0.00	0.00	0.00
	144	-46.94	751.57	-4.47	0.00	0.00	0.00
14	145	-59.83	836.03	-19.76	0.00	0.00	0.00
	146	-46.58	788.57	-15.70	0.00	0.00	0.00
	147	-59.91	777.69	-19.50	0.00	0.00	0.00
	148	-46.68	715.63	-15.38	0.00	0.00	0.00
	149	-59.62	786.20	-30.33	0.00	0.00	0.00
	150	-46.31	726.27	-28.91	0.00	0.00	0.00
	151	-59.72	765.69	-27.52	0.00	0.00	0.00
	152	-46.44	700.64	-25.40	0.00	0.00	0.00
	153	-46.44	700.64	-25.40	0.00	0.00	0.00
	154	-46.44	700.64	-25.40	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
115	-219.23	892.66	-55.13	0.00	0.00	0.00	0.00
116	-221.52	939.57	14.02	0.00	0.00	0.00	0.00
117	128.25	584.74	-44.43	0.00	0.00	0.00	0.00
118	125.96	631.65	24.72	0.00	0.00	0.00	0.00
119	-95.97	732.44	-130.00	0.00	0.00	0.00	0.00
120	10.20	638.35	-126.73	0.00	0.00	0.00	0.00
121	-103.47	885.96	96.32	0.00	0.00	0.00	0.00
122	2.71	791.88	99.59	0.00	0.00	0.00	0.00
123	-214.56	816.45	-53.61	0.00	0.00	0.00	0.00
124	-216.85	863.36	15.54	0.00	0.00	0.00	0.00
125	132.92	508.53	-42.91	0.00	0.00	0.00	0.00
126	130.63	555.44	26.25	0.00	0.00	0.00	0.00
127	-91.31	656.22	-128.47	0.00	0.00	0.00	0.00
128	14.86	562.14	-125.20	0.00	0.00	0.00	0.00
129	-98.80	809.75	97.84	0.00	0.00	0.00	0.00
130	7.37	715.66	101.11	0.00	0.00	0.00	0.00
131	-197.21	1051.86	-52.68	0.00	0.00	0.00	0.00
132	-198.98	1088.11	0.75	0.00	0.00	0.00	0.00
133	63.40	820.92	-44.66	0.00	0.00	0.00	0.00
134	61.63	857.17	8.78	0.00	0.00	0.00	0.00
135	-106.00	933.29	-108.08	0.00	0.00	0.00	0.00
136	-23.96	860.59	-105.56	0.00	0.00	0.00	0.00
137	-111.62	1048.44	61.65	0.00	0.00	0.00	0.00
138	-29.58	975.73	64.18	0.00	0.00	0.00	0.00
13	101	25.23	475.87	-15.22	0.00	0.00	0.00
102	23.44	449.72	-13.83	0.00	0.00	0.00	0.00
103	20.35	363.08	-9.28	0.00	0.00	0.00	0.00
104	21.85	401.55	-8.81	0.00	0.00	0.00	0.00
105	20.30	350.77	-10.29	0.00	0.00	0.00	0.00
106	21.79	386.16	-10.08	0.00	0.00	0.00	0.00
107	20.18	390.05	-12.39	0.00	0.00	0.00	0.00
108	21.64	435.26	-12.70	0.00	0.00	0.00	0.00
109	20.17	362.83	-12.37	0.00	0.00	0.00	0.00
110	21.63	401.24	-12.68	0.00	0.00	0.00	0.00
111	20.03	399.23	-14.82	0.00	0.00	0.00	0.00
112	21.45	446.74	-15.74	0.00	0.00	0.00	0.00
113	20.06	383.98	-13.93	0.00	0.00	0.00	0.00
114	21.50	427.67	-14.64	0.00	0.00	0.00	0.00
115	-2.77	369.52	-24.88	0.00	0.00	0.00	0.00
116	-1.71	274.55	2.89	0.00	0.00	0.00	0.00
117	45.00	572.59	-27.77	0.00	0.00	0.00	0.00
118	46.06	477.62	0.00	0.00	0.00	0.00	0.00
119	12.62	547.94	-57.44	0.00	0.00	0.00	0.00
120	27.22	609.99	-58.32	0.00	0.00	0.00	0.00
121	16.07	237.14	33.45	0.00	0.00	0.00	0.00
122	30.67	299.19	32.56	0.00	0.00	0.00	0.00
123	-4.93	327.16	-23.64	0.00	0.00	0.00	0.00
124	-3.88	232.19	4.13	0.00	0.00	0.00	0.00

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
125	42.84	530.23	-26.53	0.00	0.00	0.00	0.00
126	43.89	435.26	1.25	0.00	0.00	0.00	0.00
127	10.46	505.59	-56.20	0.00	0.00	0.00	0.00
128	25.05	567.64	-57.08	0.00	0.00	0.00	0.00
129	13.91	194.79	34.69	0.00	0.00	0.00	0.00
130	28.50	256.84	33.81	0.00	0.00	0.00	0.00
131	6.01	423.33	-24.17	0.00	0.00	0.00	0.00
132	6.83	349.95	-2.71	0.00	0.00	0.00	0.00
133	41.84	575.64	-26.34	0.00	0.00	0.00	0.00
134	42.65	502.25	-4.88	0.00	0.00	0.00	0.00
135	17.40	555.37	-48.27	0.00	0.00	0.00	0.00
136	28.68	603.32	-48.95	0.00	0.00	0.00	0.00
137	19.99	322.27	19.90	0.00	0.00	0.00	0.00
138	31.27	370.22	19.22	0.00	0.00	0.00	0.00
14	101	11.84	789.28	-20.46	0.00	0.00	0.00
102	8.26	722.46	-17.62	0.00	0.00	0.00	0.00
103	9.74	609.90	-17.11	0.00	0.00	0.00	0.00
104	5.01	628.74	-15.71	0.00	0.00	0.00	0.00
105	9.66	596.76	-17.63	0.00	0.00	0.00	0.00
106	4.92	612.31	-16.35	0.00	0.00	0.00	0.00
107	9.47	641.16	-16.51	0.00	0.00	0.00	0.00
108	4.67	667.81	-14.94	0.00	0.00	0.00	0.00
109	9.46	612.59	-16.78	0.00	0.00	0.00	0.00
110	4.66	632.10	-15.29	0.00	0.00	0.00	0.00
111	9.21	653.82	-15.32	0.00	0.00	0.00	0.00
112	4.35	683.63	-13.46	0.00	0.00	0.00	0.00
113	9.27	638.02	-15.25	0.00	0.00	0.00	0.00
114	4.43	663.88	-13.38	0.00	0.00	0.00	0.00
115	-45.24	710.26	-24.61	0.00	0.00	0.00	0.00
116	-43.53	595.45	-5.95	0.00	0.00	0.00	0.00
117	52.88	715.82	-23.60	0.00	0.00	0.00	0.00
118	54.60	601.02	-4.94	0.00	0.00	0.00	0.00
119	-13.12	842.65	-45.46	0.00	0.00	0.00	0.00
120	16.86	844.35	-45.15	0.00	0.00	0.00	0.00
121	-7.51	466.92	15.61	0.00	0.00	0.00	0.00
122	22.48	468.62	15.92	0.00	0.00	0.00	0.00
123	-45.71	644.69	-23.13	0.00	0.00	0.00	0.00
124	-43.99	529.89	-4.47	0.00	0.00	0.00	0.00
125	52.41	650.26	-22.12	0.00	0.00	0.00	0.00
126	54.13	535.45	-3.46	0.00	0.00	0.00	0.00
127	-13.59	777.09	-43.99	0.00	0.00	0.00	0.00
128	16.39	778.79	-43.68	0.00	0.00	0.00	0.00
129	-7.97	401.36	17.08	0.00	0.00	0.00	0.00
130	22.01	403.06	17.39	0.00	0.00	0.00	0.00
131	-27.41	798.14	-26.63	0.00	0.00	0.00	0.00
132	-26.08	709.42	-12.21	0.00	0.00	0.00	0.00
133	46.18	802.31	-25.87	0.00	0.00	0.00	0.00
134	47.51	713.60	-11.45	0.00	0.00	0.00	0.00

□

□

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
15	107	-36.35	462.45	-45.12	0.00	0.00	0.00
	108	-37.14	470.98	-38.84	0.00	0.00	0.00
	109	-36.41	435.44	-45.23	0.00	0.00	0.00
	110	-37.22	437.23	-38.97	0.00	0.00	0.00
	111	-37.19	472.03	-47.81	0.00	0.00	0.00
	112	-38.20	482.96	-42.19	0.00	0.00	0.00
	113	-37.02	456.88	-46.82	0.00	0.00	0.00
	114	-37.98	464.03	-40.96	0.00	0.00	0.00
	115	-190.99	638.94	-60.20	0.00	0.00	0.00
	116	-185.41	544.24	-21.26	0.00	0.00	0.00
	117	111.09	374.38	-55.07	0.00	0.00	0.00
	118	116.67	279.68	-16.13	0.00	0.00	0.00
	119	-92.45	654.70	-102.67	0.00	0.00	0.00

□

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
15	135	-3.64	896.11	-42.06	0.00	0.00	0.00
	136	19.53	897.42	-41.82	0.00	0.00	0.00
	137	0.57	614.31	3.74	0.00	0.00	0.00
	138	23.74	615.63	3.98	0.00	0.00	0.00
	101	11.17	899.15	-47.76	0.00	0.00	0.00
	102	11.79	801.82	-39.11	0.00	0.00	0.00
	103	9.56	698.19	-40.69	0.00	0.00	0.00
	104	13.19	678.09	-33.57	0.00	0.00	0.00
	105	9.38	685.43	-41.13	0.00	0.00	0.00
	106	12.97	662.14	-34.11	0.00	0.00	0.00
	107	8.94	728.75	-38.46	0.00	0.00	0.00
	108	12.42	716.29	-30.77	0.00	0.00	0.00
	109	8.90	701.17	-38.98	0.00	0.00	0.00

110	12.38	681.81	-31.43	0.00	0.00	0.00	0.00
111	8.32	741.24	-35.16	0.00	0.00	0.00	0.00
112	11.65	731.90	-26.65	0.00	0.00	0.00	0.00
113	8.45	726.07	-35.57	0.00	0.00	0.00	0.00
114	11.82	712.94	-27.16	0.00	0.00	0.00	0.00
115	-107.69	733.01	-38.14	0.00	0.00	0.00	0.00
116	-103.65	618.88	-20.65	0.00	0.00	0.00	0.00
117	128.48	790.11	-40.27	0.00	0.00	0.00	0.00
118	132.52	675.98	-22.79	0.00	0.00	0.00	0.00
119	-30.28	882.53	-58.74	0.00	0.00	0.00	0.00
120	41.88	899.98	-59.40	0.00	0.00	0.00	0.00
121	-17.05	509.01	-1.53	0.00	0.00	0.00	0.00
122	55.11	526.46	-2.18	0.00	0.00	0.00	0.00
123	-108.93	662.56	-35.09	0.00	0.00	0.00	0.00
124	-104.89	548.43	-17.61	0.00	0.00	0.00	0.00
125	127.24	719.66	-37.23	0.00	0.00	0.00	0.00
126	131.28	605.53	-19.74	0.00	0.00	0.00	0.00
127	-31.52	812.08	-55.70	0.00	0.00	0.00	0.00
128	40.64	829.53	-56.35	0.00	0.00	0.00	0.00
129	-18.29	438.56	1.52	0.00	0.00	0.00	0.00
130	53.87	456.01	0.86	0.00	0.00	0.00	0.00
131	-78.65	873.17	-49.39	0.00	0.00	0.00	0.00
132	-75.53	784.97	-35.88	0.00	0.00	0.00	0.00
133	98.48	915.99	-50.99	0.00	0.00	0.00	0.00
134	101.60	827.80	-37.48	0.00	0.00	0.00	0.00
135	-21.37	983.81	-64.64	0.00	0.00	0.00	0.00
136	34.40	997.30	-65.14	0.00	0.00	0.00	0.00
137	-11.44	703.67	-21.73	0.00	0.00	0.00	0.00
138	44.32	717.16	-22.23	0.00	0.00	0.00	0.00
16	101	-45.45	566.39	-55.74	0.00	0.00	0.00
102	-41.31	512.85	-46.95	0.00	0.00	0.00	0.00
103	-35.52	435.04	-40.64	0.00	0.00	0.00	0.00
104	-36.11	436.72	-33.23	0.00	0.00	0.00	0.00
105	-35.77	422.88	-42.15	0.00	0.00	0.00	0.00
106	-36.42	421.53	-35.12	0.00	0.00	0.00	0.00

***** END OF LATEST ANALYSIS RESULT *****

743. *****

744. LOAD LIST 201 TO 236

745. START CONCRETE DESIGN

□CONCRETE DESIGN

746. CODE BS8110

PROGRAM CODE REVISION V1.5_8110_97/1

747. UNIT MMS NEWTON

748. FYMAIN 460 ALL

C O L U M N N O. 1001 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 2726. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
36 10 MM	0.935	225	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

749. FC 40 ALL

750. MINSEC 10 ALL

|BRACED /SHORT in z E.L.z= 3900 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 3900 mm (3.8.1.3 & 5)|
|END MOMS. MZ1= 0 MZ2= 157 MY1= 0 MY2= 298|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 442 AXIAL LOAD= 881|
|DESIGNED CAP. KN METER: MOM.= 442 AXIAL CAP.= 882|

751. CLEAR 40 MEMB 1 TO 42 101 TO 124

C O L U M N N O. 1002 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 3522. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
32 12 MM	1.196	231	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

753. TRACK 2 ALL

754. DESIGN COLUMN 1001 TO 1016 2001 TO 2005 2008 TO 2016

-- PAGE NO. 32

C O L U M N N O . 1 0 0 3 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 5759. SQ. MM.

BAR CONFIGURATION	REINF. PCT.	LOAD	LOCATION
1	1.0	100	100
2	1.0	100	100
3	1.0	100	100
4	1.0	100	100
5	1.0	100	100
6	1.0	100	100
7	1.0	100	100
8	1.0	100	100
9	1.0	100	100
10	1.0	100	100
11	1.0	100	100
12	1.0	100	100
13	1.0	100	100
14	1.0	100	100
15	1.0	100	100
16	1.0	100	100
17	1.0	100	100
18	1.0	100	100
19	1.0	100	100
20	1.0	100	100
21	1.0	100	100
22	1.0	100	100
23	1.0	100	100
24	1.0	100	100
25	1.0	100	100
26	1.0	100	100
27	1.0	100	100
28	1.0	100	100
29	1.0	100	100
30	1.0	100	100
31	1.0	100	100
32	1.0	100	100
33	1.0	100	100
34	1.0	100	100
35	1.0	100	100
36	1.0	100	100
37	1.0	100	100
38	1.0	100	100
39	1.0	100	100
40	1.0	100	100
41	1.0	100	100
42	1.0	100	100
43	1.0	100	100
44	1.0	100	100
45	1.0	100	100
46	1.0	100	100
47	1.0	100	100
48	1.0	100	100
49	1.0	100	100
50	1.0	100	100
51	1.0	100	100
52	1.0	100	100
53	1.0	100	100
54	1.0	100	100
55	1.0	100	100
56	1.0	100	100
57	1.0	100	100
58	1.0	100	100
59	1.0	100	100
60	1.0	100	100
61	1.0	100	100
62	1.0	100	100
63	1.0	100	100
64	1.0	100	100
65	1.0	100	100
66	1.0	100	100
67	1.0	100	100
68	1.0	100	100
69	1.0	100	100
70	1.0	100	100
71	1.0	100	100
72	1.0	100	100
73	1.0	100	100
74	1.0	100	100
75	1.0	100	100
76	1.0	100	100
77	1.0	100	100
78	1.0	100	100
79	1.0	100	100
80	1.0	100	100
81	1.0	100	100
82	1.0	100	100
83	1.0	100	100
84	1.0	100	100
85	1.0	100	100
86	1.0	100	100
87	1.0	100	100
88	1.0	100	100
89	1.0	100	100
90	1.0	100	100
91	1.0	100	100
92	1.0	100	100
93	1.0	100	100
94	1.0	100	100
95	1.0	100	100
96	1.0	100	100
97	1.0		

12	25 MM	1.947	229	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)				

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BRACED /SHORT      in z E.L.z= 2000 mm (3.8,1.3 & 5)|
BRACED /SHORT      in y E.L.y= 2000 mm (3.8,1.3 & 5)|
END MOMS.           MZ1= 0 MZ2= 582 MY1= 0 MY2= 118|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM= 692 AXIAL LOAD= 777|
DESIGNED CAP. KN METER: MOM= 692 AXIAL CAP.= 784|

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C O L U M N N O . 1 0 0 4 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 4424. SQ. MM.

BAR CONFIGURATION	REINF. PCT.	LOAD	LOCATION
1	1.0	100	1
2	1.0	100	2
3	1.0	100	3
4	1.0	100	4
5	1.0	100	5
6	1.0	100	6
7	1.0	100	7
8	1.0	100	8
9	1.0	100	9
10	1.0	100	10
11	1.0	100	11
12	1.0	100	12
13	1.0	100	13
14	1.0	100	14
15	1.0	100	15
16	1.0	100	16
17	1.0	100	17
18	1.0	100	18
19	1.0	100	19
20	1.0	100	20
21	1.0	100	21
22	1.0	100	22
23	1.0	100	23
24	1.0	100	24
25	1.0	100	25
26	1.0	100	26
27	1.0	100	27
28	1.0	100	28
29	1.0	100	29
30	1.0	100	30
31	1.0	100	31
32	1.0	100	32
33	1.0	100	33
34	1.0	100	34
35	1.0	100	35
36	1.0	100	36
37	1.0	100	37
38	1.0	100	38
39	1.0	100	39
40	1.0	100	40
41	1.0	100	41
42	1.0	100	42
43	1.0	100	43
44	1.0	100	44
45	1.0	100	45
46	1.0	100	46
47	1.0	100	47
48	1.0	100	48
49	1.0	100	49
50	1.0	100	50
51	1.0	100	51
52	1.0	100	52
53	1.0	100	53
54	1.0	100	54
55	1.0	100	55
56	1.0	100	56
57	1.0	100	57
58	1.0	100	58
59	1.0	100	59
60	1.0	100	60
61	1.0	100	61
62	1.0	100	62
63	1.0	100	63
64	1.0	100	64
65	1.0	100	65
66	1.0	100	66
67	1.0	100	67
68	1.0	100	68
69	1.0	100	69
70	1.0	100	70
71	1.0	100	71
72	1.0	100	72
73	1.0	100	73
74	1.0	100	74
75	1.0	100	75
76	1.0	100	76
77	1.0	100	77
78	1.0	100	78
79	1.0	100	79
80	1.0	100	80
81	1.0	100	81
82	1.0	100	82
83	1.0	100	83
84	1.0	100	84
85	1.0	100	85
86	1.0	100	86
87	1.0	100	87
88	1.0	100	88
89	1.0	100	89
90	1.0	100	90
91	1.0	100	91
92	1.0	100	92
93	1.0	100	93
94	1.0	100	94
95	1.0	100	95
96	1.0	100	96
97	1.0	100	97
98	1.0	100	98
99	1.0	100	99

40	12 MM	1.495	220	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)				

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BRACED /SHORT      in z E.L.z= 2000 mm (3.8.1.3 & 5) |
BRACED /SHORT      in y E.L.y= 2000 mm (3.8.1.3 & 5) |
END MOMS.          MZ1= 0 MZ2= 469 MY1= 0 MY2= 161 |
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM= 615 AXIAL LOAD= 1007 |
DESIGNED CAP. KN METER: MOM= 615 AXIAL CAP.= 1015 |

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C O L U M N N O . 1 0 0 5 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 6363. SQ. MM.

BAR CONFIGURATION	REINF. PCT.	LOAD	LOCATION
1	1.0	100	100
2	1.0	100	100
3	1.0	100	100
4	1.0	100	100
5	1.0	100	100
6	1.0	100	100
7	1.0	100	100
8	1.0	100	100
9	1.0	100	100
10	1.0	100	100
11	1.0	100	100
12	1.0	100	100
13	1.0	100	100
14	1.0	100	100
15	1.0	100	100
16	1.0	100	100
17	1.0	100	100
18	1.0	100	100
19	1.0	100	100
20	1.0	100	100
21	1.0	100	100
22	1.0	100	100
23	1.0	100	100
24	1.0	100	100
25	1.0	100	100
26	1.0	100	100
27	1.0	100	100
28	1.0	100	100
29	1.0	100	100
30	1.0	100	100
31	1.0	100	100
32	1.0	100	100
33	1.0	100	100
34	1.0	100	100
35	1.0	100	100
36	1.0	100	100
37	1.0	100	100
38	1.0	100	100
39	1.0	100	100
40	1.0	100	100
41	1.0	100	100
42	1.0	100	100
43	1.0	100	100
44	1.0	100	100
45	1.0	100	100
46	1.0	100	100
47	1.0	100	100
48	1.0	100	100
49	1.0	100	100
50	1.0	100	100
51	1.0	100	100
52	1.0	100	100
53	1.0	100	100
54	1.0	100	100
55	1.0	100	100
56	1.0	100	100
57	1.0	100	100
58	1.0	100	100
59	1.0	100	100
60	1.0	100	100
61	1.0	100	100
62	1.0	100	100
63	1.0	100	100
64	1.0	100	100
65	1.0	100	100
66	1.0	100	100
67	1.0	100	100
68	1.0	100	100
69	1.0	100	100
70	1.0	100	100
71	1.0	100	100
72	1.0	100	100
73	1.0	100	100
74	1.0	100	100
75	1.0	100	100
76	1.0	100	100
77	1.0	100	100
78	1.0	100	100
79	1.0	100	100
80	1.0	100	100
81	1.0	100	100
82	1.0	100	100
83	1.0	100	100
84	1.0	100	100
85	1.0	100	100
86	1.0	100	100
87	1.0	100	100
88	1.0	100	100
89	1.0	100	100
90	1.0	100	100
91	1.0	100	100
92	1.0	100	100
93	1.0	100	100
94	1.0	100	100
95	1.0	100	100
96	1.0	100	100
97	1.0		

32	16 MM	2.127	225	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)				

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BRACED /SHORT      in z E.L.z= 2700 mm (3.8.1.3 & 5) |
BRACED /SHORT      in y E.L.y= 2700 mm (3.8.1.3 & 5) |
END MOMS.          MZ1= 0 MZ2= 124 MY1= 0 MY2= 694 |
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM= 805 AXIAL LOAD= 1121 |
DESIGNED CAP. KN METER: MOM= 805 AXIAL CAP.= 1123 |

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C O L U M N N O . 1 0 0 6 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 8985. SQ. MM.

BAR CONFIGURATION	REINF. PCT.	LOAD	LOCATION
1	1.0	100	100
2	1.0	100	100
3	1.0	100	100
4	1.0	100	100
5	1.0	100	100
6	1.0	100	100
7	1.0	100	100
8	1.0	100	100
9	1.0	100	100
10	1.0	100	100
11	1.0	100	100
12	1.0	100	100
13	1.0	100	100
14	1.0	100	100
15	1.0	100	100
16	1.0	100	100
17	1.0	100	100
18	1.0	100	100
19	1.0	100	100
20	1.0	100	100
21	1.0	100	100
22	1.0	100	100
23	1.0	100	100
24	1.0	100	100
25	1.0	100	100
26	1.0	100	100
27	1.0	100	100
28	1.0	100	100
29	1.0	100	100
30	1.0	100	100
31	1.0	100	100
32	1.0	100	100
33	1.0	100	100
34	1.0	100	100
35	1.0	100	100
36	1.0	100	100
37	1.0	100	100
38	1.0	100	100
39	1.0	100	100
40	1.0	100	100
41	1.0	100	100
42	1.0	100	100
43	1.0	100	100
44	1.0	100	100
45	1.0	100	100
46	1.0	100	100
47	1.0	100	100
48	1.0	100	100
49	1.0	100	100
50	1.0	100	100
51	1.0	100	100
52	1.0	100	100
53	1.0	100	100
54	1.0	100	100
55	1.0	100	100
56	1.0	100	100
57	1.0	100	100
58	1.0	100	100
59	1.0	100	100
60	1.0	100	100
61	1.0	100	100
62	1.0	100	100
63	1.0	100	100
64	1.0	100	100
65	1.0	100	100
66	1.0	100	100
67	1.0	100	100
68	1.0	100	100
69	1.0	100	100
70	1.0	100	100
71	1.0	100	100
72	1.0	100	100
73	1.0	100	100
74	1.0	100	100
75	1.0	100	100
76	1.0	100	100
77	1.0	100	100
78	1.0	100	100
79	1.0	100	100
80	1.0	100	100
81	1.0	100	100
82	1.0	100	100
83	1.0	100	100
84	1.0	100	100
85	1.0	100	100
86	1.0	100	100
87	1.0	100	100
88	1.0	100	100
89	1.0	100	100
90	1.0	100	100
91	1.0	100	100
92	1.0	100	100
93	1.0	100	100
94	1.0	100	100
95	1.0	100	100
96	1.0	100	100
97	1.0		

12	32 MM	3.191	233	EACH END
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(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

BRACED	/SHORT	in z E.L.z=	2700 mm	(3.8.1.3 & 5)
BRACED	/SHORT	in y E.L.y=	2700 mm	(3.8.1.3 & 5)
END MOMS.		M21=	0 M22= 137 Mx1=	0 MY2= 741
SLENDERNESS	MOMTS.	KNM:	MOMZ=	0 MOMY= 0
DESIGN	LOADS	KN	METER:	MOM.= 875 AXIAL LOAD= 2421
DESIGNED	CAP.	KN	METER:	MOM.= 875 AXIAL CAP.= 2431

C O L U M N N O. 1007 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 9180. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION

12 32 MM	3.191	234	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

BRACED /SHORT	in z E.L.z= 2000 mm (3.8.1.3 & 5)
BRACED /SHORT	in y E.L.y= 2000 mm (3.8.1.3 & 5)
END MOMS.	M21= 0 M22= 189 MY1= 0 MY2= 709
SLENDERNESS MOMTS.	KNM: MOMZ= 0 MOMY= 0
DESIGN LOADS KN METER: MOM.	= 894 AXIAL LOAD= 248
DESIGNED CAP. KN METER: MOM.	= 894 AXIAL CAP.= 249

C O L U M N N O. 1008 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 2947. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION

40 10 MM	1.039	225	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

BRACED /SHORT	in z E.L.z= 2000 mm (3.8.1.3 & 5)
BRACED /SHORT	in y E.L.y= 2000 mm (3.8.1.3 & 5)
END MOMS.	M21= 0 M22= 133 MY1= 0 MY2= 400
SLENDERNESS MOMTS.	KNM: MOMZ= 0 MOMY= 0
DESIGN LOADS KN METER: MOM.	= 518 AXIAL LOAD= 1210
DESIGNED CAP. KN METER: MOM.	= 518 AXIAL CAP.= 1215

C O L U M N N O. 1009 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 4534. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION

24 16 MM	1.596	224	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

BRACED /SHORT	in z E.L.z= 3650 mm (3.8.1.3 & 5)
BRACED /SHORT	in y E.L.y= 3650 mm (3.8.1.3 & 5)
END MOMS.	M21= 0 M22= 156 MY1= 0 MY2= 490
SLENDERNESS MOMTS.	KNM: MOMZ= 0 MOMY= 0
DESIGN LOADS KN METER: MOM.	= 631 AXIAL LOAD= 1046
DESIGNED CAP. KN METER: MOM.	= 631 AXIAL CAP.= 1050

C O L U M N N O. 1010 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 2956. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION

40 10 MM	1.039	230	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

BRACED /SHORT	in z E.L.z= 3650 mm (3.8.1.3 & 5)
BRACED /SHORT	in y E.L.y= 3650 mm (3.8.1.3 & 5)
END MOMS.	M21= 0 M22= 366 MY1= 0 MY2= 142
SLENDERNESS MOMTS.	KNM: MOMZ= 0 MOMY= 0
DESIGN LOADS KN METER: MOM.	= 494 AXIAL LOAD= 1057
DESIGNED CAP. KN METER: MOM.	= 494 AXIAL CAP.= 1063

C O L U M N N O. 1011 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 5084. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
40 16 MM	1.862	230	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

|BRACED /SHORT in z E.L.z= 2700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 2700 mm (3.8.1.3 & 5)|
|END MOMS. M21= 0 M22= 594 MY1= 0 MY2= 149|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 724 AXIAL LOAD= 1297|
|DESIGNED CAP. KN METER: MOM.= 724 AXIAL CAP.= 1320|

C O L U M N N O. 1012 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 6386. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
32 16 MM	2.127	219	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

|BRACED /SHORT in z E.L.z= 2000 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 2000 mm (3.8.1.3 & 5)|
|END MOMS. M21= 0 M22= 712 MY1= 0 MY2= 180|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 866 AXIAL LOAD= 1488|
|DESIGNED CAP. KN METER: MOM.= 866 AXIAL CAP.= 1491|

C O L U M N N O. 1015 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 4871. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
16	20 MM	1.662	229 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 2700 mm (3.8.1.3 & 5) |
IBRACED /SHORT in y E.L.y= 2700 mm (3.8.1.3 & 5) |
END MOMS. M21= 0 M22= 512 MY1= 0 MY2= 121 |
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 623 AXIAL LOAD= 826 |
|DESIGNED CAP. KN METER: MOM.= 623 AXIAL CAP.= 838 |

C O L U M N N O. 1016 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 550. X 550. MM,

AREA OF STEEL REQUIRED = 6219. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
20	20 MM	2.077	219 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 2000 mm (3.8.1.3 & 5) |
IBRACED /SHORT in y E.L.y= 2000 mm (3.8.1.3 & 5) |
END MOMS. M21= 0 M22= 586 MY1= 0 MY2= 205 |
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 772 AXIAL LOAD= 997 |
|DESIGNED CAP. KN METER: MOM.= 772 AXIAL CAP.= 1003 |

C O L U M N N O. 2001 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 4920. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
16	20 MM	2.011	222 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5) |
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5) |
END MOMS. M21= 363 M22= 331 MY1= 97 MY2= 100 |
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 460 AXIAL LOAD= 292 |
|DESIGNED CAP. KN METER: MOM.= 460 AXIAL CAP.= 292 |

C O L U M N N O. 2002 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 9863. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
8	40 MM	4.021	220 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5) |
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5) |
END MOMS. M21= 359 M22= 389 MY1= 244 MY2= 526 |
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 889 AXIAL LOAD= 563 |
|DESIGNED CAP. KN METER: MOM.= 889 AXIAL CAP.= 563 |

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C O L U M N N O. 2003 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 9656. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

20 25 MM 3.927 222 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. M21= 288 M22= 361 MY1= 260 MY2= 535|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 873 AXIAL LOAD= 560|
DESIGNED CAP. KN METER: MOM.= 873 AXIAL CAP.= 563

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C O L U M N N O. 2004 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 4174. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

24 16 MM 1.931 220 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. M21= 265 M22= 304 MY1= 96 MY2= 100|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 401 AXIAL LOAD= 285|
DESIGNED CAP. KN METER: MOM.= 401 AXIAL CAP.= 288

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C O L U M N N O. 2005 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 7587. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

16 25 MM 3.142 226 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. M21= 278 M22= 398 MY1= 220 MY2= 303|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 687 AXIAL LOAD= 407|
DESIGNED CAP. KN METER: MOM.= 687 AXIAL CAP.= 407

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C O L U M N N O. 2008 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 7679. SQ. MM.

BAR CONFIGURATION REINF PCT. LOAD LOCATION

16 25 MM 3.142 223 EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. M21= 268 M22= 428 MY1= 245 MY2= 299|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 710 AXIAL LOAD= 502|
DESIGNED CAP. KN METER: MOM.= 710 AXIAL CAP.= 503

C O L U M N N O. 2009 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SORE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 5791. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
12 25 MM	2.356	224	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. M21= 147 M22= 179 MY1= 327 MY2= 360|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 533 AXIAL LOAD= 315|
DESIGNED CAP. KN METER: MOM.= 533 AXIAL CAP.= 320|

C O L U M N N O. 2010 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SORE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 9021. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
12 32 MM	3.861	223	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. M21= 102 M22= 131 MY1= 553 MY2= 721|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 842 AXIAL LOAD= 674|
DESIGNED CAP. KN METER: MOM.= 842 AXIAL CAP.= 678|

C O L U M N N O. 2011 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SORE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 9287. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
12 32 MM	3.861	224	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. M21= 105 M22= 133 MY1= 568 MY2= 740|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 863 AXIAL LOAD= 673|
DESIGNED CAP. KN METER: MOM.= 863 AXIAL CAP.= 675|

C O L U M N N O. 2012 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SORE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 5651. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
12 25 MM	2.356	223	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

IBRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
IBRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
END MOMS. M21= 160 M22= 181 MY1= 303 MY2= 348|
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
DESIGN LOADS KN METER: MOM.= 522 AXIAL LOAD= 319|
DESIGNED CAP. KN METER: MOM.= 522 AXIAL CAP.= 323|

C O L U M N N O. 2013 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 3738. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
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12	20 MM	1.508	224 EACH END
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(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. MZ1= 113 MZ2= 116 MY1= 215 MY2= 258|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 371 AXIAL LOAD= 306|
|DESIGNED CAP. KN METER: MOM.= 371 AXIAL CAP.= 307|

C O L U M N N O. 2014 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 5083. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
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28	16 MM	2.252	219 EACH END
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(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. MZ1= 280 MZ2= 323 MY1= 153 MY2= 109|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 472 AXIAL LOAD= 284|
|DESIGNED CAP. KN METER: MOM.= 472 AXIAL CAP.= 285|

C O L U M N N O. 2015 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 5168. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
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28	16 MM	2.252	221 EACH END
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(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. MZ1= 238 MZ2= 308 MY1= 174 MY2= 118|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 477 AXIAL LOAD= 276|
|DESIGNED CAP. KN METER: MOM.= 477 AXIAL CAP.= 276|

C O L U M N N O. 2016 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 SQRE SIZE - 500. X 500. MM,

AREA OF STEEL REQUIRED = 3749. SQ. MM.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION
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12	20 MM	1.508	223 EACH END
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(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)

|BRACED /SHORT in z E.L.z= 6700 mm (3.8.1.3 & 5)|
|BRACED /SHORT in y E.L.y= 6700 mm (3.8.1.3 & 5)|
|END MOMS. MZ1= 91 MZ2= 114 MY1= 246 MY2= 261|
|SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 0 |
|DESIGN LOADS KN METER: MOM.= 371 AXIAL LOAD= 305|
|DESIGNED CAP. KN METER: MOM.= 371 AXIAL CAP.= 305|

*****END OF COLUMN DESIGN RESULTS*****

101J 2975.X 300.X 600 149J

B E A M N O. 1 D E S I G N R E S U L T S - F L E X U R E

LEN - 2975. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
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1	52.	3- 25 MM	0.	2975.	YES YES
2	528.	2- 50 MM	0.	2975.	YES YES
COMP.		3- 10 MM	(REQD. STEEL=	234. SQ. MM)	

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
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0.	632.5/	3382.2	142.46/	575.51	228/	221
248.	704.2/	2541.5	158.18/	488.35	228/	221
496.	766.3/	2009.8	171.29/	405.07	228/	221
744.	816.5/	1553.8	181.80/	325.69	228/	221
992.	854.6/	1154.8	189.70/	250.21	228/	221
1240.	884.4/	805.3	195.84/	179.46	220/	229
1488.	992.5/	582.1	217.89/	131.11	220/	229
1735.	1083.0/	379.0	236.04/	85.37	220/	229
1983.	1155.2/	234.0	250.30/	42.23	220/	229
2231.	1208.3/	234.0	260.66/	1.70	220/	229
2479.	1241.7/	0.0	267.13/	0.00	220/	0
2727.	1257.7/	0.0	270.23/	0.00	219/	0
2975.	1254.7/	0.0	269.64/	0.00	219/	0

B E A M N O. 1 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM -	TO	SHEAR TORSN	LOAD	LINK	NO.	SPACING	mm	C/C		
mm	kNm	S	T	SIZE	S	T	S	T		
END 1	1487	359.4	12	221226	10mm	7	7	15	193	93
1487 END 2	171.0	12	221226	10mm	2	7	11	571	199	135
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 36 mm2 EVENLY DISTRIBUTED										
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *										

103J	6500.X 300.X 600	104J
=====		
3N032 H 537.	0..TO 2977	=====
35*10c/c	80	=====
2N032 H 56.	0..TO 6500	=====
=====		
13*10c/c 83		

[illegible]

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)	
0.	1279.3/	4545.9	204.71/ 822.20	228/ 221
542.	1024.7/	3407.3	224.38/ 580.85	228/ 221
1083.	1055.0/	1753.8	230.45/ 361.43	228/ 221
1625.	1087.5/	818.4	236.91/ 182.18	220/ 229
2167.	1189.7/	265.8	257.04/ 59.88	220/ 229
2708.	1125.8/	0.0	244.51/ 0.00	220/ 0
3250.	1072.7/	0.0	233.98/ 0.00	221/ 0
3792.	1414.6/	0.0	300.01/ 0.00	221/ 0
4333.	1591.5/	330.5	332.55/ 74.45	221/ 228
4875.	1590.9/	838.2	332.44/ 186.30	221/ 228
5417.	1456.2/	1505.4	307.76/ 316.85	229/ 220
5958.	1479.9/	2735.6	312.14/ 516.26	229/ 220
6500.	1430.3/	4146.8	302.93/ 737.60	229/ 220

BEAM NO. 4 DESIGN RESULTS - SHEAR

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO mm	SHEAR kN	TORSN kNm	LOAD S	LINK T	SIZE	NO.			SPACING mm		
						S	T	S+T	S	T	S+T
END 1	2709	463.6	10	221225	10mm	20	14	35	135	199	80
2709	5418	217.3	10	228225	10mm	5	14	18	541	199	146
5416 END 2	426.6	10	228225	10mm		7	5	13	150	199	83
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 516 mm2 EVENLY DISTRIBUTED											
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *											

1	52.	3- 25 MM	0.	3525.	YES	YES
2	528.	2- 50 MM	569.	3525.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1409.0/	0.0	220/ 0
294.	1219.3/	0.0	220/ 0
588.	1004.6/	0.0	220/ 0
881.	845.5/	0.0	222/ 0
1175.	799.7/	0.0	222/ 0
1469.	725.8/	234.0	222/ 227
1763.	625.7/	246.6	222/ 227
2056.	499.6/	437.9	221/ 228
2350.	485.0/	785.7	229/ 220
2644.	486.5/	1213.7	229/ 220
2938.	481.9/	1697.1	229/ 220
3231.	472.1/	2250.4	229/ 220
3525.	459.2/	2893.8	229/ 220

B E A M N O. 7 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1762 mm	112.9 kN	220	10 mm	6	352 mm
1762	END 2	321.3 kN	220	10 mm	11	176 mm

152J	3525.X 300.X 600	106J
2No50	H 528. 569.TO 3525	
6*10c/c352		
3No25	H 52. 0.TO 3525	

OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50
3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 8 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	48.	4- 16 MM	0.	6000.	YES YES
2	533.	2- 40 MM	0.	2843.	YES NO
3	533.	2- 40 MM	2657.	6000.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
-------------------	-------------------------------------	---------------------------------	----------------

0.	424.0/	2363.9	95.50/	461.64	227/ 222
500.	431.5/	1552.6	97.19/	325.48	227/ 222
1000.	426.2/	885.2	95.99/	196.02	227/ 222
1500.	426.6/	371.7	96.09/	83.72	219/ 230
2000.	554.6/	234.0	124.90/	23.65	219/ 230
2500.	589.5/	0.0	132.77/	0.00	219/ 0
3000.	688.5/	0.0	154.86/	0.00	221/ 0
3500.	803.4/	234.0	179.07/	12.60	222/ 227
4000.	783.1/	281.5	174.82/	63.39	222/ 227
4500.	664.3/	541.6	149.63/	121.99	222/ 227
5000.	641.7/	1023.8	144.53/	224.20	230/ 219
5500.	653.5/	1689.2	147.20/	350.04	230/ 219
6000.	652.5/	2502.5	146.98/	482.59	230/ 219

B E A M N O. 8 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2000 mm	275.1 kN	222	10 mm	8	285 mm
2000	4001 mm	187.5 kN	227	10 mm	4	500 mm
1999	4000 mm	180.2 kN	227	10 mm	4	500 mm
4000	END 2	267.8 kN	219	10 mm	7	333 mm

106J		6000.X 300.X 600		107J
2No40 H 533.	0.TO 2843	2No40 H 533.2657.	TO 6000	
8*10c/c285				7*10c/c333
4No16 H 148.	0.TO 6000			

OO	OO	OO	OO	OO	OO
2T40	2T40	2T40	2T40	2T40	2T40
4T16	4T16	4T16	4T16	4T16	4T16
oooo	oooo	oooo	oooo	oooo	oooo

BEAM NO. 9 DESIGN RESULTS - FLEXURE

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	52.	3- 25 MM	0.	6500.	YES YES
2	528.	2- 50 MM	0.	3067.	YES NO
	COMP.	4- 12 MM	(REQD. STEEL=	418. SQ. MM)	
3	528.	2- 50 MM	2892.	6500.	NO YES
	COMP.	5- 12 MM	(REQD. STEEL=	548. SQ. MM)	

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	963.0/ 3685.0	211.91/ 639.71	227/ 222
542.	896.1/ 2309.7	198.25/ 453.28	227/ 222
1083.	813.0/ 1282.9	181.07/ 275.06	227/ 222
1625.	798.3/ 585.3	178.00/ 131.83	219/ 230
2167.	911.2/ 234.0	201.35/ 42.44	219/ 230
2708.	884.3/ 0.0	195.84/ 0.00	220/ 0
3250.	1036.6/ 0.0	226.77/ 0.00	222/ 0
3792.	1271.5/ 234.0	272.87/ 33.37	222/ 227
4333.	1337.9/ 496.1	285.55/ 111.73	222/ 227
4875.	1252.9/ 908.3	269.29/ 200.76	222/ 227
5417.	1166.8/ 1504.9	252.57/ 316.75	230/ 219
5958.	1257.1/ 2538.3	270.11/ 487.88	230/ 219
6500.	1330.5/ 3814.8	284.14/ 667.23	230/ 219

BEAM NO. 9 DESIGN RESULTS - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2709 mm	347.4 kN	222	10 mm	14	208 mm
2709	5418 mm	174.9 kN	227	10 mm	5	541 mm
1083	4334 mm	224.9 kN	227	10 mm	6	541 mm
4334	END 2	334.3 kN	219	10 mm	10	240 mm

107J 6500.X 300.X 600 108J

2No50 H 528.	0.TO 3067 J	2No50 H 528.2892.	TO 6500	
14*10c/c208			10*10c/c240	
3No25 H 52.	0.TO 6500 J			

OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50
3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo

BEAM NO. 10 DESIGN RESULTS - FLEXURE

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	56.	2- 32 MM	0.	6500.	YES YES
2	528.	2- 50 MM	0.	3608.	YES NO
	COMP.	3- 10 MM	(REQD. STEEL=	234. SQ. MM)	
3	528.	2- 50 MM	3433.	6500.	NO YES
	COMP.	3- 10 MM	(REQD. STEEL=	234. SQ. MM)	

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
-------------------	-------------------------------------	---------------------------------	----------------

0.	1499.6/	3361.5	315.78/	571.13	228/	221
542.	1348.8/	2133.5	287.62/	425.34	228/	221
1083.	1184.5/	1348.4	256.02/	287.54	228/	221
1625.	1149.5/	868.9	249.18/	192.66	220/	229
2167.	1088.2/	505.1	237.06/	113.76	220/	229
2708.	931.7/	234.0	205.54/	42.23	220/	229
3250.	670.3/	0.0	150.97/	0.00	220/	0
3792.	654.8/	0.0	147.47/	0.00	222/	0
4333.	750.4/	337.1	167.95/	75.92	221/	228
4875.	755.6/	710.1	169.05/	159.43	221/	228
5417.	858.5/	1332.7	190.51/	284.57	229/	220
5958.	990.5/	2183.6	217.49/	433.39	229/	220
6500.	1108.3/	3451.4	241.05/	590.20	229/	220

B E A M N O. 10 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX.	SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1626 mm	272.5	kN	221	10 mm	5	406 mm
1626	3252 mm	217.1	kN	229	10 mm	3	541 mm
3250	4334 mm	210.9	kN	229	10 mm	2	541 mm
4334	END 2	292.9	kN	220	10 mm	8	309 mm
109J		6500.X	300.X	600			110J
2No50 H 528.	0.TO 3608			2No50 H 528.3433.TO 6500			
5*10c/c406							
2No32 H 56.	0.TO 6500						

FROM -	TO	MAX.	SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2500 mm	456.3	kN	221	10 mm	19	138 mm
2500	3000 mm	191.7	kN	222	10 mm	2	249 mm
3000	3500 mm	201.7	kN	222	10 mm	2	249 mm
3500	END 2	466.3	kN	220	10 mm	120	131 mm

B E A M N O. 11 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
	mm		mm	mm	STA END

1 52. 3- 25 MM 0. 6000. YES YES

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing. *

2 533. 2- 40 MM 0. 2811. YES NO

3 453. 2- 40 MM 0. 2811. YES NO

COMP. 4- 20 MM (REQD. STEEL= 1151. SQ. MM)

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing. *

4 533. 2- 40 MM 3189. 6000. NO YES

5 453. 2- 40 MM 3189. 6000. NO YES

COMP. 2- 32 MM (REQD. STEEL= 1479. SQ. MM)

REQUIRED REINF. STEEL SUMMARY :

SECTION	REINF. STEEL (+VE/-VE)	MOMENTS (+VE/-VE)	LOAD (+VE/-VE)
(MM)	(SQ. MM)	(KN-METER)	
0.	807.1/ 4043.6	179.83/ 715.72	228/ 221
500.	904.2/ 2585.2	199.93/ 494.74	228/ 221
1000.	946.1/ 1371.8	208.48/ 291.96	228/ 221
1500.	1033.5/ 598.9	226.14/ 134.89	220/ 229
2000.	1170.7/ 234.0	253.33/ 29.85	220/ 229
2500.	1136.4/ 0.0	246.60/ 0.00	220/ 0
3000.	1086.0/ 0.0	236.63/ 0.00	221/ 0
3500.	1291.6/ 0.0	276.72/ 0.00	221/ 0
4000.	1300.6/ 326.3	278.45/ 73.50	221/ 228
4500.	1134.7/ 821.3	246.26/ 182.80	221/ 228
5000.	1035.6/ 1672.7	226.57/ 347.10	229/ 220
5500.	972.1/ 3284.8	213.76/ 554.88	229/ 220
6000.	1084.3/ 4350.9	189.40/ 780.86	229/ 220

B E A M N O. 11 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX.	SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2500 mm	456.3	kN	221	10 mm	19	138 mm
2500	3000 mm	191.7	kN	222	10 mm	2	249 mm
3000	3500 mm	201.7	kN	222	10 mm	2	249 mm
3500	END 2	466.3	kN	220	10 mm	120	131 mm

110J		6000.X 300.X 600		111J				
=====								
2No40 H	533.		10.TO 2811					20*10c/c131
19*10c/c138								
3No25 H	152.		10.TO 6000					
=====					=====			

OO		OO						
2T40		2T40		2T40				
3T25		3T25		3T25		3T25		3T25
ooo		ooo		ooo		ooo		ooo

BEAM NO. 12 DESIGN RESULTS - FLEXURE

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1 60. 2- 40 MM 0. 6500. YES YES

**Please Note: * Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*

2 528. 2- 50 MM 0. 3067. YES NO	3 428. 1- 50 MM 0. 3067. YES NO
COMP. 2- 40 MM (REQD. STEEL= 2442. SQ. MM)	
4 528. 2- 50 MM 2892. 6500. NO YES	5 428. 1- 50 MM 2892. 6500. NO YES
COMP. 2- 40 MM (REQD. STEEL= 2242. SQ. MM)	

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1982.9/ 5249.5	256.30/ 971.34	228/ 221
542.	1235.2/ 3901.6	265.88/ 685.62	228/ 221
1083.	1214.3/ 2110.0	261.83/ 421.53	228/ 221
1625.	1196.2/ 916.8	258.31/ 202.51	220/ 229
2167.	1285.5/ 240.3	275.55/ 54.11	220/ 229
2708.	1184.9/ 0.0	256.10/ 0.00	220/ 0
3250.	1465.7/ 0.0	309.52/ 0.00	221/ 0
3792.	1948.1/ 234.0	394.77/ 24.86	221/ 228
4333.	2223.1/ 628.4	439.68/ 141.54	221/ 228
4875.	2275.2/ 1293.2	447.90/ 277.04	221/ 228
5417.	2141.3/ 2180.5	426.60/ 432.89	229/ 220
5958.	2247.5/ 3835.1	443.54/ 671.52	229/ 220

6500. 2268.6/ 5062.8 446.85/ 931.78 229/ 220

BEAM NO. 12 DESIGN RESULTS - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C

END 1	1083 mm	543.4 kN	221	10 mm	12	98 mm
1083	3792 mm	216.6 kN	228	10 mm	5	541 mm
3792	END 2	496.4 kN	220	10 mm	23	123 mm

111J	6500.X 300.X 600	112J
=====		
2No50 H	528. 10.TO 3067	23*10c/c123
1No50cHc428.	10.TO 3067	
2No40 H	160. 10.TO 6500	
=====		

OO		OO						
2T50		2T50		2T50				
2T40		2T40		2T40		2T40		2T40
OO		OO		OO		OO		OO

BEAM NO. 13 DESIGN RESULTS - FLEXURE

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	46.	5- 12 MM	0.	6500.	YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	0.0/ 0.0	0.00/ 0.00	0/ 0
542.	234.0/ 0.0	32.67/ 0.00	219/ 0
1083.	268.8/ 0.0	60.53/ 0.00	219/ 0

1625.	365.1/	0.0	82.24/	0.00	219/	0
2167.	433.9/	0.0	97.74/	0.00	219/	0
2708.	475.2/	0.0	107.04/	0.00	219/	0
3250.	489.0/	0.0	110.14/	0.00	219/	0
3792.	475.2/	0.0	107.04/	0.00	219/	0
4333.	433.9/	0.0	97.74/	0.00	219/	0
4875.	365.1/	0.0	82.24/	0.00	219/	0
5417.	268.8/	0.0	60.53/	0.00	219/	0
5958.	234.0/	0.0	32.67/	0.00	219/	0
6500.	0.0/	0.0	0.00/	0.00	0/	0

B E A M N O. 13 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX.	SHEAR	LOAD	LINKS	NO.	SPACING	C/C
END 1	1625 mm	62.8	kN	226	10 mm	6	324	mm
4875	END 2	62.8	kN	226	10 mm	6	324	mm

153J 6500.X 300.X 600 154J

6*10c/c324

6*10c/c324

5No12 H 46. 0.TO 6500

5T12	5T12	5T12	5T12	5T12	5T12	5T12
oooo	oooo	oooo	oooo	oooo	oooo	oooo

B E A M N O. 14 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
	mm		mm	mm	STA
					END

1 50. 4- 20 MM 0. 6500. YES YES

2	528.	2- 50 MM	0.	3067.	YES	NO
3	528.	2- 50 MM	3433.	6500.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION	REINF	STEEL (+VE/-VE)	MOMENTS (+VE/-VE)	LOAD (+VE/-VE)
(MM)	(SQ. MM)		(KN-METER)	

0.	877.4/	2631.6	194.41/	501.47	227/	222
542.	954.5/	1714.9	210.20/	354.59	227/	222
1083.	971.0/	1034.2	213.55/	226.29	227/	222
1625.	1010.3/	593.1	221.48/	133.59	219/	230
2167.	968.4/	246.5	213.01/	55.52	219/	230
2708.	836.5/	0.0	185.95/	0.00	219/	0
3250.	623.0/	0.0	140.31/	0.00	219/	0
3792.	605.7/	0.0	136.43/	0.00	222/	0
4333.	680.8/	234.0	153.22/	27.54	222/	227
4875.	672.4/	493.9	151.42/	111.25	222/	227
5417.	673.9/	1043.3	151.74/	228.10	230/	219
5958.	633.8/	1783.9	142.75/	366.67	230/	219
6500.	538.7/	2789.9	121.32/	523.82	230/	219

B E A M N O. 14 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX.	SHEAR	LOAD	LINKS	NO.	SPACING	C/C
END 1	1084 mm	288.3	kN	222	10 mm	4	361	mm
1084	1626 mm	219.7	kN	222	10 mm	2	270	mm
1624	4875 mm	204.3	kN	227	10 mm	6	541	mm
4875	END 2	307.3	kN	219	10 mm	7	270	mm

113J 6500.X 300.X 600 114J

2No50 H 528.	0.TO 3067	2No50 H 528.3433.TO 6500	
4*10c/c361			7*10c/c270
4No20 H 50.	0.TO 6500		

OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50
4T20	4T20	4T20	4T20	4T20	4T20
oooo	oooo	oooo	oooo	oooo	oooo

=====

B E A M N O. 15 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	50.	4- 20 MM	0.	6000.	YES YES
2	528.	2- 50 MM	0.	2900.	YES NO
	COMP.	3- 10 MM (REQD. STEEL=	234. SQ. MM)		
3	528.	2- 50 MM	3100.	6000.	NO YES
	COMP.	6- 8 MM (REQD. STEEL=	280. SQ. MM)		

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	535.9/ 3465.1	120.70/ 593.09	227/ 222
500.	639.9/ 2037.1	144.13/ 409.60	227/ 222
1000.	694.0/ 1126.7	156.03/ 244.68	227/ 222
1500.	787.1/ 535.4	175.66/ 120.59	219/ 230
2000.	877.3/ 234.0	194.39/ 32.20	219/ 230
2500.	840.3/ 0.0	186.73/ 0.00	219/ 0
3000.	812.3/ 0.0	180.92/ 0.00	222/ 0
3500.	996.2/ 0.0	218.64/ 0.00	222/ 0
4000.	1044.7/ 260.3	228.38/ 58.63	222/ 227
4500.	962.1/ 645.2	211.73/ 145.32	222/ 227
5000.	868.8/ 1236.7	192.64/ 266.16	230/ 219
5500.	819.6/ 2156.2	182.44/ 429.01	230/ 219
6000.	716.1/ 3546.8	160.71/ 610.41	230/ 219

B E A M N O. 15 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
----- ----- ----- ----- ----- -----						
END 1	2000 mm	383.4 kN	222	10 mm	12	181 mm
2000	4500 mm	220.4 kN	227	10 mm	5	499 mm
4499	END 2	379.3 kN	219	10 mm	10	166 mm
----- ----- ----- ----- ----- -----						

=====

B E A M N O. 16 D E S I G N R E S U L T S - F L E X U R E

LEN - 2225. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

00	00	00		00	00
2T50	2T50	2T50		2T50	2T50
4T20	4T20	4T20		4T20	4T20
oooo	oooo	oooo		oooo	oooo

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1022.3/ 4204.7	223.89/ 749.89	227/ 222
185.	1034.3/ 3839.0	226.31/ 672.36	227/ 222
371.	1038.9/ 3484.2	227.23/ 597.15	227/ 222
556.	1035.8/ 2793.9	226.60/ 524.38	227/ 222
742.	1024.7/ 2315.4	224.39/ 454.16	227/ 222
927.	1005.6/ 1900.0	220.53/ 386.64	227/ 222
1113.	978.1/ 1533.4	214.99/ 321.98	227/ 222
1298.	942.5/ 1205.9	207.74/ 260.19	227/ 222
1483.	898.8/ 910.7	198.81/ 201.25	227/ 222
1669.	870.6/ 665.1	193.02/ 149.80	219/ 230
1854.	859.3/ 470.4	190.68/ 105.94	219/ 230
2040.	836.3/ 282.5	185.90/ 63.63	219/ 230
2225.	802.2/ 234.0	178.81/ 22.82	219/ 230

B E A M N O. 16 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1112 mm	424.1 kN	222	10 mm	9	139 mm
1112	END 2	257.9 kN	222	10 mm	4	370 mm
3No32 H 537.	0.TO 2225					
9*10c/c139						4*10c/c370
4No20 H 50.	0.TO 2225					

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1112 mm	424.1 kN	222	10 mm	9	139 mm
1112	END 2	257.9 kN	222	10 mm	4	370 mm
3No32 H 537.	0.TO 2225					
9*10c/c139						4*10c/c370
4No20 H 50.	0.TO 2225					

=====

B E A M N O. 17 D E S I G N R E S U L T S - F L E X U R E

LEN - 2050. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
	mm		mm	mm	STA END

1	60.	2- 40 MM	0.	2050.	YES YES
2	548.	3- 10 MM	0.	1013.	YES NO
3	547.	5- 12 MM	695.	2050.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION	REINF STEEL (+VE/-VE)	MOMENTS (+VE/-VE)	LOAD (+VE/-VE)
(MM)	(SQ. MM)	(KN-METER)	

0.	949.8/	234.0	209.24/	48.99	219/	230
171.	892.0/	234.0	197.40/	14.30	219/	230

342.	825.8/	0.0	183.72/	0.00	219/	0
512.	751.5/	0.0	168.19/	0.00	219/	0
683.	669.6/	0.0	150.81/	0.00	219/	0
854.	761.4/	0.0	170.27/	0.00	222/	0
1025.	908.5/	0.0	200.81/	0.00	222/	0
1196.	1050.2/	0.0	229.49/	0.00	222/	0
1367.	1186.1/	0.0	256.33/	0.00	222/	0
1537.	1315.7/	234.0	281.32/	26.41	222/	227
1708.	1438.5/	241.8	304.46/	54.46	222/	227
1879.	1554.1/	371.8	325.75/	83.75	222/	227
2050.	1662.0/	507.4	345.20/	114.28	222/	227

B E A M N O. 17 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1024 mm	238.3 kN	222	10 mm	6	204 mm
1024	END 2	193.7 kN	219	10 mm	4	341 mm

159J	2050. X 300. X 600	157J
3No10 H 548.	0.TO 10135No12 H 547. 695.TO 2050	
6*10c/c204		4*10c/c341
2No40 H 60.	0.TO 2050	

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1024 mm	238.3 kN	222	10 mm	6	204 mm
1024	END 2	193.7 kN	219	10 mm	4	341 mm

=====

B E A M N O. 18 D E S I G N R E S U L T S - F L E X U R E

LEN - 2225. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
	mm		mm	mm	STA END

1 60. 2- 40 MM 0. 2225. YES YES

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*

2 533. 2- 40 MM 0. 2225. YES YES

3 453. 2- 40 MM 0. 2225. YES YES

COMP. 4- 20 MM (REQD. STEEL= 1151. SQ. MM)

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1498.0/ 462.4	315.49/ 104.15	222/ 227
185.	1582.1/ 623.5	330.83/ 140.43	222/ 227
371.	1654.6/ 799.3	343.88/ 178.22	222/ 227
556.	1714.2/ 990.8	354.47/ 217.55	222/ 227
742.	1760.1/ 1197.1	362.52/ 258.47	222/ 227
927.	1790.7/ 1420.1	367.86/ 301.03	222/ 227
1113.	1805.0/ 1662.5	370.34/ 345.28	222/ 227
1298.	1817.9/ 1942.6	372.57/ 393.84	230/ 219
1483.	1866.7/ 2308.6	380.95/ 453.10	230/ 219
1669.	1906.2/ 2727.1	387.69/ 515.06	230/ 219
1854.	1936.7/ 3401.3	392.85/ 579.57	230/ 219
2040.	1958.1/ 3717.2	396.45/ 646.53	230/ 219
2225.	1970.7/ 4043.9	398.56/ 715.79	230/ 219

B E A M N O. 18 D E S I G N R E S U L T S - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1112 mm	213.4 kN	219	10 mm	4	370 mm
1112	END 2	379.5 kN	219	10 mm	8	158 mm
157J		2225. X 300. X 600				116J
2No40 H 533.	0. TO 2225					
4*10c/c370						8*10c/c158
2No40 H 60.	0. TO 2225					

OO	OO	OO	OO	OO	OO	OO
2T40	2T40	2T40	2T40	2T40	2T40	2T40
2T40	2T40	2T40	2T40	2T40	2T40	2T40
OO	OO	OO	OO	OO	OO	OO

B E A M N O. 19 D E S I G N R E S U L T S - FLEXURE

LEN - 3525. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	60.	2- 40 MM	0.	3525.	YES YES
2	528.	2- 50 MM	0.	3250.	YES NO
3	547.	5- 12 MM	2389.	3525.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1719.9/ 2918.9	355.47/ 541.38	233/ 224
294.	1617.3/ 2296.8	337.20/ 451.27	233/ 224
587.	1496.8/ 1783.6	315.26/ 366.62	233/ 224
881.	1359.7/ 1347.9	289.67/ 287.44	233/ 224
1175.	1248.0/ 1011.1	268.34/ 221.63	225/ 232
1469.	1117.1/ 716.3	242.79/ 160.75	225/ 232
1762.	962.3/ 459.6	211.78/ 103.51	225/ 232
2056.	785.4/ 234.0	175.31/ 49.94	225/ 232
2350.	592.1/ 234.0	133.37/ 0.03	225/ 232
2644.	381.6/ 0.0	85.96/ 0.00	225/ 0
2937.	505.2/ 0.0	113.80/ 0.00	224/ 0
3231.	662.7/ 234.0	149.26/ 46.74	224/ 233
3525.	804.3/ 467.3	179.25/ 105.25	224/ 233

B E A M N O. 19 D E S I G N R E S U L T S - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1469 mm	316.1 kN	224	10 mm	6	293 mm
1469	1763 mm	223.0 kN	224	10 mm	2	146 mm
1762	END 2	226.5 kN	225	10 mm	8	251 mm

113J_____3525.X 300.X 600_____153J_____									
=====									
2No50 H 528.	10.T0	3250				15No12 H 547.2389.	T0	3525	
6*10c/c293						8*10c/c251			
2No40 H 60.	10.T0	3525							
=====									
=====									
OO		OO		OO		OO		OO	
2T50		2T50		2T50		2T50		2T50	
2T40		2T40		2T40		2T40		2T40	
OO		OO		OO		OO		OO	
=====									
=====									

B E A M N O. 20 D E S I G N R E S U L T S - F L E X U R E

LEN - 1515. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	50.	4- 20 MM	0.	1515.	YES YES
2	528.	2- 50 MM	0.	1515.	YES YES
COMP. 6- 8 MM (REQD. STEEL= 281. SQ. MM)					

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
0.	797.0/	471.1	177.72/
126.	812.7/	604.1	180.99/
253.	823.3/	744.5	183.21/
379.	862.3/	929.1	191.30/
505.	902.4/	1130.0	199.54/
631.	939.1/	1344.6	207.05/
758.	972.3/	1574.9	213.81/
884.	1002.0/	1823.1	219.82/
1010.	1028.2/	2092.2	225.08/
1136.	1050.8/	2385.7	229.61/
1263.	1069.9/	2708.5	233.42/
1389.	1085.5/	3313.7	236.52/
1515.	1097.6/	3547.7	238.94/

B E A M N O. 20 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C

END 1	757 mm	289.4 kN	225	10 mm	4	252 mm
757	END 2	397.0 kN	225	10 mm	6	151 mm

153J	1515.X 300.X 600	109J
=====		
2No50 H 528.	0.T0 1515	
4*10c/c252		6*10c/c151
4No20 H 50.	0.T0 1515	
=====		
=====		

OO		OO		OO		OO		OO	
2T50		2T50		2T50		2T50		2T50	
4T20		4T20		4T20		4T20		4T20	
OOOO		OOOO		OOOO		OOOO		OOOO	

B E A M N O. 21 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	52.	3- 25 MM	0.	6650.	YES YES
2	528.	2- 50 MM	0.	3117.	YES NO
COMP. 4- 16 MM (REQD. STEEL= 619. SQ. MM)					

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*
3 533. 2- 40 MM 3623. 6650. NO YES
4 453. 2- 40 MM 3623. 6650. NO YES
COMP. 4- 20 MM (REQD. STEEL= 1156. SQ. MM)

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	619.3/ 3885.9	97.68/ 682.30	233/ 224
554.	606.6/ 2336.6	136.63/ 457.45	233/ 224
1108.	719.1/ 1182.2	161.34/ 255.57	233/ 224
1662.	901.5/ 484.1	199.37/ 109.03	225/ 232
2217.	1083.6/ 234.0	236.14/ 6.58	225/ 232
2771.	1094.5/ 0.0	238.31/ 0.00	225/ 0
3325.	1169.4/ 0.0	253.07/ 0.00	224/ 0
3879.	1359.6/ 0.0	289.66/ 0.00	224/ 0
4433.	1360.9/ 234.0	289.90/ 52.52	224/ 233
4988.	1182.0/ 682.0	255.53/ 153.46	224/ 233
5542.	984.7/ 1387.4	216.31/ 294.90	232/ 225
6096.	871.1/ 2582.6	193.10/ 494.36	232/ 225
6650.	782.1/ 4048.7	155.66/ 716.81	232/ 225

B E A M N O. 21 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2771 mm	424.2 kN	224	10 mm	20	145 mm
2771	4434 mm	166.3 kN	233	10 mm	3	554 mm
2217	4434 mm	237.4 kN	233	10 mm	4	554 mm
4434	END 2	419.8 kN	225	10 mm	15	158 mm
109J		6650.X 300.X 600				105J
2No50 H 528.	10.TO 3117			2No40 H 533.3623	TO 6650	
20*10c/c145					15*10c/c158	
3No25 H 152.	10.TO 6650					
OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T40	2T40	2T40	2T40
3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 22 D E S I G N R E S U L T S - F L E X U R E

LEN - 2885. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
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1 52. 3- 25 MM 0. 2885. YES YES

**Please Note:

* Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*

2 537. 3- 32 MM 0. 2885. YES YES

3 473. 3- 32 MM 0. 2885. YES YES

COMP. 3- 25 MM (REQD. STEEL= 1277. SQ. MM)

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
-------------------	------------------------------------	---------------------------------	----------------

0. 1460.9/ 4161.7 308.63/ 740.76 233/ 224

240. 1419.5/ 3695.4 300.93/ 641.91 233/ 224

481. 1365.0/ 2961.1 290.67/ 547.00 233/ 224

721. 1297.0/ 2329.6 277.76/ 456.36 233/ 224

962. 1215.9/ 1804.0 262.13/ 370.16 233/ 224

1202. 1121.4/ 1355.0 243.64/ 288.79 233/ 224

1442. 1013.9/ 966.1 222.20/ 212.55 233/ 224

1683. 948.9/ 678.6 209.05/ 152.74 225/ 232

1923. 864.4/ 430.1 191.73/ 96.87 225/ 232

2164. 758.2/ 234.0 169.59/ 43.83 225/ 232

2404. 634.9/ 0.0 143.00/ 0.00 225/ 0

2645. 497.9/ 0.0 112.15/ 0.00 225/ 0

2885. 638.1/ 0.0 143.73/ 0.00 223/ 0

B E A M N O. 22 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM -	TO	SHEAR TORSN mm	LOAD kNm	LINK S T	NO. S T	SPACING mm C/C
END 1	1442	418.9	9	224224	10mm	9 7 16 155 199 85
1442	END 2	193.8	9	224222	10mm	2 7 10 571 199 144
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 105 mm2 EVENLY DISTRIBUTED						
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *						

105J	2885.X 300.X 600	150J
3No32 H 537.	0.TO 2885	
16*10c/c 85		10*10c/c144
3No25 H 52.	0.TO 2885	

OOO	OOO	OOO	OOO	OOO	OOO
3T32	3T32	3T32	3T32	3T32	3T32
3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 23 D E S I G N R E S U L T S - F L E X U R E

LEN - 2425. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	60.	2- 40 MM	0.	2425.	YES YES
2	528.	2- 50 MM	0.	2425.	YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	551.1/	0.0	224/ 0
202.	692.6/	0.0	224/ 0
404.	830.7/	234.0	224/ 233
606.	959.3/	292.2	224/ 233
808.	1077.9/	478.8	224/ 233
1010.	1185.8/	673.2	224/ 233
1212.	1282.3/	890.4	224/ 233
1415.	1366.9/	1124.8	224/ 233
1617.	1438.8/	1378.6	224/ 233
1819.	1527.2/	1685.3	224/ 233
2021.	1624.1/	2045.6	224/ 233
2223.	1713.0/	2453.3	224/ 233
2425.	1793.6/	2923.9	224/ 233

B E A M N O. 23 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO mm	SHEAR TORSN kNm	LOAD S T	LINK SIZE	NO. S T	SPACING mm C/C

END 1	1212	183.5	9	225225	10mm	2	6	8	571	199	135
1212	END 2	337.1	9	225222	10mm	5	6	12	218	199	101

EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 581 mm2 EVENLY DISTRIBUTED
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *

150J 2425.X 300.X 600 101J

2No50/H 528.	0.TO 2425										
8*10c/c135											
2No40/H 60.	0.TO 2425										

B E A M N O. 24 D E S I G N R E S U L T S - F L E X U R E

LEN - 2885. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	50.	3- 20 MM	0.	2885.	YES YES
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REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	0.0/	0.0	0/ 0
240.	234.0/	0.0	226/ 0
481.	263.7/	0.0	226/ 0
721.	387.0/	0.0	226/ 0

962.	501.6/	0.0	112.98/	0.00	226/	0
1202.	604.7/	0.0	136.20/	0.00	226/	0
1442.	694.8/	0.0	156.18/	0.00	226/	0
1683.	772.3/	0.0	172.55/	0.00	226/	0
1923.	832.3/	0.0	185.07/	0.00	226/	0
2164.	876.1/	0.0	194.15/	0.00	226/	0
2404.	904.7/	0.0	200.02/	0.00	226/	0
2645.	918.7/	0.0	202.90/	0.00	226/	0
2885.	919.7/	0.0	203.09/	0.00	226/	0

B E A M N O. 24 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2163 mm	126.4 kN	226	10 mm	7	360 mm
2163	END 2	10.8 kN	223	10 mm	3	360 mm
Nominal links should be provided through out the length.						

152J	2885.X 300.X 600	151J
7*10c/c360		3*10c/c360
3No20 H 50.	0.TO 2885	

3T20	3T20	3T20	3T20	3T20	3T20	3T20
ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 25 D E S I G N R E S U L T S - F L E X U R E

LEN - 2425. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1 50. 3- 20 MM 0. 2425. YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	930.0/	0.0	205.19/ 0.00 223/ 0
202.	884.6/	0.0	195.89/ 0.00 223/ 0
404.	831.4/	0.0	184.89/ 0.00 223/ 0
606.	770.5/	0.0	172.19/ 0.00 223/ 0
808.	702.8/	0.0	157.88/ 0.00 223/ 0
1010.	630.2/	0.0	141.95/ 0.00 223/ 0
1212.	553.2/	0.0	124.59/ 0.00 223/ 0
1415.	470.3/	0.0	105.92/ 0.00 223/ 0
1617.	382.7/	0.0	86.19/ 0.00 223/ 0
1819.	291.1/	0.0	65.56/ 0.00 223/ 0
2021.	234.0/	0.0	44.21/ 0.00 223/ 0
2223.	234.0/	0.0	22.31/ 0.00 223/ 0
2425.	0.0/	0.0	0.00/ 0.00 0/ 0

B E A M N O. 25 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1212 mm	41.8 kN	223	10 mm	5	303 mm
1212	END 2	111.1 kN	223	10 mm	5	303 mm

151J	2425.X 300.X 600	149J
5*10c/c303		5*10c/c303
3No20 H 50.	0.TO 2425	

3T20	3T20	3T20	3T20	3T20	3T20	3T20
ooo	ooo	ooo	ooo	ooo	ooo	ooo

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B E A M N O. 26 D E S I G N R E S U L T S - F L E X U R E

LEN - 3525. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	60.	2- 40 MM	0.	3525.	YES YES
2	528.	2- 50 MM	0.	2956.	YES NO
	COMP.	3- 12 MM	(REQD. STEEL=	318. SQ. MM)	
3	548.	5- 10 MM	2389.	3525.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
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0.	1711.7/	3584.9	354.02/	618.50	234/	223
294.	1631.7/	2695.9	339.78/	510.65	234/	223
587.	1533.5/	2026.2	322.00/	407.80	234/	223
881.	1417.3/	1470.7	300.51/	310.44	234/	223
1175.	1353.5/	1065.0	288.50/	232.43	225/	232
1469.	1271.8/	719.5	272.92/	161.44	225/	232
1762.	1155.4/	420.3	250.32/	94.68	225/	232
2056.	1003.9/	234.0	220.20/	32.33	225/	232
2350.	816.5/	0.0	181.80/	0.00	225/	0
2644.	596.4/	0.0	134.33/	0.00	225/	0
2937.	777.8/	0.0	173.70/	0.00	223/	0
3231.	955.0/	234.0	210.30/	25.74	223/	234
3525.	1099.1/	387.0	239.21/	87.17	223/	234

B E A M N O. 26 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING	C/C
----- ----- ----- ----- ----- ----- ----- -----							
END 1	1762 mm	375.0 kN	223	10 mm	11	176	mm
1762	END 2	256.2 kN	226	10 mm	10	195	mm
----- ----- ----- ----- ----- ----- ----- -----							

_____ 114J _____ 3525. X 300. X 600 _____ 154J _____

2No50 H 528.	0.	TO 2956						5No10 H 548.	2389.	TO 3525
11*10c/c176								10*10c/c195		
2No40 H 160.	0.	TO 3525								
=====										

OO		OO		OO		OO		OO		OO
2T50		2T50		2T50		2T50		2T50		5T10
2T40		2T40		2T40		2T40		2T40		2T40
OO		OO		OO		OO		OO		OO
=====										

=====

B E A M N O. 27 D E S I G N R E S U L T S - F L E X U R E

LEN - 1515. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	52.	3- 25 MM	0.	1515.	YES YES
2	528.	2- 50 MM	0.	1515.	YES YES
	COMP.	5- 10 MM	(REQD. STEEL=	378. SQ. MM)	

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
-------------------	------------------------------------	---------------------------------	----------------

0.	1106.4/	383.0	240.68/	86.26	223/	234
126.	1118.2/	522.3	243.01/	117.65	223/	234
253.	1123.1/	665.3	243.96/	149.85	223/	234
379.	1120.9/	821.6	243.55/	182.85	223/	234
505.	1167.8/	1040.1	252.77/	227.48	231/	226
631.	1212.2/	1275.7	261.41/	273.69	231/	226
758.	1252.6/	1529.5	269.24/	321.26	231/	226
884.	1289.5/	1803.4	276.32/	370.07	231/	226
1010.	1322.5/	2101.4	282.62/	420.14	231/	226
1136.	1351.8/	2427.9	288.18/	471.40	231/	226
1263.	1377.6/	2788.9	293.05/	523.69	231/	226
1389.	1399.8/	3389.1	297.24/	576.99	231/	226
1515.	1418.7/	3644.8	300.77/	631.20	231/	226

B E A M N O. 27 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	757 mm	319.0 kN	226	10 mm	5	189 mm
757	END 2	432.7 kN	226	10 mm	7	126 mm
154J			1515.X 300.X 600			110J
2No50 H 528.	0.TO 1515					
5*10c/c189						7*10c/c126
3No25 H 52.	0.TO 1515					
OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50	2T50
3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 28 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
1	50.	4- 20 MM	0.	6650.	YES	YES
2	528.	2- 50 MM	0.	3117.	YES	NO
COMP.	4- 12 MM	(REQD. STEEL=	429. SQ. MM)			
3	528.	2- 50 MM	3533.	6650.	NO	YES
COMP.	3- 10 MM	(REQD. STEEL=	234. SQ. MM)			

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
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0.	605.5/	3695.9	136.39/	642.03	234/	223
554.	632.3/	2331.8	142.42/	456.70	234/	223
1108.	642.1/	1310.7	144.63/	280.39	234/	223
1662.	743.5/	643.5	166.50/	144.94	226/	231
2217.	955.7/	277.8	210.44/	62.57	226/	231
2771.	1025.6/	0.0	224.55/	0.00	226/	0
3325.	934.9/	0.0	206.21/	0.00	204/	0
3879.	1067.1/	0.0	232.87/	0.00	224/	0
4433.	1047.5/	234.0	228.95/	46.27	223/	234
4988.	881.5/	539.4	195.26/	121.48	223/	234
5542.	766.2/	1090.0	171.28/	237.41	231/	226
6096.	789.8/	2000.2	176.23/	403.47	231/	226
6650.	795.2/	3396.5	177.35/	578.56	231/	226

B E A M N O. 28 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM -	TO	SHEAR TORSN	LOAD	LINK	NO.	SPACING mm C/C
mm	kN	kNm	S	T	S	T
END 1	2771	337.7	11	223226	10mm	12 14 27 230 199 107
2771	4988	169.3	11	234226	10mm	4 12 15 554 199 146
4987	END 2	319.2	11	234220	10mm	6 8 15 243 199 104
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 350 mm2 EVENLY DISTRIBUTED						
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *						

110J 6650.X 300.X 600 106J

2No50 H 528.	10.TO 3117				2No50 H 528.3533.TO 16650				
27*10c/c107									15*10c/c104
4No20 H 150.	10.TO 6650								

OO	OO	OO	OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50
4T20	4T20	4T20	4T20	4T20	4T20	4T20	4T20	4T20	4T20
oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo

B E A M N O. 30 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1137 mm	459.1 kN	223	10 mm	10	126 mm
1137	END 2	262.0 kN	223	10 mm	4	379 mm
115J			2275.X 300.X 600			155J
2No50 H 528.	0.TO 2275					
10*10c/c126						4*10c/c379
3No25 H 52.	0.TO 2275					
OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50	2T50
3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 31 D E S I G N R E S U L T S - F L E X U R E

LEN - 2765. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
1	52.	3- 25 MM	0.	2765.	YES	YES
2	528.	2- 50 MM	22.	2765.	NO	YES
COMP.		3- 16 MM	(REQD. STEEL=	587. SQ. MM)		

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1309.7/	0.0	280.18/ 0.00
230.	1055.7/	0.0	230.58/ 0.00

461.	939.2/	0.0	207.09/	0.00	223/	0
691.	1013.5/	0.0	222.14/	0.00	223/	0
922.	1057.7/	234.0	231.00/	11.72	223/	234
1152.	1077.5/	292.4	234.93/	65.85	224/	233
1383.	1069.5/	546.6	233.33/	123.11	224/	233
1613.	1049.6/	841.2	229.37/	186.93	232/	225
1843.	1115.8/	1289.0	242.54/	276.23	232/	225
2074.	1169.0/	1803.9	253.00/	370.15	232/	225
2304.	1209.6/	2407.1	260.91/	468.24	232/	225
2535.	1237.7/	3357.2	266.35/	570.22	232/	225
2765.	1254.4/	3853.4	269.58/	675.41	232/	225

B E A M N O. 31 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1382 mm	203.6 kN	225	10 mm	5	345 mm
1382	END 2	462.7 kN	225	10 mm	14	106 mm
155J			2765.X 300.X 600			111J
2No50 H 528.	22.TO 2765					
5*10c/c345						14*10c/c106
3No25 H 52.	0.TO 2765					
OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50	2T50
3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 32 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	50.	4- 20 MM	0.	6650.	YES	YES
2	528.	2- 50 MM	0.	3117.	YES	NO
	COMP.	4- 12 MM	(REQD. STEEL=	423. SQ. MM)		
3	528.	2- 50 MM	3533.	6650.	NO	YES
	COMP.	3- 10 MM	(REQD. STEEL=	234. SQ. MM)		

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	571.4/ 3689.3	128.69/ 640.63	233/ 224
554.	604.2/ 2331.2	136.08/ 456.61	233/ 224
1108.	620.0/ 1317.2	139.64/ 281.61	233/ 224
1662.	701.8/ 631.7	157.67/ 142.29	225/ 232
2217.	921.4/ 273.9	203.44/ 61.70	225/ 232
2771.	999.8/ 0.0	219.37/ 0.00	225/ 0
3325.	908.1/ 0.0	200.72/ 0.00	204/ 0
3879.	1027.9/ 0.0	225.02/ 0.00	224/ 0
4433.	1002.2/ 234.0	219.85/ 43.13	224/ 233
4988.	831.2/ 519.4	184.84/ 116.99	224/ 233
5542.	719.6/ 1070.1	161.44/ 233.46	232/ 225
6096.	734.5/ 1965.5	164.61/ 397.69	232/ 225
6650.	731.4/ 3360.6	163.95/ 570.95	232/ 225

B E A M N O. 32 D E S I G N R E S U L T S - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2217 mm	335.3 kN	224	10 mm	11	221 mm
2217	4434 mm	229.1 kN	233	10 mm	4	554 mm
2217	4434 mm	209.6 kN	233	10 mm	4	554 mm
4434	END 2	315.9 kN	225	10 mm	9	277 mm

111J	6650.X 300.X 600	107J
2No50 H 528.	0.TO 3117	2No50 H 528.3533.TO 6650
11*10c/c221		9*10c/c277
4No20 H 50.	0.TO 6650	

OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50
4T20	4T20	4T20	4T20	4T20	4T20
oooo	oooo	oooo	oooo	oooo	oooo

B E A M N O. 33 D E S I G N R E S U L T S - FLEXURE

LEN - 5310. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	60.	2- 40 MM	0.	5310.	YES YES
2	528.	2- 50 MM	0.	5310.	YES YES
	COMP.	4- 12 MM	(REQD. STEEL=	418. SQ. MM)	

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1593.1/ 3528.8	332.83/ 606.61	233/ 224
442.	1335.7/ 2356.1	285.13/ 460.45	233/ 224
885.	1079.8/ 1516.5	235.39/ 318.87	233/ 224
1328.	818.6/ 836.1	182.24/ 185.87	233/ 224
1770.	670.4/ 399.9	150.99/ 90.06	225/ 232
2212.	492.8/ 234.0	110.98/ 6.82	225/ 232
2655.	577.8/ 234.0	130.14/ 9.40	223/ 234
3098.	890.0/ 386.4	197.01/ 87.02	223/ 234
3540.	1125.3/ 768.8	244.40/ 171.83	223/ 234
3982.	1289.3/ 1217.4	276.28/ 262.43	223/ 234
4425.	1451.7/ 1786.4	306.92/ 367.11	231/ 226
4868.	1716.6/ 2628.9	354.88/ 501.08	231/ 226
5310.	1984.3/ 3684.8	400.83/ 639.66	232/ 225

B E A M N O. 33 D E S I G N R E S U L T S - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2213 mm	332.7 kN	224	10 mm	10	245 mm
2213	2656 mm	210.1 kN	232	10 mm	2	221 mm
2654	3097 mm	193.2 kN	232	10 mm	2	221 mm
3097	END 2	315.8 kN	225	10 mm	9	276 mm

B E A M N O. 38 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO SHEAR TORSN LOAD LINK NO. SPACING mm C/C									
mm	kN	S	T	SIZE	S	T	S+T	S	T
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
END 1	2217	403.2	17	223226	10mm	14	12	27	158 199 88
2217	4434	220.8	17	234226	10mm	4	12	15	554 199 146
2217	4434	215.3	17	234221	10mm	4	12	15	554 199 146
4434	END 2	397.7	17	226221	10mm	13	12	27	170 199 92
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 361 mm2 EVENLY DISTRIBUTED									
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									

112J	6650.X 300.X 600	108J
=====		
2No50 H 528.	10.TO 3117	12No50 H 528.3533.TO16650
27*10c/c 88		127*10c/c 92
4No20 H 150.	10.TO 6650	
=====		

OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50
4T20	4T20	4T20	4T20	4T20	4T20
oooo	oooo	oooo	oooo	oooo	oooo

B E A M N O. 39 D E S I G N R E S U L T S - F L E X U R E

LEN - 5310. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
----- ----- ----- ----- ----- ----- -----						

1	52.	3- 25 MM	0.	5310.	YES	YES
2	528.	2- 50 MM	0.	2670.	YES	NO
COMP.	5- 10 MM	(REQD. STEEL=	370. SQ. MM)			
528.	2- 50 MM	2198.	5310.	NO	YES	

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	842.5/ 3636.5	187.20/ 629.43	234/ 223

442.	853.3/ 2350.4	189.43/ 459.57	234/ 223
885.	821.2/ 1435.2	182.77/ 303.85	234/ 223
1328.	743.6/ 732.8	166.51/ 164.25	234/ 223
1770.	752.0/ 311.1	168.30/ 70.08	226/ 231
2212.	671.1/ 0.0	151.15/ 0.00	226/ 0
2655.	628.9/ 0.0	141.64/ 0.00	223/ 0
3098.	908.7/ 234.0	200.85/ 5.40	223/ 234
3540.	1095.3/ 341.6	238.46/ 76.94	223/ 234
3982.	1186.8/ 709.3	256.48/ 159.27	223/ 234
4425.	1190.2/ 1161.9	257.13/ 251.61	223/ 234
4868.	1280.5/ 1899.2	274.61/ 386.49	231/ 226
5310.	1327.4/ 2878.1	283.56/ 535.89	231/ 226

B E A M N O. 39 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM - TO MAX. SHEAR LOAD LINKS NO. SPACING C/C									
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									
END 1	884 mm	398.4 kN	223	10 mm	7	147 mm			
	884	3540 mm	212.0 kN	234	10 mm	5	531 mm		
	3540	END 2	352.2 kN	226	10 mm	9	221 mm		
----- ----- ----- ----- ----- ----- ----- ----- ----- -----									

108J 5310.X 300.X 600 104J

2No50 H 528.	0.TO 2670	2No50 H 528.2198.TO 5310							
7*10c/c147						9*10c/c221			
3No25 H 52.	0.TO 5310								
=====									

OO	OO	OO	OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50
3T25	3T25	3T25	3T25	3T25	3T25	3T25	3T25	3T25	3T25
ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo

155J 2225.X 300.X 600 160J

B E A M N O. 40 D E S I G N R E S U L T S - F L E X U R E

LEN - 2225. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1 50. 4- 20 MM 0. 2225. YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
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0.	0.0/	0.00/	0.00	0/	0
185.	234.0/	0.0	25.81/	0.00	219/ 0
371.	234.0/	0.0	51.17/	0.00	219/ 0
556.	336.5/	0.0	75.80/	0.00	219/ 0
742.	441.7/	0.0	99.48/	0.00	219/ 0
927.	541.3/	0.0	121.93/	0.00	219/ 0
1113.	634.3/	0.0	142.86/	0.00	219/ 0
1298.	722.8/	0.0	162.14/	0.00	219/ 0
1483.	805.5/	0.0	179.50/	0.00	219/ 0
1669.	881.6/	0.0	195.27/	0.00	219/ 0
1854.	951.1/	0.0	209.51/	0.00	219/ 0
2040.	1014.4/	0.0	222.31/	0.00	219/ 0
2225.	1071.9/	0.0	233.82/	0.00	219/ 0

B E A M N O. 40 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C

END 1	1112 mm	139.9 kN	219	10 mm	4	370 mm
1112	END 2	58.8 kN	219	10 mm	4	370 mm

B E A M N O. 41 D E S I G N R E S U L T S - F L E X U R E

LEN - 2050. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1 50. 4- 20 MM 0. 2050. YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KN-METER)	LOAD(+VE/-VE)
-------------------	-----------------------------------	--------------------------------	---------------

0.	1033.7/	0.0	226.19/	0.00	222/ 0
171.	1054.5/	0.0	230.35/	0.00	222/ 0
342.	1070.4/	0.0	233.51/	0.00	222/ 0
512.	1081.3/	0.0	235.69/	0.00	222/ 0
683.	1087.3/	0.0	236.89/	0.00	222/ 0
854.	1088.3/	0.0	237.09/	0.00	222/ 0
1025.	1084.4/	0.0	236.30/	0.00	222/ 0
1196.	1075.5/	0.0	234.53/	0.00	222/ 0
1367.	1061.6/	0.0	231.77/	0.00	222/ 0
1537.	1042.9/	0.0	228.02/	0.00	222/ 0
1708.	1019.3/	0.0	223.30/	0.00	221/ 0
1879.	991.1/	0.0	217.61/	0.00	221/ 0
2050.	966.1/	0.0	212.54/	0.00	219/ 0

B E A M N O. 41 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	683 mm	39.2 kN	219	10 mm	3	341 mm
1366	END 2	42.2 kN	222	10 mm	3	341 mm
Nominal links should be provided through out the length.						

160J	2050.X 300.X 600	158J
3*10c/c341	3*10c/c341	
4No20 H 50.	0.TO 2050	

4T20	4T20	4T20	4T20	4T20	4T20	4T20
oooo	oooo	oooo	oooo	oooo	oooo	oooo

B E A M N O. 42 D E S I G N R E S U L T S - F L E X U R E

LEN - 2225. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
mm	mm		mm	mm	STA END

1	50.	4- 20 MM	0.	2225.	YES YES
---	-----	----------	----	-------	---------

REQUIRED REINF. STEEL SUMMARY :

SECTION	REINF STEEL(+VE/-VE)	MOMENTS (+VE/-VE)	LOAD (+VE/-VE)
(MM)	(SQ. MM)	(KN-METER)	
0.	1105.8/	0.0	240.55/ 0.00 222/ 0
185.	1045.2/	0.0	228.48/ 0.00 222/ 0
371.	978.8/	0.0	215.12/ 0.00 222/ 0
556.	906.1/	0.0	200.32/ 0.00 222/ 0

742.	827.0/	0.0	183.99/	0.00	222/ 0
927.	741.4/	0.0	166.06/	0.00	222/ 0
1113.	649.2/	0.0	146.23/	0.00	222/ 0
1298.	553.8/	0.0	124.73/	0.00	222/ 0
1483.	451.6/	0.0	101.72/	0.00	222/ 0
1669.	344.0/	0.0	77.48/	0.00	222/ 0
1854.	234.0/	0.0	52.29/	0.00	222/ 0
2040.	234.0/	0.0	26.37/	0.00	222/ 0
2225.	0.0/	0.0	0.00/	0.00	0/ 0

B E A M N O. 42 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	1112 mm	61.9 kN	222	10 mm	4	370 mm
1112	END 2	143.0 kN	222	10 mm	4	370 mm

158J	2225.X 300.X 600	156J
4*10c/c370	4*10c/c370	
4No20 H 50.	0.TO 2225	

4T20	4T20	4T20	4T20	4T20	4T20	4T20
oooo	oooo	oooo	oooo	oooo	oooo	oooo

B E A M N O. 101 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
mm	mm		mm	mm	STA END

1	50.	4- 20 MM	0.	6500.	YES YES
2	537.	2- 32 MM	0.	3551.	YES NO

3 541. 3- 25 MM 3491. 6500. NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)		LOAD (+VE/-VE)	
0.	1100.1/ 1525.4	239.42/	320.51	228/	221
542.	1001.1/ 1162.5	219.64/	251.72	228/	221
1083.	993.1/ 836.7	197.64/	186.00	228/	221
1625.	798.3/ 577.8	177.99/	130.13	220/	229
2167.	685.1/ 352.3	154.12/	79.36	220/	229
2708.	545.0/ 234.0	122.75/	33.86	220/	229
3250.	370.8/ 0.0	83.51/	0.00	220/	0
3792.	234.0/ 0.0	52.44/	0.00	221/	0
4333.	341.3/ 234.0	76.86/	27.29	221/	228
4875.	416.4/ 371.7	93.78/	83.72	221/	228
5417.	495.9/ 668.0	111.70/	150.45	229/	220
5958.	568.5/ 1018.4	128.05/	223.12	229/	220
6500.	631.3/ 1408.5	142.19/	298.86	229/	220

B E A M N O. 101 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM -	TO	SHEAR	TORSN	LOAD	LINK	NO.	SPACING	mm	C/C
mm	kN	kNm	S	T	SIZE	S	T	S	T
END 1	4875	128.9	46	221223	10mm	8	39	48	571
4875	END 2	141.7	46	228223	10mm	2	13	16	571
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 877 mm2 EVENLY DISTRIBUTED									
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *									
118J	6500.X	300.X	600	119J					
2No32 H	537.	10.	TO	3551	11	11	13No25	H	541.3491.
48*10c/c	101	11	11	11	11	11	11	16*10c/c	101
4No20 H	150.	10.	TO	6500	11	11	11	11	11

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING	C/C
END 1	3999	mm	107.2	kN	221	10	mm
1999	END 2	108.4	kN	220	10	mm	

B E A M N O. 102 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
1	46.	5- 12 MM	0.	6000.	YES	YES
2	543.	4- 20 MM	0.	3343.	YES	NO
3	543.	4- 20 MM	2657.	6000.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

B E A M N O. 102 D E S I G N R E S U L T S - S H E A R

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)		LOAD (+VE/-VE)	
0.	524.9/ 1005.2	118.23/	220.45	228/	221
500.	489.2/ 749.1	110.18/	167.69	228/	221
1000.	445.1/ 521.7	100.26/	117.50	228/	221
1500.	407.8/ 330.9	91.85/	74.52	220/	229
2000.	357.9/ 234.0	80.61/	37.15	220/	229
2500.	284.9/ 234.0	64.16/	3.46	220/	229
3000.	234.0/ 0.0	40.95/	0.00	219/	0
3500.	269.0/ 234.0	60.60/	0.93	221/	228
4000.	339.5/ 234.0	76.47/	35.08	221/	228
4500.	386.8/ 323.7	87.12/	72.91	221/	228
5000.	423.8/ 518.8	95.44/	116.84	229/	220
5500.	465.7/ 748.8	104.90/	167.61	229/	220
6000.	499.5/ 1007.7	112.50/	220.95	229/	220

119J	6000.X 300.X 600	120J
4No20 H 543. 0.TO 3343 4No20 H 543.2657.TO 6000		
11*10c/c399 11*10c/c399		
5No12 H 46. 0.TO 6000		

oooo		oooo		oooo		oooo	
4T20		4T20		4T20		4T20	
5T12		5T12		5T12		5T12	
oooo		oooo		oooo		oooo	

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B E A M N O. 103 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	50.	4- 20 MM	0.	6500.	YES YES
2	541.	3- 25 MM	0.	3009.	YES NO
3	541.	3- 25 MM	2949.	6500.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)

0.	551.4/ 1334.5	124.20/ 284.91	228/ 221
542.	505.9/ 963.6	113.94/ 212.05	228/ 221
1083.	450.5/ 631.5	101.47/ 142.24	228/ 221
1625.	385.4/ 345.2	86.81/ 77.75	220/ 229
2167.	329.0/ 234.0	74.10/ 24.52	220/ 229
2708.	239.2/ 0.0	53.89/ 0.00	220/ 0
3250.	356.3/ 0.0	80.25/ 0.00	221/ 0
3792.	517.7/ 234.0	116.61/ 24.68	221/ 228
4333.	644.3/ 294.3	145.12/ 66.29	221/ 228
4875.	741.7/ 502.5	166.12/ 113.19	221/ 228
5417.	815.9/ 731.4	181.67/ 163.95	221/ 228
5958.	904.7/ 1027.8	200.02/ 225.01	229/ 220
6500.	986.1/ 1359.2	216.60/ 289.58	229/ 220

B E A M N O. 103 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR TORSN kNm	LOAD S T	LINK SIZE	NO. S T	SPACING mm S T	C/C

END 1	5416	136.4	48	221224	10mm	9 46 55
5416 END 2	121.1	48	221224	10mm	1 9 12	571 117 90

EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 850 mm2 EVENLY DISTRIBUTED
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *

120J 6500.X 300.X 600 121J

3No25 H 541. 10.TO 3009 3No25 H 541.2949.TO 6500		55*10c/c 97 12*10c/c 90		4No20 H 50. 10.TO 6500	
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ooo	ooo	ooo	ooo	ooo	ooo	ooo
3T25	3T25	3T25	3T25	3T25	3T25	3T25
4T20	4T20	4T20	4T20	4T20	4T20	4T20
oooo	oooo	oooo	oooo	oooo	oooo	oooo

B E A M N O. 104 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	50.	3- 20 MM	0.	6500.	YES YES
2	528.	2- 50 MM	0.	4692.	YES NO
COMP. 4- 12 MM (REQD. STEEL= 400. SQ. MM)					

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)

0.	399.9/ 3666.5	0.00/ 635.79	0/ 221
542.	234.0/ 2724.7	34.01/ 514.72	228/ 221

1083.	335.1/	1967.6	75.49/	398.04	228/	221
1625.	496.7/	1353.8	111.88/	288.56	228/	221
2167.	629.5/	847.9	141.80/	188.31	228/	221
2708.	753.1/	474.7	168.53/	106.92	220/	229
3250.	881.7/	234.0	195.30/	50.78	220/	229
3792.	939.4/	234.0	207.11/	5.87	219/	230
4333.	932.7/	0.0	205.76/	0.00	219/	0
4875.	863.0/	0.0	191.43/	0.00	219/	0
5417.	750.0/	0.0	167.88/	0.00	219/	0
5958.	648.5/	0.0	146.07/	0.00	223/	0
6500.	625.1/	0.0	140.80/	0.00	221/	0

B E A M N O. 104 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM -	TO	SHEAR	TORSN	LOAD	LINK	NO.	SPACING	mm	C/C
mm	mm	kN	kNm	S	T	S	T	S	T
END 1	2709	225.7	20	221225	10mm	8	14	23	338
2709	5418	145.2	20	221225	10mm	5	14	18	541
5416	END 2	71.1	20	221225	10mm	1	5	8	571
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 578 mm2 EVENLY DISTRIBUTED									
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *									

122J 6500.X 300.X 600 123J

2No50 H 528.	10.TO 4692								
23*10c/c125									8*10c/c135
3No20 H 150.	10.TO 6500								

OO	OO	OO	OO	OO	OO	OO	OO	OO	OO
2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50	2T50
3T20	3T20	3T20	3T20	3T20	3T20	3T20	3T20	3T20	3T20
ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo

B E A M N O. 105 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT	BAR	INFO	FROM	TO	ANCHOR
	mm			mm	mm	STA
END						

1	50.	4- 20 MM	0.	6000.	YES	YES
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REQUIRED REINF. STEEL SUMMARY :

SECTION	REINF STEEL (+VE/-VE)	MOMENTS (+VE/-VE)	LOAD (+VE/-VE)
(MM)	(SQ. MM)	(KN-METER)	
0.	351.7/	0.0	79.21/
500.	554.6/	0.0	124.90/
1000.	745.8/	0.0	166.98/
1500.	921.7/	0.0	203.51/
2000.	1063.5/	0.0	232.14/
2500.	1160.0/	0.0	251.24/
3000.	1194.2/	0.0	257.91/
3500.	1171.5/	0.0	253.49/
4000.	1084.9/	0.0	236.40/
4500.	952.5/	0.0	209.79/
5000.	785.2/	0.0	175.27/
5500.	600.3/	0.0	135.20/
6000.	406.3/	0.0	91.52/

B E A M N O. 105 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING	C/C
END 1	2499	mm	98.4 kN	221	10 mm	8	357 mm
3499	END 2		95.8 kN	220	10 mm	8	357 mm

123J 6000.X 300.X 600 124J

8*10c/c357									8*10c/c357
4No20 H 50.	0.TO 6000								

4T20	4T20	4T20	4T20	4T20	4T20	4T20	4T20	4T20	4T20
oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo	oooo

B E A M N O . 106 D E S I G N R E S U L T S - F L E X U R E						
L E N -		6500. mm	F Y - 460.	F C - 40.	S I Z E - 300. X 600. mm	
L E V E L	H E I G H T mm	B A R I N F O	F R O M mm	T O mm	A N C H O R S T A E N D	
1	50.	4- 20 MM	0.	6500.	YES	YES
2	528.	2- 50 MM	1808.	6500.	NO	YES
C O M P .		3- 12 MM	(REQD. STEEL=	332. SQ. MM)		

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	663.7/	149.48/	223/ 0
542.	692.6/	155.73/	224/ 0
1083.	782.5/	174.70/	221/ 0
1625.	890.7/	197.15/	221/ 0
2167.	955.4/	210.37/	221/ 0
2708.	956.6/	210.61/	221/ 228
3250.	886.0/	196.18/	222/ 227
3792.	752.5/	168.40/	222/ 227
4333.	614.4/	138.39/	230/ 219
4875.	477.9/	107.64/	230/ 219
5417.	312.6/	70.42/	230/ 219
5958.	234.0/	28.11/	230/ 219
6500.	332.1/	0.00/	0/ 219

BEAM NO. 106 DESIGN RESULTS - SHEAR

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR	TORSN	LOAD	LINK	NO.	SPACING
mm	kN	kNm	S	T	SIZE	S T S+T S T S+T
END 1	1093	69.2	21	222226	10mm	1 5 8 571 199 135
1083	4334	171.8	21	1219226	10mm	6 17 22 541 199 146
4334 END 2	224.4	21	1219226	10mm	6 11 8 361 199 128	
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 281 mm2 EVENLY DISTRIBUTED						
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *						

124J	6500.X 300.X 600	125J
	2Nc50 H 528.1808.TO 6500	
8*10c/c135		8*10c/c128
4No20 H 50.	0.TO 6500	

[illegible]

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	1074.5/	234.34/	227/ 222
542.	1034.5/	226.34/	227/ 222
1083.	984.8/	216.34/	219/ 230
1625.	965.3/	212.38/	219/ 230
2167.	899.8/	199.01/	219/ 230
2708.	775.0/	173.12/	219/ 0
3250.	587.8/	132.40/	219/ 0
3792.	463.9/	104.48/	222/ 0
4333.	532.0/	119.83/	222/ 227
4875.	544.6/	122.67/	222/ 227
5417.	548.6/	123.56/	230/ 219
5958.	560.4/	126.22/	230/ 219
6500.	558.6/	125.82/	230/ 219

B E A M N O. 107 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR TORSN	LOAD	LINK	NO.	SPACING mm C/C
mm	kN	mm	S T SIZE	S T S+T	S T S+T
END 1 1625	175.2	21	222225	10mm	2 8 10 571 199 148
1625 END 2	194.6	21	222225	10mm	8 24 32 571 199 148
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 581 mm2 EVENLY DISTRIBUTED					
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *					

126J	6500.X 300.X 600	127J
2No40 H 533.	10.TO 3009	12No40 H 533.3491.TO 16500
10*10c/c148		11*32*10c/c148
4No20 H 150.	10.TO 6500	

FROM - TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1 3999	mm	kN	222	10 mm	11 399 mm
1999	END 2	154.2 kN	219	10 mm	11 399 mm

B E A M N O. 108 D E S I G N R E S U L T S - F L E X U R E

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
1	46.	5- 12 MM	0.	6000.	YES	YES
2	537.	2- 32 MM	0.	2843.	YES	NO
3	541.	3- 25 MM	3157.	6000.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	467.0/ 1485.9	105.18/ 313.25	227/ 222
500.	468.7/ 1080.9	105.57/ 235.61	227/ 222

1000.	459.1/	720.1	103.40/	161.56	227/ 222
1500.	439.3/	421.7	98.95/	94.97	219/ 230
2000.	427.0/	234.0	96.17/	42.88	219/ 230
2500.	369.7/	0.0	83.28/	0.00	219/ 0
3000.	259.4/	0.0	58.43/	0.00	219/ 0
3500.	370.2/	0.0	83.39/	0.00	222/ 0
4000.	434.2/	234.0	97.79/	39.23	222/ 227
4500.	453.2/	400.5	102.09/	90.22	222/ 227
5000.	472.7/	684.9	106.46/	154.09	230/ 219
5500.	487.3/	1035.9	109.75/	226.62	230/ 219
6000.	490.5/	1429.3	110.47/	302.75	230/ 219

B E A M N O. 108 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM - TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1 3999	mm	kN	222	10 mm	11 399 mm
1999	END 2	154.2 kN	219	10 mm	11 399 mm

127J	6000.X 300.X 600	128J
2No32 H 537.	0.TO 2843	3No25 H 541.3157.TO 6000
11*10c/c399		11*10c/c399
5No12 H 46.	0.TO 6000	

FROM - TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1 3999	mm	kN	222	10 mm	11 399 mm
1999	END 2	154.2 kN	219	10 mm	11 399 mm

B E A M N O. 109 D E S I G N R E S U L T S - F L E X U R E

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
0.	467.0/ 1485.9	105.18/ 313.25	227/ 222			
500.	468.7/ 1080.9	105.57/ 235.61	227/ 222			

1	50.	3- 20 MM	0.	6500.	YES	YES
2	533.	2- 40 MM	0.	3009.	YES	NO
3	537.	2- 32 MM	3491.	6500.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	453.4/ 1718.0	102.11/ 355.13	228/ 221
542.	477.1/ 1186.9	107.46/ 256.48	228/ 221
1083.	487.2/ 109.74/ 722.8	162.13/ 722.8	228/ 221
1625.	499.0/ 357.9	112.38/ 80.62	220/ 229
2167.	508.0/ 234.0	114.42/ 14.51	220/ 229
2708.	461.4/ 0.0	103.93/ 0.00	220/ 0
3250.	524.3/ 0.0	118.09/ 0.00	222/ 0
3792.	681.2/ 0.0	153.29/ 0.00	222/ 0
4333.	777.6/ 234.0	173.67/ 38.30	222/ 227
4875.	815.2/ 428.7	181.53/ 96.56	222/ 227
5417.	811.8/ 723.4	180.81/ 162.27	230/ 219
5958.	838.8/ 1137.4	186.43/ 246.79	230/ 219
6500.	851.1/ 1608.5	188.98/ 335.60	230/ 219

B E A M N O. 109 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR	TORSN	LOAD	LINK	NO.	SPACING	mm C/C
mm	kN	kNm	S T	SIZE	S T	S T	S T
END 1	4333	184.4	22	222226	10mm	7 21 29	571 199 145
2166 END 2	166.2	22	222226	10mm	7 21 29	571 199 145	
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 580 mm2 EVENLY DISTRIBUTED							
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *							

128J	6500.X 300.X 600	129J
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2No40 H 533.	10.TO 3009	12No32 H 537.3491.	TO 6500
29*10c/c145			129*10c/c145
3No20 H 50.	10.TO 6500		

OO	OO	OO	OO	OO	OO
2T40	2T40	2T40	2T32	2T32	2T32
3T20	3T20	3T20	3T20	3T20	3T20
ooo	ooo	ooo	ooo	ooo	ooo

130J	6500.X 300.X 600	131J
3No25 H 541.	10.TO 3551	1 1 1 4No20 H 543.3491.TO 6500
10*10c/c148	1 1 1 1	1 1 1 1 32*10c/c148
1 1 1 1	1 1 1 1	1 1 1 1 1
4No16 H 48.	10.TO 6500	1 1 1 1 1

ooo	ooo	ooo	ooo	ooo	ooo
3T25	3T25	3T25	3T25	4T20	4T20
4T16	4T16	4T16	4T16	4T16	4T16
ooo	ooo	ooo	ooo	ooo	ooo

BEAM NO. 111 DESIGN RESULTS - FLEXURE

LEN - 6000. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	46.	4- 12 MM	0.	6000.	YES YES
2	543.	4- 20 MM	0.	3343.	YES NO
3	543.	4- 20 MM	2657.	6000.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	441.5/ 1011.4	99.45/ 221.70	227/ 222
500.	414.7/ 761.3	93.41/ 170.24	227/ 222
1000.	379.7/ 538.7	85.53/ 121.34	227/ 222
1500.	333.6/ 337.1	75.15/ 75.93	227/ 222
2000.	289.4/ 234.0	65.18/ 38.71	219/ 230
2500.	234.0/ 234.0	50.66/ 6.85	219/ 230
3000.	234.0/ 0.0	29.76/ 0.00	208/ 0
3500.	234.0/ 234.0	50.43/ 5.44	222/ 227
4000.	292.3/ 234.0	65.83/ 36.99	222/ 227
4500.	334.8/ 322.5	75.42/ 72.64	222/ 227
5000.	380.3/ 518.3	85.65/ 116.73	230/ 219
5500.	416.7/ 735.2	93.85/ 164.75	230/ 219
6000.	444.8/ 979.8	100.19/ 215.33	230/ 219

BEAM NO. 111 DESIGN RESULTS - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	3499 mm	104.6 kN	222	10 mm	10	388 mm
2499	END 2	102.8 kN	219	10 mm	10	388 mm

131J	6000.X 300.X 600	132J
4No20 H 543.	0.TO 3343	4No20 H 543.2657.TO 6000
10*10c/c388	1	1
4No12 H 46.	0.TO 6000	1

ooo	ooo	ooo	ooo	ooo	ooo
4T20	4T20	4T20	4T20	4T20	4T20
4T12	4T12	4T12	4T12	4T12	4T12
ooo	ooo	ooo	ooo	ooo	ooo

BEAM NO. 112 DESIGN RESULTS - FLEXURE

LEN - 6500. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	48.	4- 16 MM	0.	6500.	YES YES
2	543.	4- 20 MM	0.	3009.	YES NO
3	543.	4- 20 MM	2949.	6500.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	442.1/ 1077.7	99.57/ 234.98	227/ 222
542.	425.7/ 774.4	95.88/ 173.01	227/ 222

=====

B E A M N O. 114 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. mm FC - 40. mm SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	45.	5- 10 MM	0.	6650.	YES YES
2	543.	3- 20 MM	0.	3059.	YES NO
3	543.	3- 20 MM	3591.	6650.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	302.8/ 909.1	68.19/ 200.92	233/ 224
554.	326.7/ 649.4	73.58/ 146.26	233/ 224
1108.	340.3/ 421.0	76.64/ 94.82	233/ 224
1662.	357.5/ 234.0	80.52/ 52.23	225/ 232
2217.	366.7/ 234.0	82.59/ 17.62	225/ 232
2771.	340.2/ 0.0	76.63/ 0.00	225/ 0
3325.	274.8/ 0.0	61.90/ 0.00	225/ 0
3879.	288.0/ 0.0	64.86/ 0.00	223/ 0
4433.	313.3/ 234.0	70.58/ 6.50	223/ 234
4988.	303.0/ 234.0	68.25/ 41.32	223/ 234
5542.	285.2/ 373.9	64.24/ 84.22	231/ 226
6096.	270.6/ 603.4	60.95/ 135.90	231/ 226
6650.	245.7/ 859.8	55.35/ 190.79	231/ 226

B E A M N O. 114 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM -	TO	SHEAR TORSN mm	LOAD kNm	S	T	SIZE	S	T	S+T	NO.	SPACING mm	C/C

END 1	3325	100.6	27	224225	10mm	5	16	22	571	199	145	
3325	END 2	101.0	27	224219	10mm	5	16	22	571	199	145	

EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 547 mm2 EVENLY DISTRIBUTED												
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *												

_____ 126J _____ 6650.X 300.X 600 _____ 122J _____

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3No20 H 543. | 0.TO 3059 | | | | 3No20 H 543.3591.TO 6650 | | | |

22*10c/c145 | | | | | | | | 122*10c/c145 | | | |

5No10 H | 45. | 10.TO 6650 | | | | | | | | | | | |

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ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo	ooo
3T20	3T20	3T20	3T20	3T20	3T20	3T20	3T20	3T20	3T20
5T10	5T10	5T10	5T10	5T10	5T10	5T10	5T10	5T10	5T10
ooooo	ooooo	ooooo	ooooo	ooooo	ooooo	ooooo	ooooo	ooooo	ooooo

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B E A M N O. 115 D E S I G N R E S U L T S - F L E X U R E

LEN - 5310. mm FY - 460. mm FC - 40. mm SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END

1	50.	4- 20 MM	0.	5310.	YES YES
2	543.	4- 20 MM	0.	2613.	YES NO
3	543.	4- 20 MM	1812.	5310.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	507.0/ 1120.5	114.20/ 243.46	233/ 224
442.	440.4/ 832.2	99.19/ 185.07	233/ 224
885.	367.8/ 570.6	82.83/ 128.51	233/ 224
1328.	286.7/ 331.3	64.58/ 74.62	233/ 224
1770.	234.0/ 234.0	48.68/ 28.52	225/ 232
2212.	234.0/ 0.0	28.99/ 0.00	225/ 0
2655.	287.0/ 234.0	64.64/ 6.49	223/ 234
3098.	449.0/ 234.0	101.14/ 36.75	223/ 234
3540.	589.7/ 313.4	132.81/ 70.59	224/ 233
3982.	714.9/ 476.8	160.45/ 107.39	224/ 233
4425.	831.5/ 650.2	184.91/ 146.46	224/ 233
4868.	937.4/ 843.6	206.71/ 187.43	224/ 233
5310.	1043.6/ 1058.9	228.18/ 231.23	232/ 225

100

B E A M N O . 1 1 7 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	65.	2- 50 MM	1317.	6650.	NO YES
	COMP.	4- 10 MM	(REQD. STEEL=	313. SQ. MM)	
Please Note:					
Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*					
2	537.	3- 32 MM	0.	3581.	YES NO
3	473.	3- 32 MM	0.	3581.	YES NO
	COMP.	2- 40 MM	(REQD. STEEL=	1627. SQ. MM)	

B E A M N O. 120 D E S I G N R E S U L T S - F L E X U R E									
LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm									
LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END				

1	65.	2- 50 MM	1317.	6650.	NO YES				
	COMP.	5- 10 MM	(REQD. STEEL=	340. SQ. MM)					
*Please Note:									
Max/Min Reinforcement Sizes Do Not Permit Acceptable Bar Spacing.*									
2	537.	3- 32 MM	0.	3581.	YES NO				
3	473.	3- 32 MM	0.	3581.	YES NO				
	COMP.	2- 40 MM	(REQD. STEEL=	1676. SQ. MM)					

REQUIRED REINF. STEEL SUMMARY :

SECTION REINF STEEL(+VE/-VE) MOMENTS(+VE/-VE) LOAD(+VE/-VE)									
(MM) (SQ. MM) (KN-METER)									
0.	1267.4/	4534.0	0.00/	819.69	0/ 224				
554.	418.5/	3685.1	0.00/	639.74	0/ 224				
1108.	0.0/	2382.9	0.00/	464.55	0/ 224				
1662.	0.0/	1397.0	0.00/	296.71	0/ 224				
2217.	258.5/	618.2	58.22/	139.23	233/ 224				
2771.	659.1/	234.0	148.45/	18.07	225/ 232				
3325.	1121.0/	0.0	243.55/	0.00	226/ 0				
3879.	1536.5/	0.0	322.55/	0.00	226/ 0				
4433.	1891.4/	0.0	385.17/	0.00	226/ 0				
4988.	2297.1/	0.0	451.32/	0.00	222/ 0				
5542.	2704.7/	0.0	511.90/	0.00	221/ 0				
6096.	3355.7/	234.0	569.91/	0.00	221/ 0				
6650.	3606.9/	340.3	623.16/	0.00	221/ 0				

B E A M N O. 120 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM -	TO	SHEAR	TORSN	LOAD	LINK	NO.		SPACING mm		C/C	
mm		kN	kNm	S	T	SIZE	S	T	S	T	S+T

END 1	3325	327.0	38	224224	10mm		13	22	36	249	147 92
3325	END 2	103.2	38	224219	10mm		5	22	28	571	147 115

EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 467 mm2 EVENLY DISTRIBUTED											
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *											

128J										6650.X 300.X 600										124J									
=====																													
3No32 H 537.										10.TO 3581																			
36*10c/c 92																													
2No50 H										165.1317										TO 6650									
=====																													
28*10c/c115																													
=====																													
																				</									

B E A M N O. 121 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR TORSN	LOAD	LINK	NO.	SPACING mm	C/C
mm	kN	S T	SIZE	S T	S T	S T
END 1	2655	161.8	45	226226	10mm	4 21 26 571 126 102
2654	END 2	311.2	45	226219	10mm	10 21 31 256 126 83
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 633 mm2 EVENLY DISTRIBUTED						
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *						

124J	5310.X 300.X 600	120J
26*10c/c102		
2No50 H 65. 0.TO 5310		
2No40 H 533.1844.TO 5310		
31*10c/c 83		

2T50	2T50	2T50	2T50	2T50	2T50	2T50
OO	OO	OO	OO	OO	OO	OO

B E A M N O. 122 D E S I G N R E S U L T S - F L E X U R E

LEN - 5040. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	50.	3- 20 MM	0.	5040.	YES	YES
2	541.	3- 25 MM	0.	5040.	YES	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	918.8/ 1259.8	202.91/ 270.62	234/ 223
420.	802.5/ 1003.1	178.88/ 220.04	234/ 223
840.	683.0/ 765.3	153.68/ 171.08	234/ 223

1260.	563.0/	552.6	126.81/	124.45	234/ 223
1680.	440.7/	363.6	99.26/	81.89	226/ 231
2100.	310.6/	234.0	69.96/	43.64	226/ 231
2520.	234.0/	234.0	36.65/	8.66	225/ 232
2940.	234.0/	234.0	27.45/	5.21	223/ 234
3360.	243.6/	234.0	54.88/	45.65	223/ 234
3780.	353.4/	397.2	79.60/	89.46	231/ 226
4200.	463.0/	612.5	104.28/	137.97	231/ 226
4620.	565.2/	850.2	127.29/	188.80	231/ 226
5040.	662.1/	1109.4	149.12/	241.26	231/ 226

B E A M N O. 122 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR LINKS AS FOLLOWS

FROM - TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2519 mm	121.8 kN	223	10 mm	8
2519	END 2	126.3 kN	226	10 mm	8

133J	5040.X 300.X 600	129J
3No25 H 541. 0.TO 5040		
8*10c/c359		
3No20 H 50. 0.TO 5040		

OOO	OOO	OOO	OOO	OOO	OOO
3T25	3T25	3T25	3T25	3T25	3T25
3T20	3T20	3T20	3T20	3T20	3T20
OOO	OOO	OOO	OOO	OOO	OOO

B E A M N O. 123 D E S I G N R E S U L T S - F L E X U R E

LEN - 6650. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA	END
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1	45.	5- 10 MM	0.	6650.	YES	YES
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2	543.	3- 20 MM	0.	3059.	YES	NO
3	543.	3- 20 MM	3591.	6650.	NO	YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	295.4/ 894.7	66.53/ 197.97	234/ 223
554.	317.6/ 634.1	71.54/ 142.82	234/ 223
1108.	329.5/ 403.4	74.22/ 90.87	234/ 223
1662.	349.2/ 234.0	78.65/ 48.72	226/ 231
2217.	356.7/ 234.0	80.35/ 13.61	226/ 231
2771.	328.6/ 0.0	74.01/ 0.00	226/ 0
3325.	261.5/ 0.0	58.91/ 0.00	226/ 0
3879.	315.9/ 0.0	71.15/ 0.00	224/ 0
4433.	343.4/ 234.0	77.34/ 11.03	223/ 234
4988.	335.3/ 234.0	75.53/ 46.24	223/ 234
5542.	315.5/ 393.2	71.06/ 88.56	231/ 226
6096.	303.1/ 624.3	68.28/ 140.63	231/ 226
6650.	280.5/ 884.6	63.17/ 195.90	231/ 226

B E A M N O. 123 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

FROM - TO	SHEAR TORSN	LOAD	LINK	NO.	SPACING	mm C/C
mm	KN	S T	SIZE	S T	S T	S T
END 1 3325	101.5	31	223226	10mm	5 18 25	571 180 133
3325 END 2	101.7	31	223221	10mm	5 18 25	571 180 133
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 597 mm2 EVENLY DISTRIBUTED						
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *						

129J	6650.X 300.X 600	125J
3No20 H 543.	10.TO 3059	3No20 H 543.3591.TO 6650
25*10c/c133		125*10c/c133
5No10 H 145.	10.TO 6650	

ooo	ooo	ooo	ooo	ooo	ooo
3T20	3T20	3T20	3T20	3T20	3T20
5T10	5T10	5T10	5T10	5T10	5T10
ooooo	ooooo	ooooo	ooooo	ooooo	ooooo

B E A M N O. 124 D E S I G N R E S U L T S - F L E X U R E

LEN - 5310. mm FY - 460. FC - 40. SIZE - 300. X 600. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	50.	4- 20 MM	0.	5310.	YES YES
2	543.	4- 20 MM	0.	2613.	YES NO
3	543.	4- 20 MM	1812.	5310.	NO YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	549.1/ 1178.4	123.67/ 254.84	234/ 223
442.	480.1/ 884.0	108.13/ 195.76	234/ 223
885.	405.1/ 615.0	91.24/ 138.53	234/ 223
1328.	321.7/ 372.8	72.46/ 83.96	234/ 223
1770.	244.5/ 234.0	55.08/ 36.23	226/ 231
2212.	234.0/ 0.0	34.88/ 0.00	226/ 0
2655.	254.6/ 234.0	57.35/ 0.20	224/ 233
3098.	419.7/ 234.0	94.54/ 31.00	224/ 233
3540.	563.3/ 290.2	126.87/ 65.36	223/ 234
3982.	690.1/ 455.9	155.19/ 102.69	223/ 234
4425.	809.5/ 631.7	180.33/ 142.29	223/ 234
4868.	918.3/ 826.1	202.81/ 183.79	223/ 234
5310.	1031.3/ 1047.1	225.71/ 228.88	231/ 226

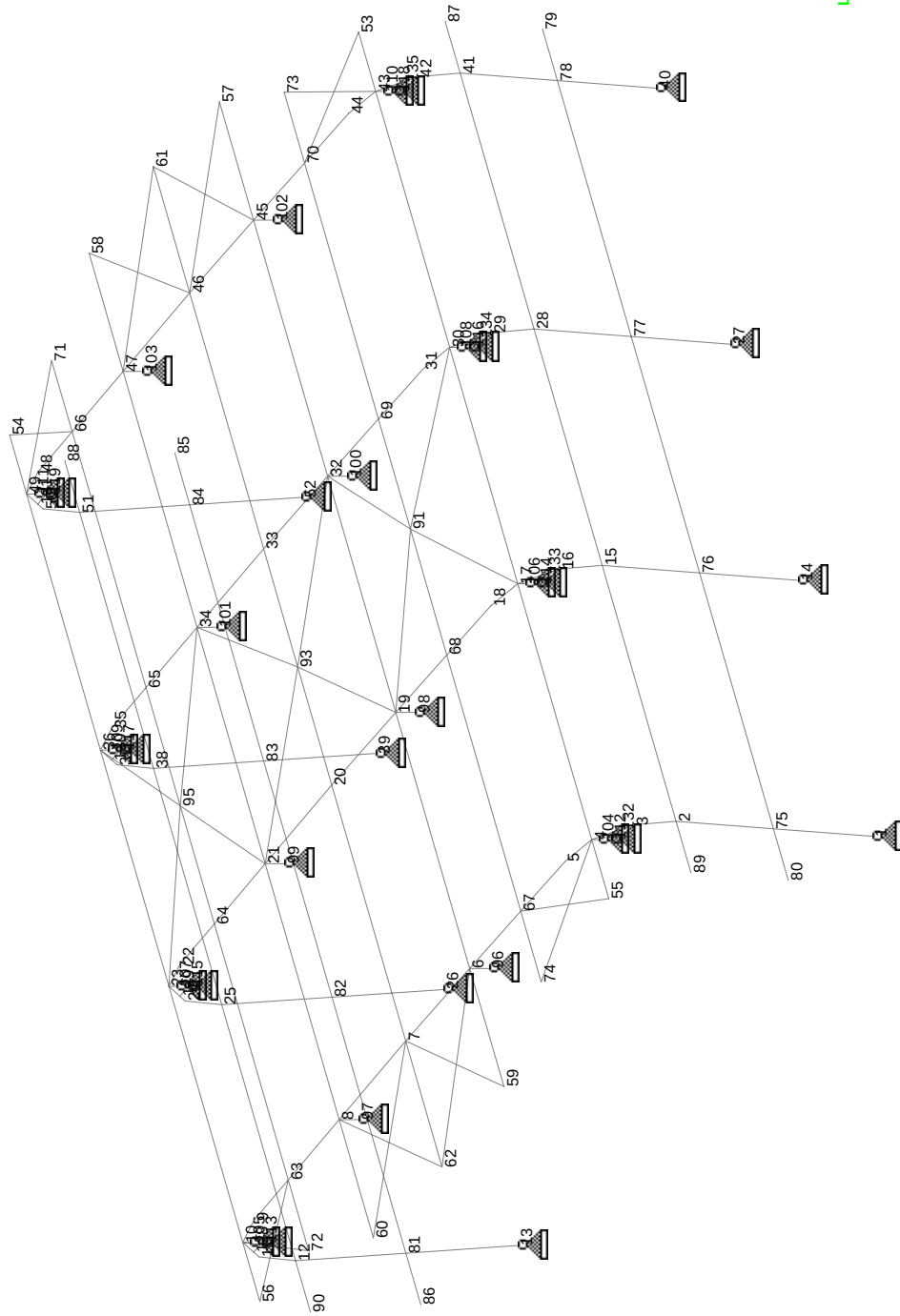
B E A M N O. 124 D E S I G N R E S U L T S - S H E A R

PROVIDE SHEAR AND TORSIONAL LINKS AS FOLLOWS

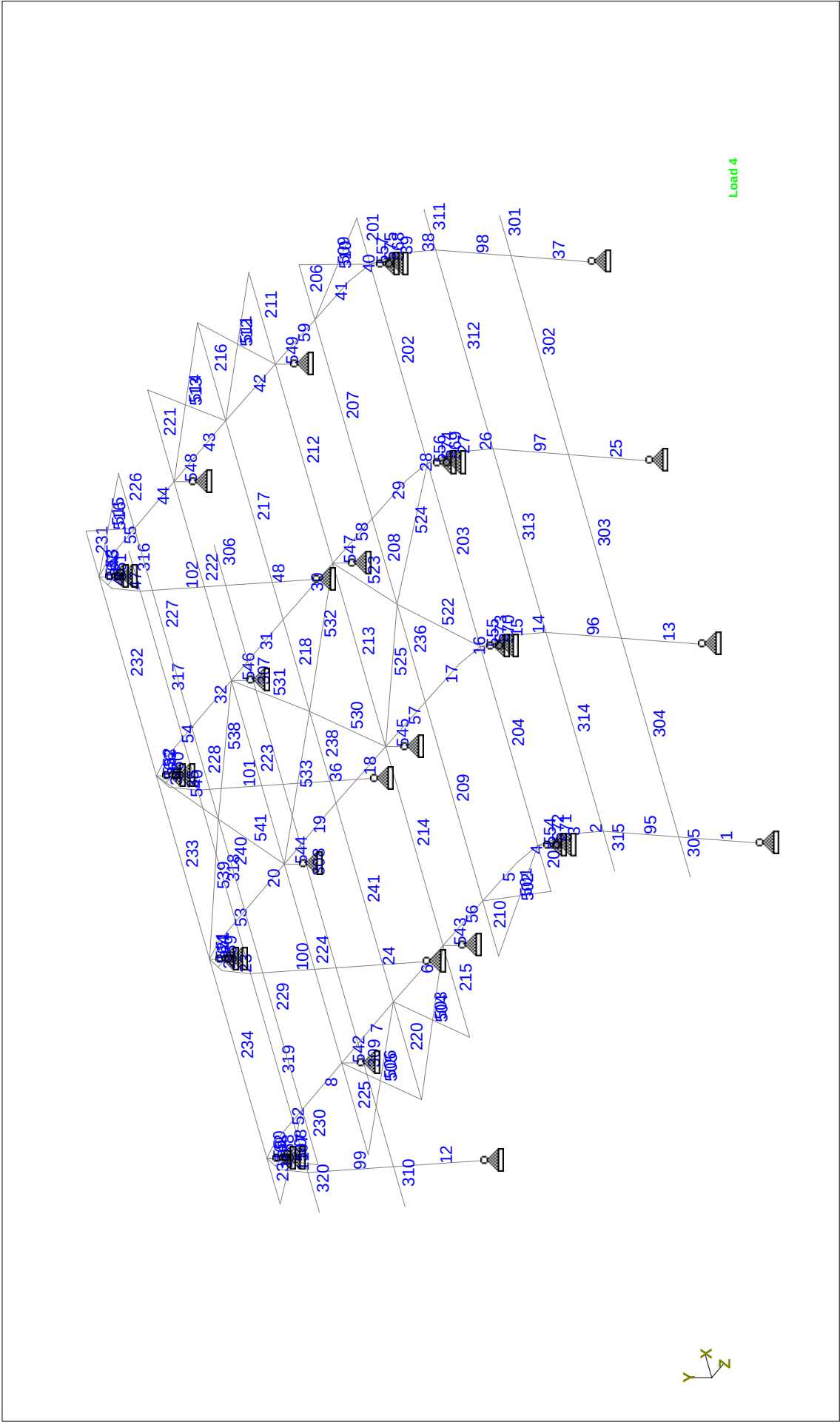
FROM - TO	SHEAR TORSN	LOAD	LINK	NO.	SPACING	mm C/C
mm	KN	kNm	S T	SIZE	S T	S T
END 1 2655	135.0	41	223231	10mm	4 19 23	571 138 111
2654 END 2	107.4	41	223221	10mm	4 19 23	571 138 111
EXTRA PERIPHERAL LONGITUDINAL TORSION STEEL: 762 mm2 EVENLY DISTRIBUTED						
* TORSIONAL RIGIDITY SHOULD CONFORM TO CL.2.4.3 - BS8110 *						



Load 4



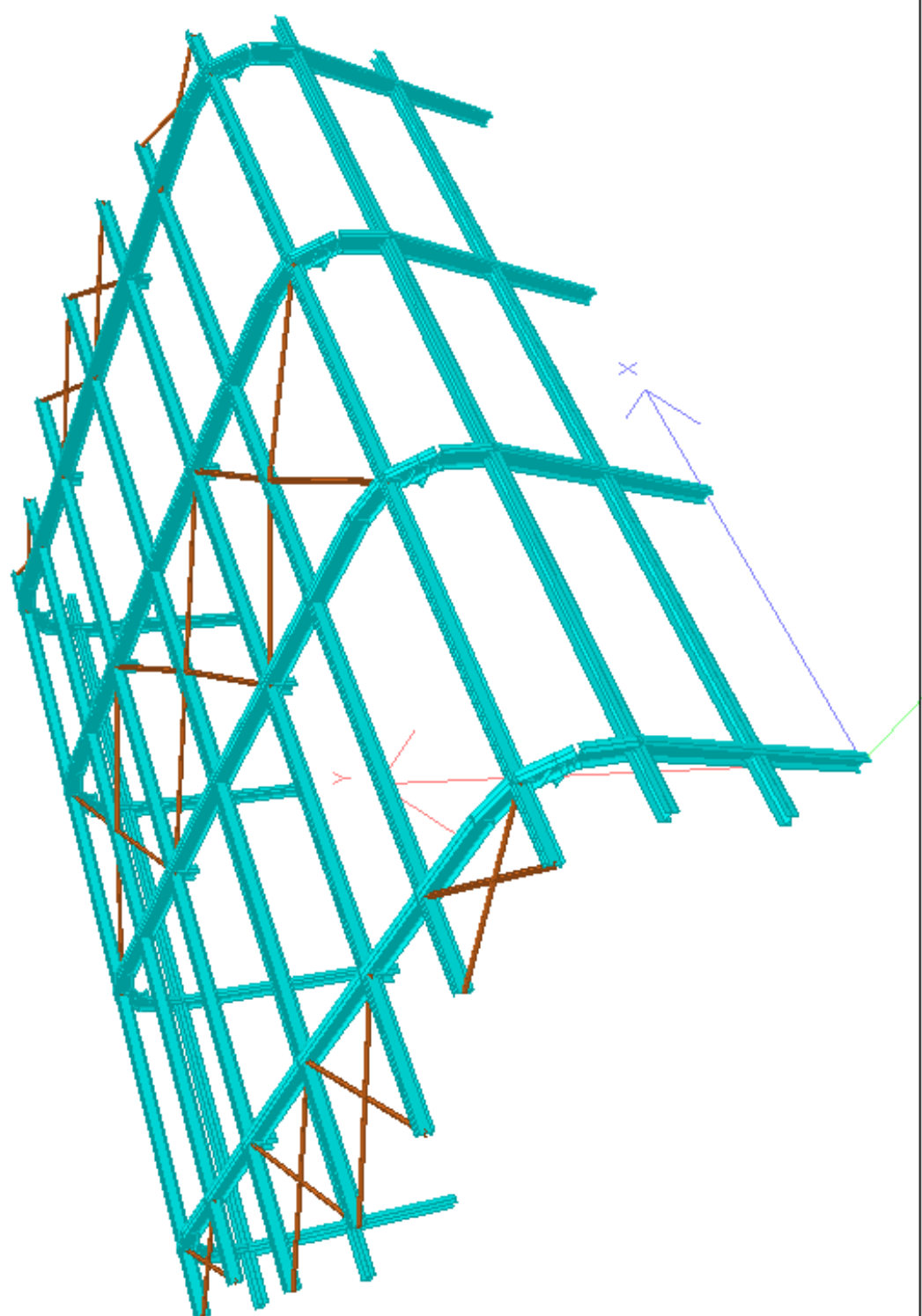
Node No.



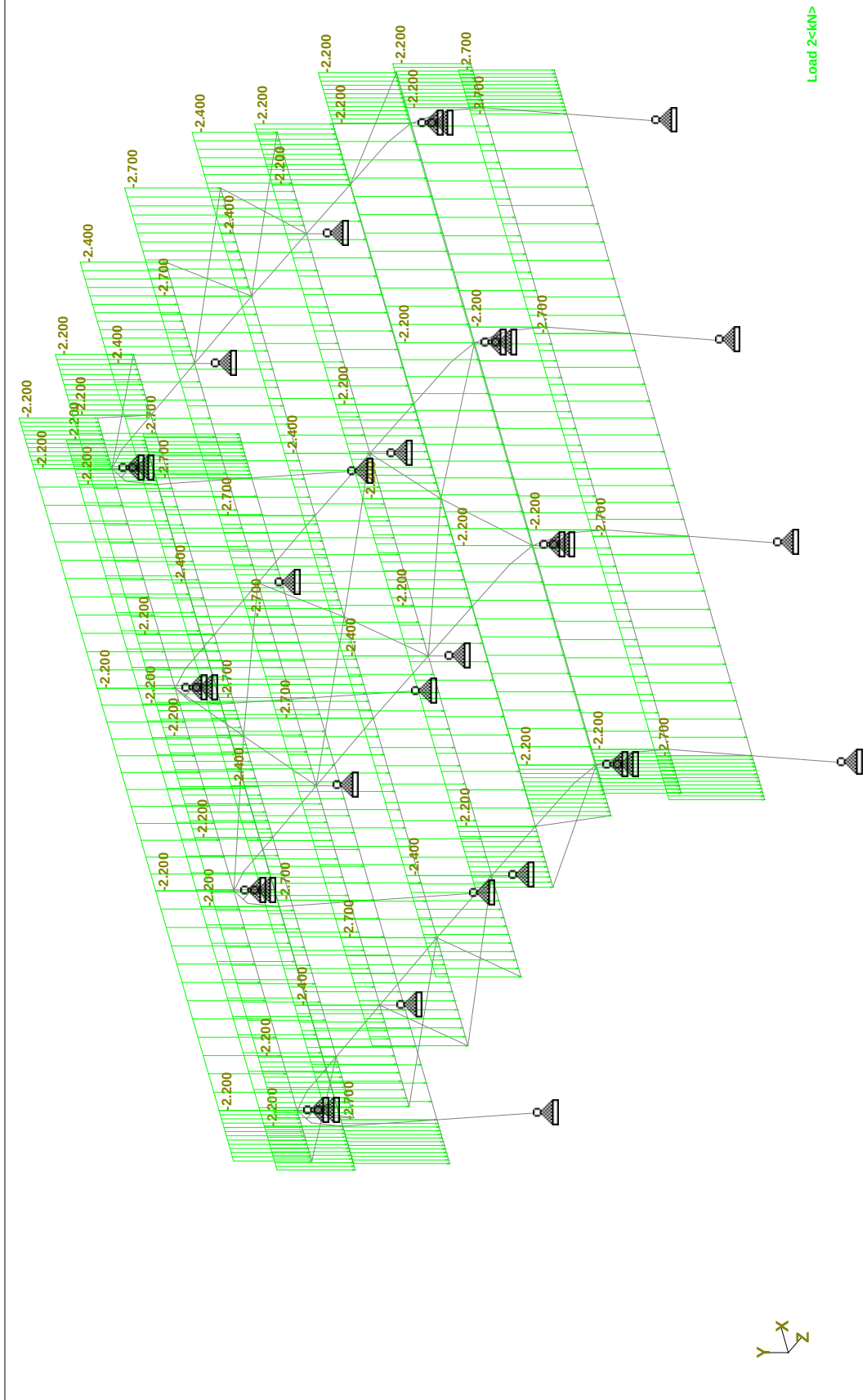
X
Y Z

Load 4

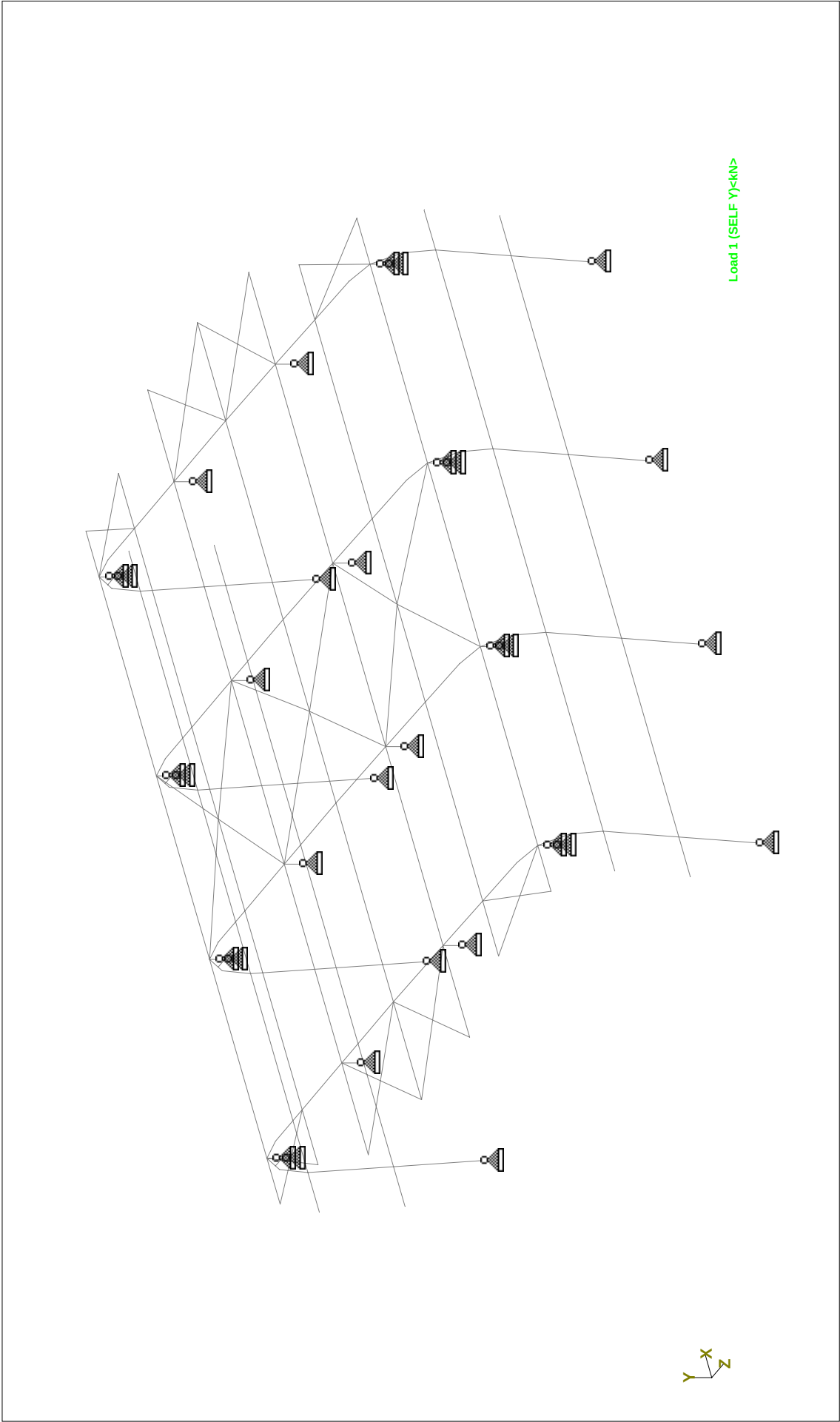
Beam No.



3D Rendered View



Load 2 Dead Load



Load 1 SelfWeight

